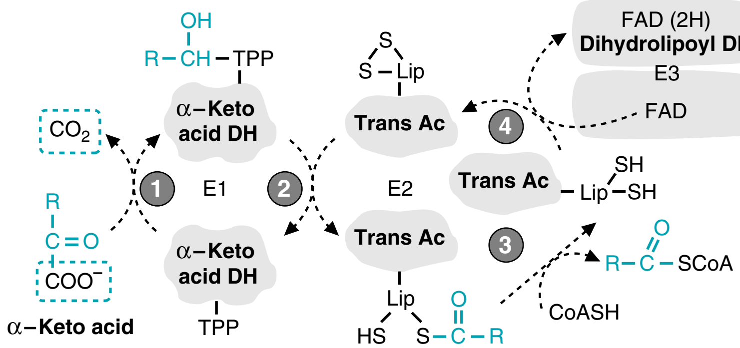


Fig. 20.9. Mechanism of α-keto acid dehydrogenase complexes (including α-ketoglutarate dehydrogenase, pyruvate dehydrogenase and the branched-chain α-keto acid dehydrogenase complex). **R** represents the portion of the α-ketoacid that begins with the β carbon. In α-ketoglutarate, R is CH_2-CH_2-COOH. In pyruvate, R is CH_3. The individual steps in the oxidative decarboxylation of α-keto acids are catalyzed by three different subunits: E_1, α-ketoacid decarboxylase (α-ketoglutarate decarboxylase); E_2, transacylase (trans-succinylase), and E_3, dihydrolipoyl dehydrogenase. Circle 1: Thiamine pyrophosphate (TPP) on E_1 decarboxylates the α-ketoacid and forms a covalent intermediate with the remaining portion. Circle 2: The acyl portion of the α-keto acid is transferred by TPP on E_1 to lipoate on E_2, which is a transacylase. Circle 3: E_2 transfers the acyl group from lipoate to CoASH. This process has reduced the lipoyl disulfide bond to sulfhydryl groups (dihydrolipoyl). Circle 4: E_3, dihydrolipoyl dehydrogenase (DH) transfers the electrons from reduced lipoate to its tightly bound FAD molecule, thereby oxidizing lipoate back to its original disulfide form. Circle 5: The electrons are then transferred from FAD(2H) to NAD^+ to form NADH.

FAD; it transfers electrons from reduced lipoate to NAD^+. The collection of 3 enzyme activities into one huge complex enables the product of one enzyme to be transferred to the next enzyme without loss of energy. Complex formation also increases the rate of catalysis because the substrates for E_2 and E_3 remain bound to the enzyme complex.

1. THIAMINE PYROPHOSPHATE IN THE α-KETOGLUTARATE DEHYDROGENASE COMPLEX

Thiamine pyrophosphate is synthesized from the vitamin thiamine by the addition of pyrophosphate (see Fig. 8.11). The pyrophosphate group binds magnesium, which binds to amino acid side chains on the enzyme. This binding is relatively weak for a coenzyme, so thiamine turns over rapidly in the body, and a deficiency can develop rapidly in individuals on a thiamine-free or low thiamine diet.

The general function of thiamine pyrophosphate is the cleavage of a carbon-carbon bond next to a keto group. In the α-ketoglutarate, pyruvate, and branched chain α-keto acid dehydrogenase complexes, the functional carbon on the thiazole ring forms a covalent bond with the α-keto carbon, thereby cleaving the bond between the α-keto carbon and the adjacent carboxylic acid group (see Fig. 8.11 for the mechanism of this reaction). Thiamine pyrophosphate is also a coenzyme for transketolase in the pentose phosphate pathway, where it similarly cleaves the carbon-carbon bond next to a keto group. In thiamine deficiency, α-ketoglutarate, pyruvate, and other α-keto acids accumulate in the blood.

2. LIPOATE

Lipoate is a coenzyme found only in α-keto acid dehydrogenase complexes. It is synthesized in the human from carbohydrate and amino acids, and does not require

Q: The $E^{0\prime}$ for FAD accepting electrons is −0.20 (see Table 19.4). The $E^{0\prime}$ for NAD^+ accepting electrons is −0.32. Thus, transfer of electrons from FAD(2H) to NAD^+ is energetically unfavorable. How do the α-keto acid dehydrogenase complexes allow this electron transfer to occur?

In **Al Martini's** heart failure, which is caused by a dietary deficiency of the vitamin thiamine, pyruvate dehydrogenase, α-ketoglutarate dehydrogenase, and the branched chain α-keto acid dehydrogenase complexes are less functional than normal. Because heart muscle, skeletal muscle, and nervous tissue have a high rate of ATP production from the NADH produced by the oxidation of pyruvate to acetyl CoA and of acetyl CoA to CO_2 in the TCA cycle, these tissues present with the most obvious signs of thiamine deficiency.

In Western societies, gross thiamine deficiency is most often associated with alcoholism. The mechanism for active absorption of thiamine is strongly and directly inhibited by alcohol. Subclinical deficiency of thiamine from malnutrition or anorexia may be common in the general population and is usually associated with multiple vitamin deficiencies.

A: The $E^{0\prime}$ values were calculated in a test tube under standard conditions. When FAD is bound to an enzyme, as it is in the α-keto acid dehydrogenase complexes, amino acid side chains can alter its $E^{0\prime}$ value. Thus, the transfer of electrons from the bound FAD(2H) to NAD^+ in dihydrolipoyl dehydrogenase is actually energetically favorable.

Arsenic poisoning is caused by the presence of a large number of different arsenious compounds that are effective metabolic inhibitors. Acute accidental or intentional arsenic poisoning requires high doses and involves arsenate (AsO_4^{2-}) and arsenite (AsO^{2-}). Arsenite, which is 10 times more toxic than arsenate, binds to neighboring sulfhydryl groups, such as those in dihydrolipoate and in nearby cysteine pairs (vicinal) found in α-keto acid dehydrogenase complexes and in succinic dehydrogenase. Arsenate weakly inhibits enzymatic reactions involving phosphate, including the enzyme glyceraldehyde 3-P dehydrogenase in glycolysis (see Chapter 22). Thus both aerobic and anaerobic ATP production can be inhibited. The low doses of arsenic compounds found in water supplies are a major public health concern, but are associated with increased risk of cancer rather than direct toxicity.

a vitamin precursor. Lipoate is attached to the transacylase enzyme through its carboxyl group, which is covalently bound to the terminal $-NH_2$ of a lysine in the protein (Fig. 20.10). At its functional end, lipoate contains a disulfide group that accepts electrons when it binds the acyl fragment of α-ketoglutarate. It can thus act like a long flexible $-CH_2-$ arm of the enzyme that reaches over to the decarboxylase to pick up the acyl fragment from thiamine and transfer it to the active site containing bound CoASH. It then swings over to dihydrolipoyl dehydrogenase to transfer electrons from the lipoyl sulfhydryl groups to FAD.

3. FAD AND DIHYDROLIPOYL DEHYDROGENASE

FAD on dihydrolipoyl dehydrogenase accepts electrons from the lipoyl sulfhydryl groups and transfers them to bound NAD^+. FAD thus accepts and transfers electrons without leaving its binding site on the enzyme. The direction of the reaction is favored by interactions of FAD with groups on the enzyme, which change its reduction potential and by the overall release of energy from cleavage and oxidation of α-ketoglutarate.

III. ENERGETICS OF THE TCA CYCLE

Like all metabolic pathways, the TCA cycle operates with an overall net negative $\Delta G^{0\prime}$ (Fig 20.11). The conversion of substrates to products is, therefore, energetically favorable. However, some of the reactions, such as the malate dehydrogenase reaction, have a positive value.

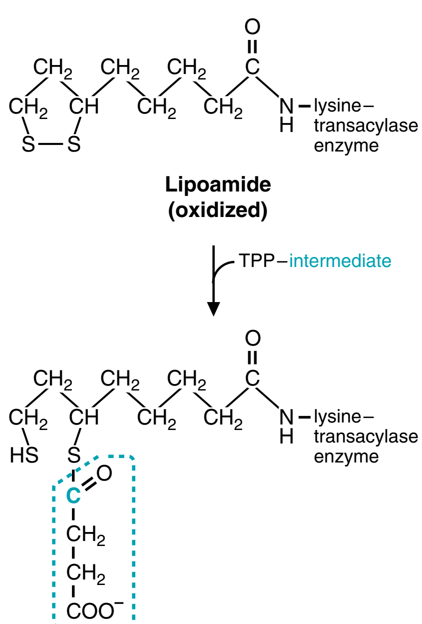


Fig. 20.10. Function of lipoate. Lipoate is attached to the ϵ-amino group on the lysine side chain of the tranacylase enzyme (E_2). The oxidized lipoate disulfide form is reduced as it accepts the acyl group from thiamine pyrophosphate (TPP) attached to E_1. The example shown is for the α-ketoglutarate dehydrogenase complex.

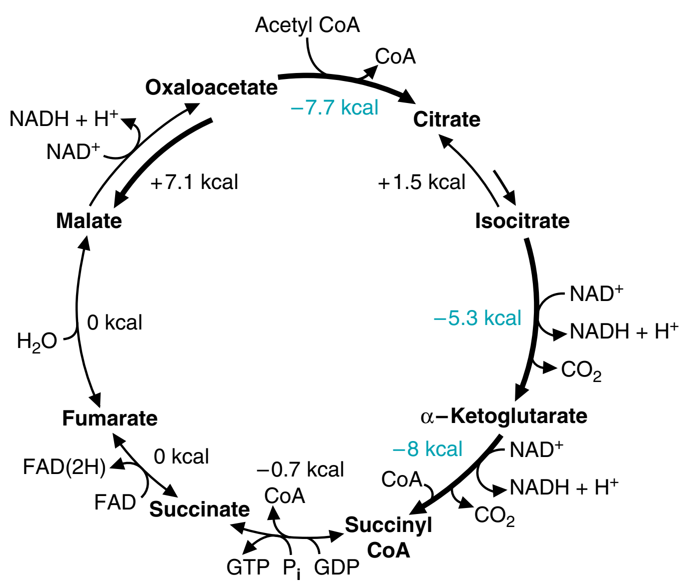


Fig. 20.11. Approximate $\Delta G^{0\prime}$ values for the reactions in the TCA cycle, given for the forward direction. The reactions with large negative $\Delta G^{0\prime}$ values are shown in blue. The standard free energy ($\Delta G^{0\prime}$) refers to the free energy change for conversion of 1 mole of substrate to 1 mole of product under standard conditions (see Chapter 19).

A. Overall Efficiency of the TCA Cycle

The reactions of the TCA cycle are extremely efficient in converting energy in the chemical bonds of the acetyl group to other forms. The total amount of energy available from the acetyl group is about 228 kcal/mole (the amount of energy that could be released from complete combustion of 1 mole of acetyl groups to CO_2 in a bomb calorimeter). The products of the TCA cycle (NADH, FAD(2H), and GTP) contain about 207 kcal (Table 20.1). Thus, the TCA cycle reactions are able to conserve about 90% of the energy available from the oxidation of acetyl CoA.

B. Thermodynamically and Kinetically Reversible and Irreversible reactions

Three reactions in the TCA cycle have large negative values for $\Delta G^{0\prime}$ that strongly favor the forward direction: the reactions catalyzed by citrate synthase, isocitrate dehydrogenase, and α-ketoglutarate dehydrogenase (see Fig. 20.11). Within the TCA cycle, these reactions are physiologically irreversible for two reasons: the products do not rise to high enough concentrations under physiological conditions to overcome the large negative $\Delta G^{0\prime}$ values, and the enzymes involved catalyze the reverse reaction very slowly. These reactions make the major contribution to the overall negative $\Delta G^{0\prime}$ for the TCA cycle, and keep it going in the forward direction.

In contrast to these irreversible reactions, the reactions catalyzed by aconitase and malate dehydrogenase have a positive $\Delta G^{0\prime}$ for the forward direction, and are thermodynamically and kinetically reversible. Because aconitase is rapid in both directions, equilibrium values for the concentration ratio of products to substrates is maintained, and the concentration of citrate is about 20 times that of isocitrate. The accumulation of citrate instead of isocitrate facilitates transport of excess citrate to the cytosol, where it can provide a source of acetyl CoA for pathways like fatty acid and cholesterol synthesis. It also allows citrate to serve as an inhibitor of citrate synthase when flux through isocitrate dehydrogenase is decreased. Likewise, the equilibrium constant of the malate dehydrogenase reaction favors the accumulation of malate over oxaloacetate, resulting in a low oxaloacetate concentration that is influenced by the $NADH/NAD^+$ ratio. Thus, there is a net flux of oxaloacetate towards malate in the liver during fasting (due to fatty acid oxidation, which raises the $NADH/NAD^+$ ratio), and malate can then be transported out of the mitochondria to provide a substrate for gluconeogenesis.

IV. REGULATION OF THE TCA CYCLE

The oxidation of acetyl CoA in the TCA cycle and the conservation of this energy as NADH and FAD(2H) is essential for generation of ATP in almost all tissues in the body. In spite of changes in the supply of fuels, type of fuels in the blood, or rate of ATP utilization, cells maintain ATP homeostasis (a constant level of ATP). The rate of the TCA cycle, like that of all fuel oxidation pathways, is principally regulated to correspond to the rate of the electron transport chain, which is regulated by the ATP/ADP ratio and the rate of ATP utilization (see Chapter 21). The major sites of regulation are shown in Fig 20.12.

Two major messengers feed information on the rate of ATP utilization back to the TCA cycle: (a) the phosphorylation state of ATP, as reflected in ATP and ADP levels, and (b) the reduction state of NAD^+, as reflected in the ratio of $NADH/NAD^+$. Within the cell, even within the mitochondrion, the total adenine nucleotide pool (AMP, ADP, plus ATP) and the total NAD pool (NAD^+ plus NADH) are relatively constant. Thus, an increased rate of ATP utilization results in a small decrease of ATP concentration and an increase of ADP. Likewise, increased NADH oxidation to NAD^+ by the electron transport chain increases the rate of pathways producing NADH. Under normal physiological conditions, the TCA cycle and other

Table 20.1. Energy Yield of the TCA Cycle

kcal/mole	
3 NADH: 3 × 53	= 159
1 FAD(2H)	= 41
1 GTP	= 7
Sum	= 207

Chapter 19 explains the values given for energy yield from NADH and FAD(2H).

The net standard free energy change for the TCA cycle, $\Delta G^{0\prime}$, can be calculated from the sum of the $\Delta G^{0\prime}$ values for the individual reactions. The $\Delta G^{0\prime}$, −13 kcal, is the amount of energy lost as heat. It can be considered the amount of energy spent to ensure that oxidation of the acetyl group to CO_2 goes to completion. This value is surprisingly small. However, oxidation of NADH and FAD(2H) in the electron transport chain helps to make acetyl oxidation more energetically favorable and pull the TCA cycle forward.

Otto Shape had difficulty losing weight because human fuel utilization is too efficient. His adipose tissue fatty acids are being converted to acetyl CoA, which is being oxidized in the TCA cycle, thereby generating NADH and FAD(2H). The energy in these compounds is used for ATP synthesis from oxidative phosphorylation. If his fuel utilization were less efficient and his ATP yield were lower, he would have to oxidize much greater amounts of fat to get the ATP he needs for exercise.

As **Otto Shape** exercises, his myosin ATPase hydrolyzes ATP to provide the energy for movement of myofibrils. The decrease of ATP and increase of ADP stimulates the electron transport chain to oxidize more NADH and FAD(2H). The TCA cycle is stimulated to provide more NADH and FAD(2H) to the electron transport chain. The activation of the TCA cycle occurs through a decrease of the $NADH/NAD^+$ ratio, an increase of ADP concentration, and an increase of Ca^{2+}. Although regulation of the transcription of genes for TCA cycle enzymes is too slow to respond to changes of ATP demands during exercise, the number and size of mitochondria increase during training. Thus, Otto Shape is increasing his capacity for fuel oxidation as he trains.

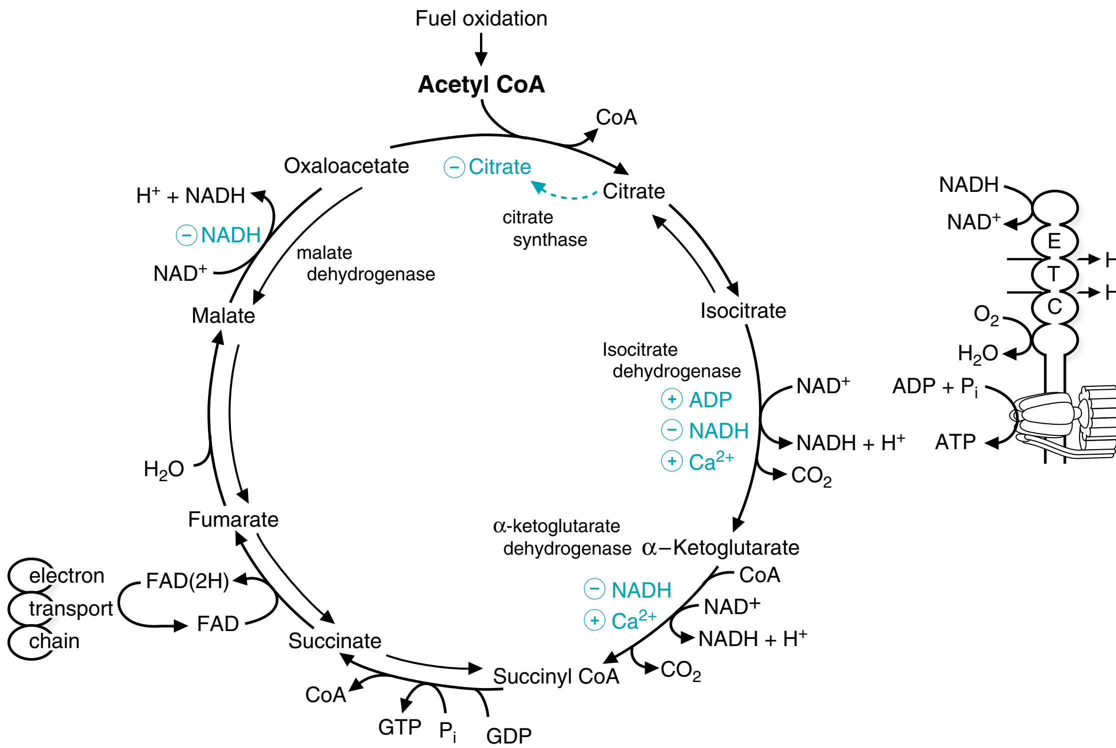


Fig. 20.12. Major regulatory interactions in the TCA cycle. The rate of ATP hydrolysis controls the rate of ATP synthesis, which controls the rate of NADH oxidation in the electron transport chain (ETC). All NADH and FAD(2H) produced by the cycle donate electrons to this chain (shown on the right). Thus, oxidation of acetyl CoA in the TCA cycle can go only as fast as electrons from NADH enter the electron transport chain, which is controlled by the ATP and ADP content of the cells. The ADP and NADH concentrations feed information on the rate of oxidative phosphorylation back to the TCA cycle. Isocitrate dehydrogenase (DH), α-ketoglutarate dehydrogenase (DH), and malate dehydrogenase (DH) are inhibited by increased NADH concentration. The NADH/NAD^+ ratio changes the concentration of oxaloacetate. Citrate is a product inhibitor of citrate synthase. ADP is an allosteric activator of isocitrate dehydrogenase. During muscular contraction, increased Ca^{2+} concentrations activate isocitrate DH and α-ketoglutarate dehydrogenase (as well as pyruvate dehydrogenase).

oxidative pathways respond so rapidly to increased ATP demand that the ATP concentration does not significantly change.

A. Regulation of Citrate Synthase

The principles of pathway regulation are summarized in Table 20.2. In pathways subject to feedback regulation, the first step of the pathway must be regulated so that

Table 20.2. Generalizations on the Regulation of Metabolic Pathways

1. Regulation matches function. The type of regulation use depends on the function of the pathway. Tissue-specific isozymes may allow the features of regulatory enzymes to match somewhat different functions of the pathway in different tissues.
2. Regulation of metabolic pathways occurs at rate-limiting steps, the slowest steps, in the pathway. These are reactions in which a small change of rate will affect the flux through the whole pathway.
3. Regulation usually occurs at the first committed step of a pathway or at metabolic branchpoints. In human cells, most pathways are interconnected with other pathways and have regulatory enzymes for every branchpoint.
4. Regulatory enzymes often catalyze physiologically irreversible reactions. These are also the steps that differ in biosynthetic and degradative pathways.
5. Many pathways have "feedback" regulation, that is, the endproduct of the pathway controls the rate of its own synthesis. Feedback regulation may involve inhibition of an early step in the pathway (feedback inhibition) or regulation of gene transcription.
6. Human cells use compartmentation to control access of substrate and activators or inhibitors to different enzymes.
7. Hormonal regulation integrates responses in pathways requiring more than one tissue. Hormones generally regulate fuel metabolism by:
   a. Changing the phosphorylation state of enzymes.
   b. Changing the amount of enzyme present by changing its rate of synthesis (often induction or repression of mRNA synthesis) or degradation.
   c. Changing the concentration of an activator or inhibitor.

precursors flow into alternate pathways if product is not needed. Citrate synthase, which is the first enzyme of the TCA cycle, is a simple enzyme that has no allosteric regulators. Its rate is controlled principally by the concentration of oxaloacetate, its substrate, and the concentration of citrate, a product inhibitor, competitive with oxaloacetate.(see Fig. 20.12). The malate-oxaloacetate equilibrium favors malate, so the oxaloacetate concentration is very low inside the mitochondrion, and is below the $K_{m,app}$ (see Chapter 9, section I.A.4) of citrate synthase. When the NADH/NAD^+ ratio decreases, the ratio of oxaloacetate to malate increases. When isocitrate dehydrogenase is activated, the concentration of citrate decreases, thus relieving the product inhibition of citrate synthase. Thus, both increased oxaloacetate and decreased citrate levels regulate the response of citrate synthase to conditions established by the electron transport chain and oxidative phosphorylation. In the liver, the NADH/NAD^+ ratio helps determine whether acetyl CoA enters the TCA cycle or goes into the alternate pathway for ketone body synthesis.

B. Allosteric Regulation of Isocitrate Dehydrogenase

Another generalization that can be made about regulation of metabolic pathways is that it occurs at the enzyme that catalyzes the rate-limiting (slowest) step in a pathway (see Table 20.2). Isocitrate dehydrogenase is considered one of the rate-limiting steps of the TCA cycle, and is allosterically activated by ADP and inhibited by NADH (Fig. 20.13). In the absence of ADP, the enzyme exhibits positive cooperativity; as isocitrate binds to one subunit, other subunits are converted to an active conformation (see Chapter 9, section III.A on allosteric enzymes). In the presence of ADP, all of the subunits are in their active conformation, and isocitrate binds more readily. Consequently, the $K_{m,app}$ (the $S_{0.5}$) shifts to a much lower value. Thus, at the concentration of isocitrate found in the mitochondrial matrix, a small change in the concentration of ADP can produce a large change in the rate of the isocitrate dehydrogenase reaction. Small changes in the concentration of the product, NADH, and of the cosubstrate, NAD^+, also affect the rate of the enzyme more than they would a nonallosteric enzyme.

C. Regulation of α-Ketoglutarate Dehydrogenase

The α-ketoglutarate dehydrogenase complex, although not an allosteric enzyme, is product-inhibited by NADH and succinyl CoA, and may also be inhibited by GTP (see Fig. 20.12). Thus, both α-ketoglutarate dehydrogenase and isocitrate dehydrogenase respond directly to changes in the relative levels of ADP and hence the rate at which NADH is oxidized by electron transport. Both of these enzymes are also activated by Ca^{2+}. In contracting heart muscle, and possibly other muscle tissues, the release of Ca^{2+} from the sarcoplasmic reticulum during muscle contraction may provide an additional activation of these enzymes when ATP is being rapidly hydrolyzed.

D. Regulation of TCA Cycle Intermediates

Regulation of the TCA cycle serves two functions: it ensures that NADH is generated fast enough to maintain ATP homeostasis and it regulates the concentration of TCA cycle intermediates. For example, in the liver, a decreased rate of isocitrate dehydrogenase increases citrate concentration, which stimulates citrate efflux to the cytosol. A number of regulatory interactions occur in the TCA cycle, in addition to those mentioned above, that control the levels of TCA intermediates and their flux into pathways that adjoin the TCA cycle.

V. PRECURSORS OF ACETYL CoA

Compounds enter the TCA cycle as acetyl CoA or as an intermediate that can be converted to malate or oxaloacetate. Compounds that enter as acetyl CoA are

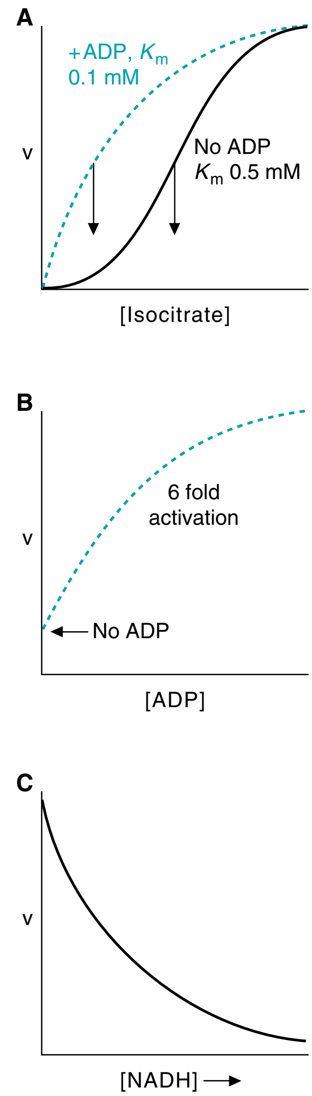


Fig. 20.13. Allosteric regulation of isocitrate dehydrogenase (ICDH). Isocitrate dehydrogenase has eight subunits, and two active sites. Isocitrate, NAD^+, and NADH bind in the active site; ADP and Ca^{2+} are activators and bind to separate allosteric sites. **A.** A graph of velocity versus isocitrate concentration shows positive cooperativity (sigmoid curve) in the absence of ADP. The allosteric activator ADP changes the curve into one closer to a rectangular hyperbola, and decreases the K_m ($S_{0.5}$) for isocitrate. **B.** The allosteric activation by ADP is not an all-or-nothing response. The extent of activation by ADP depends on its concentration. **C.** Increases in the concentration of product, NADH, decrease the velocity of the enzyme through effects on the allosteric activation.

Acetate (acetic acid) is present in the diet, and can be produced from the oxidation of ethanol. Roman soldiers carried vinegar, a dilute solution of acetic acid. The acidity of the vinegar made it a relatively safe source of drinking water because many kinds of pathogenic bacteria do not grow well in acid solutions. The acetate, which is activated to acetyl CoA, provided an excellent fuel for muscular exercise.

oxidized to CO_2. Compounds that enter as TCA cycle intermediates replenish intermediates that have been used in biosynthetic pathways, such as gluconeogenesis or heme synthesis, but cannot be fully oxidized to CO_2.

A. Sources of Acetyl CoA

Acetyl CoA serves as a common point of convergence for the major pathways of fuel oxidation. It is generated directly from the β-oxidation of fatty acids and degradation of the ketone bodies β-hydroxybutyrate and acetoacetate (Fig. 20.14). It is also formed from acetate, which can arise from the diet or from ethanol oxidation. Glucose and other carbohydrates enter glycolysis, a pathway common to all cells, and are oxidized to pyruvate. The amino acids alanine and serine are also converted to pyruvate. Pyruvate is oxidized to acetyl CoA by the pyruvate dehydrogenase complex. A number of amino acids, such as leucine and isoleucine are also oxidized to acetyl CoA. Thus, the final oxidation of acetyl CoA to CO_2 in the TCA cycle is the last step in all the major pathways of fuel oxidation.

B. Pyruvate Dehydrogenase Complex

The pyruvate dehydrogenase complex (PDC) oxidizes pyruvate to acetyl CoA, thus linking glycolysis and the TCA cycle. In the brain, which is dependent on the oxidation of glucose to CO_2 to fulfill its ATP needs, regulation of the PDC is a life and death matter.

1. STRUCTURE OF PDC

PDC belongs to the α-ketoacid dehydrogenase complex family and, thus, shares structural and catalytic features with the α-ketoglutarate dehydrogenase complex and the branched chain α-ketoacid dehydrogenase complex (Fig. 20.15). It contains the same three basic types of catalytic subunits: (1) pyruvate decarboxylase subunits that bind thiamine-pyrophosphate (E_1); (2) transacetylase subunits that bind lipoate (E_2), and (3) dihyrolipoyl dehydrogenase subunits that bind FAD (E_3) (see Fig. 20.9). Although the E_1 and E_2 enzymes in PDC are relatively specific for pyruvate, the same dihydrolipoyl dehydrogenase participates in all of the α-ketoacid dehydrogenase

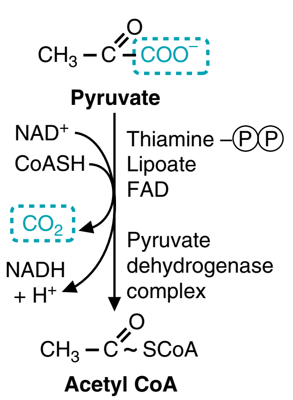


Fig. 20.15. Pyruvate dehydrogenase complex (PDC) catalyzes oxidation of the α-ketoacid pyruvate to acetyl CoA.

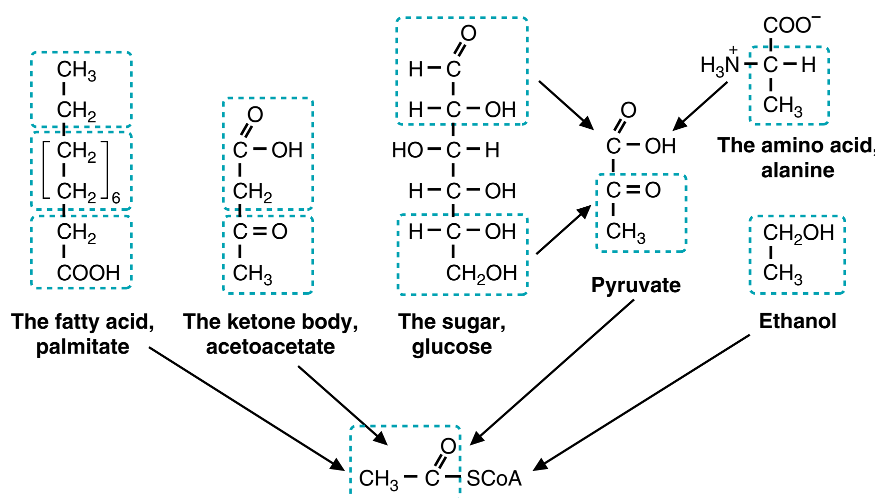


Fig. 20.14. Origin of the acetyl group from various fuels. Acetyl CoA is derived from the oxidation of fuels. The portions of fatty acids, ketone bodies, glucose, pyruvate, the amino acid alanine, and ethanol that are converted to the acetyl group of acetyl CoA are shown in blue.

complexes. In addition to these three types of subunits, the PDC complex contains one additional catalytic subunit, protein X, which is a transacetylase. Each functional component of the PDC complex is present in multiple copies (e.g., bovine heart PDC has 30 subunits of E_1, 60 subunits of E_2, and 6 subunits each of E_3 and X). The E_1 enzyme is itself a tetramer of two different types of subunits, α and β.

2. REGULATION OF PDC

PDC activity is controlled principally through phosphorylation by pyruvate dehydrogenase kinase, which inhibits the enzyme, and dephosphorylation by pyruvate dehydrogenase phosphatase, which activates it (Fig. 20.16). Pyruvate dehydrogenase kinase and pyruvate dehydrogenase phosphatase are regulatory subunits within the PDC complex and act only on the complex. PDC kinase transfers a phosphate from ATP to specific serine hydroxyl (ser-OH) groups on pyruvate decarboxylase (E_1). PDC phosphatase removes these phosphate groups by hydrolysis. Phosphorylation of just one serine on the PDC E_1 α subunit can decrease its activity by over 99%. PDC kinase is present in complexes as tissue-specific isozymes that vary in their regulatory properties.

PDC kinase is, itself, inhibited by ADP and pyruvate. Thus, when rapid ATP utilization results in an increase of ADP, or when activation of glycolysis increases pyruvate levels, PDC kinase is inhibited, and PDC remains in an active, nonphosphorylated form. PDC phosphatase requires Ca^{2+} for full activity. In the heart, increased intramitochondrial Ca^{2+} during rapid contraction activates the phosphatase, thereby increasing the amount of active, nonphosphorylated PDC.

PDC is also regulated through inhibition by its products, acetyl CoA and NADH. This inhibition is stronger than regular product inhibition because their binding to

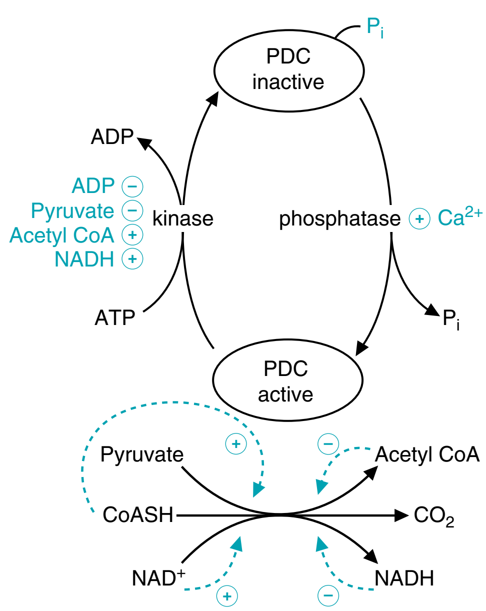


Fig. 20.16. Regulation of pyruvate dehydrogenase complex (PDC). PDC kinase, a subunit of the enzyme, phosphorylates PDC at a specific serine residue, thereby converting PDC to an inactive form. The kinase is inhibited by ADP and pyruvate. PDC phosphatase, another subunit of the enzyme, removes the phosphate, thereby activating PDC. The phosphatase is activated by Ca^{2+}. When the substrates, pyruvate and CoASH, are bound to PDC, the kinase activity is inhibited and PDC is active. When the products acetyl CoA and NADH bind to PDC, the kinase activity is stimulated, and the enzyme is phosphorylated to the inactive form. E_1 and the kinase exist as tissue-specific isozymes with overlapping tissue specificity, and somewhat different regulatory properties.

Deficiencies of the pyruvate dehydrogenase complex (PDC) are among the most common inherited diseases leading to lacticacidemia and, like pyruvate carboxylase deficiency, are grouped into the category of Leigh's disease. In its severe form, PDC deficiency presents with overwhelming lactic acidosis at birth, with death in the neonatal period. In a second form of presentation, the lactic academia is moderate, but there is profound psychomotor retardation with increasing age. In many cases, concomitant damage to the brain stem and basal ganglia lead to death in infancy. The neurological symptoms arise because the brain has a very limited ability to use fatty acids as a fuel, and is, therefore, dependent on glucose metabolism for its energy supply.

The most common PDC genetic defects are in the gene for the α subunit of E_1. The E_1 α-gene is X-linked. Because of its importance in central nervous system metabolism, pyruvate dehydrogenase deficiency is a problem in both males and females, even if the female is a carrier. For this reason, it is classified as an X-linked dominant disorder.

PDC stimulates its phosphorylation to the inactive form. The substrates of the enzyme, CoASH and NAD^+, antagonize this product inhibition. Thus, when an ample supply of acetyl CoA for the TCA cycle is already available from fatty acid oxidation, acetyl CoA and NADH build up and dramatically decrease their own further synthesis by PDC.

PDC can also be rapidly activated through a mechanism involving insulin, which plays a prominent role in adipocytes. In many tissues, insulin may, slowly over time, increase the amount of pyruvate dehydrogenase complex present.

The rate of other fuel oxidation pathways that feed into the TCA cycle is also increased when ATP utilization increases. Insulin, other hormones and diet control the availability of fuels for these oxidative pathways.

VI. TCA CYCLE INTERMEDIATES AND ANAPLEROTIC REACTIONS

A. TCA Cycle Intermediates are Precursors for Biosynthetic Pathways

Pyruvate, citrate, α-ketoglutarate and malate, ADP, ATP, and phosphate (as well as many other compounds) have specific transporters in the inner mitochondrial membrane that transport compounds between the mitochondrial matrix and cytosol in exchange for a compound of similar charge. In contrast, CoASH, acetyl CoA, other CoA derivatives, NAD^+ and NADH, and oxaloacetate, are not transported at a metabolically significant rate. To obtain cytosolic acetyl CoA, many cells transport citrate to the cytosol, where it is cleaved to acetyl CoA and oxaloacetate by citrate lyase.

The intermediates of the TCA cycle serve as precursors for a variety of different pathways present in different cell types (Fig. 20.17). This is particularly important in the central metabolic role of the liver. The TCA cycle in the liver is often called an "open cycle" because there is such a high efflux of intermediates. After a high carbohydrate meal, citrate efflux and cleavage to acetyl CoA provides acetyl units for cytosolic fatty acid synthesis. During fasting, gluconeogenic precursors are converted to malate, which leaves the mitochondria for cytosolic gluconeogenesis. The liver also uses TCA cycle intermediates to synthesize carbon skeletons of amino acids. Succinyl CoA may be removed from the TCA cycle to form heme in cells of the liver and bone marrow. In the brain, α-ketoglutarate is converted to glutamate and then to γ-aminobutyric acid (GABA), a neurotransmitter. In skeletal muscle, α-ketoglutarate is converted to glutamine, which is transported through the blood to other tissues.

B. Anaplerotic Reactions

Removal of any of the intermediates from the TCA cycle removes the 4 carbons that are used to regenerate oxaloacetate during each turn of the cycle. With depletion of oxaloacetate, it is impossible to continue oxidizing acetyl CoA. To enable the TCA

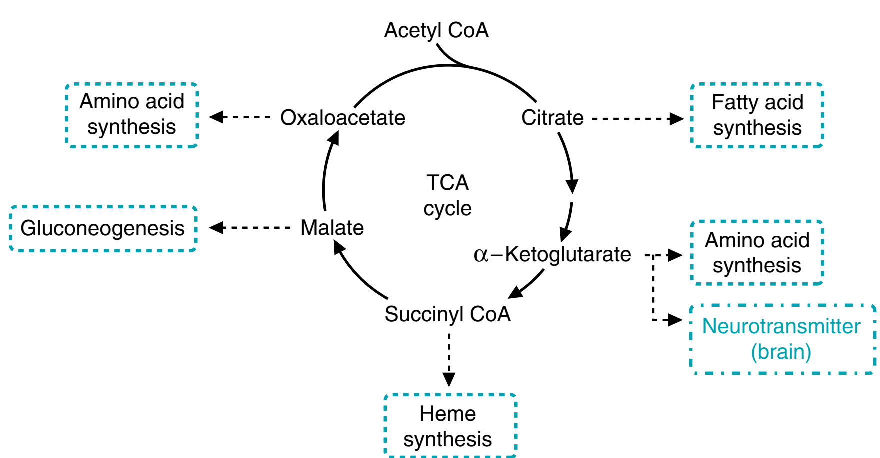


Fig. 20.17. Efflux of intermediates from the TCA cycle. In the liver, TCA cycle intermediates are continuously withdrawn into the pathways of fatty acid synthesis, amino acid synthesis, gluconeogenesis, and heme synthesis. In brain, α-ketoglutarate is converted to glutamate and GABA, both neurotransmitters.

cycle to keep running, cells have to supply enough four-carbon intermediates from degradation of carbohydrate or certain amino acids to compensate for the rate of removal. Pathways or reactions that replenish the intermediates of the TCA cycle are referred to as anaplerotic ("filling up").

1. PYRUVATE CARBOXYLASE IS A MAJOR ANAPLEROTIC ENZYME

Pyruvate carboxylase is one of the major anaplerotic enzymes in the cell. It catalyzes the addition of CO_2 to pyruvate to form oxaloacetate (Fig. 20.18). Like most carboxylases, pyruvate carboxylase contains biotin, which forms a covalent intermediate with CO_2 in a reaction requiring ATP and Mg^{2+} (see Fig. 8.12, Chap. 8). The activated CO_2 is then transferred to pyruvate to form the carboxyl group of oxaloacetate.

Pyruvate carboxylase is found in many tissues, such as liver, brain, adipocytes, and fibroblasts, where its function is anaplerotic. Its concentration is high in liver and kidney cortex, where there is a continuous removal of oxaloacetate and malate from the TCA cycle to enter the gluconeogenic pathway.

Pyruvate carboxylase is activated by acetyl CoA and inhibited by high concentrations of many acyl CoA derivatives. As the concentration of oxaloacetate is depleted through the efflux of TCA cycle intermediates, the rate of the citrate synthase reaction decreases and acetyl CoA concentration rises. The acetyl CoA then activates pyruvate carboxylase to synthesize more oxaloacetate.

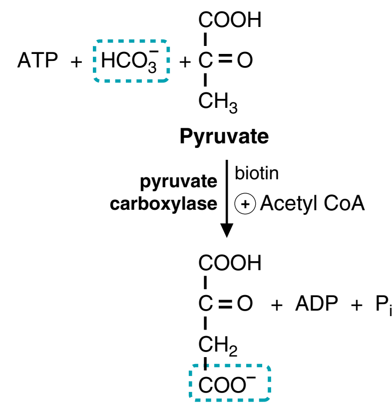


Fig. 20.18. Pyruvate carboxylase reaction. Pyruvate carboxylase adds a carboxyl group from bicarbonate (which is in equilibrium with CO_2) to pyruvate to form oxaloacetate. Biotin is used to activate and transfer the CO_2. The energy to form the covalent biotin–CO_2 complex is provided by the high-energy phosphate bond of ATP, which is cleaved in the reaction. The enzyme is activated by acetyl CoA.

2. AMINO ACID DEGRADATION FORMS TCA CYCLE INTERMEDIATES

The pathways for oxidation of many amino acids convert their carbon skeletons into 5- and 4-carbon intermediates of the TCA cycle that can regenerate oxaloacetate (Fig 20.19). Alanine and serine carbons can enter through pyruvate carboxylase (see Fig.20.19, circle 1). In all tissues with mitochondria (except for, surprisingly, the liver), oxidation of the two branched chain amino acids isoleucine and valine to succinyl CoA forms a major anaplerotic route (see Fig.20.19, circle 3). In the liver, other compounds forming propionyl CoA (e.g., methionine, thymine and odd-chain length or branched fatty acids) also enter the TCA cycle as succinyl CoA. In most tissues, glutamine is taken up from the blood, converted to glutamate, and then oxidized to α-ketoglutarate, forming another major anaplerotic route (see Fig.20.19, circle 2). However, the TCA cycle cannot be resupplied with intermediates by even chain length fatty acid oxidation, or ketone body oxidation, which forms only acetyl CoA. In the TCA cycle, two carbons are lost from citrate before succinyl CoA is formed, and, therefore, there is no net conversion of acetyl carbon to oxaloacetate.

Pyruvate carboxylase deficiency is one of the genetic diseases grouped together under the clinical manifestations of Leigh's disease (subacute necrotizing encephalopathy). In the mild form, the patient presents early in life with delayed development and a mild-to-moderate lactic acidemia. Patients who survive are severely mentally retarded, and there is a loss of cerebral neurons. In the brain, pyruvate carboxylase is present in the astrocytes, which use TCA cycle intermediates to synthesize glutamine. This pathway is essential for neuronal survival. The major cause of the lactic acidemia is that cells dependent on pyruvate carboxylase for an anaplerotic supply of oxaloacetate cannot oxidize pyruvate in the TCA cycle (because of low oxaloacetate levels), and the liver cannot convert pyruvate to glucose (because the pyruvate carboxylase reaction is required for this pathway to occur), so the excess pyruvate is converted to lactate.

Biotin is a vitamin. A deficiency of biotin is very rare in humans because it is required in such small amounts and is synthesized by intestinal bacteria. However, an interesting case of biotin deficiency arose in a man eating a diet composed principally of peanuts and raw egg whites. Egg whites contain a biotin binding protein, avidin. Since he did not denature avidin by cooking the egg whites, it depleted his diet of biotin.

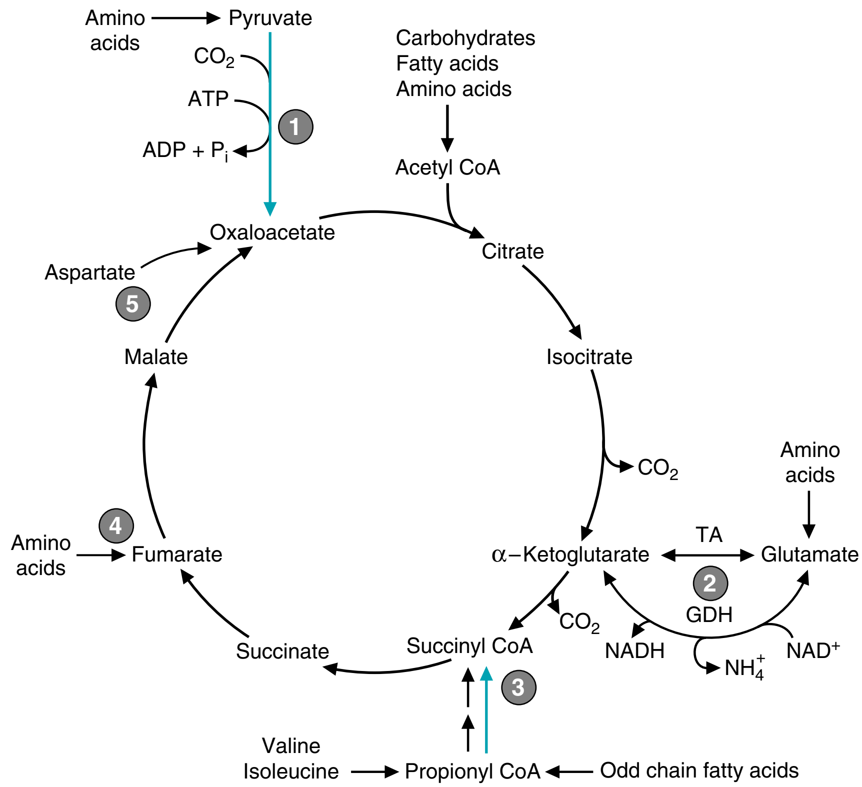


Fig. 20.19. Major anaplerotic pathways of the TCA cycle. 1 and 3 (blue arrows) are the two major anabolic pathways. (1) Pyruvate carboxylase (2) Glutamate is reversibly converted to α-ketoglutarate by transaminases (TA) and glutamate dehydrogenase (GDH) in many tissues. (3) The carbon skeletons of valine and isoleucine, a 3-carbon unit from odd chain fatty acid oxidation, and a number of other compounds enter the TCA cycle at the level of succinyl CoA. Other amino acids are also degraded to fumarate (4) and oxaloacetate (5), principally in the liver.

CLINICAL COMMENTS

Otto Shape. Otto Shape is experiencing the benefits of physical conditioning. A variety of functional adaptations in the heart, lungs, vascular system, and skeletal muscle occur in response to regular graded exercise. The pumping efficiency of the heart increases, allowing a greater cardiac output with fewer beats per minute and at a lower rate of oxygen utilization. The lungs extract a greater percentage of oxygen from the inspired air, allowing fewer respirations per unit of activity. The vasodilatory capacity of the arterial beds in skeletal muscle increases, promoting greater delivery of oxygen and fuels to exercising muscle. Concurrently, the venous drainage capacity in muscle is enhanced, ensuring that lactic acid will not accumulate in contracting tissues. These adaptive changes in physiological responses are accompanied by increases in the number, size, and activity of skeletal muscle mitochondria, along with the content of TCA cycle enzymes and components of the electron transport chain. These changes markedly enhance the oxidative capacity of exercising muscle.

In skeletal muscle and other tissues, ATP is generated by anaerobic glycolysis when the rate of aerobic respiration is inadequate to meet the rate of ATP utilization. Under these circumstances, the rate of pyruvate production exceeds the cell's capacity to oxidize NADH in the electron transport chain, and hence, to oxidize pyruvate in the TCA cycle. The excess pyruvate is reduced to lactate. Because lactate is an acid, its accumulation affects the muscle and causes pain and swelling.

Ann O'Rexia. Ann O'Rexia is experiencing fatigue for a number of reasons. She has iron deficiency anemia, which affects both iron-containing hemoglobin in her red blood cells, iron in aconitase and succinic dehydrogenase, as well as iron in the heme proteins of the electron transport chain. She may also be experiencing the consequences of multiple vitamin deficiencies, including thiamine, riboflavin, and niacin (the vitamin precursor of NAD^+). It is less likely, but possible, that she also has subclinical deficiencies of pantothenate (the precursor of CoA) or biotin. Because of this, Ann's muscle must use glycolysis as their primary source of energy, which results in sore muscles.

Riboflavin deficiency generally occurs in conjunction with other water-soluble vitamin deficiencies. The classic deficiency symptoms are cheilosis (inflammation of the corners of the mouth), glossitis (magenta tongue), and seborrheic ("greasy") dermatitis. It is also characterized by sore throat, edema of the pharyngeal and oral mucus membranes, and normochromic, normocytic anemia associated with pure red cell cytoplasia of the bone marrow. However, it is not known whether the glossitis and dermatitis are actually due to multiple vitamin deficiencies.

Al Martini. Al Martini presents a second time with an alcohol-related high output form of heart failure sometimes referred to as "wet" beriberi, or as the "beriberi heart" (see Chapter 9). The term "wet" refers to the fluid retention which may eventually occur when left ventricular contractility is so compromised that cardiac output, although initially relatively "high," cannot meet the "demands" of the peripheral vascular beds, which have dilated in response to the thiamine deficiency.

The cardiomyopathy is directly related to a reduction in the normal biochemical function of the vitamin thiamine in heart muscle. Inhibition of the α-keto acid dehydrogenase complexes causes accumulation of α-keto acids in heart muscle (and in blood), resulting in a chemically-induced cardiomyopathy. Impairment of two other functions of thiamine may also contribute to the cardiomyopathy. Thiamine pyrophosphate serves as the coenzyme for transketolase in the pentose phosphate pathway, and pentose phosphates accumulate in thiamine deficiency. In addition, thiamine triphosphate (a different coenzyme form) may function in Na^+ conductance channels.

Immediate treatment with large doses (50–100 mg) of intravenous thiamine may produce a measurable decrease in cardiac output and increase in peripheral vascular resistance as early as 30 minutes after the initial injection. Dietary supplementation of thiamine is not as effective because ethanol consumption interferes with thiamine absorption. Because ethanol also affects the absorption of most water-soluble vitamins, or their conversion to the coenzyme form, **Al Martini** was also given a bolus containing a multivitamin supplement.

BIOCHEMICAL COMMENTS

Compartmentation of Mitochondrial Enzymes. The mitochondrion forms a structural, functional, and regulatory compartment within the cell. The inner mitochondrial membrane is impermeable to anions and cations, and compounds can cross the membrane only on specific transport proteins. The enzymes of the TCA cycle, therefore, have more direct access to products of the previous reaction in the pathway than they would if these products were able to diffuse throughout the cell. Complex formation between enzymes also restricts access to pathway intermediates. Malate dehydrogenase and citrate synthase may form a loosely associated complex. The multienzyme pyruvate dehydrogenase and α-ketoglutarate dehydrogenase complexes are examples of substrate channeling by tightly bound enzymes; only the transacylase enzyme has access to the thiamine-bound intermediate of the reaction, and only lipoamide dehydrogenase has access to reduced lipoic acid.

Riboflavin has a wide distribution in foods, and small amounts are present as coenzymes in most plant and animal tissues. Eggs, lean meats, milk, broccoli, and enriched breads and cereals are especially good sources. A portion of our niacin requirement can be met by synthesis from tryptophan. Meat (especially red meat), liver, legumes, milk, eggs, alfalfa, cereal grains, yeast, and fish are good sources of niacin and tryptophan.

Beri-beri, now known to be caused by thiamine deficiency, was attributed to lack of a nitrogenous component in food by Takaki, a Japanese surgeon, in 1884. In 1890, Eijkman, a Dutch physician working in Java, noted that the polyneuritis associated with beri-beri could be prevented by rice bran that had been removed during polishing. Thiamine is present in the bran portion of grains, and abundant in pork and legumes. In contrast to most vitamins, milk and milk products, seafood, fruits, and vegetables are NOT good sources of thiamine.

Thiamine

"Now polished rice isn't nice",
Said the Dutchman Eijkman.
Whole grains are a far better
Source of thiamine.
For beri-beri is very, very
Hard on your nerves, you see.
Polyneuritis and an enlarged heart
May both accompany
A very bad diet, a very sad diet
A diet thiamine-free.
And many who dine, only on wine
Or consume brandy, whiskey or gin
May never recover, if you don't discover
They can't absorb thiamine.
Wernicke-Korsakoff describe the signs
And the confusion in the minds
Of patients with this deficiency.
So good doctors remember, try to recall
Before you charge your fee,
To give an injection, *im* or *iv*
Of this vitamin B.

—revised from an anonymous author

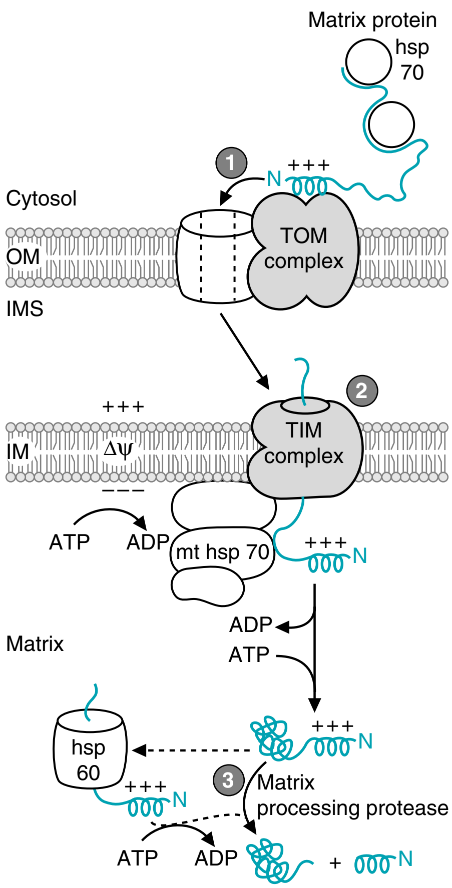


Fig. 20.20. Model for the import of nuclear-encoded proteins into the mitochondrial matrix. The matrix preprotein with its positively charged N-terminal presequence is shown in blue. Abbreviations: OM, outer mitochondrial membrane; IMS, intramembrane space; IM, inner mitochondrial membrane; TOM, translocases of the outer mitochondrial membrane; TIM, translocases of the inner mitochondrial membrane; mthsp70, mitochondrial heat shock protein 70.

Compartmentation plays an important role in regulation. The close association between the rate of the electron transport chain and the rate of the TCA cycle is maintained by their mutual access to the same pool of NADH and NAD^+ in the mitochondrial matrix. NAD^+, NADH, CoASH, and acyl CoA derivatives have no transport proteins and cannot cross the mitochondrial membrane. Thus, all of the dehydrogenases compete for the same NAD^+ molecules, and are inhibited when NADH rises. Likewise, accumulation of acyl CoA derivatives (e.g., acetyl CoA) within the mitochondrial matrix affects other CoA-utilizing reactions, either by competing at the active site or limiting CoASH availability.

Import of Nuclear Encoded Proteins. All mitochondrial matrix proteins, such as the TCA cycle enzymes, are encoded by the nuclear genome. They are imported into the mitochondrial matrix as unfolded proteins that are pushed and pulled through channels in the outer and inner mitochondrial membranes (Fig. 20.20). Proteins destined for the mitochondrial matrix have a targeting N-terminal presequence of about 20 amino acids that includes several positively charged amino acid residues. They are synthesized on free ribosomes in the cytosol and maintain an unfolded conformation by binding to hsp70 chaperonins. This basic presequence binds to a receptor in a TOM complex (**t**ranslocators of the **o**uter **m**embrane) (see Fig. 20.20, step 1). The TOM complexes consist of channel proteins, assembly proteins and receptor proteins with different specificities (e.g., TOM20 binds the matrix protein presequence). Negatively charged acidic residues on the receptors and in the channel pore assist in translocation of the matrix protein through the channel, presequence first.

The matrix preprotein is translocated across the inner membrane through a TIM complex (**t**ranslocases of the **i**nner **m**embrane) (see Fig. 20.20, step 2). Insertion of the preprotein into the TIM channel is driven by the potential difference across the membrane, $\Delta\psi$. Mitochondrial hsp70 (mthsp70), which is bound to the matrix side of the TIM complex, binds the incoming preprotein and may "ratchet" it through the membrane. ATP is required for binding of mthsp70 to the TIM complex and again for the subsequent dissociation of the mthsp70 and the matrix preprotein. In the matrix, the preprotein may require another heat shock protein, hsp60, for proper folding. The final step in the import process is cleavage of the signal sequence by a matrix processing protease (see Fig. 20.20, step 3).

Proteins of the inner mitochondrial membrane are imported through a similar process, using TOM and TIM complexes containing different protein components.

Suggested References

Robinson, BH. Lactic acidemia: Disorders of pyruvate carboxylase and pyruvate dehydrogenase. In: Scriver CR, Beudet AL, Sly WS, Valle D, eds: The Metabolic and Molecular Bases of Inherited Disease. Vol. I. 8th Ed. New York: McGraw-Hill, 2001:4451–4480.

REVIEW QUESTIONS—CHAPTER 20

1. Which of the following coenzymes is unique to α-ketoacid dehydrogenase complexes?

   (A) NAD^+
   (B) FAD
   (C) GDP
   (D) H_2O
   (E) Lipoic acid

2. A patient diagnosed with thiamine deficiency exhibited fatigue and muscle cramps. The muscle cramps have been related to an accumulation of metabolic acids. Which of the following metabolic acids is most likely to accumulate in a thiamine deficiency?

   (A) Isocitric acid
   (B) Pyruvic acid
   (C) Succinic acid
   (D) Malic acid
   (E) Oxaloacetic acid

3. Succinate dehydrogenase differs from all other enzymes in the TCA cycle in that it is the only enzyme that displays which of the following characteristics?

   (A) It is embedded in the inner mitochondrial membrane.
   (B) It is inhibited by NADH.
   (C) It contains bound FAD.
   (D) It contains Fe-S centers.
   (E) It is regulated by a kinase.

4. During exercise, stimulation of the tricarboxylic acid cycle results principally from which of the following?

   (A) Allosteric activation of isocitrate dehydrogenase by increased NADH
   (B) Allosteric activation of fumarase by increased ADP
   (C) A rapid decrease in the concentration of four carbon intermediates
   (D) Product inhibition of citrate synthase
   (E) Stimulation of the flux through a number of enzymes by a decreased $NADH/NAD^+$ ratio

5. Coenzyme A is synthesized from which of the following vitamins?

   (A) Niacin
   (B) Riboflavin
   (C) Vitamin A
   (D) Pantothenate
   (E) Vitamin C

21 Oxidative Phosphorylation and Mitochondrial Function

Most cells are dependent on oxidative phosphorylation for ATP homeostasis. The ability to generate ATP depends on O_2 and an intact inner mitochondrial membrane. During oxygen deprivation from ischemia (a low blood flow), an inability to generate energy from the electron transport chain results in an increased permeability of this membrane and mitochondrial swelling. Mitochondrial swelling is a key element in the pathogenesis of irreversible cell injury leading to cell lysis and death (necrosis).

*Energy from fuel oxidation is converted to the high-energy phosphate bonds of adenosine triphosphate (ATP) by the process of **oxidative phosphorylation.** Most of the energy from oxidation of fuels in the TCA cycle and other pathways is conserved in the form of the reduced electron-accepting coenzymes, NADH and FAD(2H). The **electron transport chain** oxidizes **NADH** and **FAD(2H)**, and donates the electrons to O_2, which is reduced to H_2O (Fig. 21.1). Energy from reduction of O_2 is used for phosphorylation of adenosine diphosphate (ADP) to ATP by **ATP synthase (F_0F_1ATPase)**. The net yield of oxidative phosphorylation is approximately 2.5 moles of ATP per mole of NADH oxidized, or 1.5 moles of ATP per mole of FAD(2H) oxidized.*

***Chemiosmotic model of ATP synthesis**. The **chemiosmotic model** explains how energy from transport of electrons to O_2 is transformed into the high-energy phosphate bond of ATP (see Fig. 21.1). Basically, the electron transport chain contains three large **protein complexes (I, III, and IV)** that span the inner mitochondrial membrane. As electrons pass through these complexes in a series of oxidation–reduction reactions, protons are transferred from the mitochondrial matrix to the cytosolic side of the inner mitochondrial membrane. The pumping of protons generates an **electrochemical gradient** (Δp) across the membrane composed of the membrane potential and the proton gradient. ATP synthase contains a proton pore through the inner mitochondrial membrane and a catalytic headpiece that protrudes into the matrix. As protons are driven*

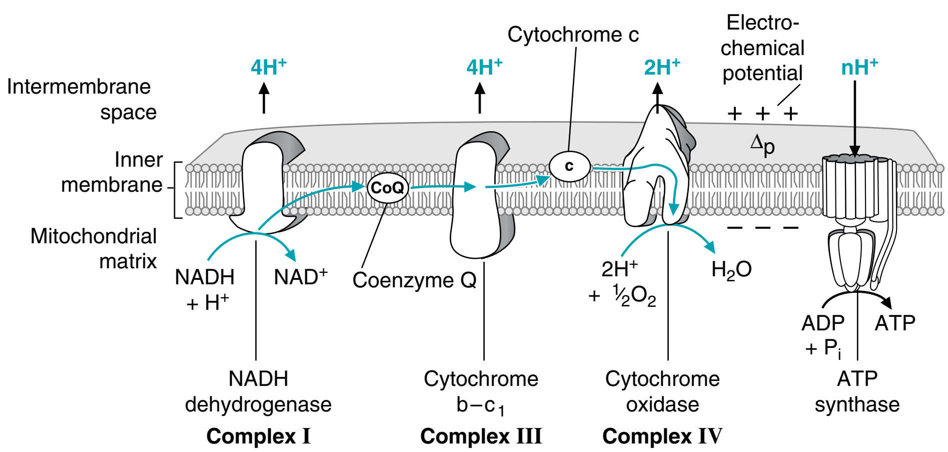


Fig. 21.1. Oxidative phosphorylation. Blue arrows show the path of electron transport from NADH to O_2. As electrons pass through the chain, protons are pumped from the mitochondrial matrix to the intermembrane space, thereby establishing an electochemical potential gradient, Δp, across the inner mitochondrial membrane. The positive and negative charges on the membrane denote the membrane potential ($\Delta\psi$). Δp drives protons into the matrix through a pore in ATP synthase, which uses the energy to form ATP from ADP and Pi.

into the matrix through the pore, they change the conformation of the headpiece, which releases ATP from one site and catalyzes formation of ATP from ADP and Pi at another site.

Deficiencies of electron transport*. In cells, complete transfer of electrons from NADH and FAD(2H) through the chain to O_2 is necessary for ATP generation. Impaired transfer through any complex can have pathologic consequences. Fatigue can result from iron-defeciency* ***anemia****, which decreases* ***Fe*** *for* ***Fe-S centers*** *and* ***cytochromes****.* ***Cytochrome c_1 oxidase,*** *which contains the* ***O_2 binding site,*** *is inhibited by* ***cyanide****.* ***Mitochondrial DNA (mtDNA),*** *which is maternally inherited, encodes some of the subunits of the electron transport chain complexes and ATP synthase.* ***Oxphos diseases*** *are caused by* ***mutations*** *in* ***nuclear DNA*** *or* ***mt DNA*** *that decrease mitochondrial capacity for oxidative phosphorylation.*

Regulation of oxidative phosphorylation*. The rate of the electron transport chain is* ***coupled*** *to the rate of ATP synthesis by the transmembrane electochemical gradient. As ATP is used for energy-requiring processes and ADP levels increase, proton influx through the ATP synthase pore generates more ATP, and the electron transport chain responds to restore Δp. In* ***uncoupling,*** *protons return to the matrix by a mechanism that bypasses the ATP synthase pore, and the energy is released as heat.* ***Proton leakage****,* ***chemical uncouplers,*** *and regulated* ***uncoupling proteins*** *increase our metabolic rate and heat generation.*

Mitochondria and cell death*. Although oxidative phosphorylation is a mitochondrial process, most ATP utilization occurs outside of the mitochondrion. ATP synthesized from oxidative phosphorylation is actively transported from the matrix to the intermembrane space by* ***adenine nucleotide translocase (ANT)****.* ***Porins form voltage-dependent anion channels (VDAC)*** *through the outer mitochondrial membrane for the diffusion of H_2O, ATP metabolites, and other ions. Under certain types of stress, ANT, VDAC, and other proteins form a nonspecific open channel known as the* ***mitochondrial permeability transition pore****. This pore is associated with events that lead rapidly to* ***necrotic cell death.***

THE WAITING ROOM

Cora Nari was recovering uneventfully from her heart attack 1 month earlier (see Chapter 19), when she won the Georgia State lottery. When she heard her number announced over television, she experienced crushing chest pain, grew short of breath, and passed out. She regained consciousness as she was being rushed to the hospital emergency room.

On initial examination, her blood pressure was extremely high and her heart rhythm irregular. Her blood levels of CK-MB and TnI (troponin I) were elevated. An electrocardiogram showed unequivocal evidence of severe lack of oxygen (ischemia) in the muscles of the anterior and lateral walls of her heart. Life support measures including nasal oxygen were initiated. An intravenous drip of nitroprusside, a vasodilating agent, was started in an effort to reduce her hypertension. After her blood pressure was well controlled, a decision was made to administer intravenous tissue plasminogen activator (TPA) in an attempt to break up any intracoronary artery blood clots in vessels supplying the ischemic myocardium (thrombolytic therapy).

Cora Nari is experiencing a second myocardial infarction. Ischemia (a low blood flow) has caused hypoxia (low levels of oxygen) in her heart muscle, resulting in inadequate generation of ATP for the maintenance of low intracellular Na^+ and Ca^{2+} levels (see Chapter 19). As a consequence, cells have become swollen and the cytosolic proteins creatine kinase (MB isoform) and troponin (heart isoform) have leaked into the blood. (See Ann Jeina, Chapters 6 and 7).

An ^{123}I thyroid uptake and scan performed on **X.S. Teefore** confirmed that his hyperthyroidism was the result of Graves disease (see Chapter 19). Graves disease, also known as diffuse toxic goiter, is an autoimmune genetic disorder caused by the generation of human thyroid-stimulating immunoglobulins. These immunoglobulins stimulate enlargement of the thyroid gland (goiter) and excess secretion of the thyroid hormones, T_3 and T_4. As a consequence, Mr. Teefore's heat intolerance and sweating were growing worse with time.

Ivy Sharer, an intravenous drug abuser, appeared to be responding well to her multidrug regimens to treat pulmonary tuberculosis and AIDS (see Chapters 11, 12, 15, and 16). In the past 6 weeks, however, she has developed increasing weakness in her extremities to the point that she has difficulty carrying light objects or walking. Physical examination indicates a diffuse proximal and distal muscle weakness associated with muscle atrophy. The muscles are neither painful on motion nor tender to compression. The blood level of the muscle enzymes, creatine phosphokinase (CK) and aldolase, are normal. An electromyogram (EMG) revealed a generalized reduction in the muscle action potentials, suggestive of a primary myopathic process. Proton spectroscopy of her brain and upper spinal cord showed no anatomic or biochemical abnormalities. The diffuse and progressive skeletal muscle weakness was out of proportion to that expected from her AIDS or her tuberculosis. This information led her physicians to consider the possibility that her skeletal muscle dysfunction might be drug induced.

I. OXIDATIVE PHOSPHORYLATION

Generation of ATP from oxidative phosphorylation requires an electron donor (NADH or FAD(2H)), an electron acceptor (O_2), an intact inner mitochondrial membrane that is impermeable to protons, all the components of the electron transport chain, and ATP synthase. It is regulated by the rate of ATP utilization.

Arlyn Foma, who has a follicular type non-Hodgkin's lymphoma, was being treated with the anthracycline drug doxorubicin (see Chapter 16). During the course of his treatment, he developed biventricular heart failure. Although doxorubicin is a highly effective anticancer agent against a wide variety of human tumors, its clinical use is limited by a specific, cumulative, dose-dependent cardiotoxicity. Impairment of mitochondrial function may play a major role in this toxicity. Doxorubicin binds to cardiolipin, a lipid component of the inner membrane of mitochondria, where it might directly affect components of oxidative phosphorylation. Doxorubicin inhibits succinate oxidation, inactivates cytochrome oxidase, interacts with CoQ, affects ion pumps, and inhibits ATP synthase, resulting in decreased ATP levels and mildly swollen mitochondria. It decreases the ability of the mitochondria to sequester Ca^{2+} and increases free radicals (highly reactive single-electron forms) leading to damage of the mitochondrial membrane (see Chapter 24). It also might affect heart function indirectly through other mechanisms.

A. Overview of Oxidative Phosphorylation

Our understanding of oxidative phosphorylation is based on the chemiosmotic hypothesis, which proposes that the energy for ATP synthesis is provided by an electrochemical gradient across the inner mitochondrial membrane. This electrochemical gradient is generated by the components of the electron transport chain, which pump protons across the inner mitochondrial membrane as they sequentially accept and donate electrons (see Fig. 21.1). The final acceptor is O_2, which is reduced to H_2O.

1. ELECTRON TRANSFER FROM NADH TO O_2

In the electron transport chain, electrons donated by NADH or FAD(2H) are passed sequentially through a series of electron carriers embedded in the inner mitochondrial membrane. Each of the components of the electron transfer chain is oxidized as it accepts an electron, and then reduced as it passes the electrons to the next member of the chain. From NADH, electrons are transferred sequentially through NADH dehydrogenase (complex I), CoQ (coenzyme Q), the cytochrome b-c_1 complex (complex III), cytochrome c, and finally cytochrome c oxidase (complex IV). NADH dehydrogenase, the cytochrome b-c_1 complex and cytochrome c oxidase are each multisubunit protein complexes that span the inner mitochondrial membrane. CoQ is a lipid soluble quinone that is not protein-bound and is free to diffuse in the lipid membrane. It transports electrons from complex I to complex III and is an intrinsic part of the proton pumps for each of these complexes. Cytochome c is a small protein in the inner membrane space that transfers electrons from the b–c_1

complex to cytochrome oxidase. The terminal complex, cytochrome c oxidase, contains the binding site for O_2. As O_2 accepts electrons from the chain, it is reduced to H_2O.

2. THE ELECTROCHEMICAL POTENTIAL GRADIENT

At each of the three large membrane-spanning complexes in the chain, electron transfer is accompanied by proton pumping across the membrane. There is an energy drop of approximately 16 kcal in reduction potential as electrons pass through each of these complexes, which provides the energy required to move protons against a concentration gradient. The membrane is impermeable to protons, so they cannot diffuse through the lipid bilayer back into the matrix. Thus, in actively respiring mitochondria, the intermembrane space and cytosol may be approximately 0.75 pH units lower than the matrix.

The transmembrane movement of protons generates an electrochemical gradient with two components: the membrane potential (the external face of the membrane is charged positive relative to the matrix side) and the proton gradient (the inter membrane space has a higher proton concentration and is therefore more acidic than the matrix) (Fig. 21.2). The electrochemical gradient is sometimes called the proton motive force because it is the energy pushing the protons to re-enter the matrix to equilibrate on both sides of the membrane. The protons are attracted to the more negatively charged matrix side of the membrane, where the pH is more alkaline.

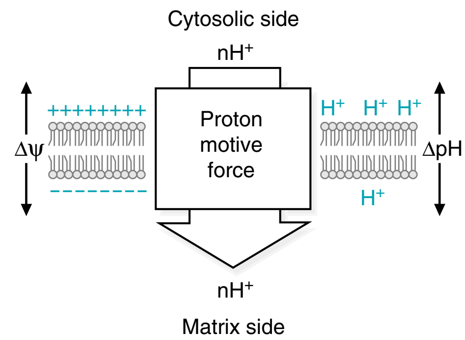


Fig. 21.2. Proton motive force (electrochemical gradient) across the inner mitochondrial membrane. The proton motive force consists of a membrane potential, $\Delta\psi$, and a proton gradient, denoted by ΔpH for the difference in pH across the membrane. The electrochemical potential is called the proton motive force because it represents the potential energy driving protons to return to the more negatively charged alkaline matrix.

3. ATP SYNTHASE

ATP synthase (F_0F_1ATPase), the enzyme that generates ATP, is a multisubunit enzyme containing an inner membrane portion (F_0) and a stalk and headpiece (F_1) that project into the matrix (Fig. 21.3). The 12 **c** subunits in the membrane form a rotor that is attached to a central asymmetric shaft composed of the ε and γ subunits. The headpiece is composed of three $\alpha\beta$ subunit pairs. Each β subunit contains a catalytic site for ATP synthesis. The headpiece is held stationary by a $\boldsymbol{\delta}$ subunit attached to a long **b** subunit connected to subunit **a** in the membrane.

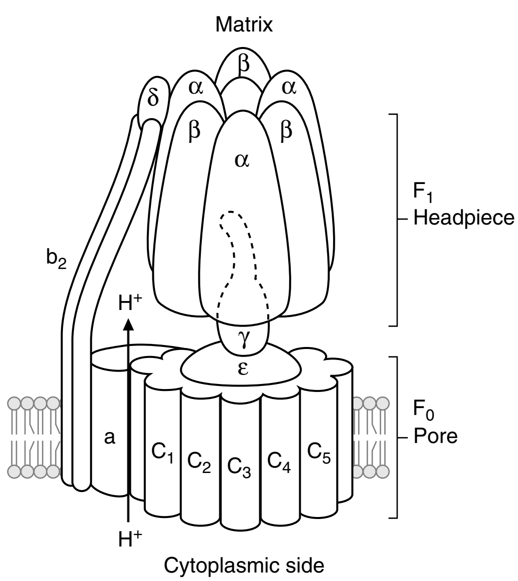


Fig. 21.3. ATP synthase (F_0F_1ATPase).

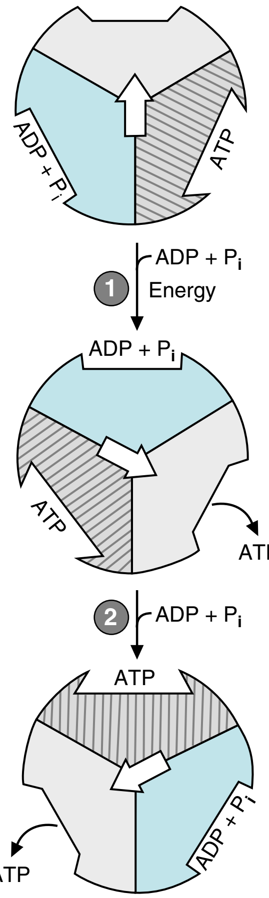


Fig. 21.4. Binding change mechanism for ATP synthesis. The three αβ subunit pairs of the ATP synthase headpiece have binding sites that can exist in three different conformations, depending on the position of the γ stalk subunit. Step 1: When ADP + Pi bind to an open site and the proton influx rotates the γ spindle (represented by the arrow), the conformation of the subunits change and ATP is released from one site. (ATP dissociation is, thus, the energy-requiring step). Bound ADP and Pi combine to form ATP at another site. Step 2: As the ADP + Pi bind to the new open site, and the γ shaft rotates, the conformations of the sites change again, and ATP is released. ADP and Pi combine to form another ATP.

FMN, like FAD, is synthesized from the vitamin riboflavin. It contains the electron-accepting flavin ring structure, but not the adenosine monophosphate (AMP) portion of FAD (see Fig. 19.10). Severe riboflavin deficiency decreases the ability of mitochondria to generate ATP from oxidative phosphorylation due to the lack of FMN in the electron carriers.

The influx of protons through the proton channel turns the rotor. The proton channel is formed by the **c** subunits on one side and the **a** subunit on the other side. Although continuous, it has two offset portions, one portion directly open to the intermembrane space and one portion directly open to the matrix. In the current model, each **c** subunit contains a glutamyl carboxyl group that extends into the proton channel. As this carboxyl group accepts a proton from the intermembrane space, the **c** subunit rotates into the hydrophobic lipid membrane. The rotation exposes a different proton-containing **c** subunit to the portion of the channel directly open to the matrix side. Because the matrix has a lower proton concentration, the glutamyl carboxylic acid group releases a proton into the matrix portion of the channel. Rotation is completed by an attraction between the negatively charged glutamyl residue and a positively charged arginyl group on the **a** subunit.

According to the binding change mechanism, as the asymmetric shaft rotates to a new position, it forms different binding associations with the αβ subunits (Fig. 21.4). The new position of the shaft alters the conformation of one β subunit so that it releases a molecule of ATP and another subunit spontaneously catalyzes synthesis of ATP from inorganic phosphate, one proton, and ADP. Thus, energy from the electrochemical gradient is used to change the conformation of the ATP synthase subunits so that the newly synthesized ATP is released. Twelve **c** subunits are hypothesized, and it takes 12 protons to complete one turn of the rotor and synthesize three ATP.

B. Oxidation–Reduction Components of the Electron Transport Chain

Electron transport to O_2 occurs via a series of oxidation–reduction steps in which each successive component of the chain is reduced as it accepts electrons and oxidized as it passes electrons to the next component of the chain. The oxidation–reduction components of the chain include flavin mononucleotide (FMN), Fe-S centers, CoQ, and Fe in the cytochromes b, c_1, c, a, and a_3. Cu is also a component of cytochromes a and a_3 (Fig. 21.5). With the exception of CoQ, all of these electron acceptors are tightly bound to the protein subunits of the carriers.

The reduction potential of each complex of the chain is at a lower energy level than the previous complex, so that energy is released as electrons pass through each complex. This energy is used to move protons against their concentration gradient, so that they become concentrated on the cytosolic side of the inner membrane.

1. NADH DEHYDROGENASE

NADH dehydrogenase is an enormous 42-subunit complex that contains a binding site for NADH, several FMN and iron-sulfur (Fe-S) center binding proteins, and binding sites for CoQ (see Fig 21.5). An FMN accepts two electrons from NADH and is able to pass single electrons to the Fe-S centers (Fig. 21.6). Fe-S centers, which are able to delocalize single electrons into large orbitals, transfer electrons to and from CoQ. Fe-S centers are also present in other enzyme systems, such as other proteins, which transfer electrons to CoQ, in the cytochrome b–c_1 complex, and in aconitase in the TCA cycle.

2. SUCCINATE DEHYDROGENASE AND OTHER FLAVOPROTEINS

In addition to NADH dehydrogenase, succinic dehydrogenase and other flavoproteins in the inner mitochondrial membrane also pass electrons to CoQ (see Fig. 21.5). Succinate dehydrogenase is part of the TCA cycle. ETF-CoQ oxidoreductase accepts electrons from ETF (**e**lectron **t**ransferring **f**lavoprotein), which acquires them from fatty acid oxidation and other pathways. Both of these flavoproteins have Fe-S centers. α-Glycerophosphate dehydrogenase is a flavoprotein that is part of a shuttle for reoxidizing cytosolic NADH.

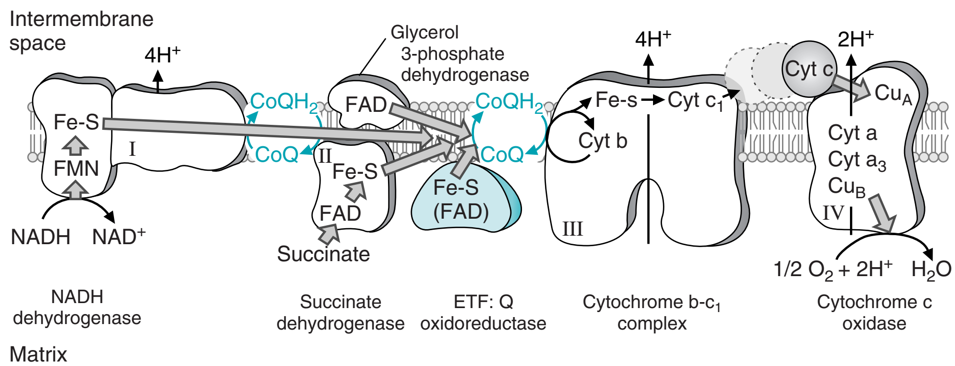


Fig. 21.5. Components of the electron transfer chain. NADH dehydrogenase (complex I) spans the membrane and has a proton pumping mechanism involving CoQ. The electrons go from CoQ to cytochrome b–c1 complex (complex III), and electron transfer does NOT involve complex II. Succinate dehydrogenase (complex II), glycerol 3-phosphate dehydrogenase, and ETF:Q oxidoreductase (shown in blue) all transfer electrons to CoQ, but do not span the membrane and do not have a proton pumping mechanism. As CoQ accepts protons from the matrix side, it is converted to QH_2. Electrons are transferred from complex III to complex IV (cytochrome c oxidase) by cytochrome c, a small cytochrome in the intermembrane space that has reversible binding sites on the b–c_1 complex and cytochrome c oxidase.

The free energy drop between NADH and CoQ of approximately −13 to −14 kcal is able to support movement of four protons. However, the FAD in succinate dehydrogenase (as well as ETF-CoQ oxidoreductase and α-glycerophosphate dehydrogenase) is at roughly the same energy level as CoQ, and there is no energy released as they transfer electrons to CoQ. These proteins do not span the membrane and consequently do not have a proton pumping mechanism.

3. COENZYME Q

CoQ is the only component of the electron transport chain that is not protein bound. The large hydrophobic side chain of 10 isoprenoid units (50 carbons) confers lipid solubility, and CoQ is able to diffuse through the lipids of the inner mitochondrial membrane (Fig. 21.7). When the oxidized quinone form accepts a single electron, it forms a free radical (a compound with a single electron in an orbital). The transfer of single electrons makes it the major site for generation of toxic oxygen free radicals in the body (see Chapter 24).

The long side chain of CoQ has 10 of the 5-carbon isoprenoid units, and is sometimes called CoQ_{10}. It is also called ubiquinone (the quinone found everywhere) because quinones with similar structures are found in all plants and animals. CoQ can be synthesized in the human from precursors derived from carbohydrates and fat. The long isoprenoid side chain is formed in the pathway that produces the isoprenoid precursors of cholesterol. CoQ_{10} is sometimes prescribed for patients recovering from a myocardial infarction, in an effort to increase their exercise capacity.

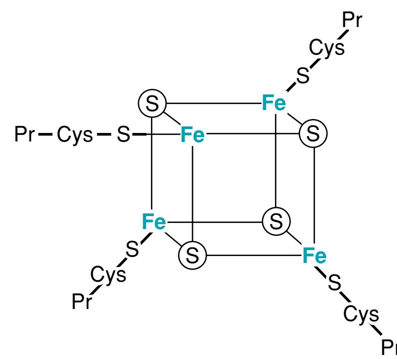


Fig. 21.6. Fe_4S_4 center. In Fe-S centers, the Fe is chelated to free sulfur (S) atoms, and to cysteine sulfhydryl groups on proteins. Other Fe-S centers contain Fe_2S_2. The protein subunits are sometimes called non-heme iron proteins. When these proteins are treated with acid, the free sulfur produces hydrogen sulfide (H_2S)—the familiar smell of rotten eggs.

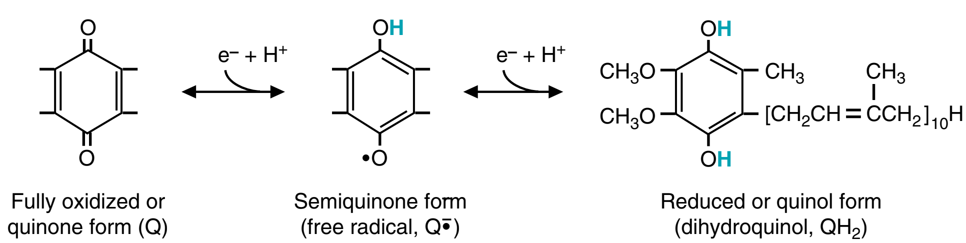


Fig. 21.7. Coenzyme Q contains a quinone with a long lipophilic side chain comprising 10 isoprenoid units (thus, it is sometimes called CoQ_{10}.) CoQ can accept one electron (e^-) to become the half-reduced form, or 2 e^- to become fully reduced.

The semiquinone can accept a second electron and two protons from the matrix side of the membrane to form the fully reduced quinone. The mobility of CoQ in the membrane, its ability to accept one or two electrons, and its ability to accept and donate protons enable it to participate in the proton pumps for both complexes I and III as it shuttles electrons between them (see Section I.C.).

4. CYTOCHROMES

The remainder of the components in the electron transport chain are cytochromes (see Fig. 21.5). Each cytochrome is a protein that contains a bound heme (i.e., an Fe atom bound to a porphyrin nucleus similar in structure to the heme in hemoglobin) (Fig. 21.8).

Because of differences in the protein component of the cytochromes and small differences in the heme structure, each heme has a different reduction potential. The cytochromes of the b-c_1 complex have a higher energy level than those of cytochrome oxidase (a and a_3). Thus, energy is released by electron transfer between complexes III and IV. The iron atoms in the cytochromes are in the Fe^{3+} state. As they accept an electron, they are reduced to Fe^{2+}. As they are reoxidized to Fe^{3+}, the electrons pass to the next component of the electron transport chain.

Although iron deficiency anemia is characterized by decreased levels of hemoglobin and other iron-containing proteins in the blood, the iron-containing cytochromes and Fe-S centers of the electron transport chain in tissues such as skeletal muscle are affected as rapidly. Fatigue in iron deficiency anemia, in patients such as **Ann O'Rexia** (see Chapter 16), results, in part, from the lack of electron transport for ATP production.

5. COPPER (Cu^+) AND THE REDUCTION OF OXYGEN

The last cytochrome complex is cytochrome oxidase, which passes electrons from cytochrome c to O_2 (see Fig. 21.5). It contains cytochromes a and a_3 and the oxygen binding site. A whole oxygen molecule, O_2, must accept four electrons to be reduced to 2 H_2O. Bound copper (Cu^+) ions in the cytochrome oxidase complex facilitate the collection of the four electrons and the reduction of O_2.

Cytochrome oxidase has a much lower K_m for O_2 than myoglobin (the heme-containing intracellular oxygen carrier) or hemoglobin (the heme-containing oxygen transporter in the blood). Thus, O_2 is "pulled" from the erythrocyte to myoglobin, and from myoglobin to cytochrome oxidase, where it is reduced to H_2O.

The iron in the heme in hemoglobin, unlike the iron in the heme of cytochromes, never changes its oxidation state (it is Fe^{2+} in hemoglobin). If the iron in hemoglobin were to become oxidized (Fe^{3+}), the oxygen-binding capacity of the molecule would be lost. Normally, the protein structures binding the heme either protect the iron from oxidation (such as the globin proteins), or allow oxidation to occur (such as happens in the cytochromes). However, in hemoglobin M, a rare hemoglobin variant found in the human population, a tyrosine is substituted for the histidine at position F8 in the normal hemoglobin A. This tyrosine stabilizes the Fe^{3+} form of heme, and these subunits cannot bind oxygen. This is a lethal condition if homozygous.

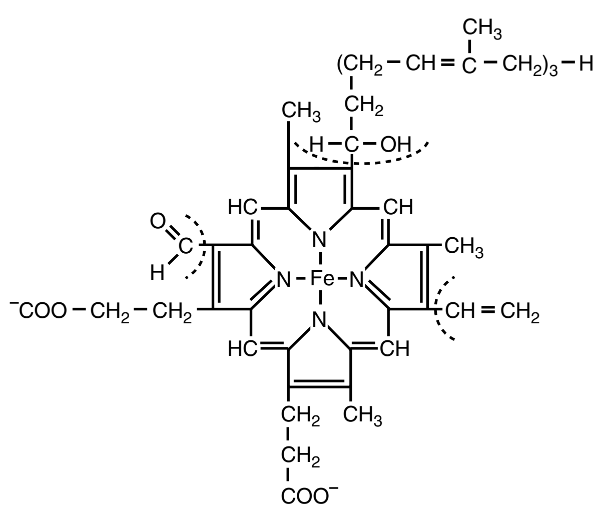


Fig. 21.8. Heme A. Heme A is found in cytochromes a and a_3. Cytochromes are proteins containing a heme chelated with an iron atom. Hemes are derivatives of protoporphyrin IX. Each cytochrome has a heme with different modifications of the side chains (indicated with dashed lines), resulting in a slightly different reduction potential and, consequently, a different position in the sequence of electron transfer.

C. Pumping of Protons

One of the tenets of the chemiosmotic theory is that energy from the oxidation–-reduction reactions of the electron transport chain is used to transport protons from the matrix to the intermembrane space. This proton pumping is generally facilitated by the vectorial arrangement of the membrane spanning complexes. Their structure allows them to pick up electrons and protons on one side of the membrane and release protons on the other side of the membrane as they transfer an electron to the next component of the chain. The direct physical link between proton movement and electron transfer can be illustrated by an examination of the Q cycle for the b-c_1 complex (Fig. 21.9). The Q cycle involves a double cycle of CoQ reduction and oxidation. CoQ accepts two protons at the matrix side together with two electrons; it then releases protons into the intermembrane space while donating one electron back to another component of the cytochrome b-c_1 complex and one to cytochrome c.

The mechanism for pumping protons at the NADH dehydrogenase complex is not well understood, but it involves a Q cycle in which the Fe-S centers and FMN might participate. However, transmembrane proton movement at cytochrome c oxidase probably involves direct transport of the proton through a series of bound water molecules or amino acid side chains in the protein, a mechanism that has been described as a "proton wire."

The significance of the direct link between the electron transfer and proton movement is that one cannot occur without the other. Thus, when protons are not being used for ATP synthesis, the proton gradient and the membrane potential build up. This "proton back-pressure" controls the rate of proton pumping, which controls electron transport and O_2 consumption.

D. Energy Yield from the Electron Transport Chain

The overall free energy release from oxidation of NADH by O_2 is approximately −53 kcal, and from FAD(2H), it is approximately −41 kcal. This $\Delta G^{0\prime}$ is so negative

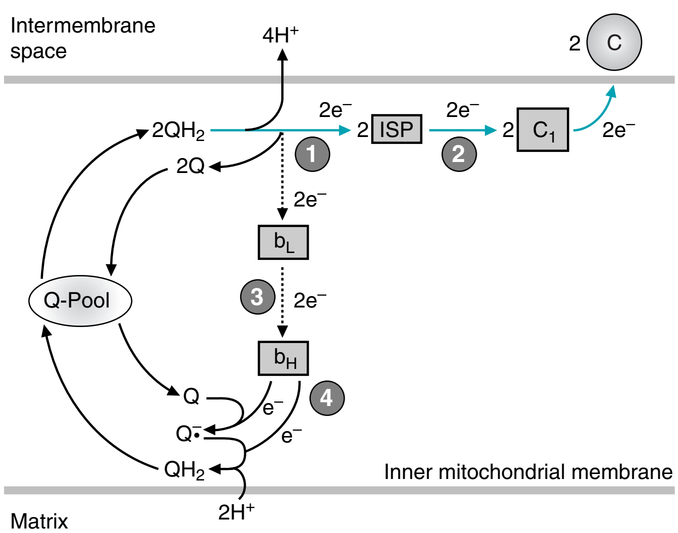


Fig. 21.9. The proton motive Q cycle for the b–c_1 complex. (1) From 2 QH_2, electrons go down two different paths: one path is through an FeS center protein (ISP) toward cytochrome c (shown with blue arrows). Another path is "backward" to one of the b cytochromes, shown with dashed arrows. (2) Electrons are transferred from ISP through cytochrome c_1. Cytochrome c, which is in the intermembrane space, binds to the b–c_1 complex to accept an electron. (3) Returning electrons go through another b cytochrome and are directed toward the matrix. (4) At the matrix side, electrons and $2H^+$ are accepted by Q. Q, Coenzyme Q; $Q\bullet^-$, CoQ semiquinone; QH_2, CoQ hydroquinone.

Electron Transport Chain
Transferring 2 electrons
To Coenzyme Q
Takes Fe-S proteins
And riboflavin, too.

Transferring an electron
Down the E.T. chain
Takes Fe 3 to Fe 2
And back to 3 again.

Transferring 4 electrons
To an oxygen
Takes some Cu^{++} ions
And Fe porphyrin.

Transferring 2 electrons
From NADH to O_2
Pumps just 10 protons
That's the best that it can do

Transferring 2 electrons
From reduced FAD
Pumps only 6 protons
And makes less ATP.

—C.M. Smith

Cora Nari has a lack of oxygen in the anterior and lateral walls of her heart caused by severe ischemia (lack of blood flow) resulting from clots formed at the site of ruptured atherosclerotic plaques. The limited availability of O_2 to act as an electron acceptor will decrease proton pumping and generation of an electrochemical potential gradient across the inner mitochondrial membrane. As a consequence, the rate of ATP generation in her heart will decrease, thereby triggering events that lead to irreversible cell injury.

Intravenous nitroprusside rapidly lowers elevated blood pressure through its direct vasodilating action. Fortunately, it was only required in **Cora Nari**'s case for several hours. During prolonged infusions of 24 to 48 hours or more, nitroprusside is converted to cyanide, an inhibitor of the cytochrome c oxidase complex. Because small amounts of cyanide are detoxified in the liver by conversion to thiocyanate, which is excreted in the urine, the conversion of nitroprusside to cyanide can be monitored by following blood thiocyanate levels.

that the chain is never reversible; we never synthesize oxygen from H_2O. It is so negative that it drives NADH and FAD(2H) formation from the pathways of fuel oxidation, such as the TCA cycle and glycolysis, to completion.

Overall, each NADH donates two electrons, equivalent to the reduction of ½ of an O_2 molecule. A generally (but not universally) accepted estimate of the stoichiometry of ATP synthesis is that four protons are pumped at complex I, four protons at complex III, and two at complex IV. With four protons translocated for each ATP synthesized, an estimated 2.5 ATPs are formed for each NADH oxidized and 1.5 ATPs for each of the other FAD(2H)-containing flavoproteins that donate electrons to CoQ. (This calculation neglects proton requirements for the transport of phosphate and substrates from the cytosol, as well as the basal proton leak.) Thus, only approximately 30% of the energy available from NADH and FAD(2H) oxidation by O_2 is used for ATP synthesis. Some of the remaining energy in the electrochemical potential is used for the transport of anions and Ca^{2+} into the mitochondrion. The remainder of the energy is released as heat. Consequently, the electron transport chain is also our major source of heat.

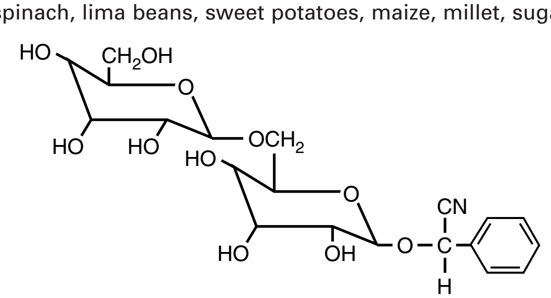
E. Respiratory Chain Inhibition and Sequential Transfer

In the cell, electron flow in the electron transport chain must be sequential from NADH or a flavoprotein all the way to O_2 to generate ATP (see Fig. 21.5). In the absence of O_2, there is no ATP generated from oxidative phosphorylation because electrons back up in the chain. Even complex I cannot pump protons to generate the electrochemical gradient, because every molecule of CoQ already has electrons that it cannot pass down the chain without an O_2 to accept them at the end. The action of the respiratory chain inhibitor cyanide, which binds to cytochrome oxidase, is similar to that of anoxia; it prevents proton pumping by all three complexes. Complete inhibition of the b-c_1 complex prevents pumping at cytochrome

Cyanide binds to the Fe^{3+} in the heme of the cytochrome aa_3 component of cytochrome c oxidase and prevents electron transport to O_2. Mitochondrial respiration and energy production cease, and cell death rapidly occurs. The central nervous system is the primary target for cyanide toxicity. Acute inhalation of high concentrations of cyanide (e.g., smoke inhalation during a fire) provokes a brief central nervous system stimulation rapidly followed by convulsion, coma, and death. Acute exposure to lower amounts can cause lightheadedness, breathlessness, dizziness, numbness, and headaches.

Cyanide is present in the air as hydrogen cyanide (HCN), in soil and water as cyanide salts (e.g., NaCN), and in foods as cyanoglycosides. Most of the cyanide in the air usually comes from automobile exhaust. Examples of populations with potentially high exposures include active and passive smokers, people who are exposed to house or other building fires, residents who live near cyanide- or thiocyanate-containing hazardous waste sites, and workers involved in a number of manufacturing processes (e.g., photography or pesticide application.)

Cyanoglycosides such as amygdalin are present in edible plants such as almonds, pits from stone fruits (e.g., apricots, peaches, plums, cherries), sorghum, cassava, soybeans, spinach, lima beans, sweet potatoes, maize, millet, sugar cane, and bamboo shoots.



Amygdalin, a cyanoglycoside

HCN is released from cyanoglycosides by β-glucosidases present in the plant or in intestinal bacteria. Small amounts are inactivated in the liver principally by rhodanase, which converts it to thiocyanate.

In the United States, toxic amounts have been ingested as ground apricot pits, either due to health food promotion or as a treatment for cancer. The drug Laetrile (amygdalin) was used as a cancer therapeutic agent, although it was banned in the United States because it was ineffective and potentially toxic. Commercial fruit juices made from unpitted fruit could provide toxic amounts of cyanide, particularly in infants or children. In countries in which cassava is a dietary staple, improper processing results in retention of its high cyanide content at potentially toxic levels.

oxidase because there is no donor of electrons; it prevents pumping at complex I because there is no electron acceptor. Although complete inhibition of any one complex inhibits proton pumping at all of the complexes, partial inhibition of proton pumping can occur when only a fraction of the molecules of a complex contain bound inhibitor. The partial inhibition results in a partial decrease of the maximal rate of ATP synthesis.

Q: A decreased activity of the electron transport chain can result from inhibitors as well as from mutations in mtDNA and nuclear DNA. Why does an impairment of the electron transport chain result in lactic acidosis?

II. OXPHOS DISEASES

Clinical diseases involving components of oxidative phosphorylation (referred to as OXPHOS diseases) are among the most commonly encountered degenerative diseases. The clinical pathology may be caused by gene mutations in either mitochondrial DNA (mtDNA) or nuclear DNA (nDNA) that encode proteins required for normal oxidative phosphorylation.

Oxidative phosphorylation (OXPHOS) is responsible for producing most of the ATP that our cells require. The genes responsible for the polypeptides that comprise the OXPHOS complexes within the mitochondria are located within either the nuclear DNA (nDNA) or the mitochondrial DNA (mtDNA). A broad spectrum of human disorders (the OXPHOS diseases) may result from genetic mutations or nongenetic alterations (spontaneous mutations) in either the nDNA or the mtDNA. Increasingly, such changes appear to be responsible for at least some aspects of common disorders, such as Parkinson's disease, dilated and hypertrophic cardiomyopathies, diabetes mellitus, Alzheimer disease, depressive disorders, and a host of less well-known clinical entities.

A. Mitochondrial DNA and OXPHOS Diseases

The mtDNA is a small 16,569 nucleotide pair, double-stranded, circular DNA. It encodes 13 subunits of the complexes involved in oxidative phosphorylation: 7 of the 42 subunits of complex I (NADH dehydrogenase complex), 1 of the 11 subunits of complex III (cytochrome b-c_1 complex), 3 of 13 of the subunits of complex IV (cytochrome oxidase), and two subunits of the F_0 portion ATP–synthase complex. In addition, mtDNA encodes the necessary components for translation of its mRNA: a large and small rRNA and 22 tRNAs. Mutations in mtDNA have been identified as deletions, duplications, or point mutations (Table 21.1).

The genetics of mutations in mtDNA are defined by *maternal inheritance*, *replicative segregation*, *threshold expression*, *a high mtDNA mutation rate*, and the *accumulation of somatic mutations with age*. The *maternal inheritance* pattern reflects the exclusive transmission of mtDNA from the mother to her children. The egg contains approximately 300,000 molecules of mtDNA packaged into mitochondria. These are

Table 21.1 Examples of OXPHOS Diseases Arising from mtDNA Mutations

Syndrome	Characteristic Symptoms	mtDNA Mutation
I. mtDNA rearrangements in which genes are deleted or duplicated.		
Kearns-Sayre syndrome	Onset before 20 years of age, characterized by opthalmoplegia, atypical retinitis pigmentosa, mitochondrial myopathy, and one of the following: cardiac conduction defect, cerebellar syndrome, or elevated CSF proteins.	Deletion of contiguous segments of tRNA and OXPHOS polypeptides, or duplication mutations consisting of tandemly arranged normal mtDNA and an mtDNA with a deletion mutation.
Pearson syndrome	Systemic disorder of oxidative phosphorylation that predominantly affects bone marrow	(same as above)
II. mtDNA point mutations in tRNA or ribosomal RNA genes		
MERRF (**m**yoclonic **e**pilepsy and **r**agged-**r**ed **f**iber disease)	Progressive myoclonic epilepsy, a mitochondrial myopathy with ragged-red fibers, and a slowly progressive dementia. Onset of symptoms: late childhood to adult	$tRNA^{lys}$
MELAS (mitochondrial **m**yopathy, **e**ncephalo-myopathy, **l**actic **a**cidosis, and **s**troke-like episodes)	Progressive neurodengenerative disease characterized by stoke-like episodes first occurring between 5 and 15 years of age and a mitochondrial myopathy	80-90% mutations in $tRNA^{leu}$
III. mtDNA missense mutations in OXPHOS polypeptides		
Leigh disease (subacute necrotizing encephaolpathy)	Mean age of onset, 1.5–5 years; clinical manifestations included optic atrophy, opththalmoplegia, nystagmus, respiratory abnormalities, ataxia, hypotonia, spasticity, and developmental delay or regression.	7–20 % of cases have mutations in Fo subunits of F_0-F_1-ATPase.
LHON (**L**eber **h**ereditary **o**ptic **n**europathy)	Late onset, acute optic atrophy.	90% of European and Asian cases result from mutation in NADH dehydrogenase.

retained during fertilization, whereas those of the sperm do not enter the egg or are lost. Usually, some mitochondria are present with the mutant mtDNA and some with normal (wild-type) DNA. As cells divide during mitosis and meiosis mitochondria replicate by fission, but various amounts of mitochondria with mutant and wild-type DNA are distributed to each daughter cell (*replicative segregation*). Thus, any cell can have a mixture of mitochondria, each with mutant or wild-type mtDNAs (heteroplasmy). The mitotic and meiotic segregation of the heteroplasmic mtDNA mutation results in variable oxidative phosphorylation deficiencies between patients with the same mutation, and even among a patient's own tissues.

The disease pathology usually becomes worse with age, because a small amount of normal mitochondria might confer normal function and exercise capacity while the patient is young. As the patient ages, somatic (spontaneous) mutations in mtDNA accumulate from the generation of free radicals within the mitochondria (see Chapter 24). These mutations frequently become permanent, partly because mtDNA does not have access to the same repair mechanisms available for nuclear DNA (*high mutation rate*). Even in normal individuals, somatic mutations result in a decline of oxidative phosphorylation capacity with age (*accumulation of somatic mutations with age*). At some stage, the ATP-generating capacity of a tissue falls below the tissue-specific threshold for normal function (*threshhold expression*). In general, symptoms of these defects would appear in one or more of the tissues with the highest ATP demands: nervous tissue, heart, skeletal muscle, and kidney.

B. Other Genetic Disorders of Oxidative Phosphorylation

Genetic mutations also have been reported for mitochondrial proteins encoded by nuclear DNA. Most of the estimated 1,000 proteins required for oxidative phosphorylation are encoded by nuclear DNA, whereas mtDNA encodes only 13 subunits of the oxidative phosphorylation complexes (including ATP synthase). Nuclear DNA encodes the additional 70 or more subunits of the oxidative phosphorylation complexes, as well as adenine nucleotide translocase (ANT) and other anion translocators. Coordinate regulation of expression of nuclear and mtDNA, import of proteins into the mitochondria, assembly of the complexes, and regulation of mitochondrial fission are nuclear encoded.

Nuclear DNA mutations differ from mtDNA mutations in several important respects. These mutations do not show a pattern of maternal inheritance but are usually autosomal recessive. The mutations are uniformly distributed to daughter cells and therefore are expressed in all tissues containing the allele for a particular tissue-specific isoform. However, phenotypic expression still will be most apparent in tissues with high ATP requirements.

III. COUPLING OF ELECTRON TRANSPORT AND ATP SYNTHESIS

The electrochemical gradient couples the rate of the electron transport chain to the rate of ATP synthesis. Because electron flow requires proton pumping, electron flow cannot occur faster than protons are used for ATP synthesis (coupled oxidative phosphorylation) or returned to the matrix by a mechanism that short circuits the ATP synthase pore (uncoupling).

The nuclear respiratory factors (NRF-1 and NRF-2) are nuclear transcription factors that bind to and activate promotor regions of the nuclear genes encoding subunits of the respiratory chain complexes, including cytochrome c. They also activate the transcription of the nuclear gene for the mitochondrial transcription factor (mTF)-A. The protein product of this gene translocates into the mitochondrial matrix, where it stimulates transcription and replication of the mitochondrial genome.

A: The effect of inhibition of electron transport is an impaired oxidation of pyruvate, fatty acids, and other fuels. In many cases, the inhibition of mitochondrial electron transport results in higher than normal levels of lactate and pyruvate in the blood and an increased lactate/pyruvate ratio. NADH oxidation requires the completed transfer of electrons from NADH to O_2, and a defect anywhere along the chain will result in the accumulation of NADH and decrease of NAD^+. The increase in $NADH/NAD^+$ inhibits pyruvate dehydrogenase and causes the accumulation of pyruvate. It also increases the conversion of pyruvate to lactate, and elevated levels of lactate appear in the blood. A large number of genetic defects of the proteins in respiratory chain complexes have, therefore, been classified together as "congenital lactic acidosis."

A patient experienced spontaneous muscle jerking (myoclonus) in her mid-teens, and her condition progressed over 10 years to include debilitating myoclonus, neurosensory hearing loss, dementia, hypoventilation, and mild cardiomyopathy. Energy metabolism was affected in the central nervous system, heart, and skeletal muscle, resulting in lactic acidosis. A history indicated that the patient's mother, her grandmother, and two maternal aunts had symptoms involving either nervous or muscular tissue (clearly a case of maternal inheritance). However, no other relative had identical symptoms. The symptoms and history of the patient are those of myoclonic epileptic ragged red fiber disease (MERRF). The affected tissues (central nervous system and muscle) are two of the tissues with the highest ATP requirements. Most cases of MERRF are caused by a point mutation in mitochondrial $tRNA^{lys}$ ($mtRNA^{lys}$). The mitochondria, obtained by muscle biopsy, are enlarged and show abnormal patterns of cristae. The muscle tissue also shows ragged red fibers.

Q: How does shivering generate heat?

A. Regulation through Coupling

As ATP chemical bond energy is used by energy-requiring reactions, ADP and Pi concentrations increase. The more ADP present to bind to the ATP synthase, the greater will be proton flow through the ATP synthase pore, from the intermembrane space to the matrix. Thus, as ADP levels rise, proton influx increases, and the electrochemical gradient decreases (Fig 21.10). The proton pumps of the electron transport chain respond with increased proton pumping and electron flow to maintain the electrochemical gradient. The result is increased O_2 consumption. The increased oxidation of NADH in the electron transport chain and the increased concentration of ADP stimulate the pathways of fuel oxidation, such as the TCA cycle, to supply more NADH and FAD(2H) to the electron transport chain. For example, during exercise, we use more ATP for muscle contraction, consume more oxygen, oxidize more fuel (which means burn more calories), and generate more heat from the electron transport chain. If we rest, and the rate of ATP utilization decreases, proton influx decreases, the electrochemical gradient increases, and proton "back-pressure" decreases the rate of the electron transport chain. NADH and FAD(2H) cannot be oxidized as rapidly in the electron transport chain, and consequently, their build-up inhibits the enzymes that generate them.

The system is poised to maintain very high levels of ATP at all times. In most tissues, the rate of ATP utilization is nearly constant over time. However, in skeletal muscles, the rates of ATP hydrolysis change dramatically as the muscle goes from rest to rapid contraction. Even under these circumstances, ATP concentration decreases by only approximately 20% because it is so rapidly regenerated. In the heart, Ca^{2+} activation of TCA cycle enzymes provides an extra push to NADH generation, so that neither ATP nor NADH levels fall as ATP demand is increased. The electron transport chain has a very high capacity and can respond very rapidly to any increase in ATP utilization.

A: Shivering results from muscular contraction, which increases the rate of ATP hydrolysis. As a consequence of proton entry for ATP synthesis, the electron transport chain is stimulated. Oxygen consumption increases, as does the amount of energy lost as heat by the electron transport chain.

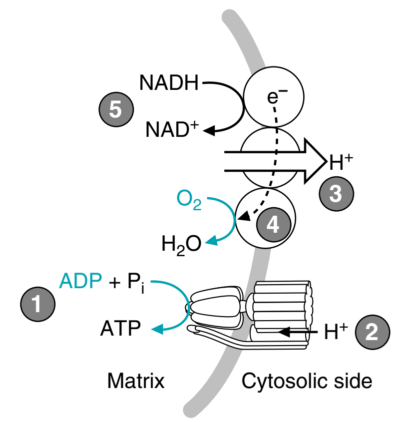


Fig. 21.10. The concentration of ADP (or the phosphate potential -[ATP]/[ADP][P_i]) controls the rate of oxygen consumption. (1) ADP is phosphorylated to ATP by ATP synthase. (2) The release of the ATP requires proton flow through ATP synthase into the matrix. (3) The use of protons from the intermembrane space for ATP synthesis decreases the proton gradient. (4) As a result, the electron transport chain pumps more protons, and oxygen is reduced to H_2O. (5) As NADH donates electrons to the electron transport chain, NAD^+ is regenerated and returns to the TCA cycle or other NADH-producing pathways.

B. Uncoupling ATP Synthesis from Electron Transport

When protons leak back into the matrix without going through the ATP synthase pore, they dissipate the electrochemical gradient across the membrane without generating ATP. This phenomenon is called "uncoupling" oxidative phosphorylation. It occurs with chemical compounds, known as uncouplers, and it occurs physiologically with uncoupling proteins that form proton conductance channels through the membrane. Uncoupling of oxidative phosphorylation results in increased oxygen consumption and heat production as electron flow and proton pumping attempt to maintain the electrochemical gradient.

1. CHEMICAL UNCOUPLERS OF OXIDATIVE PHOSPHORYLATION

Chemical uncouplers, also known as proton ionophores, are lipid-soluble compounds that rapidly transport protons from the cytosolic to the matrix side of the inner mitochondrial membrane (Fig. 21.11). Because the proton concentration is higher in the intermembrane space than in the matrix, uncouplers pick up protons from the intermembrane space. Their lipid solubility enables them to diffuse through the inner mitochondrial membrane while carrying protons and release these

A skeletal muscle biopsy performed on **Ivy Sharer** indicated proliferation of subsarcolemmal mitochondria with degeneration of muscle fibers (ragged-red fibers) in approximately 55% of the total fibers observed. An analysis of mitochondrial (mtDNA) indicated no genetic mutations but did show a moderate quantitative depletion of mtDNA.

Ivy Sharer's AIDS was being treated with zidovudine (AZT), which also can act as an inhibitor of the mitochondrial DNA polymerase (polymerase γ). A review of the drugs' potential adverse effects showed that, rarely, it may cause varying degrees of mtDNA depletion in different tissues, including skeletal muscle. The depletion may cause a severe mitochondrial myopathy, including "ragged-red fiber" accumulation within the skeletal muscle cells associated with ultrastructural abnormalities in their mitochondria.

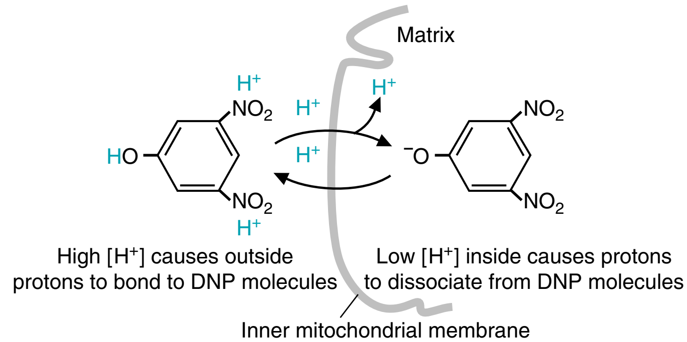


Fig. 21.11. Dinitrophenol (DNP) is lipid soluble and can therefore diffuse across the membrane. It has a dissociable proton with a pK_a near 7.2. Thus, in the intermembrane space where $[H^+]$ is high (pH low), DNP picks up a proton, which it carries across the membrane. At the lower proton concentration of the matrix, the H^+ dissociates. As a consequence, cells cannot maintain their electrochemical gradient or synthesize ATP. DNP was once recommended in the United States as a weight loss drug, based on the principle that decreased [ATP] and increased electron transport stimulate fuel oxidation. However, several deaths resulted from its use.

protons on the matrix side. The rapid influx of protons dissipates the electrochemical potential gradient; therefore, the mitochondria are unable to synthesize ATP. Eventually, mitochondrial integrity and function are lost.

2. UNCOUPLING PROTEINS AND THERMOGENESIS

Uncoupling proteins (UCPs) form channels through the inner mitochondrial membrane that are able to conduct protons from the intermembrane space to the matrix, thereby short-circuiting ATP synthase.

UCP1 (thermogenin) is associated with heat production in brown adipose tissue. The major function of brown adipose tissue is nonshivering thermogenesis, whereas the major function of white adipose tissue is the storage of triacylglycerols in white lipid droplets. The brown color arises from the large number of mitochondria that participate. Human infants, who have little voluntary control over their environment and may kick their blankets off at night, have brown fat deposits along the neck, the breastplate, between the scapulae, and around the kidneys to protect them from cold. However, there is very little brown fat in most adults.

In response to cold, sympathetic nerve endings release norepinephrine, which activates a lipase in brown adipose tissue that releases fatty acids from triacylglycerols (Fig 21.12). Fatty acids serve as a fuel for the tissue (i.e., are oxidized to generate the electrochemical potential gradient and ATP) and participate directly in the proton conductance channel by activating UCP1 along with reduced CoQ. When UCP1 is activated by purine nucleotides, fatty acids, and CoQ, it transports protons from the cytosolic side of the inner mitochondrial membrane back into the mitochondrial matrix without ATP generation. Thus, it partially uncouples oxidative phosphorylation and generates additional heat.

The uncoupling proteins exist as a family of proteins: UCP1 (thermogenin) is expressed in brown adipose tissue; UCP2 is found in most cells, UCP3 is found principally in skeletal muscle; UCP4 and UCP5 are found in the brain. These are highly regulated proteins that, when activated, increase the amount of energy from fuel oxidation that is being released as heat. Because they affect metabolic efficiency, differences in the level of UCPs (particularly skeletal muscle UCP3) may contribute to the tendency toward obesity in some individuals or populations. UCPs also may decrease the amount of reduced CoQ available to form oxygen free radicals, thereby decreasing mitochondrial and cell injury.

Salicylate, which is a degradation product of aspirin in the human, is lipid soluble and has a dissociable proton. In high concentrations, as in salicylate poisoning, salicylate is able to partially uncouple mitochondria. The decline of ATP concentration in the cell and consequent increase of AMP in the cytosol stimulates glycolysis. The overstimulation of the glycolytic pathway (see Chapter 22) results in increased levels of lactic acid in the blood and a metabolic acidosis. Fortunately, **Dennis Veere** did not develop this consequence of aspirin poisoning (see Chapter 4).

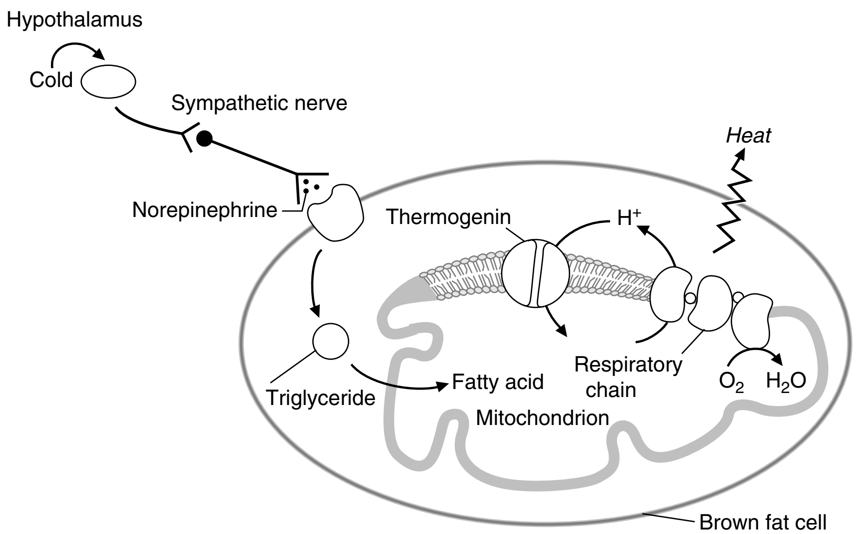


Fig. 21.12. Brown fat is a tissue specialized for nonshivering thermogenesis. Cold or excessive food intake stimulates the release of norepinephrine from the sympathetic nerve endings. As a result, a lipase is activated that releases fatty acids for oxidation. The proton conductance protein, thermogenin, is activated, and protons are brought into the matrix. This stimulates the electron transport chain, which increases its rate of NADH and FAD(2H) oxidation and produces more heat.

3. PROTON LEAK AND RESTING METABOLIC RATE

A low level of proton leak across the inner mitochondrial membrane occurs in our mitochondria all of the time, and our mitochondria thus are normally partially uncoupled. It has been estimated that more than 20% of our resting metabolic rate is the energy expended to maintain the electrochemical gradient dissipated by our basal proton leak (also referred to as global proton leak). Some of the proton leak results from permeability of the membrane associated with proteins embedded in the lipid bilayer. An unknown amount may result from uncoupling proteins.

IV. TRANSPORT THROUGH INNER AND OUTER MITOCHONDRIAL MEMBRANES

Most of the newly synthesized ATP that is released into the mitochondrial matrix must be transported out of the mitochondria, where it is used for energy-requiring processes such as active ion transport, muscle contraction, or biosynthetic reactions. Likewise, ADP, phosphate, pyruvate, and other metabolites must be transported into the matrix. This requires transport of compounds through both the inner and outer mitochondrial membranes.

The inner and outer membranes differ substantially in their lipid content. The inner mitochondrial membrane is 22% cardiolipin and contains almost no cholesterol. The outer membrane resembles the cell membrane; it is less than 3% cardiolipin and approximately 45% cholesterol.

A. Transport through the Inner Mitochondrial Membrane

The inner mitochondrial membrane forms a tight permeability barrier to all polar molecules, including ATP, ADP, P_i, anions such as pyruvate, and cations such as Ca^{2+}, H^+, and K^+. Yet the process of oxidative phosphorylation depends on rapid and continuous transport of many of these molecules across the inner mitochondrial membrane (Fig. 21.13). Ions and other polar molecules are transported across the inner mitochondrial membrane by specific protein translocases that nearly balance charge during the transport process. Most of the exchange transport is a form of active transport that generally uses energy from the electrochemical potential gradient, either the membrane potential or the proton gradient.

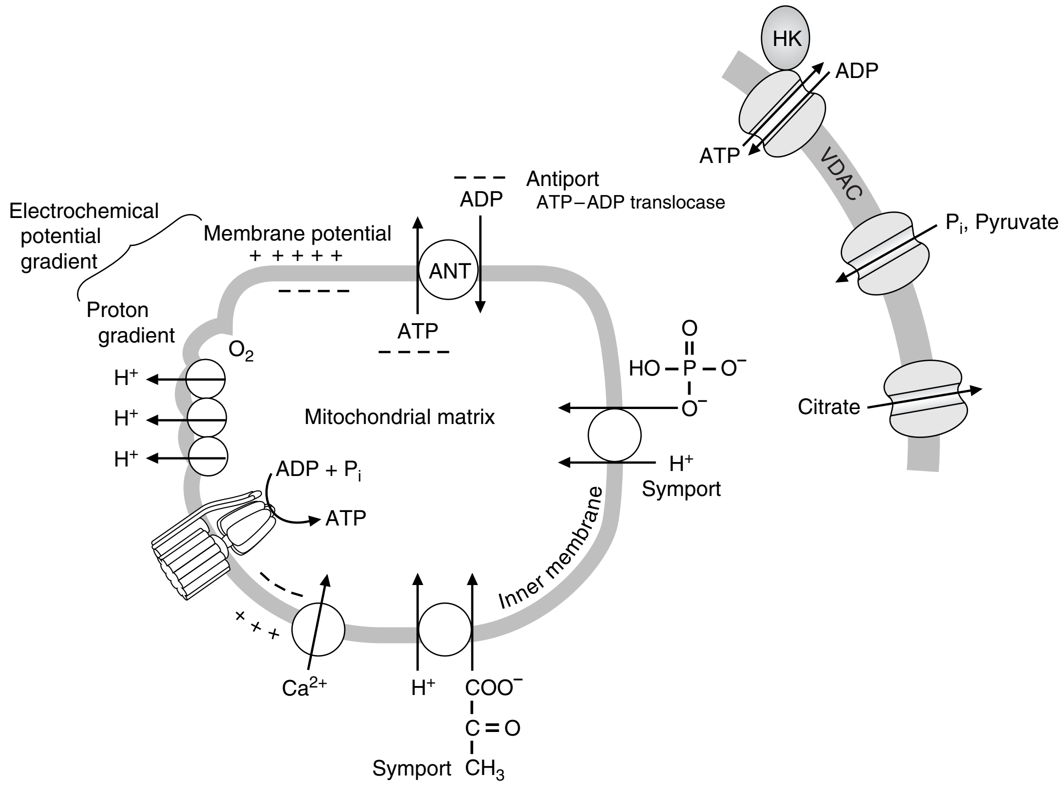


Fig. 21.13. Transport of compounds across the inner and outer mitochondrial membranes. The electrochemical potential gradient drives the transport of ions across the inner mitochondrial membrane on specific translocases. Each translocase is composed of specific membrane-spanning helices that bind only specific compounds (ANT; adenine nucleotide translocase). In contrast, the outer membrane contains relatively large unspecific pores called VDAC (voltage-dependent anion channels) through which a wide range of ions diffuse. These bind cytosolic proteins such as hexokinase (HK), which enables HK to have access to newly exported ATP.

ANT is an antiport, an exchange protein that translocates one ion in exchange for a molecule of similar charge. In contrast, the phosphate transporter and the pyruvate transporter are symports, which are translocases that co-transport two molecules of opposite charge. The Ca^{2+} channel is called a uniporter because no other ions are involved. All of the metabolites entering or leaving the TCA cycle are transported across the inner mitochondrial membrane by specific transport proteins. This includes the dicarboxylate transporter (phosphate–malate exchange), the tricarboxylate transporter (citrate–malate exchange), the aspartate-glutamate transporter, and the malate-α-ketoglutarate transporter.

Investigators reported finding antibodies against cardiac ATP-ADP translocase in an individual who died of a viral cardiomyopathy. How could these antibodies result in death?

ATP-ADP translocase (also called ANT for **a**denine **n**ucleotide **t**ranslocase) transports ATP formed in the mitochondrial matrix to the intermembrane space in a specific 1:1 exchange for ADP produced from energy-requiring reactions outside of the mitochondria (see Fig. 21.13). Because ATP contains four negative charges and ADP only three, the exchange is promoted by the electrochemical potential gradient, because the net effect is the transport of one negative charge from the matrix to the cytosol. Similar antiports exist for most metabolic anions. In contrast, inorganic phosphate and pyruvate are transported into the mitochondrial matrix on specific transporters called symports together with a proton. A specific transport protein for Ca^{2+} uptake, called the Ca^{2+} uniporter, is driven by the electrochemical potential gradient, which is negatively charged on the matrix side of the membrane relative to the cytosolic side.

B. Transport through the Outer Mitochondrial Membrane

Whereas the inner mitochondrial membrane is highly impermeable, the outer mitochondrial membrane is permeable to compounds with a molecular weight up to approximately 6,000 daltons because it contains large nonspecific pores called voltage-dependent anion channels (VDAC) that are formed by mitochondrial porins (see Fig. 21.13). Unlike most transport proteins, which are membrane-spanning helices with specific binding sites, VDACs are composed of porin homodimers that form a β-barrel with a relatively large nonspecific water-filled pore through the center. These channels are "open" at low transmembrane potential, with a preference for anions such as phosphate, chloride, pyruvate, citrate, and adenine nucleotides.

A: As ATP is hydrolyzed during muscular contraction, ADP is formed. This ADP must exchange into the mitochondria on ATP-ADP translocase to be converted back to ATP. Inhibition of ATP-ADP translocase results in rapid depletion of the cytosol ATP levels and loss of cardiac contractility.

VDACs thus facilitate translocation of these anions between the intermembrane space and the cytosol. A number of cytosolic kinases, such as the hexokinase that initiates glycolysis, bind to the cytosolic side of the channel, where they have ready access to newly synthesized ATP.

C. The Mitochondrial Permeability Transition Pore

The mitochondrial permeability transition involves the opening of a large nonspecific pore (called MPTP, the **m**itochondrial **p**ermeability **t**ransition **p**ore) through the inner mitochondrial membrane and outer membranes at sites where they form a junction (Fig. 21.14). The basic components of the mitochondrial permeability transition pore are adenine nucleotide translocase (ANT), the voltage-dependant anion channel (VDAC), and cyclophilin D (which is a cis-trans isomerase for the proline peptide bond). Normally ANT is a closed pore that functions specifically in a 1:1 exchange of matrix ATP for ADP in the intermembrane space. However, increased mitochondrial matrix Ca^{2+}, excess phosphate, or ROS (reactive oxygen species that form oxygen or oxygen–nitrogen radicals) can activate opening of the pore. Conversely, ATP on the cytosolic side of the pore (and possibly a pH below 7.0) and a membrane potential across the inner membrane protect against pore opening. Opening of the MPTP can be triggered by ischemia (hypoxia), which results in a temporary lack of O_2 for maintaining the proton gradient and ATP synthesis. When the proton gradient is not being generated by the electron transport chain, ATP synthase runs backward and hydrolyzes ATP in an attempt to restore the gradient, thus rapidly depleting cellular levels of ATP. As ATP is hydrolyzed to ADP, the ADP is converted to adenine, and the nucleotide pool is no longer able to protect against pore opening. This can lead to a downward spiral of cellular events. A lack of ATP for maintaining the low intracellular Ca^{2+} can contribute to pore opening. When the MPTP pore opens, protons will flood in, and maintaining a proton gradient becomes impossible. Anions and cations enter the matrix, mitochondrial swelling ensues, and the mitochondria become irreversibly damaged. The result is cell lysis and death (necrosis).

In addition to a normal role in movement of molecules across the mitochondrial outer membrane, VDACs appear to have roles in processes leading to cell death. These roles are influenced by proteins that bind to the VDACs. By forming a component of the mitochondrial permeability transition pore, VDACs may contribute to events leading to cell necrosis. Alternatively, binding of pro-apoptotic or anti-apoptotic members of the Bcl-2 family may change the permeability of the outer membrane so as to either favor or block events leading to programmed cell death (apoptosis).

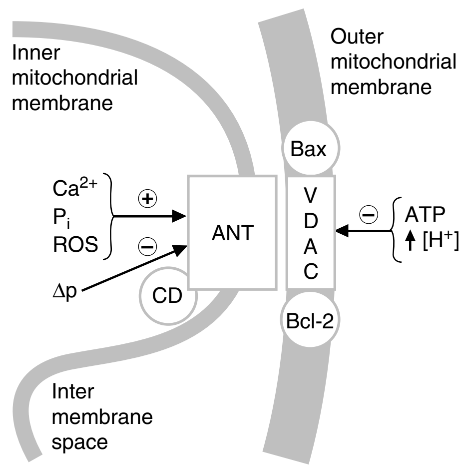


Fig. 21.14. The mitochondrial permeability transition pore (MPTP). In the MPTP, ANT is thought to complex with VDAC. The conformation of ANT is regulated by cyclophilin D (CD), and Ca^{2+}. VDACs bind a number of proteins, including Bcl2 and Bax, which regulate apoptosis. The change to an open pore is activated by Ca^{2+}, depletion of adenine nucleotides, and oxygen radicals (ROS) that alter SH groups. It is inhibited by the electrochemical potential gradient (Δp), by cytosolic ATP, and by a low cytosolic pH.

CLINICAL COMMENTS

Cora Nari. Thrombolysis stimulated by intravenous recombinant tissue plasminogen activator (TPA) restored O_2 to **Cora Nari's** heart muscle and successfully decreased the extent of ischemic damage. The rationale for the use of TPA within 4 to 6 hours after the onset of a myocardial infarction relates

As infusion of TPA lysed the clot blocking blood flow to **Cora Nari**'s heart, oxygenated blood was reintroduced into the ischemic heart. Although oxygen may rapidly restore the capacity to generate ATP, it often increases cell death, a phenomenon called ischemia–reperfusion injury.

During ischemia, a number of factors may protect heart cells against ireversible injury and cell death until oxygen is reintroduced. The stimulation of anaerobic glycolysis in the cytosol generates ATP without oxygen as glucose is converted to lactic acid. Lactic acid decreases cytosolic pH. Both cytosolic ATP and a lowering of the pH protect against opening of the MPTP. In addition, Ca^{2+} uptake by mitochondria requires a membrane potential, and it is matrix Ca^{2+} that activates opening of the MPTP. Thus, depending on the severity of the ischemic insult, the MPTP may not open, or may open and reseal, until oxygen is reintroduced. Then, depending on the sequence of events, reestablishment of the proton gradient, mitochondrial uptake of Ca^{2+}, or an increase of pH above 7.0 may activate the MPTP before the cell has recovered. In addition, the reintroduction of O_2 generates oxygen free radicals, particularly through free radical forms of CoQ in the electron transport chain. These also may open the MPTP. The role of free radicals in ischemia–reperfusion injury is covered in more detail in Chapter 24.

to the function of the normal intrinsic fibrinolytic system (see Chapter 45). This system is designed to dissolve unwanted intravascular clots through the action of the enzyme plasmin, a protease that digests the fibrin matrix within the clot. TPA stimulates the conversion of plasminogen to its active form, plasmin. The result is a lysis of the thrombus and improved blood flow through the previously obstructed vessel, allowing fuels and oxygen to reach the heart cells. The human TPA protein used in Mrs. Nari is produced by recombinant DNA technology (see Chapter 17). This treatment rapidly restored oxygen supply to the heart.

In addition to increased transcription of genes encoding TCA cycle enzymes and certain other enzymes of fuel oxidation, thyroid hormones increase the level of UCP2 and UCP3. In hyperthyroidism, the efficiency with which energy is derived from the oxidation of these fuels is significantly less than normal. As a consequence of the increased rate of the electron transport chain, hyperthyroidism results in increased heat production. Patients with hyperthyroidism, such as **X.S. Teefore,** complain of constantly feeling hot and sweaty.

X.S. Teefore. Mr. Teefore could be treated with antithyroid drugs, by subtotal resection of the thyroid gland, or with radioactive iodine. Successful treatment normalizes thyroid hormone secretion, and all of the signs, symptoms, and metabolic alterations of hyperthyroidism quickly subside.

Ivy Sharer. In the case of **Ivy Sharer,** a diffuse myopathic process was superimposed on her AIDS and her pulmonary tuberculosis, either of which could have caused progressive weakness. In addition, she could have been suffering from a congenital mtDNA myopathy, symptomatic only as she ages. A systematic diagnostic process, however, finally led her physician to conclude that her myopathy was caused by a disorder of oxidative phosphorylation induced by her treatment with zidovudine (AZT). Fortunately, when AZT was discontinued, Ivy's myopathic symptoms gradually subsided. A repeat skeletal muscle biopsy performed 4 months later showed that her skeletal muscle cell mtDNA had been restored to normal and that she had experienced a reversible drug-induced disorder of oxidative phosphorylation.

BIOCHEMICAL COMMENTS

Mitochondria and Apoptosis The loss of mitochondrial integrity is a major route initiating apoptosis (see Chapter 18, section V). The intermembrane space contains procaspases −2, −3, and −9, which are proteolytic enzymes that are in the zymogen form (i.e., must be proteolytically cleaved to be active). It also contains apoptosis initiating factor (AIF) and caspase-activated DNAase (CAD). Cytochrome c, which is loosely bound to the outer mitochondrial membrane, may also enter the intermembrane space when the electrochemical potential gradient is lost. The release of cytochrome c and the other proteins into the cytosol initiates apoptosis.

But how are cytochrome c and the other proteins released? The VDAC pore is not large enough to allow the passage of proteins. A number of theories have been proposed, each supported and contradicted by experimental evidence. One is that Bax (a member of the Bcl-2 family of proteins that forms an ion channel in the outer mitochondrial membrane) allows the entry of ions into the intermembrane space, causing swelling of this space and rupture of the outer mitochondrial membrane. Another theory is that Bax and VDAC (which is known to bind Bax and other Bcl-2 family members) combine to form an extremely large pore, much larger than formed by either alone. Finally, it is possible that the MPTP or ANT participate in rupture of the outer membrane, but close in a way that still provides the energy for apoptosis.

Cytochrome c allosterically activates cytosolic apoptosis activating factor (Apaf-1), which activates the initiator caspase-9. Caspase-9, in turn, activates effector caspases-3, -6, and -7 through proteolytic cleavage, which then degrade cytoplasmic proteins. AIF, which has a nuclear targeting sequence, is transported into the nucleus, where it initiates chromatin condensation and degradation. Caspase-2 and caspase-9 also activate CAD, which migrates to the nucleus and hydrolyzes bonds in nuclear DNA.

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REVIEW QUESTIONS—CHAPTER 21

1. Consider the following experiment. Carefully isolated liver mitochondria are incubated in the presence of a limiting amount of malate. Three minutes after adding the substrate, cyanide is added, and the reaction is allowed to proceed for another 7 minutes. At this point, which of the following components of the electron transfer chain will be in an oxidized state?

   (A) Complex I
   (B) Complex II
   (C) Complex III
   (D) Coenzyme Q
   (E) Cytochrome C

2. Consider the following experiment. Carefully isolated liver mitochondria are placed in a weakly buffered solution. Malate is added as an energy source, and an increase in oxygen consumption confirms that the electron transfer chain is functioning properly within these organelles. Valinomycin and potassium are then added to the mitochondrial suspension. Valinomycin is a drug that allows potassium ions to freely cross the inner mitochondrial membrane. What is the effect of valinomycin on the proton motive force that had been generated by the oxidation of malate?

   (A) The proton motive force will be reduced to a value of zero
   (B) There will be no change in the proton motive force
   (C) The proton motive force will be increased
   (D) The proton motive force will be decreased, but to a value greater than zero
   (E) The proton motive force will be decreased to a value less than zero

3. Dinitrophenol acts as an uncoupler of oxidative phosphorylation by which of the following mechanisms?

   (A) Activating the H^+-ATPase
   (B) Activating coenzyme Q
   (C) Blocking proton transport across the inner mitochondrial membrane
   (D) Allowing for proton exchange across the inner mitochondrial membrane
   (E) Enhancing oxygen transport across the inner mitochondrial membrane

4. A 25-year-old female presents with chronic fatigue. A series of blood tests are ordered, and the results suggest that her red blood cell count is low because of iron deficiency anemia. Such a deficiency would lead to fatigue because of which of the following?

   (A) Her decrease in Fe-S centers is impairing the transfer of electrons in the electron transport chain.
   (B) She is not producing as much H_2O in the electron transport chain, leading to dehydration, which has resulted in fatigue.
   (C) Iron forms a chelate with NADH and FAD(2H) that is necessary for them to donate their electrons to the electron transport chain.
   (D) Iron acts as a cofactor for α-ketoglutarate DH in the TCA cycle, a reaction required for the flow of electrons through the electron transport chain.
   (E) Iron accompanies the protons that are pumped from the mitochondrial matrix to the cytosolic side of the inner mitochondrial membrane. Without iron, the proton gradient cannot be maintained to produce adequate ATP.

5. Which of the following would be expected for a patient with an OXPHOS disease?

   (A) A high ATP:ADP ratio in the mitochondria
   (B) A high NADH:NAD^+ ratio in the mitochondria
   (C) A deletion on the X chromosome
   (D) A high activity of complex II of the electron transport chain
   (E) A defect in the integrity of the inner mitochondrial membrane

22 Generation of ATP from Glucose: Glycolysis

***Glucose** is the universal fuel for human cells. Every cell type in the human is able to generate adenosine triphosphate (ATP) from glycolysis, the pathway in which glucose is oxidized and cleaved to form pyruvate. The importance of glycolysis in our fuel economy is related to the availability of glucose in the blood, as well as the ability of glycolysis to generate ATP in both the presence and absence of O_2. Glucose is the major sugar in our diet and the sugar that circulates in the blood to ensure that all cells have a continuous fuel supply. The brain uses glucose almost exclusively as a fuel.*

***Glycolysis** begins with the phosphorylation of glucose to glucose 6-phosphate (**glucose-6-P**) by **hexokinase** (HK). In subsequent steps of the pathway, one glucose-6-P molecule is oxidized to two **pyruvate** molecules with generation of two molecules of **NADH** (Fig. 22.1). A net generation of two molecules of ATP occurs through direct transfer of **high-energy phosphate** from intermediates of the pathway to ADP (**substrate level phosphorylation**).*

*Glycolysis occurs in the **cytosol** and generates cytosolic NADH. Because NADH cannot cross the inner mitochondrial membrane, its reducing equivalents are transferred to the electron transport chain by either the **malate-aspartate shuttle** or the **glycerol 3-phosphate shuttle** (see Fig. 22.1). Pyruvate is then oxidized completely to CO_2 by pyruvate dehydrogenase and the TCA cycle. Complete **aerobic oxidation** of glucose to CO_2 can generate approximately **30 to 32 moles of ATP per mole of glucose**.*

*When cells have a limited supply of oxygen (e.g., kidney medulla), or few or no mitochondria (e.g., the red cell), or greatly increased demands for ATP (e.g., skeletal muscle during high-intensity exercise), they rely on **anaerobic glycolysis** for generation of ATP. In anaerobic glycolysis, **lactate dehydrogenase** oxidizes the NADH generated from glycolysis by reducing pyruvate to **lactate** (Fig. 22.2). Because O_2 is not required to reoxidize the NADH, the pathway is referred to as anaerobic. The energy yield from anaerobic glycolysis (2 moles of ATP per mole of glucose) is much lower than the yield from aerobic oxidation. The lactate (lactic acid) is released into the blood. Under pathologic conditions that cause **hypoxia,** tissues may generate enough lactic acid to cause **lactic acidemia**.*

*In each cell, glycolysis is regulated to ensure that **ATP homeostasis** is maintained, without using more glucose than necessary. In most cell types, **hexokinase (HK)**, the first enzyme of glycolysis, is inhibited by glucose 6-phosphate (see Fig. 22.1). Thus, glucose is not taken up and phosphorylated by a cell unless glucose-6-P enters a metabolic pathway, such as glycolysis or glycogen synthesis. The control of glucose-6-P entry into glycolysis occurs at phosphofructokinase-1(**PFK-1**), the rate-limiting enzyme of the pathway. PFK-1 is **allosterically inhibited** by **ATP** and **allosterically activated** by **AMP**. AMP increases in the cytosol as ATP is hydrolyzed by energy-requiring reactions.*

For glucose 6-phosphate and other sugar phosphoesters, the phosphate group will be denoted with "P," as in glucose-6-P.

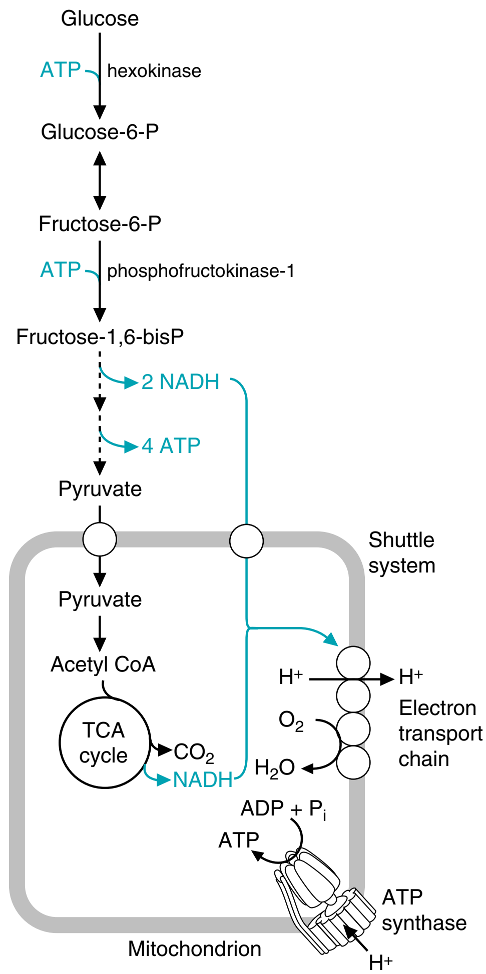


Fig. 22.1. Overview of glycolysis and the TCA cycle.

Glycolysis has functions in addition to ATP production. For example, in liver and adipose tissue, this pathway generates pyruvate as a precursor for ***fatty acid biosynthesis****. Glycolysis also provides precursors for the synthesis of compounds such as amino acids and 5-carbon sugar phosphates.*

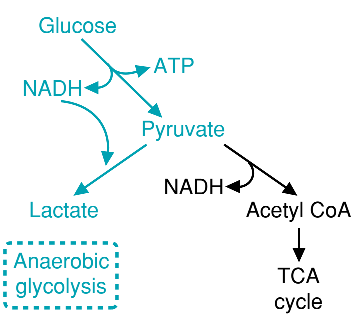


Fig. 22.2. Anaerobic glycolysis (shown in blue). The conversion of glucose to lactate generates 2 ATP from substrate-level phosphorylation. Because there is no net generation of NADH, there is no need for O_2, and, thus, the pathway is anaerobic.

THE WAITING ROOM

Lopa Fusor is a 68-year-old woman who is admitted to the hospital emergency room with very low blood pressure (80/40 mm Hg) caused by an acute hemorrhage from a previously diagnosed ulcer of the stomach. Lopa's bleeding stomach ulcer has reduced her effective blood volume severely enough to compromise her ability to perfuse (deliver blood to) her tissues. She is, therefore, a "low perfuser." She is also known to have chronic obstructive pulmonary disease (COPD) as a result of 42 years of smoking two packs of cigarettes per day. Her respiratory rate is rapid and labored, her skin is cold and clammy, and her lips are slightly blue (cyanotic). She appears anxious and moderately confused.

As appropriate emergency measures are taken to stabilize her and elevate her blood pressure, blood is sent for immediate blood typing and cross-matching, so that blood transfusions can be started. A battery of laboratory tests are ordered, including venous hemoglobin, hematocrit, and lactate levels, and arterial blood pH, partial pressures of oxygen (pO_2) and carbon dioxide (pCO_2), bicarbonate, and oxygen saturation. Results show that the hemorrhaging and COPD have resulted in hypoxemia, with decreased oxygen delivery to her tissues, and both a respiratory and metabolic acidosis.

Otto Shape, a 26-year-old medical student, had gained weight during his first sedentary year in medical school. During his second year, he began watching his diet, jogging for an hour 4 times each week, and playing tennis twice a week. He has decided to compete in a 5-km race. To prepare for the race, he begins training with wind sprints, bouts of alternately running and walking.

Ivan Applebod is a 56-year-old morbidly obese accountant (see Chapters 1–3). He decided to see his dentist because he felt excruciating pain in his teeth when he ate ice cream. He really likes sweets and keeps hard candy in his pocket. The dentist noted from Mr. Applebod's history that he had numerous cavities as a child in his baby teeth. At this visit, the dentist found cavities in two of Mr.Applebod's teeth.

The hematocrit (the percentage of the volume of blood occupied by packed red blood cells) and hemoglobin content (g hemoglobin in 100 mL blood) are measured to determine whether the oxygen-carrying capacity of the blood is adequate. They can be decreased by conditions that interfere with erythropoiesis (synthesis of red blood cells in bone marrow), such as iron deficiency. They also can be decreased during chronic bleeding, but not during immediate acute hemorrhage, if interstitial fluid replaces the lost blood volume and dilutes out the red blood cells. The pCO_2 and pO_2 are the partial pressures of CO_2 and O_2 in the blood. The pO_2 and oxygen saturation determine whether adequate oxygen is available for tissues. Measurement of the pCO_2 and bicarbonate can distinguish between a metabolic and a respiratory acidosis (see Chapter 4).

I. GLYCOLYSIS

Glycolysis is one of the principle pathways for generating ATP in cells and is present in all cell types. The central role of glycolysis in fuel metabolism is related to its ability to generate ATP with, and without, oxygen. The oxidation of glucose to pyruvate generates ATP from substrate-level phosphorylation (the transfer of phosphate from high-energy intermediates of the pathway to ADP) and NADH. Subsequently, the pyruvate may be oxidized to CO_2 in the TCA cycle and ATP generated from electron transfer to oxygen in oxidative phosphorylation. However, if the pyruvate and NADH from glycolysis are converted to lactate (anaerobic glycolysis), ATP can be generated in the absence of oxygen, via substrate-level phosphorylation.

Glucose is readily available from our diet, internal glycogen stores, and the blood. Carbohydrate provides 50% or more of the calories in most diets, and glucose is the major carbohydrate. Other dietary sugars, such as fructose and galactose, are oxidized by conversion to intermediates of glycolysis. Glucose is stored in cells as glycogen, which can provide an internal source of fuel for glycolysis in emergency situations (e.g., decreased supply of fuels and oxygen during ischemia, a low blood flow). Insulin and other hormones maintain blood glucose at a constant level (glucose homeostasis), thereby ensuring that glucose is always available to cells that depend on glycolysis for generation of ATP.

In addition to serving as an anaerobic and aerobic source of ATP, glycolysis is an anabolic pathway that provides biosynthetic precursors. For example, in liver and adipose tissue, this pathway generates pyruvate as a precursor for fatty acid biosynthesis. Glycolysis also provides precursors for the synthesis of compounds such as amino acids and ribose-5-phosphate, the precursor of nucleotides. The integration of glycolysis with other anabolic pathways is discussed in Chapter 36.

After a high-carbohydrate meal, glucose is the major fuel for almost all tissues. Exceptions include intestinal mucosal cells, which transport glucose from the gut into the blood, and cells in the proximal convoluted tubule of the kidney, which return glucose from the renal filtrate to the blood. During fasting, the brain continues to oxidize glucose because it has a limited capacity for the oxidation of fatty acids or other fuels. Cells also continue to use glucose for the portion of their ATP generation that must be met by anaerobic glycolysis, due to either a limited oxygen supply or a limited capacity for oxidative phosphorylation (e.g., the red blood cell).

A. The Reactions of Glycolysis

The glycolytic pathway, which cleaves 1 mole of glucose to 2 moles of the 3-carbon compound pyruvate, consists of a preparative phase and an ATP-generating phase. In the initial preparative phase of glycolysis, glucose is phosphorylated

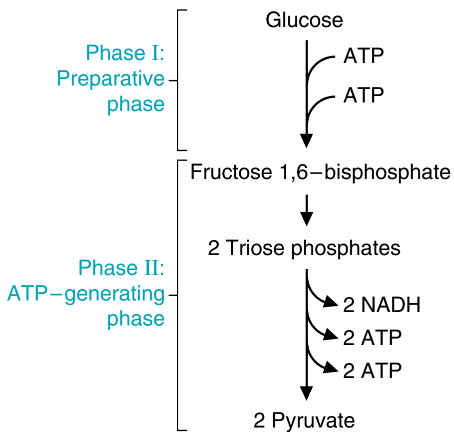


Fig. 22.3. Phases of the glycolytic pathway.

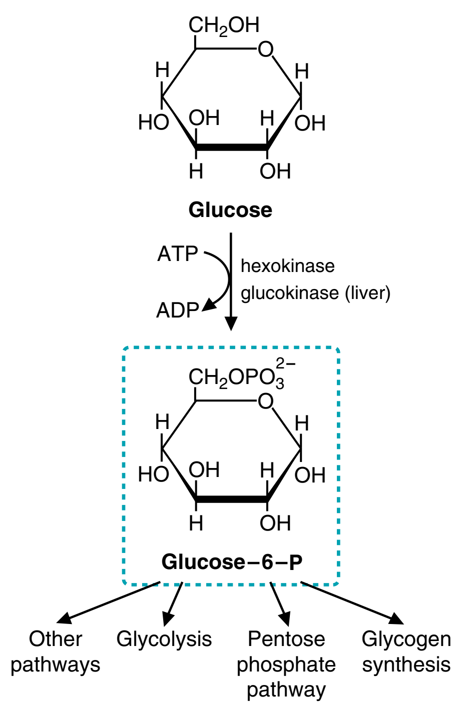


Fig. 22.4. Glucose 6-phosphate metabolism.

Hexokinases, other kinases, and many other enzymes that catalyze reactions involving the hydrolysis of ATP require Mg^{2+}. The Mg^{2+} forms a complex with the phosphate groups of ATP. Kinases also require K^{+}.

twice by ATP and cleaved into two triose phosphates (Fig. 22.3). The ATP expenditure in the beginning of the preparative phase is sometimes called "priming the pump," because this initial utilization of 2 moles of ATP/ mole of glucose results in the production of 4 moles of ATP/mole of glucose in the ATP-generating phase.

In the ATP-generating phase, glyceraldehyde 3-phosphate (a triose phosphate) is oxidized by NAD^+ and phosphorylated using inorganic phosphate. The high-energy phosphate bond generated in this step is transferred to ADP to form ATP. The remaining phosphate is also rearranged to form another high-energy phosphate bond that is transferred to ADP. Because there were 2 moles of triose phosphate formed, the yield from the ATP-generating phase is 4 ATP and 2 NADH. The result is a net yield of 2 moles of ATP, 2 moles of NADH, and 2 moles of pyruvate per mole of glucose.

1. CONVERSION OF GLUCOSE TO GLUCOSE 6-PHOSPHATE

Glucose metabolism begins with transfer of a phosphate from ATP to glucose to form glucose-6-P (Fig. 22.4). Phosphorylation of glucose commits it to metabolism within the cell because glucose-6-P cannot be transported back across the plasma membrane. The phosphorylation reaction is irreversible under physiologic conditions because the reaction has a high negative $\Delta G^{0\prime}$. Phosphorylation does not, however, commit glucose to glycolysis.

Glucose-6-P is a branchpoint in carbohydrate metabolism. It is a precursor for almost every pathway that uses glucose, including glycolysis, the pentose phosphate pathway, and glycogen synthesis. From the opposite point of view, it also can be generated from other pathways of carbohydrate metabolism, such as glycogenolysis (breakdown of glycogen), the pentose phosphate pathway, and gluconeogenesis (the synthesis of glucose from non-carbohydrate sources).

Hexokinases, the enzymes that catalyze the phosphorylation of glucose, are a family of tissue-specific isoenzymes that differ in their kinetic properties. The isoenzyme found in liver and β cells of the pancreas has a much higher K_m than other hexokinases and is called glucokinase. In many cells, some of the hexokinase is bound to porins in the outer mitochondrial membrane (voltage-dependent anion channels; see Chapter 21), which gives these enzymes first access to newly synthesized ATP as it exits the mitochondria.

2. CONVERSION OF GLUCOSE-6-P TO THE TRIOSE PHOSPHATES

In the remainder of the preparative phase of glycolysis, glucose-6-P is isomerized to fructose 6-phosphate (fructose-6-P), again phosphorylated, and subsequently cleaved into two 3-carbon fragments (Fig 22.5). The isomerization, which positions a keto group next to carbon 3, is essential for the subsequent cleavage of the bond between carbons 3 and 4.

The next step of glycolysis, phosphorylation of fructose-6-P to fructose 1,6-bisphosphate (fructose-1,6-bisP) by phosphofructokinase-1 (PFK-1), is generally considered the first committed step of the pathway. This phosphorylation requires ATP and is thermodynamically and kinetically irreversible. Therefore, PFK-1 irrevocably commits glucose to the glycolytic pathway. PFK-1 is a regulated enzyme in cells, and its regulation controls the entry of glucose into glycolysis. Like hexokinase, it exists as tissue-specific isoenzymes whose regulatory properties match variations in the role of glycolysis in different tissues.

Fructose-1,6-bisP is cleaved into two phosphorylated 3-carbon compounds (triose phosphates) by aldolase (see Fig. 22.5). Dihydroxyacetone phosphate (DHAP) is isomerized to glyceraldehyde 3-phosphate (glyceraldehyde-3-P), which is a triose phosphate. Thus, for every mole of glucose entering glycolysis, 2 moles of glyceraldehyde-3-P continue through the pathway.

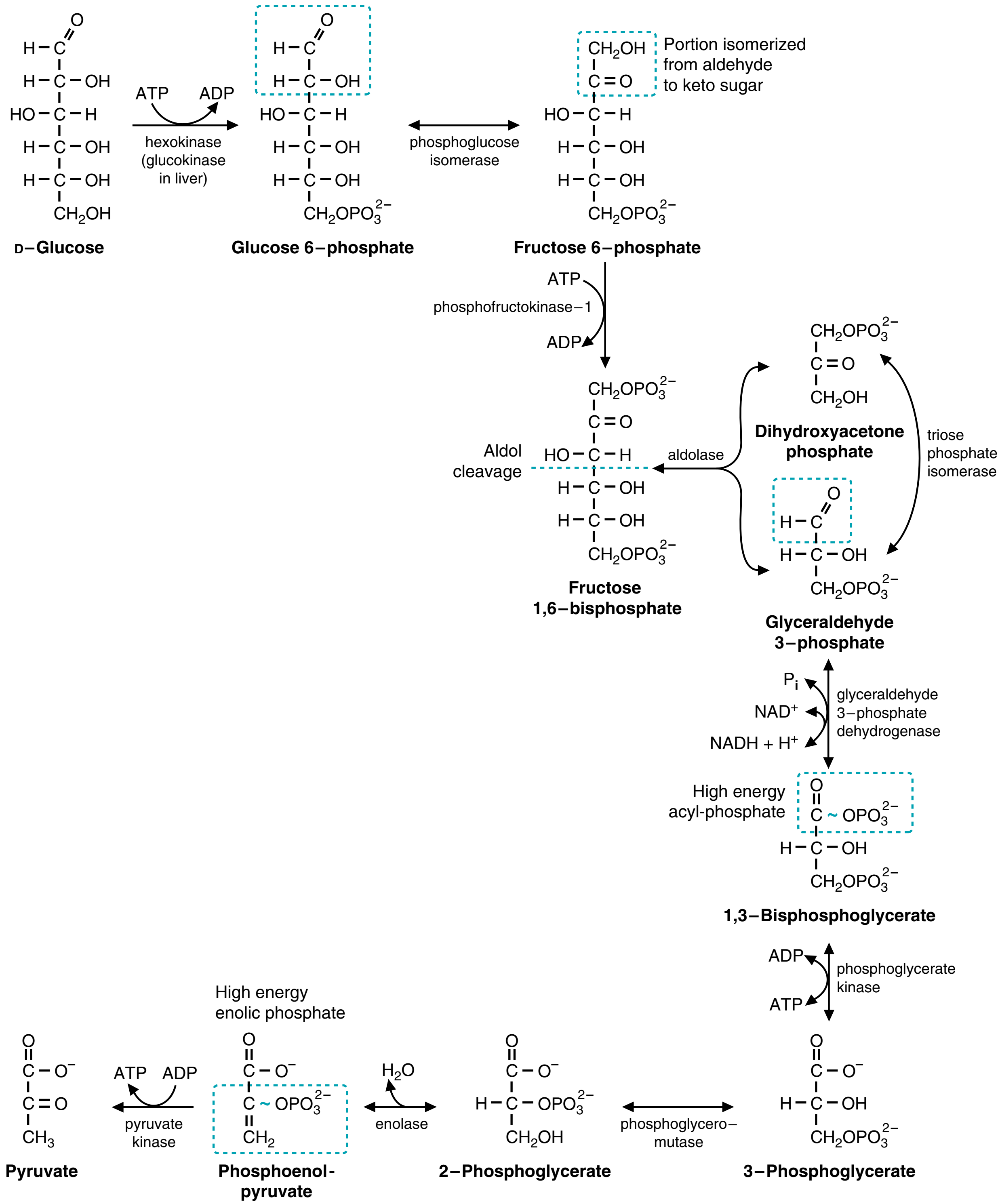


Fig. 22.5. Reactions of glycolysis. High-energy phosphates are shown in blue.

Aldolase is named for the mechanism of the forward reaction, which is an aldol cleavage, and the mechanism of the reverse reaction, which is an aldol condensation. The enzyme exists as tissue-specific isoenzymes, which all catalyze the cleavage of fructose 1,6-bisphosphate but differ in their specificities for fructose 1-P. The enzyme uses a lysine residue at the active site to form a covalent bond with the substrate during the course of the reaction. Inability to form this covalent linkage inactivates the enzyme.

3. OXIDATION AND SUBSTRATE LEVEL PHOSPHORYLATION

In the next part of the glycolytic pathway, glyceraldehyde-3-P is oxidized and phosphorylated so that subsequent intermediates of glycolysis can donate phosphate to ADP to generate ATP. The first reaction in this sequence, catalyzed by glyceraldehyde-3-P dehydrogenase, is really the key to the pathway (see Fig. 22.5). This enzyme oxidizes the aldehyde group of glyceraldehyde-3-P to an enzyme-bound carboxyl group and transfers the electrons to NAD^+ to form NADH. The oxidation step is dependent on a cysteine residue at the active site of the enzyme, which forms a high-energy thioester bond during the course of the reaction. The high-energy intermediate immediately accepts an inorganic phosphate to form the high-energy acyl phosphate bond in 1,3-bisphosphoglycerate, releasing the product from the cysteine residue on the enzyme. This high-energy phosphate bond is the start of substrate-level phosphorylation (the formation of a high-energy phosphate bond where none previously existed, without the utilization of oxygen).

In the next reaction, the phosphate in this bond is transferred to ADP to form ATP by 3-phosphoglycerate kinase. The energy of the acyl phosphate bond is high enough (~13 kcal/mole) so that transfer to ADP is an energetically favorable process. 3-phosphoglycerate is also a product of this reaction.

To transfer the remaining low-energy phosphoester on 3-phosphoglycerate to ADP, it must be converted into a high-energy bond. This conversion is accomplished by moving the phosphate to the second carbon (forming 2-phosphoglycerate) and then removing water to form phosphoenolpyruvate (PEP). The enolphosphate bond is a high-energy bond (its hydrolysis releases approximately 14 kcal/mole of energy), so the transfer of phosphate to ADP by pyruvate kinase is energetically favorable (see Fig. 22.5). This final reaction converts PEP to pyruvate.

Kinases transfer a phosphate from ATP to another compound. Hexokinase transfers a phosphate to glucose or another hexose to form a hexose phosphate. 3-Phosphoglycerate kinase is named for the reaction that is the reverse of glycolysis, transfer of phosphate from ATP to 3-phosphoglycerate to form 1,3-bisphosphoglycerate. Pyruvate kinase is also named for the reverse reaction (phosphorylation of pyruvate by ATP), although this direction does not occur under physiologic conditions.

4. SUMMARY OF THE GLYCOLYTIC PATHWAY

The overall net reaction in the glycolytic pathway is:

$$\text{Glucose} + 2NAD^+ + 2P_i + 2ADP \rightarrow 2\text{Pyruvate} + 2NADH + 4H^+ + 2ATP + 2H_2O$$

The pathway occurs with an overall negative $\Delta G^{0'}$ of approximately –22 kcal. Therefore, it cannot be reversed without the expenditure of energy.

B. Oxidative Fates of Pyruvate and NADH

The NADH produced from glycolysis must be continuously reoxidized back to NAD^+ to provide an electron acceptor for the glyceraldehyde-3-P dehydrogenase reaction and prevent product inhibition. Without oxidation of this NADH, glycolysis cannot continue. There are two alternate routes for oxidation of cytosolic NADH (Fig. 22.6). One route is aerobic, involving shuttles that transfer reducing equivalents across the mitochondrial membrane and ultimately to the electron transport chain and oxygen (see Fig. 22.6A). The other route is anaerobic (without the use of oxygen). In anaerobic glycolysis, NADH is reoxidized in the cytosol by lactate dehydrogenase, which reduces pyruvate to lactate (see Fig. 22.6B).

The fate of pyruvate depends on the route used for NADH oxidation. If NADH is reoxidized in a shuttle system, pyruvate can be used for other pathways, one of which is oxidation to acetyl-CoA and entry into the TCA cycle for complete oxidation. Alternatively, in anaerobic glycolysis, pyruvate is reduced to lactate and diverted away from other potential pathways. Thus, the use of the shuttle systems allows for more ATP to be generated than by anaerobic glycolysis by both oxidizing the cytoplasmically derived NADH in the electron transport chain and by allowing pyruvate to be oxidized completely to CO_2.

The reason that shuttles are required for the oxidation of cytosolic NADH by the electron transport chain is that the inner mitochondrial membrane is impermeable

The confusion experienced by **Lopa Fusor** in the emergency room is caused by an inadequate delivery of oxygen to the brain. Neurons have very high ATP requirements, and most of this ATP is provided by aerobic oxidation of glucose to pyruvate in glycolysis, and pyruvate oxidation to CO_2 in the TCA cycle. The brain has little or no capacity to oxidize fatty acids, and, therefore, its glucose consumption is high (approximately 125–150 g/day in the adult). Its oxygen demands are also high. If cerebral oxygen supply were completely interrupted, the brain would last only 10 seconds. The only reason consciousness lasts longer during anoxia or asphyxia is that there is still some oxygen in the lungs and in circulating blood. A decrease of blood flow to approximately ½ of the normal rate results in a loss of consciousness.

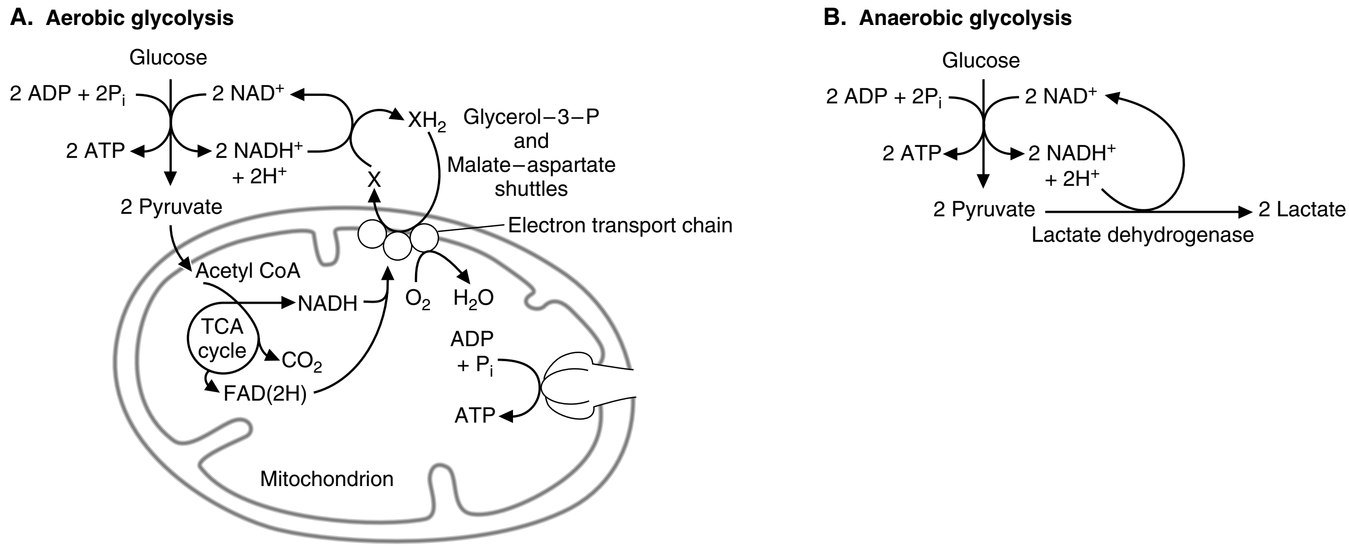


Fig. 22.6. Alternate fates of pyruvate. **A.** The pyruvate produced by glycolysis enters mitochondria and is oxidized to CO_2 and H_2O. The reducing equivalents in NADH enter mitochondria via a shuttle system. **B.** Pyruvate is reduced to lactate in the cytosol, thereby using the reducing equivalents in NADH.

to NADH, and no transport protein exists that can directly translocate NADH across this membrane. Consequently, NADH is reoxidized to NAD^+ in the cytosol by a reaction that transfers the electrons to DHAP in the glycerol 3-phosphate (glycerol-3-P) shuttle and oxaloacetate in the malate–aspartate shuttle. The NAD^+ that is formed in the cytosol returns to glycolysis while glycerol-3-P or malate carry the reducing equivalents that are ultimately transferred across the inner mitochondrial membrane. Thus, these shuttles transfer electrons and not NADH per se.

1. GLYCEROL 3–PHOSPHATE SHUTTLE

The glycerol 3–phosphate shuttle is the major shuttle in most tissues. In this shuttle, cytosolic NAD^+ is regenerated by cytoplasmic glycerol 3-phosphate dehydrogenase, which transfers electrons from NADH to DHAP to form glycerol 3-phosphate (Fig. 22.7). Glycerol 3-phosphate then diffuses through the outer mitochondrial membrane to the inner mitochondrial membrane, where the electrons are donated to a membrane-bound flavin adenive dinucleofide (FAD)-containing glycerophosphate dehydrogenase. This enzyme, like succinate dehydrogenase, ultimately donates electrons to CoQ, resulting in an energy yield of approximately 1.5 ATP from oxidative phosphorylation. Dihydroxyacetone phosphate returns to the cytosol to continue the shuttle. The sum of the reactions in this shuttle system is simply:

$$NADH_{cytosol} + H^+ + FAD_{mitochondria} \rightarrow NAD^+{}_{cytosol} + FAD(2H)_{mitochondria}$$

2. MALATE–ASPARTATE SHUTTLE

Many tissues contain both the glycerol-3-P shuttle and the malate–aspartate shuttle. In the malate–aspartate shuttle (Fig. 22.8), cytosolic NAD^+ is regenerated by cytosolic malate dehydrogenase, which transfers electrons from NADH to cytosolic oxaloacetate to form malate. Malate is transported across the inner mitochondrial membrane by a specific translocase, which exchanges malate for α-ketoglutarate. In the matrix, malate is oxidized back to oxaloacetate by mitochondrial malate dehydrogenase, and NADH is generated. This NADH can donate electrons to the electron transport chain with generation of approximately 2.5 moles of ATP per mole of NADH. The newly formed oxaloacetate cannot pass back through the inner mitochondrial membrane under physiologic conditions, so aspartate is used to

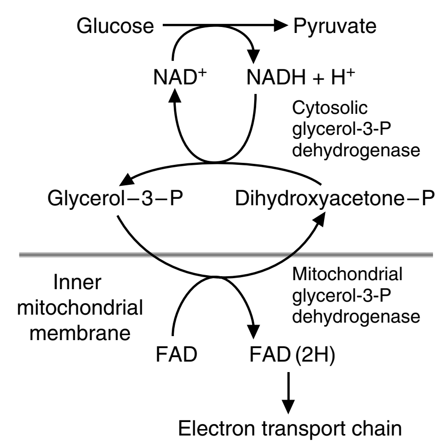


Fig. 22.7. Glycerol 3-phosphate shuttle. Because NAD^+ and NADH cannot cross the mitochondrial membrane, shuttles transfer the reducing equivalents into mitochondria. Dihydroxyacetone phosphate (DHAP) is reduced to glycerol-3-P by cytosolic glycerol 3-P dehydrogenase, using cytosolic NADH produced in glycolysis. Glycerol-3-P then reacts in the inner mitochondrial membrane with mitochondrial glycerol-3-P dehydrogenase, which transfers the electrons to FAD and regenerates DHAP, which returns to the cytosol. The electron transport chain transfers the electrons to O_2, which generates approximately 1.5 ATP for each FAD(2H) that is oxidized.

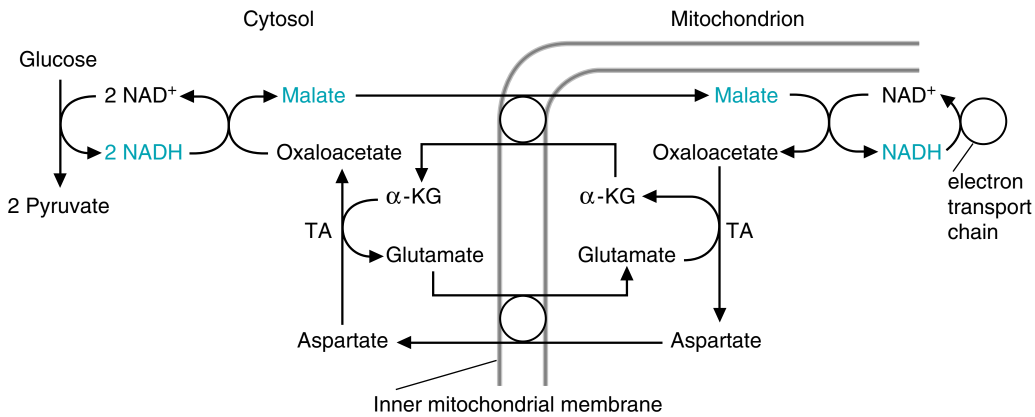


Fig. 22.8. Malate–aspartate shuttle. NADH produced by glycolysis reduces oxaloacetate (OAA) to malate, which crosses the mitochondrial membrane and is reoxidized to OAA. The mitochondrial NADH donates electrons to the electron transport chain, with 2.5 ATPs generated for each NADH. To complete the shuttle, oxaloacetate must return to the cytosol, although it cannot be directly transported on a translocase. Instead, it is transaminated to aspartate, which is then transported out to the cytosol, where it is transaminated back to oxaloacetate. The translocators exchange compounds in such a way that the shuttle is completely balanced. TA = transamination reaction. α-KG = α-ketoglutarate.

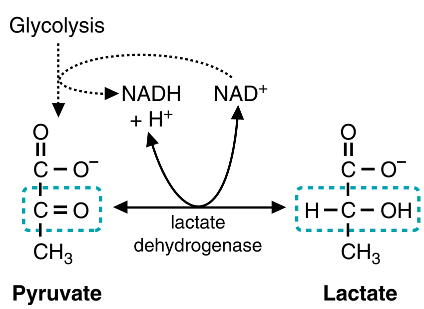


Fig. 22.9. Lactate dehydrogenase reaction. Pyruvate, which may be produced by glycolysis, is reduced to lactate. The reaction, which occurs in the cytosol, requires NADH and is catalyzed by lactate dehydrogenase. This reaction is readily reversible.

Q: What are the energy-generating steps as pyruvate is completely oxidized to carbon dioxide to generate 12.5 molecules of ATP per pyruvate?

return the oxaloacetate carbon skeleton to the cytosol. In the matrix, transamination reactions transfer an amino group to oxaloacetate to form aspartate, which is transported out to the cytosol (using an aspartate/glutamate exchange translocase) and converted back to oxaloacetate through another transamination reaction. The sum of all the reactions of this shuttle system is simply:

$$NADH_{cytosol} + NAD^+_{matrix} \rightarrow NAD^+_{cytosol} + NADH_{matrix}.$$

C. Anaerobic Glycolysis

When the oxidative capacity of a cell is limited (e.g., the red blood cell, which has no mitochondria), the pyruvate and NADH produced from glycolysis cannot be oxidized aerobically. The NADH is therefore oxidized to NAD^+ in the cytosol by reduction of pyruvate to lactate. This reaction is catalyzed by lactate dehydrogenase (LDH) (Fig. 22.9). The net reaction for anaerobic glycolysis is:

$$\text{Glucose} + 2\ \text{ADP} + 2\ P_i \rightarrow 2\ \text{Lactate} + 2\ \text{ATP} + 2\ H_2O + 2\ H^+$$

1. ENERGY YIELD OF AEROBIC VERSUS ANAEROBIC GLYCOLYSIS

In both aerobic and anaerobic glycolysis, each mole of glucose generates 2 moles of ATP, 2 of NADH and 2 of pyruvate. The energy yield from anaerobic glycolysis (glucose to 2 lactate) is only 2 moles of ATP per mole of glucose, as the NADH is recycled to NAD^+ by reducing pyruvate to lactate. Neither the NADH nor pyruvate produced is thus used for further energy generation. However, when oxygen is available, and cytosolic NADH can be oxidized via a shuttle system, pyruvate can also enter the mitochondria and be completely oxidized to CO_2 via PDH and the TCA cycle. The oxidation of pyruvate via this route generates roughly 12.5 moles of ATP per mole of pyruvate. If the cytosolic NADH is oxidized by the glycerol 3-P shuttle, approximately 1.5 moles of ATP are produced per NADH. If, instead, the NADH is oxidized by the malate–aspartate shuttle, approximately 2.5 moles are produced. Thus, the two NADH molecules produced during glycolysis can lead to 3 to 5 molecules of ATP being produced, depending on which shuttle system is used to transfer the reducing equivalents. Because each pyruvate produced can give rise to 12.5 molecules of ATP, altogether 30 to 32 molecules of ATP can be produced from one mole of glucose oxidized to carbon dioxide.

In response to the hypoxemia caused by **Lopa Fusor's** COPD, she has increased hypoxia-inducible factor-1 (HIF-1) in her tissues. HIF-1 is a gene transcription factor found in tissues throughout the body (including brain, heart, kidney, lung, liver, pancreas, skeletal muscle, and white blood cells) that plays a homeostatic role in coordinating tissue responses to hypoxia. Each tissue will respond with a subset of the following changes. HIF-1 increases transcription of the genes for many of the glycolytic enzymes, including PFK-1, enolase, phosphoglycerate kinase, and lactate dehydrogenase. HIF-1 also increases synthesis of a number of proteins that enhance oxygen delivery to tissues, including erythropoietin, which increases the generation of red blood cells in bone marrow; vascular endothelial growth factor, which regulates angiogenesis (formation of blood vessels); and inducible nitric oxide synthase, which synthesizes nitric oxide, a vasodilator. As a consequence, Mrs. Fusor was able to maintain hematocrit and hemoglobin levels that were on the high side of the normal range, and her tissues had an increased capacity for anaerobic glycolysis.

To produce the same amount of ATP per unit time from anaerobic glycolysis as from the complete aerobic oxidation of glucose to CO_2, anaerobic glycolysis must occur approximately 15 times faster, and use approximately 15 times more glucose. Cells achieve this high rate of glycolysis by expressing high levels of glycolytic enzymes. In certain skeletal muscles and in most cells during hypoxic crises, high rates of glycolysis are associated with rapid degradation of internal glycogen stores to supply the required glucose-6-P.

2. ACID PRODUCTION IN ANAEROBIC GLYCOLYSIS

Anaerobic glycolysis results in acid production in the form of H^+. Glycolysis forms pyruvic acid, which is reduced to lactic acid. At an intracellular pH of 7.35, lactic acid dissociates to form the carboxylate anion, lactate, and H^+ (the pKa for lactic acid is 3.85). Lactate and the H^+ are both transported out of the cell into interstitial fluid by a transporter on the plasma membrane and eventually diffuse into the blood. If the amount of lactate generated exceeds the buffering capacity of the blood, the pH drops below the normal range, resulting in lacticacidosis (see Chapter 4).

3. TISSUES DEPENDENT ON ANAEROBIC GLYCOLYSIS

Many tissues, including red and white blood cells, the kidney medulla, the tissues of the eye, and skeletal muscles, rely on anaerobic glycolysis for at least a portion of their ATP requirements (Table 22.1). Tissues (or cells) that are heavily dependent on anaerobic glycolysis usually have a low ATP demand, high levels of glycolytic enzymes, and few capillaries, such that oxygen must diffuse over a greater distance to reach target cells. The lack of mitochondria, or the increased rate of glycolysis, is often related to some aspect of cell function. For example, the mature red blood cell has no mitochondria because oxidative metabolism might interfere with its function in transporting oxygen bound to hemoglobin. Some of the lactic acid generated by anaerobic glycolysis in skin is secreted in sweat, where it acts as an antibacterial agent. Many large tumors use anaerobic glycolysis for ATP production, and lack capillaries in their core.

In tissues with some mitochondria, both aerobic and anaerobic glycolysis occur simultaneously. The relative proportion of the two pathways depends on the mitochondrial oxidative capacity of the tissue and its oxygen supply and may vary between cell types within the same tissue because of cell distance from the capillaries. When a cell's energy demand exceeds the capacity of the rate of the electron transport chain and oxidative phosphorylation to produce ATP, glycolysis is activated, and the increased NADH/NAD^+ ratio will direct excess pyruvate into lactate. Because under these conditions pyruvate dehydrogenase, the TCA cycle, and the electron transport chain are operating as fast as they can, anaerobic glycolysis is meeting the need for additional ATP.

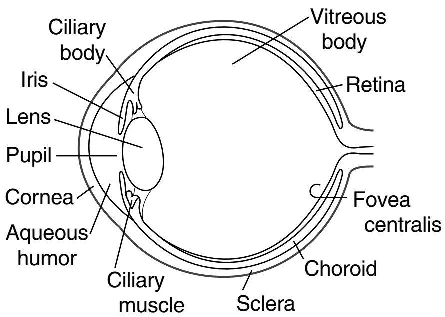
The dental caries in **Ivan Applebod's** mouth were caused principally by the low pH generated from lactic acid production by oral bacteria. Below a pH of 5.5, decalcification of tooth enamel and dentine occurs. Lactobacilli and *S. mutans* are major contributors to this process because almost all of their energy is derived from the conversion of glucose or fructose to lactic acid, and they are able to grow well at the low pH generated by this process. Mr. Applebod's dentist explained that bacteria in his dental plaque could convert all the sugar in his candy into acid in less than 20 minutes. The acid is buffered by bicarbonate and other buffers in saliva, but saliva production decreases in the evening. Thus, the acid could dissolve the hydroxyapatite in his tooth enamel during the night.

Table 22.1. Major Tissue Sites of Lactate Production in a Resting Man. An average 70-kg man consumes about 300 g of carbohydrate per day.

Daily Lactate Production (g/day)	
Total lactate production	115
Red blood cells	29
Skin	20
Brain	17
Skeletal muscle	16
Renal medulla	15
Intestinal muscosa	8
Other tissues	10

A: In the complete oxidation of pyruvate to carbon dioxide, four steps generate NADH (pyruvate dehydrogenase, isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, and malate dehydrogenase). One step generates FAD(2H) (succinate dehydrogenase), and one substrate level phosphorylation (succinate thiokinase). Thus, because each NADH generates 2.5 ATPs, the overall contribution by NADH is 10 ATP molecules. The FAD(2H) generates an additional 1.5 ATP, and the substrate-level phosphorylation provides one more. Therefore, 10 + 1.5 + 1 = 12.5 molecules of ATP.

The tissues of the eye are also partially dependent on anaerobic glycolysis.



The eye contains cells that transmit or focus light, and these cells cannot, therefore, be filled with opaque structures such as mitochondria, or densely packed capillary beds. The corneal epithelium generates most of its ATP aerobically from its few mitochondria but still metabolizes some glucose anaerobically. Oxygen is supplied by diffusion from the air. The lens of the eye is composed of fibers that must remain birefringent to transmit and focus light, so mitochondria are nearly absent. The small amount of ATP required (principally for ion balance) can readily be generated from anaerobic glycolysis even though the energy yield is low. The lens is able to pick up glucose and release lactate into the vitreous body and aqueous humor. It does not need oxygen and has no use for capillaries.

Lactate dehydrogenase (LDH) is a tetramer composed of A subunits (also called M for skeletal muscle form) and B subunits (also called H for heart). Different tissues produce different amounts of the two subunits, which then combine randomly to form five different tetramers (M_4, M_3H_1, M_2H_2, M_1H_3, and H_4). These isoenzymes differ only slightly in their properties, with the kinetic properties of the M_4 form facilitating conversion of pyruvate to lactate in skeletal muscle and the H_4 form facilitating conversion of lactate to pyruvate in the heart.

4. FATE OF LACTATE

Lactate released from cells undergoing anaerobic glycolysis is taken up by other tissues (primarily the liver, heart, and skeletal muscle) and oxidized back to pyruvate. In the liver, the pyruvate is used to synthesize glucose (gluconeogenesis), which is returned to the blood. The cycling of lactate and glucose between peripheral tissues and liver is called the Cori cycle (Fig. 22.10).

In many other tissues, lactate is oxidized to pyruvate, which is then oxidized to CO_2 in the TCA cycle. Although the equilibrium of the lactate dehydrogenase reaction favors lactate production, flux occurs in the opposite direction if NADH is being rapidly oxidized in the electron transport chain (or being used for gluconeogenesis):

$$\text{Lactate} + \text{NAD}^+ \rightarrow \text{Pyruvate} + \text{NADH} + \text{H}^+$$

The heart, with its huge mitochondrial content and oxidative capacity, is able to use lactate released from other tissues as a fuel. During an exercise such as bicycle riding, lactate released into the blood from skeletal muscles in the leg might be used by resting skeletal muscles in the arm. In the brain, glial cells and astrocytes produce lactate, which is used by neurons or released into the blood.

II. OTHER FUNCTIONS OF GLYCOLYSIS

Glycolysis, in addition to providing ATP, generates precursors for biosynthetic pathways (Fig. 22.11). Intermediates of the pathway can be converted to ribose 5-phosphate, the sugar incorporated into nucleotides such as ATP. Other sugars, such as UDP-glucose, mannose, and sialic acid, are also formed from intermediates of glycolysis. Serine is synthesized from 3-phosphoglycerate, and alanine from pyruvate. The backbone of triacylglycerols, glycerol 3-phosphate, is derived from dihydroxyacetone phosphate in the glycolytic pathway.

The liver is the major site of biosynthetic reactions in the body. In addition to those pathways mentioned previously, the liver synthesizes fatty acids from the pyruvate generated by glycolysis. It also synthesizes glucose from lactate, glycerol 3-phosphate, and amino acids in the gluconeogenic pathway, which is principally a reversal of glycolysis. Consequently, in liver, many of the glycolytic enzymes exist as isoenzymes with properties suited for these functions.

The bisphosphoglycerate shunt is a "side reaction" of the glycolytic pathway in which 1,3-bis-phosphoglycerate is converted to 2,3-bis-phosphoglycerate (2,3-BPG). Red blood cells form 2,3-BPG to serve as an allosteric inhibitor of oxygen binding to heme (see Chapter 44). 2,3-BPG reenters the glycolytic pathway via dephosphorylation to 3-phosphoglycerate. 2,3-BPG also functions as a coenzyme in the conversion of 3-phosphoglycerate to 2-phosphoglycerate by the glycolytic

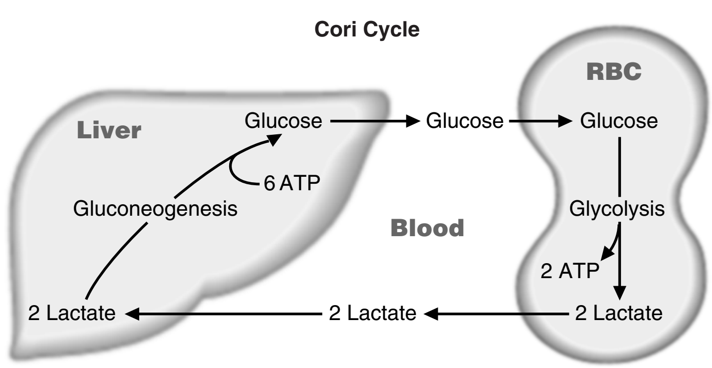


Fig. 22.10. Cori cycle. Glucose, produced in the liver by gluconeogenesis, is converted by glycolysis in muscle, red blood cells, and many other cells, to lactate. Lactate returns to the liver and is reconverted to glucose by gluconeogenesis.

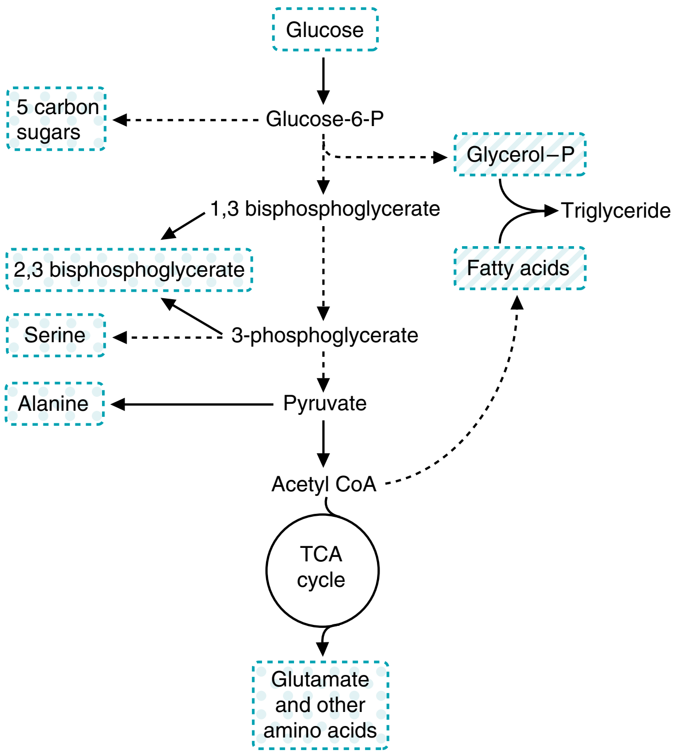


Fig. 22.11. Biosynthetic functions of glycolysis. Compounds formed from intermediates of glycolysis are shown in blue. These pathways are discussed in subsequent chapters of the book. Dotted lines indicate that more than one step is required for the conversion shown in the figure.

enzyme phosphoglyceromutase. Because 2,3-BPG is not depleted by its role in this catalytic process, most cells need only very small amounts.

III. REGULATION OF GLYCOLYSIS BY THE NEED FOR ATP

One of the major functions of glycolysis is the generation of ATP, and, therefore, the pathway is regulated to maintain ATP homeostasis in all cells. Phosphofructokinase-1 (PFK-1) and pyruvate dehydrogenase (PDH), which links glycolysis and the TCA cycle, are both major regulatory sites that respond to feedback indicators of the rate of ATP utilization (Fig. 22.12). The supply of glucose-6-P for glycolysis is tissue dependent and can be regulated at the steps of glucose transport into cells, glycogenolysis (the degradation of glycogen to form glucose), or the rate of glucose phosphorylation by hexokinase isoenzymes. Other regulatory mechanisms integrate the ATP-generating role of glycolysis with its anabolic roles.

All of the regulatory enzymes of glycolysis exist as tissue-specific isoenzymes, which alter the regulation of the pathway to match variations in conditions and needs in different tissues. For example, in the liver, an isoenzyme of pyruvate kinase introduces an additional regulatory site in glycolysis that contributes to the inhibition of glycolysis when the reverse pathway, gluconeogenesis, is activated.

A. Relationship between ATP, ADP, and AMP Concentrations

The AMP levels within the cytosol provide a better indicator of the rate of ATP utilization than the ATP concentration itself (Fig. 22.13). The concentration of AMP

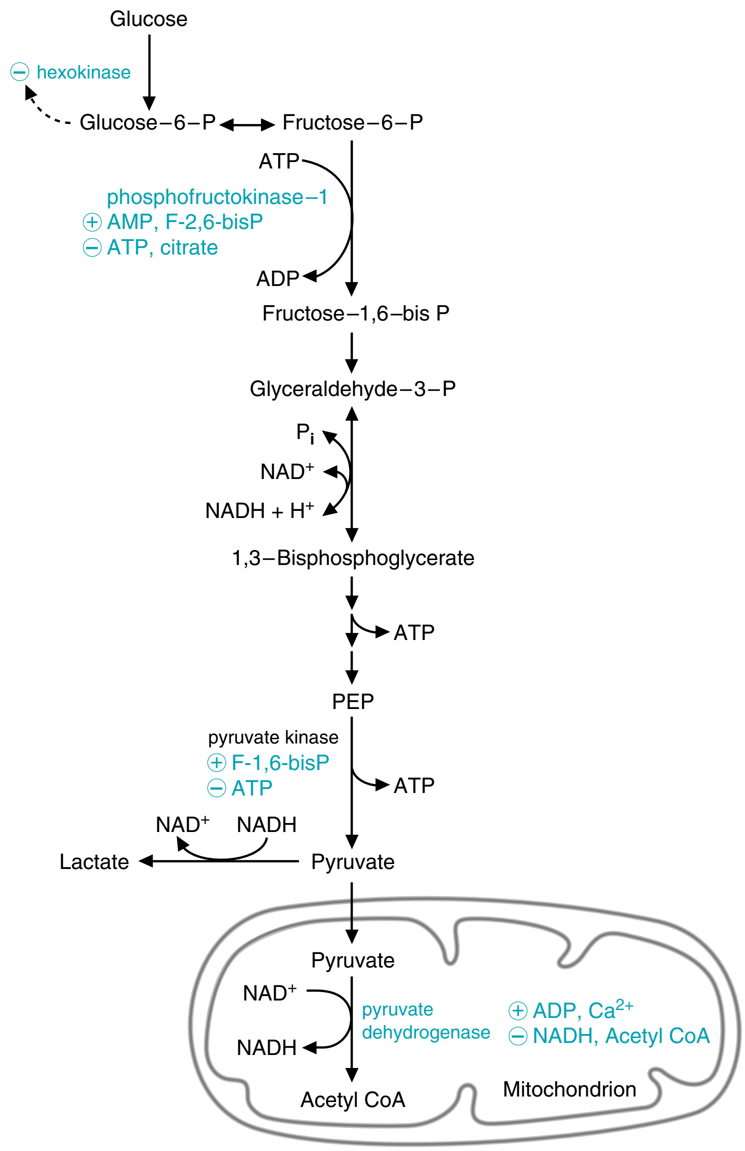


Fig. 22.12. Major sites of regulation in the glycolytic pathway. Hexokinase and phosphofructokinase-1 are the major regulatory enzymes in skeletal muscle. The activity of pyruvate dehydrogenase in the mitochondrion determines whether pyruvate is converted to lactate or to acetyl CoA. The regulation shown for pyruvate kinase only occurs for the liver (L) isoenzyme.

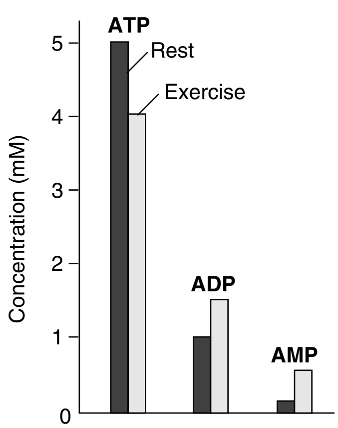


Fig. 22.13. Changes in ATP, ADP, and AMP concentrations in skeletal muscle during exercise. The concentration of ATP decreases by only approximately 20% during exercise, and the concentration of ADP rises. The concentration of AMP, produced by the adenylate kinase reaction, increases manyfold and serves as a sensitive indicator of decreasing ATP levels.

in the cytosol is determined by the equilibrium position of the adenylate kinase reaction.

$$2\ ADP \xleftrightarrow{\text{adenylate kinase}} AMP + ATP$$

The equilibrium is such that hydrolysis of ATP to ADP in energy-requiring reactions increases both the ADP and AMP contents of the cytosol. However, ATP is present in much higher quantities than AMP or ADP, so that a small decrease of ATP concentration in the cytosol causes a much larger percentage increase in the small AMP pool. In skeletal muscles, for instance, ATP levels are approximately 5 mM and decrease by no more than 20% during strenuous exercise (see Fig. 22.13). At the same time, ADP levels may increase by 50%, and AMP levels, which are in the micromolar range, increase by 300%. AMP activates a number of metabolic pathways, including glycolysis, glycogenolysis, and fatty acid oxidation (particularly in muscle tissues), to ensure that ATP homeostasis is maintained.

B. Regulation of Hexokinases

Hexokinases exist as tissue-specific isoenzymes whose regulatory properties reflect the role of glycolysis in different tissues. In most tissues, hexokinase is a low-K_m enzyme with a high affinity for glucose (see Chapter 9). It is inhibited by physiologic concentrations of its product, glucose-6-P (see Fig. 22.12). If glucose-6-P does not enter glycolysis or another pathway, it accumulates and decreases the activity of hexokinase. In the liver, the isoenzyme glucokinase is a high-K_m enzyme that is not readily inhibited by glucose-6-P. Thus, glycolysis can continue in liver even when energy levels are high so that anabolic pathways, such as the synthesis of the major energy storage compounds, glycogen and fatty acids, can occur.

C. Regulation of PFK-1

Phosphofructokinase-1 (PFK-1) is the rate-limiting enzyme of glycolysis and controls the rate of glucose-6-P entry into glycolysis in most tissues. PFK-1 is an allosteric enzyme that has a total of six binding sites: two are for substrates (Mg-ATP and fructose-6-P) and four are allosteric regulatory sites (see Fig. 22.12). The allosteric regulatory sites occupy a physically different domain on the enzyme than the catalytic site. When an allosteric effector binds, it changes the conformation at the active site and may activate or inhibit the enzyme (see also Chapter 9). The allosteric sites for PFK-1 include an inhibitory site for MgATP, an inhibitory site for citrate and other anions, an allosteric activation site for AMP, and an allosteric activation site for fructose 2,6-bisphosphate (fructose-2,6-bisP) and other bisphosphates. Several different tissue-specific isoforms of PFK-1 are affected in different ways by the concentration of these substrates and allosteric effectors, but all contain these four allosteric sites.

1. ALLOSTERIC REGULATION OF PFK-1 BY AMP AND ATP

ATP binds to two different sites on the enzyme, the substrate binding site and an allosteric inhibitory site. Under physiologic conditions in the cell, the ATP concentration is usually high enough to saturate the substrate binding site and inhibit the enzyme by binding to the ATP allosteric site. This effect of ATP is opposed by AMP, which binds to a separate allosteric activator site (Figure 22.14). For most of the PFK-1 isoenzymes, the binding of AMP increases the affinity of the enzyme for fructose 6-P (e.g., shifts the kinetic curve to the left). Thus, increases in AMP concentration can greatly increase the rate of the enzyme (see Fig. 22.14), particularly when fructose-6-P concentrations are low.

2. REGULATION OF PFK-1 BY FRUCTOSE 2,6-BISPHOSPHATE

Fructose-2,6-bisP is also an allosteric activator of PFK-1 that opposes the ATP inhibition. Its effect on the rate of activity of PFK-1 is qualitatively similar to that of AMP, but it has a separate binding site. Fructose-2,6-bisP is NOT an intermediate

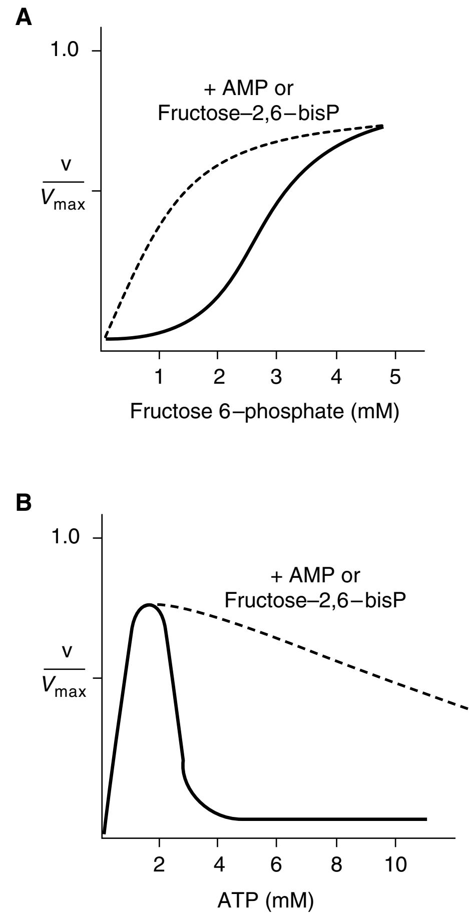


Fig. 22.14. Regulation of PFK-1 by AMP, ATP and fructose-2,6-bisP. **A.** AMP and fructose 2,6-bisphosphate activate PFK-1. **B.** ATP increases the rate of the reaction at low concentrations, but allosterically inhibits the enzyme at high concentrations.

Otto Shape has started high-intensity exercise that will increase the production of lactate in his exercising skeletal muscles. In skeletal muscles, the amount of aerobic versus anaerobic glycolysis that occurs varies with intensity of the exercise, with duration of the exercise, with the type of skeletal muscle fiber involved, and with the level of training. Human skeletal muscles are usually combinations of type I fibers (called fast glycolytic fibers, or white muscle fibers) and type IIb fibers (called slow oxidative fibers, or red muscle fibers). The designation of fast or slow refers to their rate of shortening, which is determined by the isoenzyme of myosin ATPase present. Compared with glycolytic fibers, oxidative fibers have a higher content of mitochondria and myoglobin, which gives them a red color. The gastrocnemius, a muscle in the leg used for running, has a high content of type IIb fibers. However, these fibers will still produce lactate during sprints when the ATP demand exceeds their oxidative capacity.

PFK-1 exists as a group of tissue-specific isoenzymes whose regulatory features match the role of glycolysis in different tissues. Three different types of PFK-1 isoenzyme subunits exist: M (muscle), L (liver), and C. The three subunits show variable expression in different tissues, with some tissues having more than one type. For example, mature human muscle expresses only the M subunit, the liver expresses principally the L subunit, and erythrocytes express both the M and the L subunits. The C subunit is present in highest levels in platelets, placenta, kidney, and fibroblasts but is relatively common to most tissues. Both the M and L subunits are sensitive to AMP and ATP regulation, but the C subunits are much less so. Active PFK-1 is a tetramer, composed of four subunits. Within muscle, the M4 form predominates but within tissues that express multiple isoenzymes of PFK-1 heterotetramers can form that have full activity.

Under ischemic conditions, AMP levels within the heart rapidly increase because of the lack of ATP production via oxidative phosphorylation. The increase in AMP levels activates an AMP-dependent protein kinase (protein kinase B), which phosphorylates the heart isoenzyme of PFK-2 to activate its kinase activity. This results in increased levels of fructose-2,6-bisP, which activates PFK-1 along with AMP such that the rate of glycolysis can increase to compensate for the lack of ATP production via aerobic means.

During **Cora Nari**'s myocardial infarction (see Chapter 20), her heart had a limited supply of oxygen and blood-borne fuels. The absence of oxygen for oxidative phosphorylation would decrease the levels of ATP and increase those of AMP, an activator of PFK-1 and the AMP-dependent protein kinase, resulting in a compensatory increase of anaerobic glycolysis and lactate production. However, obstruction of a vessel leading to her heart would decrease lactate removal, resulting in a decrease of intracellular pH. Under these conditions, at very low pH levels, glycolysis is inhibited and unable to compensate for the lack of oxidative phosphorylation.

of glycolysis but is synthesized by an enzyme that phosphorylates fructose 6-phosphate at the 2 position. The enzyme is therefore named phosphofructokinase-2 (PFK-2); it is a bifunctional enzyme with two separate domains, a kinase domain and a phosphatase domain. At the kinase domain, fructose-6-P is phosphorylated to fructose-2,6-bisP and at the phosphatase domain, fructose-2,6-bisP is hydrolyzed back to fructose-6-P. PFK-2 is regulated through changes in the ratio of activity of the two domains. For example, in skeletal muscles, high concentrations of fructose-6-P activate the kinase and inhibit the phosphatase, thereby increasing the concentration of fructose-2,6-bisP and activating glycolysis.

PFK-2 also can be regulated through phosphorylation by serine/threonine protein kinases. The liver isoenzyme contains a phosphorylation site near the amino terminal that decreases the activity of the kinase and increases the phosphatase activity. This site is phosphorylated by the cAMP-dependent protein kinase (protein kinase A) and is responsible for decreased levels of liver fructose-2,6-bisP during fasting conditions (as modulated by circulating glucagon levels, which is discussed in detail in Chapters 26 and 31). The cardiac isoenzyme contains a phosphorylation site near the carboxy terminal that can be phosphorylated in response to adrenergic activators of contraction (such as norepinephrine) and by increased AMP levels. Phosphorylation at this site increases the kinase activity and increases fructose-2,6-bisP levels, thereby contributing to the activation of glycolysis.

3. ALLOSTERIC INHIBITION OF PFK-1 AT THE CITRATE SITE

The function of the citrate–anion allosteric site is to integrate glycolysis with other pathways. For example, the inhibition of PFK-1 by citrate may play a role in decreasing glycolytic flux in the heart during the oxidation of fatty acids.

D. Regulation of Pyruvate Kinase

Pyruvate kinase exists as tissue-specific isoenzymes. The form present in brain and muscle contains no allosteric sites, and pyruvate kinase does not contribute to the regulation of glycolysis in these tissues. However, the liver isoenzyme can be inhibited through phosphorylation by the cAMP-dependent protein kinase, and by a number of allosteric effectors that contribute to the inhibition of glycolysis during fasting conditions. These allosteric effectors include activation by fructose-1,6-bisP, which ties the rate of pyruvate kinase to that of PFK-1, and inhibition by ATP, which signifies high energy levels.

E. Pyruvate Dehydrogenase Regulation and Glycolysis

Pyruvate dehydrogenase is also regulated principally by the rate of ATP utilization (see Chapter 20) through rapid phosphorylation to an inactive form. Thus, in a normal respiring cell, with an adequate supply of O_2, glycolysis and the TCA cycle are activated together, and glucose can be completely oxidized to CO_2. However, when tissues do not have an adequate supply of O_2 to meet their ATP demands, the increased NADH/NAD$^+$ ratio inhibits pyruvate dehydrogenase, but AMP activates glycolysis. A proportion of the pyruvate will then be reduced to lactate to allow glycolysis to continue.

IV. LACTIC ACIDEMIA

Lactate production is a normal part of metabolism. In the absence of disease, elevated lactate levels in the blood are associated with anaerobic glycolysis during exercise. In lactic acidosis, lactic acid accumulates in blood to levels that significantly affect the pH (lactate levels greater than 5 mM and a decrease of blood pH below 7.2).

Lactic acidosis generally results from a greatly increased NADH/NAD$^+$ ratio in tissues (Fig.22.15). The increased NADH concentration prevents pyruvate oxidation in the TCA cycle and directs pyruvate to lactate. To compensate for the decreased ATP production from oxidative metabolism, PFK-1, and, therefore, the entire glycolytic pathway is activated. For example, consumption of high amounts of alcohol, which is rapidly oxidized in the liver and increases NADH levels, can result in a lactic acidosis. Hypoxia in any tissue increases lactate production as cells attempt to compensate for a lack of O_2 for oxidative phosphorylation.

A number of other problems that interfere either with the electron transport chain or pyruvate oxidation in the TCA cycle result in lactic acidemia (see Fig.22.15). For example, OXPHOS diseases (inherited deficiencies in subunits of complexes in the electron transport chain, such as MERFF) increase the NADH/NAD$^+$ ratio and

Q: Lactate and pyruvate are in equilibrium in the cell, and the ratio of lactate to pyruvate reflects the NADH/NAD$^+$ ratio. Both acids are released into blood, and the normal ratio of lactate to pyruvate in blood is approximately 25:1. This ratio can provide a useful clinical diagnostic tool. Because lactic acidemia can be the result of a number of problems, such as hypoxia, MERFF, thiamine deficiency, and pyruvate dehdyrogenase deficiency, under which of these conditions would you expect the *lactate/pyruvate ratio* in blood to be much greater than normal?

Lopa Fusor had a decreased arterial pO_2 and elevated arterial pCO_2 caused by underperfusion of her lungs. The elevated CO_2 content resulted in an increase of H_2CO_3 and acidity of the blood (see Chapter 4). The decreased O_2 delivery to tissues resulted in increased lactate production from anaerobic glycolysis, and an elevation of serum lactate to 10 times normal levels. The reduction in her arterial pH to 7.18 (reference range, 7.35–7.45) resulted, therefore, from both a mild respiratory acidosis (elevated pCO_2) and a more profound metabolic acidosis (elevated serum lactate level).

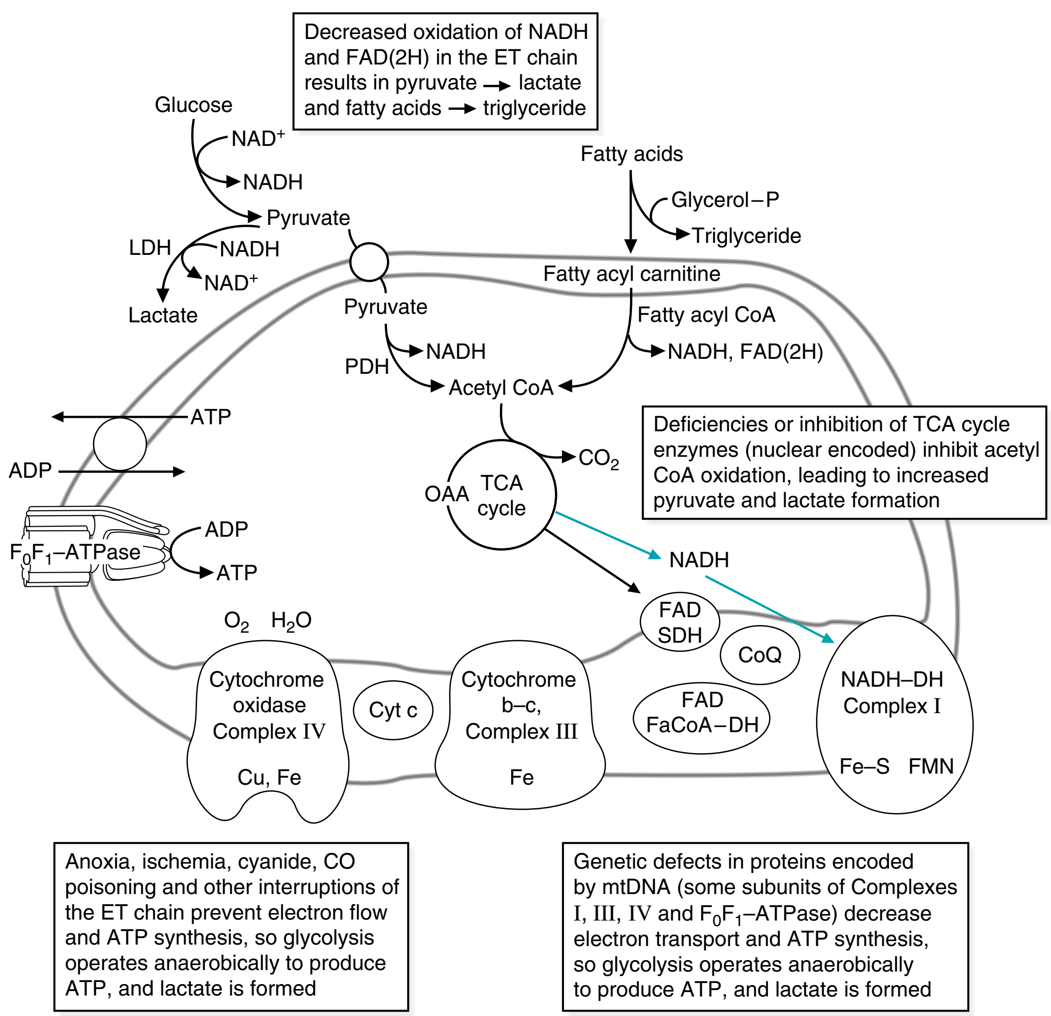


Fig. 22.15. Pathways leading to lactic acidemia.

A: Hypoxia and inherited deficiencies of subunits in the electron transport chain impair NADH oxidation, resulting in a higher NADH/NAD^+ ratio in the cell, and, therefore, a higher lactate/pyruvate ratio in blood. In contrast, conditions that cause lactic acidemia as a result of defects in the enzymes of pyruvate metabolism (thiamine deficiency or pyruvate dehydrogenase deficiency) would increase both pyruvate and lactate in the blood and have little effect on the ratio.

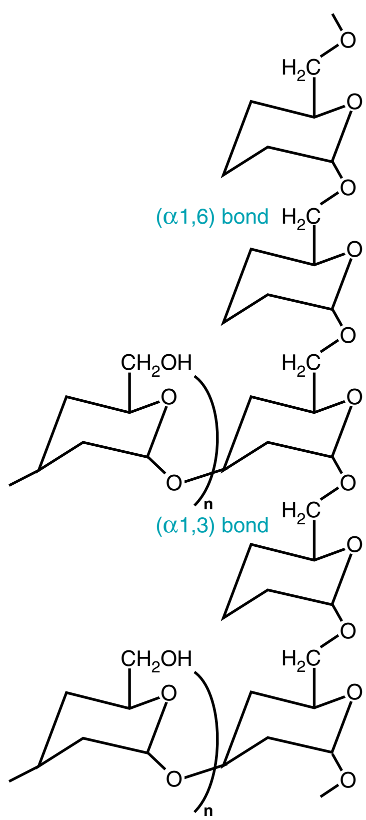


Fig. 22.16. General structure of dextran. Glucosyl residues are linked by α-1,3, α-1,6, and some α-1,4 bonds.

inhibit PDH (see Chapter 21). Impaired PDH activity from an inherited deficiency of E_1 (the decarboxylase subunit of the complex), or from severe thiamine deficiency, increases blood lactate levels (see Chapter 20). Pyruvate carboxylase deficiency also can result in lactic acidosis (see Chapter 20), because of an accumulation of pyruvate.

Lactic acidosis can also result from inhibition of lactate utilization in gluconeogenesis (e.g., hereditary fructose intolerance, which is due to a defective aldolase gene). If other pathways that use glucose-6-P are blocked, glucose-6-P can be shunted into glycolysis and lactate production (e.g., glucose 6-phosphatase deficiency).

CLINICAL COMMENTS

Lopa Fusor was admitted to he hospital with severe hypotension caused by an acute hemorrhage. Her plasma lactic acid level was elevated and her arterial pH was low. The underlying mechanism for Ms. Fusor's derangement in acid-base balance is a severe reduction in the amount of oxygen delivered to her tissues for cellular respiration (hypoxemia). Several concurrent processes contributed to this oxygen lack. The first was her severely reduced blood pressure caused by a brisk hemorrhage from a bleeding gastric ulcer. The blood loss led to hypoperfusion and, therefore, reduced delivery of oxygen to her tissues. The marked reduction in the number of red blood cells in her circulation caused by blood loss further compromised oxygen delivery. The preexisting chronic obstructive pulmonary disease (COPD) added to her hypoxemia by decreasing her ventilation, and, therefore, the transfer of oxygen to her blood (low pO_2). In addition, her COPD led to retention of carbon dioxide (high pCO_2), which caused a respiratory acidosis because the retained CO_2 interacted with water to form carbonic acid (H_2CO_3), which dissociates to H^+ and bicarbonate.

In skeletal muscles, lactate production occurs when the need for ATP exceeds the capacity of the mitochondria for oxidative phosphorylation. Thus, increased lactate production accompanies an increased rate of the TCA cycle. The extent to which skeletal muscles use aerobic versus anaerobic glycolysis to supply ATP varies with the intensity of exercise. At low-intensity exercise, the rate of ATP utilization is lower, and fibers can generate this ATP from oxidative phosphorylation, with the complete oxidation of glucose to CO_2. However, when **Otto Shape** sprints, a high-intensity exercise, the ATP demand exceeds the rate at which the electron transport chain and TCA cycle can generate ATP from oxidative phosphorylation. The increased AMP level signals the need for additional ATP and stimulates PFK-1. The NADH/NAD^+ ratio directs the increase in pyruvate production toward lactate. The fall in pH causes muscle fatigue and pain. As he trains, the amount of mitochondria and myoglobin will increase in his skeletal muscle fibers, and these fibers will rely less on anaerobic glycolysis.

Ivan Applebod had two sites of dental caries: one on a smooth surface and one in a fissure. The decreased pH resulting from lactic acid production by lactobacilli, which grow anaerobically within the fissure, is a major cause of fissure caries. *Streptococs mutans (S. mutans)* plays a major role in smooth surface caries because it secretes dextran, an insoluble polysaccharide, which forms the base for plaque. *S. mutans* contains dextran-sucrase, a glucosyltransferase that transfers glucosyl units from dietary sucrose (the glucose-fructose disaccharide in sugar and sweets) to form the α(1→6) and α(1→3) linkages between the glucosyl units in dextran (Fig. 22.16). Dextran-sucrase is specific for sucrose and does not catalyze the polymerization of free glucose, or glucose from other disaccharides or polysaccharides. Thus sucrose is responsible for the cariogenic potential of candy. The sticky water-insoluble dextran mediates the attachment of *S. mutans* and other bacteria to the tooth surface. This also keeps the acids produced from these bacteria close to the enamel surface. Fructose from sucrose is converted to intermediates of glycolysis and is rapidly metabolized to lactic acid. Other bacteria present in the plaque produce different acids from anaerobic metabolism, such as acetic acid and formic acid. The decrease in pH that results initiates demineralization of the hydroxyapatite of the tooth enamel. **Ivan Applebod's** caries in his baby teeth could have been caused by sucking on bottles containing fruit juice. The sugar in fruit juice is also sucrose, and babies who fall asleep with a bottle of fruit juice in their mouth may develop caries. Rapid decay of these baby teeth can harm the development of their permanent teeth.

BIOCHEMICAL COMMENTS

How is the first high-energy bond created in the glycolytic pathway? This is the work of the glyceraldehyde 3-phosphate dehydrogenase reaction, which converts glyceraldehyde-3-P to 1,3 bisphosphglycerate. This reaction can be considered to be two separate half reactions, the first being the oxidation of glyceraldehyde-3-P to 3-phosphoglycerate, and the second the addition of inorganic phosphate to 3-phosphoglycerate to produce 1,3 bisphosphoglycerate. The $\Delta G^{0\prime}$ for the first reaction is approximately -12 kcal/mole; for the second reaction, it is approximately $+12$ kcal/mole. Thus, although the first half reaction is extremely favorable, the second half reaction is unfavorable and would not proceed under cellular conditions. So how does the enzyme help this reaction to proceed? This is accomplished through the enzyme forming a covalent bond with the substrate, using an essential cysteine residue at the active site to form a high-energy thioester linkage during the course of the reaction

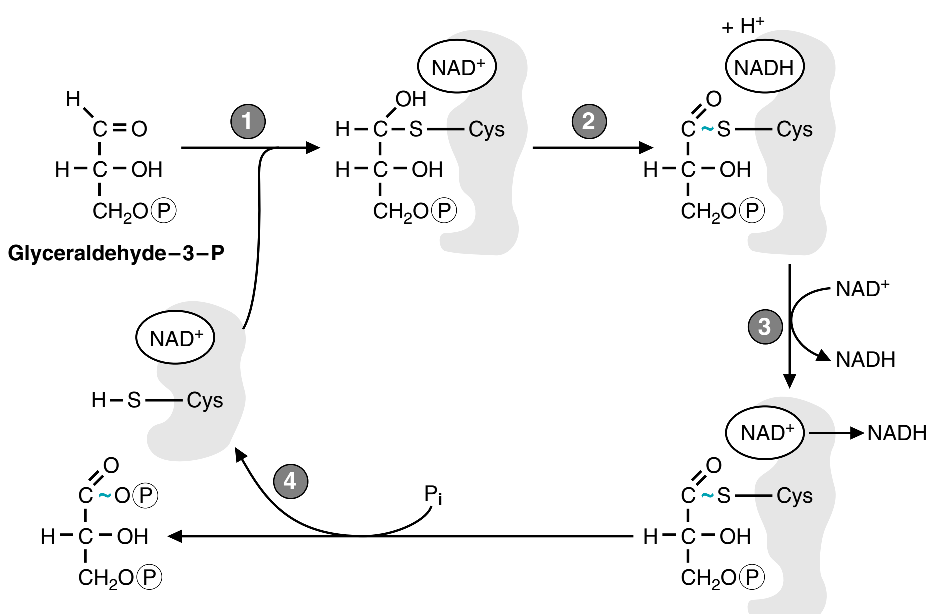


Fig. 22.17. Mechanism of the glyceraldehyde 3-phosphate dehydrogenase reaction. 1. The enzyme forms a covalent linkage with the substrate, using a cysteine group at the active site. The enzyme also contains bound NAD^+ close to the active site. 2. The substrate is oxidized, forming a high-energy thioester linkage (in blue), and NADH. 3. NADH has a low affinity for the enzyme and is replaced by a new molecule of NAD^+. 4. Inorganic phosphate attacks the thioester linkage, releasing the product 1,3 bisphosphoglycerate, and regenerating the active enzyme in a form ready to initiate another reaction.

(Fig. 22.17). Thus, the energy that would be released as heat in the oxidation of glyceraldehyde-3-P to 3-phosphoglycerate is conserved in the thioester linkage that is formed (such that the $\Delta G^{0\prime}$ of the formation of the thioester intermediate from glyceraldehyde-3-P is close to zero). Then, replacement of the sulfur with inorganic phosphate to form the final product, 1,3 bisphosphoglycerate, is relatively straightforward, as the $\Delta G^{0\prime}$ for that conversion is also close to zero, and the acylphosphate bond retains the energy from the oxidation of the aldehyde. This is one example of how covalent catalysis by an enzyme can result in the conservation of energy between different bond types.

Suggested References

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REVIEW QUESTIONS—CHAPTER 22

1. A major role of glycolysis is which of the following?

    (A) To synthesize glucose
    (B) To generate energy
    (C) To produce FAD(2H)
    (D) To synthesize glycogen
    (E) To use ATP to generate heat

2. Starting with glyceraldehyde 3-phosphate and synthesizing one molecule of pyruvate, the net yield of ATP and NADH would be which of the following?

    (A) 1 ATP, 1 NADH
    (B) 1 ATP, 2 NADH
    (C) 1 ATP, 4 NADH
    (D) 2 ATP, 1 NADH
    (E) 2 ATP, 2 NADH
    (F) 2 ATP, 4 NADH
    (G) 3 ATP, 1 NADH
    (H) 3 ATP, 2 NADH
    (I) 3 ATP, 4 NADH

3. When glycogen is degraded, glucose 1-phosphate is formed. Glucose 1-phosphate can then be isomerized to glucose 6-phosphate. Starting with glucose 1-phosphate, and ending with 2 molecules of pyruvate, what is the net yield of glycolysis, in terms of ATP and NADH formed?

    (A) 1 ATP, 1 NADH
    (B) 1 ATP, 2 NADH
    (C) 1 ATP, 3 NADH
    (D) 2 ATP, 1 NADH
    (E) 2 ATP, 2 NADH
    (F) 2 ATP, 3 NADH
    (G) 3 ATP, 1 NADH
    (H) 3 ATP, 2 NADH
    (I) 3 ATP, 3 NADH

4. Which of the following statements correctly describes an aspect of glycolysis?

   (A) ATP is formed by oxidative phosphorylation.
   (B) 2 ATP are used in the beginning of the pathway.
   (C) Pyruvate kinase is the rate-limiting enzyme.
   (D) One pyruvate and three CO_2 are formed from the oxidation of one glucose molecule.
   (E) The reactions take place in the matrix of the mitochondria.

5. How many moles of ATP are generated by the complete aerobic oxidation of 1 mole of glucose to 6 moles of CO_2?

   (A) 2–4
   (B) 10–12
   (C) 18–22
   (D) 30–32
   (E) 60–64

23 Oxidation of Fatty Acids and Ketone Bodies

Fatty acids are a major fuel for humans and supply our energy needs between meals and during periods of increased demand, such as exercise. During overnight fasting, fatty acids become the major fuel for cardiac muscle, skeletal muscle, and liver. The liver converts fatty acids to ketone bodies (acetoacetate and β-hydroxybutyrate), which also serve as major fuels for tissues (e.g., the gut). The brain, which does not have a significant capacity for fatty acid oxidation, can use ketone bodies as a fuel during prolonged fasting.

The route of metabolism for a fatty acid depends somewhat on its chain length. Fatty acids are generally classified as ***very-long-chain length fatty acids*** *(greater than C20),* ***long-chain fatty acids*** *(C12–C20),* ***medium-chain fatty acids*** *(C6–C12), and* ***short-chain fatty acids*** *(C4).*

ATP is generated from oxidation of fatty acids in the pathway of ***β-oxidation****. Between meals and during overnight fasting,* ***long-chain fatty acids*** *are released from adipose tissue triacylglycerols. They circulate through blood bound to* ***albumin*** *(Fig. 23.1). In cells, they are converted to* ***fatty acyl CoA*** *derivatives by* ***acyl CoA synthetases****. The activated acyl group is transported into the mitochondrial matrix bound to* ***carnitine,*** *where fatty acyl CoA is regenerated. In the pathway of* ***β-oxidation****, the fatty acyl group is sequentially oxidized to yield FAD(2H), NADH, and acetyl CoA. Subsequent oxidation of NADH and FAD(2H) in the electron transport chain, and oxidation of acetyl CoA to* CO_2 *in the TCA cycle, generates ATP from oxidative phosphorylation.*

Many fatty acids have structures that require variations of this basic pattern. Long-chain fatty acids that are ***unsaturated fatty acids*** *generally require additional isomerization and oxidation–reduction reactions to rearrange their double bonds during β-oxidation. Metabolism of water-soluble* ***medium-chain-length fatty acids*** *does not require carnitine and occurs only in liver.* ***Odd-chain-length fatty acids*** *undergo β-oxidation to the terminal three-carbon* ***propionyl CoA****, which enters the TCA cycle as succinyl CoA.*

Fatty acids that do not readily undergo mitochondrial β-oxidation are oxidized first by alternate routes that convert them to more suitable substrates or to urinary excretion products. ***Excess fatty acids*** *may undergo microsomal* ***ω-oxidation,*** *which converts them to* ***dicarboxylic acids*** *that appear in urine.* ***Very-long-chain fatty acids*** *(both straight chain and* ***branched fatty acids*** *such as phytanic acid) are whittled down to size in peroxisomes.* ***Peroxisomal α- and β-oxidiation*** *generates hydrogen peroxide (*$\boldsymbol{H_2O_2}$*), NADH, acetyl CoA, or propionyl CoA and a short- to medium-chain-length acyl CoA. The acyl CoA products are transferred to mitochondria to complete their metabolism.*

In the liver, much of the acetyl CoA generated from fatty acid oxidation is converted to the ketone bodies, ***acetoacetate*** *and* ***β-hydroxybutyrate,*** *which enter the blood (see Fig. 23.1). In other tissues, these ketone bodies are converted to acetyl*

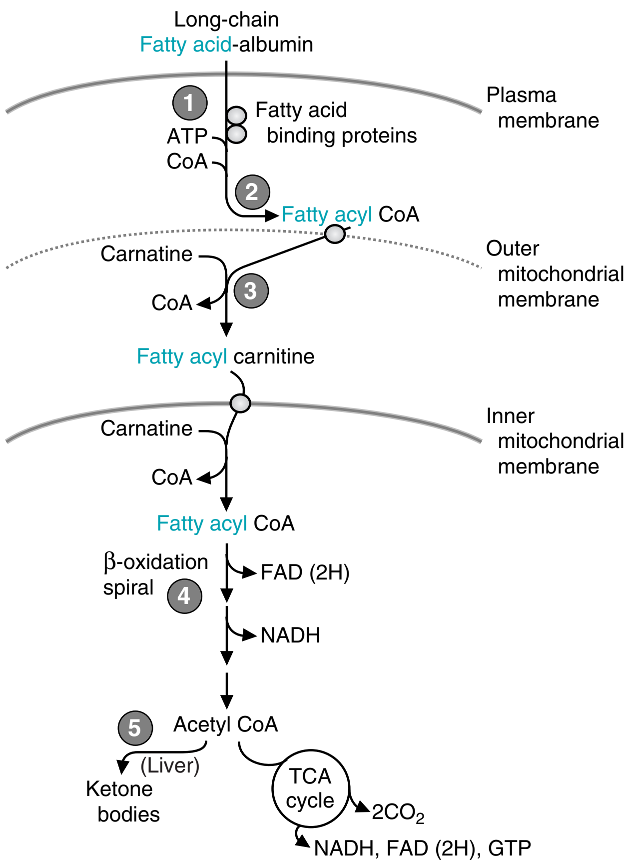


Fig. 23.1. Overview of mitochondrial long-chain fatty acid metabolism. (1) Fatty acid binding proteins (FaBP) transport fatty acids across the plasma membrane and bind them in the cytosol. (2) Fatty acyl CoA synthetase activates fatty acids to fatty acyl CoAs. (3) Carnitine transports the activated fatty acyl group into mitochondria. (4) β-oxidation generates NADH, FAD(2H), and acetyl CoA (5) In the liver, acetyl CoA is converted to ketone bodies

CoA, which is oxidized in the TCA cycle. The liver synthesizes ketone bodies but cannot use them as a fuel.

The ***rate of fatty acid oxidation*** *is linked to the rate of NADH, FAD(2H), and acetyl CoA oxidation, and, thus, to the rate of oxidative phosphorylation and* ***ATP utilization****. Additional regulation occurs through* ***malonyl CoA****, which inhibits formation of the fatty acyl carnitine derivatives. Fatty acids and ketone bodies are used as a fuel when their level increases in the blood, which is determined by hormonal regulation of adipose tissue lipolysis.*

THE WAITING ROOM

Otto Shape was disappointed that he did not place in his 5-km race and has decided that short-distance running is probably not right for him. After careful consideration, he decides to train for the marathon by running 12 miles three times per week. He is now 13 pounds over his ideal weight, and he plans on losing this weight while studying for his Pharmacology finals. He considers a variety of dietary supplements to increase his endurance and selects one containing carnitine, CoQ, pantothenate, riboflavin, and creatine.

Lofata Burne is a 16-year-old girl. Since age 14 months she has experienced recurrent episodes of profound fatigue associated with vomiting and increased perspiration, which required hospitalization. These episodes occurred only if she fasted for more than 8 hours. Because her mother gave her food late at night and woke her early in the morning for breakfast, Lofata's physical and mental development had progressed normally.

On the day of admission for this episode, Lofata had missed breakfast, and by noon she was extremely fatigued, nauseated, sweaty, and limp. She was unable to hold any food in her stomach and was rushed to the hospital, where an infusion of glucose was started intravenously. Her symptoms responded dramatically to this therapy.

Her initial serum glucose level was low at 38 mg/dL (reference range for fasting serum glucose levels = 70–100). Her blood urea nitrogen (BUN) level was slightly elevated at 26 mg/dL (reference range = 8–25) as a result of vomiting, which led to a degree of dehydration. Her blood levels of liver transaminases were slightly elevated, although her liver was not palpably enlarged. Despite elevated levels of free fatty acids (4.3 mM) in the blood, blood ketone bodies were below normal.

The liver transaminases measured in the blood are aspartate aminotransferase (AST), which was formerly called serum glutamate-oxaloacetate transaminase (SGOT), and alanine aminotransferase (ALT), which was formerly called serum glutamate pyruvate transaminase (SGPT). Elevation of liver enzymes reflects damage of the liver plasma membrane.

Di Abietes, a 27-year-old woman with type 1 diabetes mellitus, had been admitted to the hospital in a ketoacidotic coma a year ago (see Chapter 4). She had been feeling drowsy and had been vomiting for 24 hours before that admission. At the time of admission, she was clinically dehydrated, her blood pressure was low, and her breathing was deep and rapid (Kussmaul breathing). Her pulse was rapid, and her breath had the odor of acetone. Her arterial blood pH was 7.08 (reference range, 7.36–7.44), and her blood ketone body levels were 15 mM (normal is approximately 0.2 mM for a person on a normal diet).

I. FATTY ACIDS AS FUELS

The fatty acids oxidized as fuels are principally long-chain fatty acids released from adipose tissue triacylglycerol stores between meals, during overnight fasting, and during periods of increased fuel demand (e.g., during exercise). Adipose tissue triacylglycerols are derived from two sources; dietary lipids and triacylglycerols synthesized in the liver. The major fatty acids oxidized are the long-chain fatty acids, palmitate, oleate, and stearate, because they are highest in dietary lipids and are also synthesized in the human.

Between meals, a decreased insulin level and increased levels of insulin counterregulatory hormones (e.g., glucagon) activate lipolysis, and free fatty acids are transported to tissues bound to serum albumin. Within tissues, energy is derived from oxidation of fatty acids to acetyl CoA in the pathway of β-oxidation. Most of the enzymes involved in fatty acid oxidation are present as 2-3 isoenzymes, which have different but overlapping specificities for the chain length of the fatty acid. Metabolism of unsaturated fatty acids, odd-chain-length fatty acids, and medium-chain-length fatty acids requires variations of this basic pattern. The acetyl CoA produced from fatty acid oxidation is principally oxidized in the TCA cycle or converted to ketone bodies in the liver.

During Otto's distance running (a moderate-intensity exercise), decreases in insulin and increases in insulin counterregulatory hormones, such as epinephrine and norepinephrine, increase adipose tissue lipolysis. Thus, his muscles are being provided with a supply of fatty acids in the blood that they can use as a fuel.

Lofata Burne developed symptoms during fasting, when adipose tissue lipolysis was elevated. Under these circumstances, muscle tissue, liver, and many other tissues are oxidizing fatty acids as a fuel. After overnight fasting, approximately 60 to 70% of our energy supply is derived from the oxidation of fatty acids.

A. Characteristics of Fatty Acids Used as Fuels

Fat constitutes approximately 38% of the calories in the average North American diet. Of this, more than 95% of the calories are present as triacylglycerols (3 fatty acids esterified to a glycerol backbone). During ingestion and absorption, dietary triacylglycerols are broken down into their constituents and then reassembled for transport to adipose tissue in chylomicrons (see Chapter 2). Thus, the fatty acid composition of adipose triacylglycerols varies with the type of food consumed.

The most common dietary fatty acids are the saturated long-chain fatty acids palmitate (C16) and stearate (C18), the monounsaturated fatty acid oleate (C18:1), and the polyunsaturated essential fatty acid, linoleate (C18:2) (To review fatty acid nomenclature, consult Chapter 5). Animal fat contains principally saturated and monounsaturated long-chain fatty acids, whereas vegetable oils contain linoleate and some longer-chain and polyunsaturated fatty acids. They also contain smaller amounts of branched-chain and odd-chain-length fatty acids. Medium-chain-length fatty acids are present principally in dairy fat (e.g., milk and butter), maternal milk, and vegetable oils.

Adipose tissue triacylglycerols also contain fatty acids synthesized in the liver, principally from excess calories ingested as glucose. The pathway of fatty acid synthesis generates palmitate, which can be elongated to form stearate, and unsaturated to form oleate. These fatty acids are assembled into triacylglycerols and transported to adipose tissue as the lipoprotein VLDL (very-low-density lipoprotein).

B. Transport and Activation of Long-Chain Fatty Acids

Long-chain fatty acids are hydrophobic and water insoluble. In addition, they are toxic to cells because they can disrupt the hydrophobic bonding between amino acid side chains in proteins. Consequently, they are transported in the blood and in cells bound to proteins.

1. CELLULAR UPTAKE OF LONG-CHAIN FATTY ACIDS

During fasting and other conditions of metabolic need, long-chain fatty acids are released from adipose tissue triacylglycerols by lipases. They travel in the blood bound in the hydrophobic binding pocket of albumin, the major serum protein (see Fig. 23.1).

Fatty acids enter cells both by a saturable transport process and by diffusion through the lipid plasma membrane. A fatty acid binding protein in the plasma membrane facilitates transport. An additional fatty acid binding protein binds the fatty acid intracellularly and may facilitate its transport to the mitochondrion. The free fatty acid concentration in cells is, therefore, extremely low.

2. ACTIVATION OF LONG-CHAIN FATTY ACIDS

Fatty acids must be activated to acyl CoA derivatives before they can participate in β-oxidation and other metabolic pathways (Fig. 23.2). The process of activation involves an acyl CoA synthetase (also called a thiokinase) that uses ATP energy to form the fatty acyl CoA thioester bond. In this reaction, the β bond of ATP is cleaved to form a fatty acyl AMP intermediate and pyrophosphate (PPi). Subsequent cleavage of PPi helps to drive the reaction.

The acyl CoA synthetase that activates long-chain fatty acids, 12 to 20 carbons in length, is present in three locations in the cell: the endoplasmic reticulum, outer mitochondrial membranes, and peroxisomal membranes (Table 23.1). This enzyme has no activity toward C22 or longer fatty acids, and little activity below C12. In contrast, the synthetase for activation of very-long-chain fatty acids is present in peroxisomes, and the medium-chain-length fatty acid activating enzyme is present only in the mitochondrial matrix of liver and kidney cells.

3. FATES OF FATTY ACYL COAS

Fatty acyl CoA formation, like the phosphorylation of glucose, is a prerequisite to metabolism of the fatty acid in the cell (Fig. 23.3). The multiple locations of the long-chain acyl CoA synthetase reflects the location of different metabolic routes taken by fatty acyl CoA derivatives in the cell (e.g., triacylglycerol and phospholipid synthesis

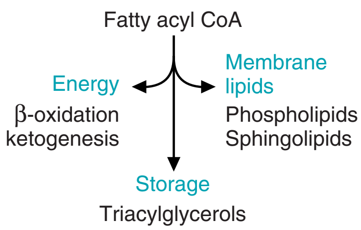


Fig. 23.3. Major metabolic routes for long-chain fatty acyl CoAs. Fatty acids are activated to acyl CoA compounds for degradation in mitochondrial β-oxidation, or incorporation into triacylglycerols or membrane lipids. When β-oxidation is blocked through an inherited enzyme deficiency, or metabolic regulation, excess fatty acids are diverted into triacylglycerol synthesis.

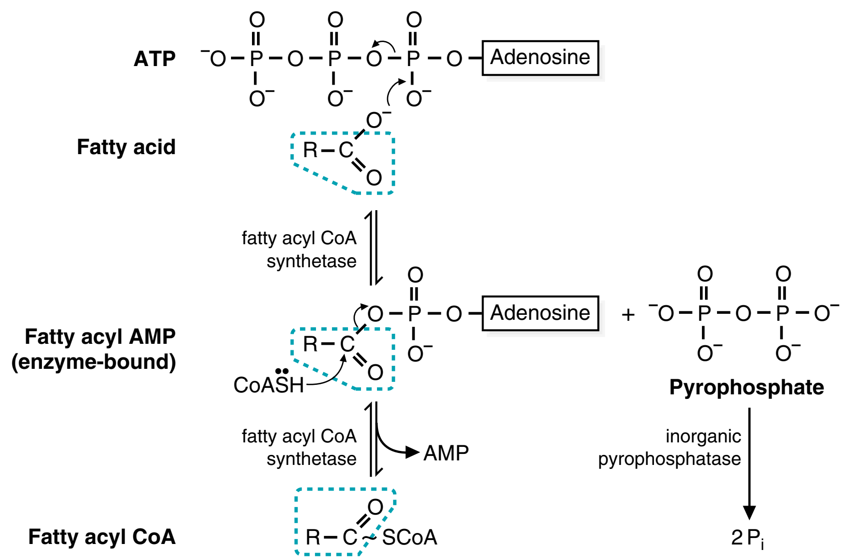


Fig. 23.2. Activation of a fatty acid by a fatty acyl CoA synthetase. The fatty acid is activated by reacting with ATP to form a high-energy fatty acyl AMP and pyrophosphate. The AMP is then exchanged for CoA. Pyrophosphate is cleaved by a pyrophosphatase.

Table 23.1. Chain-Length Specificity of Fatty Acid Activation and Oxidation Enzymes

Enzyme	Chain Length	Comments
Acyl CoA synthetases		
Very Long Chain	14–26	Only found in peroxisomes
Long Chain	12–20	Enzyme present in membranes of ER, mitochondria, and peroxisomes to facilitate different metabolic routes of acyl CoAs.
Medium Chain	6–12	Exists as many variants, present only in mitochondrial matrix of kidney and liver. Also involved in xenobiotic metabolism.
Acetyl	2–4	Present in cytoplasm and possibly mitochondrial matrix
Acyltransferases		
CPTI	12–16	Although maximum activity is for fatty acids 12–16 carbons long, it also acts on many smaller acyl CoA derivatives
Medium Chain (Octanoylcarnitine transferase)	6–12	Substrate is medium-chain acyl CoA derivatives generated during peroxisomal oxidation.
Carnitine:acetyl transferase	2	High level in skeletal muscle and heart to facilitate use of acetate as a fuel
Acyl CoA Dehydrogenases		
VLCAD	14–20	Present in inner mitochondrial membrane
LCAD MCAD	12–18 4–12	Members of same enzyme family, which also includes acyl CoA dehydrogenases for carbon skeleton of branched-chain amino acids.
SCAD	4–6	
Other enzymes		
Enoyl CoA hydratase, Short-chain	>4	Also called crotonase. Activity decreases with increasing chain length.
L-3-Hydroxyacyl CoA dehydrogenase, Short-Chain	4–16	Activity decreases with increasing chain length
Acetoacetyl CoA thiolase	4	Specific for acetoacetyl CoA
Trifunctional Protein	12–16	Complex of long-chain enoyl hydratase, acyl CoA dehydrogenase and a thiolase with broad specificity. Most active with longer chains.

in the endoplasmic reticulum, oxidation and plasmalogen synthesis in the peroxisome, and β-oxidation in mitochondria). In the liver and some other tissues, fatty acids that are not being used for energy generation are re-incorporated (re-esterified) into triacylglycerols.

4. TRANSPORT OF LONG-CHAIN FATTY ACIDS INTO MITOCHONDRIA

Carnitine serves as the carrier that transports activated long chain fatty acyl groups across the inner mitochondrial membrane (Fig. 23.4). Carnitine acyl transferases are able to reversibly transfer an activated fatty acyl group from CoA to the hydroxyl group of carnitine to form an acylcarnitine ester. The reaction is reversible, so that the fatty acyl CoA derivative can be regenerated from the carnitine ester.

Carnitine:palmitoyltransferase I (CPTI; also called carnitine acyltransferase I, CATI), the enzyme that transfers long-chain fatty acyl groups from CoA to carnitine, is located on the outer mitochondrial membrane (Fig. 23.5). Fatty acylcarnitine crosses the inner mitochondrial membrane with the aid of a translocase. The fatty acyl group is transferred back to CoA by a second enzyme, carnitine:palmitoyltransferase II (CPTII or CATII). The carnitine released in this reaction returns to the cytosolic side of the mitochondrial membrane by the same translocase that brings fatty acylcarnitine to the matrix side. Long-chain fatty acyl CoA, now located within the mitochondrial matrix, is a substrate for β-oxidation.

Carnitine is obtained from the diet or synthesized from the side chain of lysine by a pathway that begins in skeletal muscle, and is completed in the liver. The reactions use S-adenosylmethionine to donate methyl groups, and vitamin C (ascorbic acid) is also required for these reactions. Skeletal muscles have a

A number of inherited diseases in the metabolism of carnitine or acylcarnitines have been described. These include defects in the following enzymes or systems: the transporter for carnitine uptake into muscle; CPT I; carnitine-acylcarnitine translocase; and CPTII. Classical CPTII deficiency, the most common of these diseases, is characterized by adolescent to adult onset of recurrent episodes of acute myoglobinuria precipitated by prolonged exercise or fasting. During these episodes, the patient is weak, and may be somewhat hypoglycemic with diminished ketosis (hypoketosis), but metabolic decompensation is not severe. Lipid deposits are found in skeletal muscles. CPK levels, and long-chain acylcarnitines are elevated in the blood. CPTII levels in fibroblasts are approximately 25% of normal. The remaining CPTII activity probably accounts for the mild effect on liver metabolism. In contrast, when CPTII deficiency has presented in infants, CPT II levels are below 10% of normal, the hypoglycemia and hypoketosis are severe, hepatomegaly occurs from the triacylglycerol deposits, and cardiomyopathy is also present.

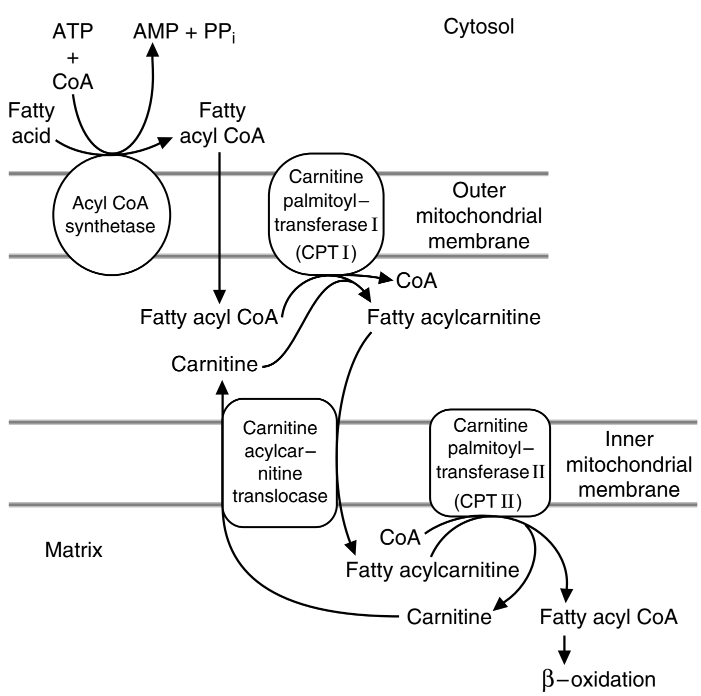
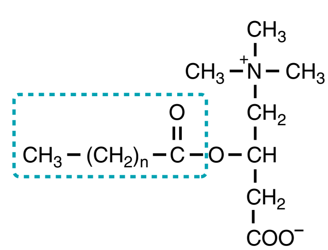


Fig. 23.5. Transport of long-chain fatty acids into mitochondria. The fatty acyl CoA crosses the outer mitochondrial membrane. Carnitine palmitoyl transferase I in the outer mitochondrial membrane transfers the fatty acyl group to carnitine and releases CoASH. The fatty acyl carnitine is translocated into the mitochondrial matrix as carnitine moves out. Carnitine palmitoyl transferase II on the inner mitochondrial membrane transfers the fatty acyl group back to CoASH, to form fatty acyl CoA in the matrix.



Fatty acylcarnitine

Fig. 23.4. Structure of fatty acylcarnitine. Carnitine: palmitoyl transferases catalyze the reversible transfer of a long-chain fatty acyl group from the fatty acyl CoA to the hydroxyl group of carnitine. The atoms in the dashed box originate from the fatty acyl CoA.

Otto Shape's power supplement contains carnitine. However, his body can synthesize enough carnitine to meet his needs, and his diet contains carnitine. Carnitine deficiency has been found only in infants fed a soy-based formula that was not supplemented with carnitine. His other supplements likewise probably provide no benefit, but are designed to facilitate fatty acid oxidation during exercise. Riboflavin is the vitamin precursor of FAD, which is required for acyl CoA dehydrogenases and ETFs. CoQ is synthesized in the body, but it is the recipient in the electron transport chain for electrons passed from complexes I and II and the ETFs. Some reports suggest that supplementation with pantothenate, the precursor of CoA, improves performance.

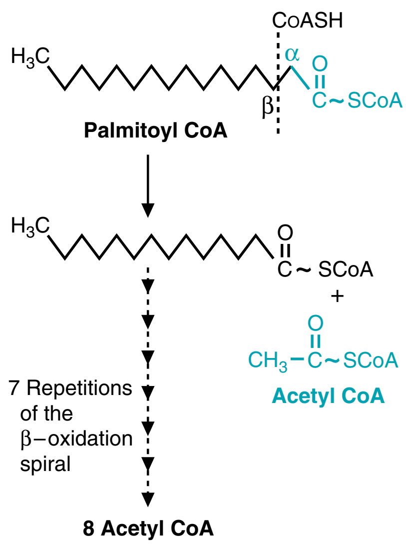


Fig. 23.6. Overview of β-oxidation. Oxidation at the β-carbon is followed by cleavage of the α—β bond, releasing acetyl CoA and a fatty acyl CoA that is two carbons shorter than the original. The carbons cleaved to form acetyl CoA are shown in blue. Successive spirals of β-oxidation completely cleave an even-chain fatty acyl CoA to acetyl CoA.

high-affinity uptake system for carnitine, and most of the carnitine in the body is stored in skeletal muscle.

C. β-Oxidation of Long-Chain Fatty Acids

The oxidation of fatty acids to acetyl CoA in the β-oxidation spiral conserves energy as FAD(2H) and NADH. FAD(2H) and NADH are oxidized in the electron transport chain, generating ATP from oxidative phosphorylation. Acetyl CoA is oxidized in the TCA cycle or converted to ketone bodies.

1. THE β-OXIDATION SPIRAL

The fatty acid β-oxidation pathway sequentially cleaves the fatty acyl group into 2-carbon acetyl CoA units, beginning with the carboxyl end attached to CoA (Fig. 23.6). Before cleavage, the β-carbon is oxidized to a keto group in two reactions that generate NADH and FAD(2H); thus, the pathway is called β-oxidation. As each acetyl group is released, the cycle of β-oxidation and cleavage begins again, but each time the fatty acyl group is 2 carbons shorter.

There are four types of reactions in the β-oxidation pathway (Fig. 23.7). In the first step, a double bond is formed between the β- and α-carbons by an acyl CoA dehydrogenase that transfers electrons to FAD. The double bond is in the trans

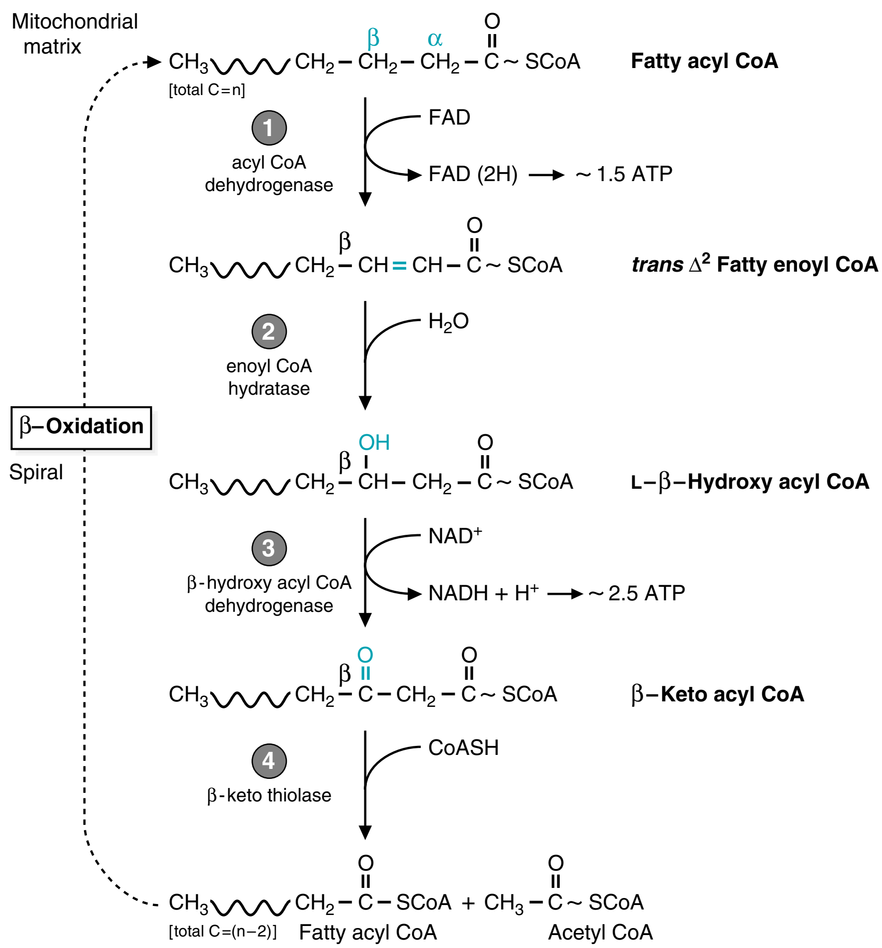


Fig. 23.7. Steps of β-oxidation . The four steps are repeated until an even-chain fatty acid is completely converted to acetyl CoA. The FAD(2H) and NADH are reoxidized by the electron transport chain, producing ATP.

configuration (a Δ^2-*trans* double bond). In the next step, an OH from water is added to the β-carbon, and an H from water is added to the β-carbon. The enzyme is called an enoyl hydratase (hydratases add the elements of water, and "ene" in a name denotes a double bond). In the third step of β-oxidation, the hydroxyl group on the β-carbon is oxidized to a ketone by a hydroxyacyl CoA dehydrogenase. In this reaction, as in the conversion of most alcohols to ketones, the electrons are transferred to NAD^+ to form NADH. In the last reaction of the sequence, the bond between the β- and α-carbons is cleaved by a reaction that attaches CoASH to the β-carbon, and acetyl CoA is released. This is a thiolytic reaction (lysis refers to breakage of the bond, and thio refers to the sulfur), catalyzed by enzymes called β-ketothiolases. The release of two carbons from the carboxyl end of the original fatty acyl CoA produces acetyl CoA and a fatty acyl CoA that is two carbons shorter than the original.

The shortened fatty acyl CoA repeats these four steps until all of its carbons are converted to acetyl CoA. β-Oxidation is, thus, a spiral rather than a cycle. In the last spiral, cleavage of the four-carbon fatty acyl CoA (butyryl CoA) produces two acetyl CoA. Thus, an even chain fatty acid such as palmitoyl CoA, which has 16 carbons, is cleaved seven times, producing 7 FAD(2H), 7 NADH, and 8 acetyl CoA.

The β-oxidation spiral uses the same reaction types seen in the TCA cycle when succinate is converted to oxaloacetate.

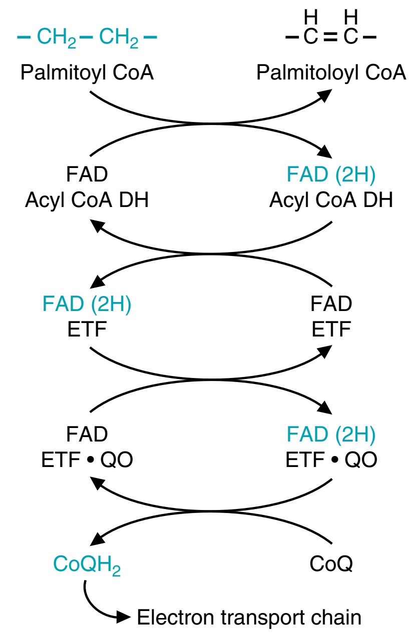


Fig. 23.8. Transfer of electrons from acyl CoA dehydrogenase to the electron transport chain. Abbreviations: ETF, electron-transferring flavoprotein; ETF-QO, electron-transferring flavoprotein–Coenzyme Q oxidoreductase.

2. ENERGY YIELD OF β-OXIDATION

Like the FAD in all flavoproteins, FAD(2H) bound to the acyl CoA dehydrogenases is oxidized back to FAD without dissociating from the protein (Fig. 23.8). Electron transfer flavoproteins (ETF) in the mitochondrial matrix accept electrons from the enzyme-bound FAD(2H) and transfer these electrons to ETF-QO (**e**lectron **t**ransfer **f**lavoprotein -Co**Q** oxidoreductase) in the inner mitochondrial membrane. ETF-QO, also a flavoprotein, transfers the electrons to CoQ in the electron transport chain. Oxidative phosphorylation thus generates approximately 1.5 ATP for each FAD(2H) produced in the β-oxidation spiral.

The total energy yield from the oxidation of 1 mole of palmityl CoA to 8 moles of acetyl CoA is therefore 28 moles of ATP: 1.5 for each of the 7 FAD(2H), and 2.5 for each of the 7 NADH. To calculate the energy yield from oxidation of 1 mole of palmitate, two ATP need to be subtracted from the total because two high-energy phosphate bonds are cleaved when palmitate is activated to palmityl CoA.

Q: What is the total ATP yield for the oxidation of 1 mole of palmitic acid to carbon dioxide and water?

3. CHAIN LENGTH SPECIFITY IN β-OXIDATION

The four reactions of β-oxidation are catalyzed by sets of enzymes that are each specific for fatty acids with different chain lengths (see Table 23.1). The acyl dehydrogenases, which catalyze the first step of the pathway, are part of an enzyme family that have four different ranges of specificity. The subsequent steps of the spiral use enzymes specific for long- or short-chain enoyl CoAs. Although these enzymes are structurally distinct, their specificity overlaps to some extent.

After reviewing **Lofata Burne's** previous hospital records, a specialist suspected that Lofata's medical problems were caused by a disorder in fatty acid metabolism. A battery of tests showed that Lofata's blood contained elevated levels of several partially oxidized medium-chain fatty acids, such as octanoic acid (8:0) and 4-decenoic acid (10:1, Δ4). A urine specimen showed an increase in organic acid metabolites of medium-chain fatty acids containing 6 to 10 carbons, including medium-chain acylcarnitine derivatives. The profile of acylcarnitine species in the urine was characteristic of a genetically determined medium-chain acyl CoA dehydrogenase (MCAD) deficiency. In this disease, long-chain fatty acids are metabolized by β-oxidation to a medium-chain-length acyl CoA, such as octanoyl CoA. Because further oxidation of this compound is blocked in MCAD deficiency, the medium chain acyl group is transferred back to carnitine. These acylcarnitines are water soluble and appear in blood and urine. The specific enzyme deficiency was demonstrated in cultured fibroblasts from Lofata's skin as well as in her circulating monocytic leukocytes.

In LCAD deficiency, fatty acylcarnitines accumulate in the blood. Those containing 14 carbons predominate. However, these do not appear in the urine.

A: Palmitic acid is 16 carbons long, with no double bonds, so it requires 7 oxidation spirals to be completely converted to acetyl-CoA. After 7 spirals, there are 7 FAD(2H), 7 NADH, and 8 acetyl-CoA. Each NADH yields 2.5 ATP, each FAD(2H) yields 1.5 ATP, and each acetyl-CoA yields 10 ATP as it is processed around the TCA cycle. This then yields 17.5 + 10.5 + 80.5 = 108 ATP. However, activation of palmitic acid to palmityl-CoA requires two high-energy bonds, so the net yield is 108 – 2, or 106 moles of ATP.

Linoleate, although high in the diet, cannot be synthesized in the human and is an essential fatty acid. It is required for formation of arachidonate, which is present in plasma lipids, and is used for eicosanoid synthesis. Therefore, only a portion of the linoleate pool is rapidly oxidized.

As the fatty acyl chains are shortened by consecutive cleavage of two acetyl units, they are transferred from enzymes that act on longer chains to those that act on shorter chains. Medium- or short-chain fatty acyl CoAs that may be formed from dietary fatty acids, or transferred from peroxisomes, enter the spiral at the enzyme most active for fatty acids of their chain length

4. OXIDATION OF UNSATURATED FATTY ACIDS

Approximately one half of the fatty acids in the human diet are unsaturated, containing cis double bonds, with oleate (C18:1, Δ^9) and linoleate (18:2,$\Delta^{9,12}$) being the most common. In β-oxidation of saturated fatty acids, a trans double bond is created between the 2nd and 3rd (α and β) carbons. For unsaturated fatty acids to undergo the β-oxidation spiral, their *cis* double bonds must be isomerized to *trans* double bonds that will end up between the 2nd and 3rd carbons during β-oxidation, or the double bond must be reduced. The process is illustrated for the polyunsaturated fatty acid linoleate in Fig. 23.9. Linoleate undergoes β-oxidation until one double bond is between carbons 3 and 4 near the carboxyl end of the fatty acyl chain, and the other is between carbons 6 and 7. An isomerase moves the double bond from the 3,4 position so that it is *trans* and in the 2,3 position, and β-oxidation continues. When a conjugated pair of double bonds is formed (two double bonds separated by one single bond) at positions 2 and 4, an NADPH-dependent reductase reduces the pair to one trans double bond at position 3. Then isomerization and β-oxidation resume.

In oleate (C18:1, Δ^9), there is only one double bond between carbons 9 and 10. It is handled by an isomerization reaction similar to that shown for the double bond at position 9 of linoleate.

5. ODD-CHAIN-LENGTH FATTY ACIDS

Fatty acids containing an odd number of carbon atoms undergo β-oxidation, producing acetyl CoA, until the last spiral, when five carbons remain in the fatty acyl CoA. In this case, cleavage by thiolase produces acetyl CoA and a three-carbon fatty acyl CoA, propionyl CoA (Fig. 23.10). Carboxylation of propionyl CoA yields methylmalonyl CoA, which is ultimately converted to succinyl CoA in a vitamin B12–dependent reaction (Fig. 23.11). Propionyl CoA also arises from the oxidation of branched chain amino acids.

The propionyl CoA to succinyl CoA pathway is a major anaplerotic route for the TCA cycle and is used in the degradation of valine, isoleucine, and a number of other compounds. In the liver, this route provides precursors of oxaloacetate, which is converted to glucose. Thus, this small proportion of the odd-carbon-number fatty acid chain can be converted to glucose. In contrast, the acetyl CoA formed from β-oxidation of even-chain-number fatty acids in the liver either enters the TCA cycle, where it is principally oxidized to CO_2, or is converted to ketone bodies.

The medium-chain-length acyl CoA synthetase has a broad range of specificity for compounds of approximately the same size that contain a carboxyl group, such as drugs (salicylate, from aspirin metabolism, and valproate, which is used to treat epileptic seizures), or benzoate, a common component of plants. Once the drug acyl CoA is formed, the acyl group is conjugated with glycine to form a urinary excretion product. With certain disorders of fatty acid oxidation, medium- and short-chain fatty acylglycines may appear in the urine, together with acylcarnitines or dicarboxylic acids.

D. Oxidation of Medium-Chain-Length Fatty Acids

Dietary medium-chain-length fatty acids are more water soluble than long-chain fatty acids and are not stored in adipose triacylglyce. After a meal, they enter the blood and pass into the portal vein to the liver. In the liver, they enter the mitochondrial matrix by the monocarboxylate transporter and are activated to acyl CoA derivatives in the mitochondrial matrix (see Fig. 23.1). Medium-chain-length acyl CoAs, like long-chain acyl CoAs, are oxidized to acetyl CoA via the β-oxidation spiral. Medium-chain acyl CoAs also can arise from the peroxisomal oxidation pathway.

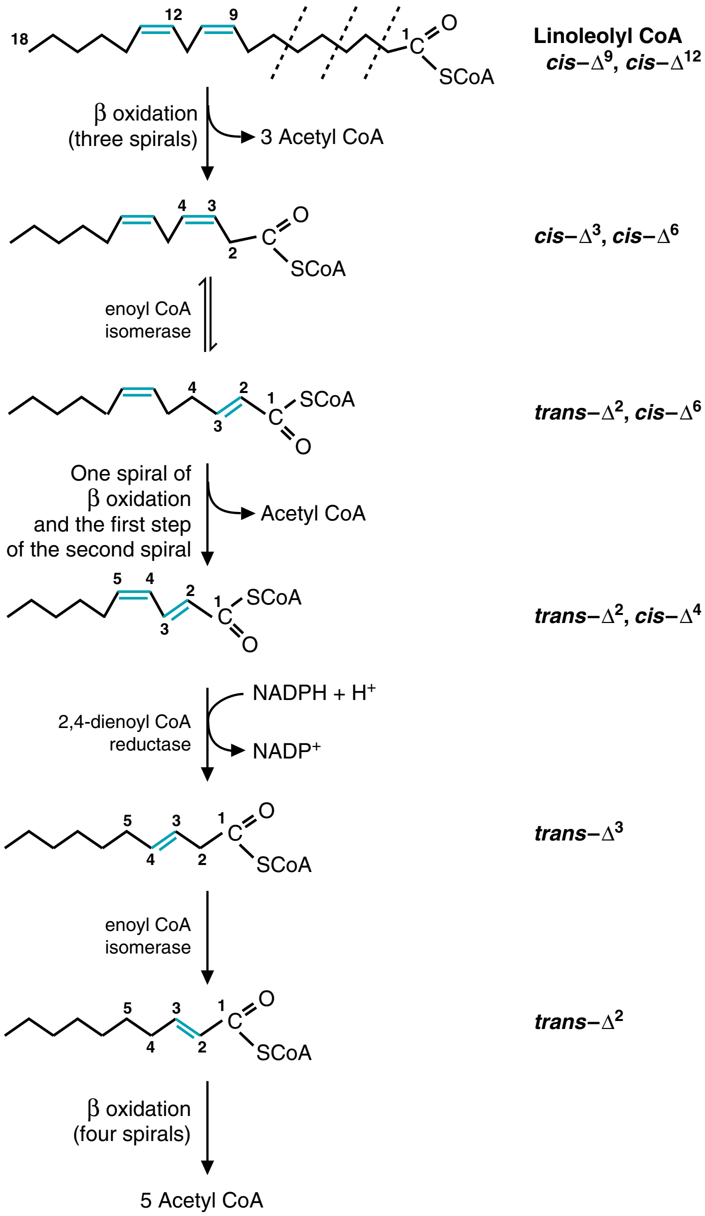


Fig. 23.9. Oxidation of linoleate. After three spirals of β-oxidation (dashed lines), there is now a 3,4 cis double bond and a 6,7 cis double bond. The 3,4 cis double bond is isomerized to a 2,3-trans double bond, which is in the proper configuration for the normal enzymes to act. One spiral of β-oxidation occurs, plus the first step of a second spiral. A reductase that uses NADPH now converts these two double bonds (between carbons 2 and 3 and carbons 4 and 5) to one double bond between carbons 3 and 4 in a trans configuration. The isomerase (which can act on double bonds that are in either the cis or the trans configuration) moves this double bond to the 2,3-trans position, and β-oxidation can resume.

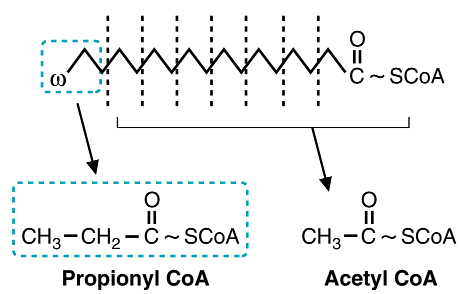


Fig. 23.10. Formation of propionyl CoA from odd-chain fatty acids. Successive spirals of β-oxidation cleave each of the bonds marked with dashed lines, producing acetyl CoA except for the three carbons at the ω-end, which produce propionyl CoA.

E. Regulation of β-Oxidation

Fatty acids are used as fuels principally when they are released from adipose tissue triacylglycerols in response to hormones that signal fasting or increased demand. Many tissues, such as muscle and kidney, oxidize fatty acids completely to CO_2 and H_2O. In these tissues, the acetyl CoA produced by β-oxidation enters the TCA cycle. The FAD(2H) and the NADH from β-oxidation and the TCA cycle are

As **Otto Shape** runs, his skeletal muscles increase their use of ATP and their rate of fuel oxidation. Fatty acid oxidation is accelerated by the increased rate of the electron transport chain. As ATP is used and AMP increases, an AMP-dependent protein kinase acts to facilitate fuel utilization and maintain ATP homeostasis. Phosphorylation of acetyl CoA carboxylase results in a decreased level of malonyl CoA and increased activity of carnitine: palmitoyl CoA transferase I. At the same time, AMP-dependent protein kinase facilitates the recruitment of glucose transporters into the plasma membrane of skeletal muscle, thereby increasing the rate of glucose uptake. AMP and hormonal signals also increase the supply of glucose 6-P from glycogenolysis. Thus, his muscles are supplied with more fuel, and all the oxidative pathways are accelerated.

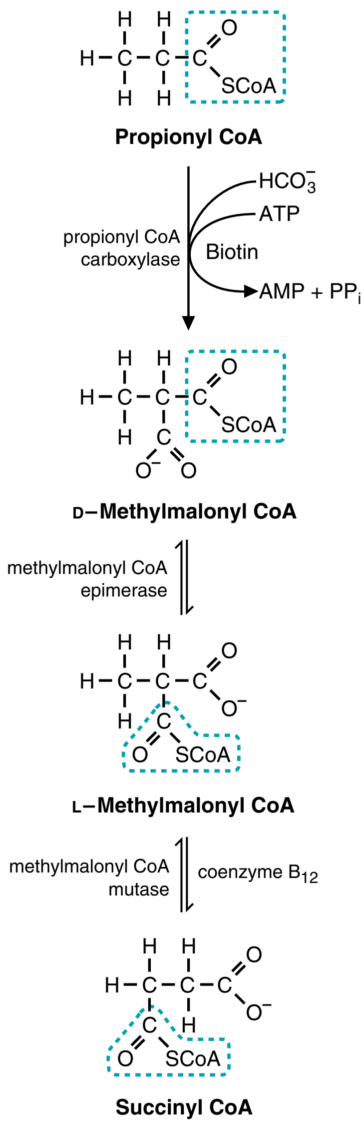


Fig. 23.11. Conversion of propionyl CoA to succinyl CoA. Succinyl CoA, an intermediate of the TCA cycle, can form malate, which can be converted to glucose in the liver through the process of gluconeogenesis. Certain amino acids also form glucose by this route (see Chapter 39).

reoxidized by the electron transport chain, and ATP is generated. The process of β-oxidation is regulated by the cells' requirements for energy (i.e., by the levels of ATP and NADH), because fatty acids cannot be oxidized any faster than NADH and FAD(2H) are reoxidized in the electron transport chain.

Fatty acid oxidation also may be restricted by the mitochondrial CoASH pool size. Acetyl CoASH units must enter the TCA cycle or another metabolic pathway to regenerate CoASH required for formation of the fatty acyl CoA derivative from fatty acyl carnitine.

An additional type of regulation occurs at carnitine:palmitoyltransferase I (CPTI). Carnitine:palmitoyltransferase I is inhibited by malonyl CoA, which is synthesized in the cytosol of many tissues by acetyl CoA carboxylase (Fig. 23.12). Acetyl CoA carboxylase is regulated by a number of different mechanisms, some of which are tissue dependent. In skeletal muscles and liver, it is inhibited when phosphorylated by protein kinase B, an AMP-dependent protein kinase. Thus, during exercise when AMP levels increase, AMP-dependent protein kinase phosphorylates acetyl CoA carboxylase, which becomes inactive. Consequently, malonyl CoA levels decrease, carnitine:palmitoyltransferase I is activated, and the β-oxidation of fatty acids is able to restore ATP homeostasis and decrease AMP levels. In liver, in addition to the regulation by the AMP-dependent protein kinase acetyl CoA carboxylase is activated by insulin-dependent mechanisms, which promotes the conversion of malonyl CoA to palmitate in the fatty acid synthesis pathway. Thus, in the liver, malonyl CoA inhibition of CPTI prevents newly synthesized fatty acids from being oxidized.

β-oxidation is strictly an aerobic pathway, dependent on oxygen, a good blood supply, and adequate levels of mitochondria. Tissues that lack mitochondria, such

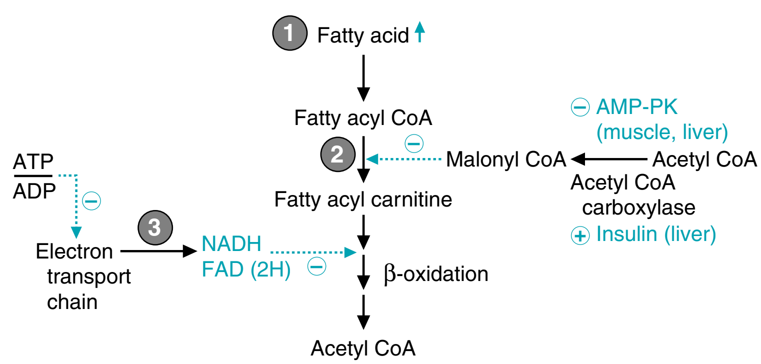


Fig. 23.12. Regulation of β-oxidation. (1) Hormones control the supply of fatty acids in the blood. (2) Carnitine:palmitoyl transferase I is inhibited by malonyl CoA, which is synthesized by acetyl CoA carboxylase (ACC). AMP-PK is the AMP-dependent protein kinase. (3) The rate of ATP utilization controls the rate of the electron transport chain, which regulates the oxidative enzymes of β-oxidation and the TCA cycle.

as red blood cells, cannot oxidize fatty acids by β-oxidation. Fatty acids also do not serve as a significant fuel for the brain. They are not used by adipocytes, whose function is to store triacylglycerols to provide a fuel for other tissues. Those tissues that do not use fatty acids as a fuel, or use them only to a limited extent, are able to use ketone bodies instead.

II. ALTERNATE ROUTES OF FATTY ACID OXIDATION

Fatty acids that are not readily oxidized by the enzymes of β-oxidation enter alternate pathways of oxidation, including peroxisomal β- and α-oxidation and microsomal ω-oxidation. The function of these pathways is to convert as much as possible of the unusual fatty acids to compounds that can be used as fuels or biosynthetic precursors, and to convert the remainder to compounds that can be excreted in bile or urine. During prolonged fasting, fatty acids released from adipose triacylglycerols may enter the ω-oxidation or peroxisomal β-oxidation pathway, even though they have a normal composition. These pathways not only use fatty acids, but they act on xenobiotic carboxylic acids that are large hydrophobic molecules resembling fatty acids.

Xenobiotic: a term used to cover all organic compounds that are foreign to an organism. This can also include naturally occurring compounds that are administered by alternate routes or at unusual concentrations. Drugs can be considered xenobiotics.

A. Peroxisomal Oxidation of Fatty Acids

A small proportion of our diet consists of very-long-chain fatty acids (20 or more carbons) or branched-chain fatty acids arising from degradative products of chlorophyll. Very-long-chain fatty acid synthesis also occurs within the body, especially in cells of the brain and nervous system, which incorporate them into the sphingolipids of myelin. These fatty acids are oxidized by peroxisomal β- and α-oxidation pathways, which are essentially chain-shortening pathways.

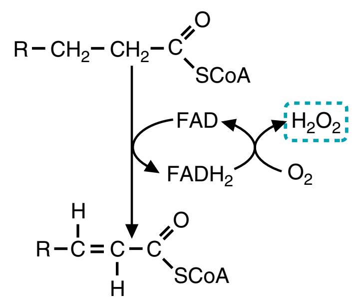


Fig. 23.13. Oxidation of fatty acids in peroxisomes. The first step of β-oxidation is catalyzed by an FAD-containing oxidase. The electrons are transferred from FAD(2H) to O_2, which is reduced to hydrogen peroxide (H_2O_2).

1. VERY-LONG-CHAIN FATTY ACIDS

Very-long-chain fatty acids of 24 to 26 carbons are oxidized exclusively in peroxisomes by a sequence of reactions similar to mitochondrial β-oxidation in that they generate acetyl CoA and NADH. However, the peroxisomal oxidation of straight-chain fatty acids stops when the chain reaches 4 to 6 carbons in length. Some of the long-chain fatty acids also may be oxidized by this route.

The long-chain fatty acyl CoA synthetase is present in the peroxisomal membrane, and the acyl CoA derivatives enter the peroxisome by a transporter that does not require carnitine. The first enzyme of peroxisomal β-oxidation is an oxidase, which donates electrons directly to molecular oxygen and produces hydrogen peroxide (H_2O_2) (Fig.23.13). (In contrast, the first enzyme of mitochondrial β-oxidation is a dehydrogenase that contains FAD and transfers the electrons to the electron transport chain via ETF.) Thus, the first enzyme of peroxisomal oxidation is not linked to energy production. The three remaining steps of β-oxidation are catalyzed by enoyl-CoA hydratase, hydroxyacyl CoA dehydrogenase, and thiolase, enzymes with activities similar to those found in mitochondrial β-oxidation, but coded for by different genes. Thus, one NADH and one acetyl CoA are generated for each turn of the spiral. The peroxisomal β-oxidation spiral continues generating acetyl CoA until a medium-chain acyl CoA, which may be as short as butyryl CoA, is produced (Fig. 23.14).

Within the peroxisome, the acetyl groups can be transferred from CoA to carnitine by an acetylcarnitine transferase, or they can enter the cytosol. A similar reaction converts medium-chain-length acyl CoAs and the short-chain butyryl CoA to acyl carnitine derivatives. These acylcarnitines diffuse from the peroxisome to the mitochondria, pass through the outer mitochondrial membrane, and are transported through the inner mitochondrial membrane via the carnitine translocase system.

A number of inherited deficiencies of peroxisomal enzymes have been described. Zellweger's syndrome, which results from defective peroxisomal biogenesis, leads to complex developmental and metabolic phenotypes affecting principally the liver and the brain. One of the metabolic characteristics of these diseases is an elevation of C26:0, and C26:1 fatty acid levels in plasma. Refsum's disease is caused by a deficiency in a single peroxisomal enzyme, the phytanoyl CoA hydroxylase that carries out α-oxidation of phytanic acid. Symptoms include retinitis pigmentosa, cerebellar ataxia, and chronic polyneuropathy. Because phytanic acid is obtained solely from the diet, placing patients on a low–phytanic acid diet has resulted in marked improvement.

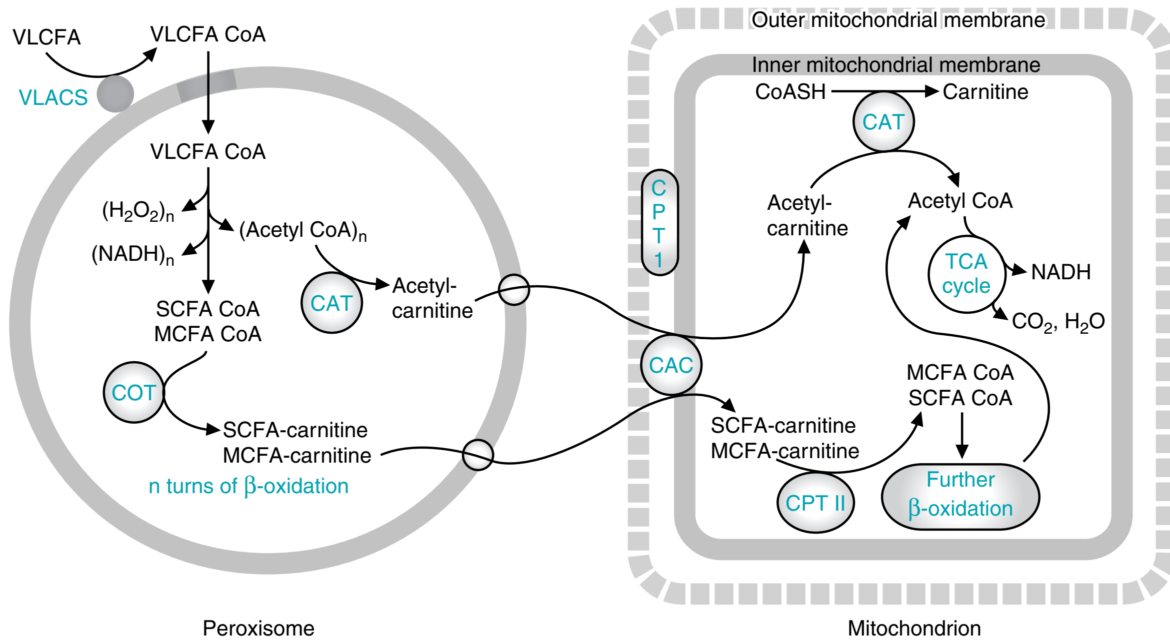


Fig. 23.14. Chain-shortening by peroxisomal β-oxidation. Abbreviations: VLCFA, very-long-chain fatty acyl; VLACS, very-long-chain acyl-CoA synthetase; MCFA, medium-chain fatty acyl; SCFA, short-chain fatty acyl; CAT, carnitine:acetyltransferase; COT, carnitine:octanoyltransferase; CAC: carnitine:acylcarnitine carrier; CPT1, carnitine: palmitoyltransferase 1; CPT2, carnitine: palmityltransferase 2; OMM, outer mitochondrial membrane; IMM, inner mitochondrial membrane. Very-long-chain fatty acyl CoAs and some long-chain fatty acyl CoAs are oxidized in peroxisomes through n cycles of β-oxidation to the stage of a short- to medium-chain fatty acyl CoA. These short to medium fatty acyl CoAs are converted to carnitine derivatives by COT or CAT in the peroxisomes. In the mitochondria, SCFA-carnitine are converted back to acyl CoA derivatives by either CPT2 or CAT.

They are converted back to acyl CoAs by carnitine: acyltransferases appropriate for their chain length and enter the normal pathways for β-oxidation and acetyl CoA metabolism. The electrons from NADH and acetyl CoA can also pass from the peroxisome to the cytosol. The export of NADH-containing electrons occurs through use of a shuttle system similar to those described for NADH electron transfer into the mitochondria.

Peroxisomes are present in almost every cell type and contain many degradative enzymes, in addition to fatty acyl CoA oxidase, that generate hydrogen peroxide. H_2O_2 can generate toxic free radicals. Thus, these enzymes are confined to peroxisomes, where the H_2O_2 can be neutralized by the free radical defense enzyme, catalase. Catalase converts H_2O_2 to water and O_2.

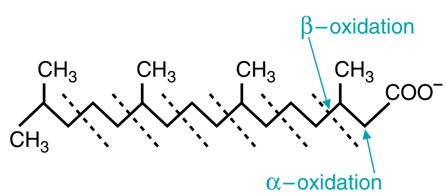


Fig. 23.15. Oxidation of phytanic acid. A peroxisomal α-hydroxylase oxidizes the α-carbon, and its subsequent oxidation to a carboxyl group releases the carboxyl carbon as CO_2. Subsequent spirals of peroxisomal β-oxidation alternately release propionyl and acetyl CoA. At a chain length of approximately 8 carbons, the remaining branched fatty acid is transferred to mitochondria as a medium-chain carnitine derivative.

2. LONG-CHAIN BRANCHED-CHAIN FATTY ACIDS

Two of the most common branched-chain fatty acids in the diet are phytanic acid and pristanic acid, which are degradation products of chlorophyll and thus are consumed in green vegetables (Fig.23.15). Animals do not synthesize branched-chain fatty acids. These two multi-methylated fatty acids are oxidized in peroxisomes to the level of a branched C8 fatty acid, which is then transferred to mitochondria. The pathway thus is similar to that for the oxidation of straight very-long-chain fatty acids.

Phytanic acid, a multi-methylated C20 fatty acid, is first oxidized to pristanic acid using the α-oxidation pathway (see Fig.23.15). Phytanic acid hydroxylase introduces a hydroxyl group on the α-carbon, which is then oxidized to a carboxyl group with release of the original carboxyl group as CO_2. By shortening the fatty acid by one carbon, the methyl groups will appear on the α-carbon rather than the β-carbon during the β-oxidation spiral, and can no longer interfere with oxidation of the β-carbon. Peroxisomal β-oxidation thus can proceed normally, releasing propionyl CoA and acetyl CoA with alternate turns of the spiral. When a medium chain length of approximately eight carbons is reached, the fatty acid is transferred to the mitochondrion as a carnitine derivative, and β-oxidation is resumed.

B. ω-Oxidation of Fatty Acids

Fatty acids also may be oxidized at the ω-carbon of the chain (the terminal methyl group) by enzymes in the endoplasmic reticulum (Fig. 23.16). The ω-methyl group is first oxidized to an alcohol by an enzyme that uses cytochrome P450, molecular oxygen, and NADPH. Dehydrogenases convert the alcohol group to a carboxylic acid. The dicarboxylic acids produced by ω-oxidation can undergo β-oxidation, forming compounds with 6 to 10 carbons that are water-soluble. Such compounds may then enter blood, be oxidized as medium-chain fatty acids, or be excreted in urine as medium-chain dicarboxylic acids.

The pathways of peroxisomal α and β-oxidation, and microsomal ω-oxidation, are not feedback regulated. These pathways function to decrease levels of water-insoluble fatty acids or of xenobiotic compounds with a fatty acid–like structure that would become toxic to cells at high concentrations. Thus, their rate is regulated by the availability of substrate.

III. METABOLISM OF KETONE BODIES

Overall, fatty acids released from adipose triacylglycerols serve as the major fuel for the body during fasting. These fatty acids are completely oxidized to CO_2 and H_2O by some tissues. In the liver, much of the acetyl CoA generated from β-oxidation of fatty acids is used for synthesis of the ketone bodies acetoacetate and β-hydroxybutyrate, which enter the blood (Fig. 23.17). In skeletal muscles and other

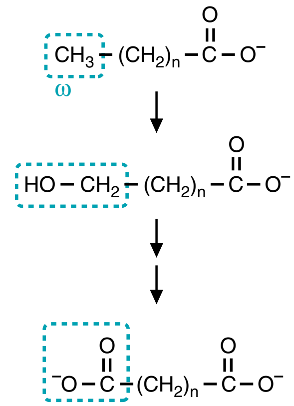


Fig. 23.16. ω-Oxidation of fatty acids converts them to dicarboxylic acids.

Normally, ω-oxidation is a minor process. However, in conditions that interfere with β-oxidation (such as carnitine deficiency or deficiency in an enzyme of β-oxidation), ω-oxidation produces dicarboxylic acids in increased amounts. These dicarboxylic acids are excreted in the urine.

Lofata Burne was excreting dicarboxylic acids in her urine, particularly adipic acid (which has 6 carbons) and suberic acid (which has 8 carbons).

–OOC—CH2—CH2—CH2—CH2—COO–
Adipic acid
–OOC—CH2—CH2—CH2—CH2—CH2—CH2—COO–Suberic acid

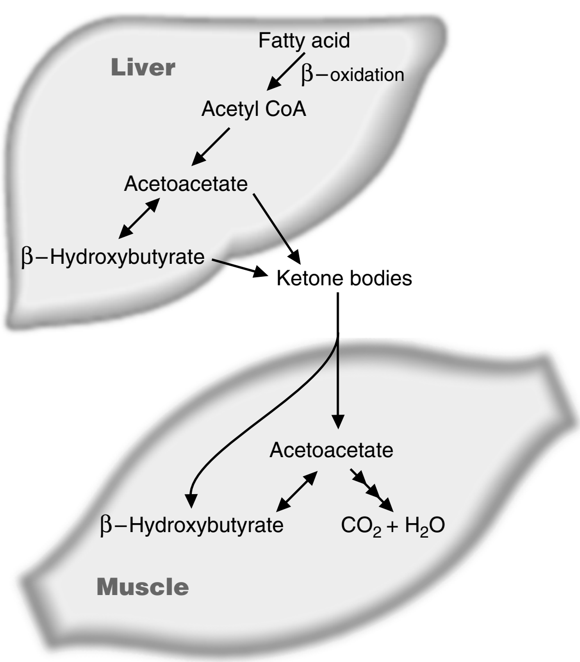


Fig. 23.17. The ketone bodies, acetoacetate and β-hydroxybutyrate, are synthesized in the liver. Their principle fate is conversion back to acetyl CoA and oxidation in the TCA cycle in other tissues.

tissues, these ketone bodies are converted back to acetyl CoA, which is oxidized in the TCA cycle with generation of ATP. An alternate fate of acetoacetate in tissues is the formation of cytosolic acetyl CoA.

A. Synthesis of Ketone Bodies

In the liver, ketone bodies are synthesized in the mitochondrial matrix from acetyl CoA generated from fatty acid oxidation (Fig. 23.18). The thiolase reaction of fatty acid oxidation, which converts acetoacetyl CoA to two molecules of acetyl CoA, is a reversible reaction, although formation of acetoacetyl-CoA is not the favored direction. It can, thus, when acetyl-CoA levels are high, generate acetoacetyl CoA

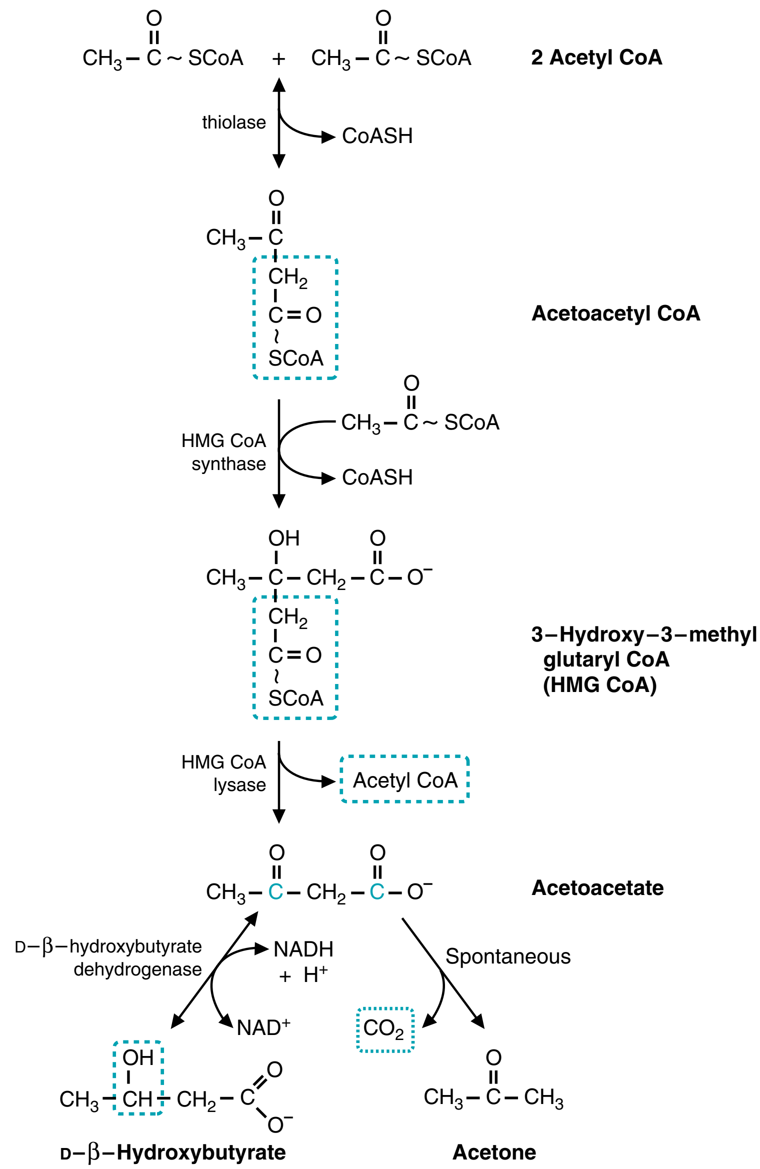


Fig. 23.18. Synthesis of the ketone bodies acetoacetate, β-hydroxybutyrate, and acetone. The portion of HMG-CoA shown in blue is released as acetyl CoA, and the remainder of the molecule forms acetoacetate. Acetoacetate is reduced to β-hydroxybutyrate or decarboxylated to acetone. Note that the dehydrogenase that interconverts acetoacetate and β-hydroxybutyrate is specific for the D-isomer. Thus, it differs from the dehydrogenases of β-oxidation, which act on 3-hydroxy acyl CoA derivatives and is specific for the L-isomer.

for ketone body synthesis. The acetoacetyl CoA will react with acetyl CoA to produce 3-hydroxy-3-methylglutaryl CoA (HMG-CoA). The enzyme that catalyzes this reaction is HMG-CoA synthase. In the next reaction of the pathway, HMG-CoA lyase catalyzes the cleavage of HMG-CoA to form acetyl CoA and acetoacetate.

Acetoacetate can directly enter the blood or it can be reduced by β-hydroxybutyrate dehydrogenase to β-hydroxybutyrate, which enters the blood (see Fig. 23.18). This dehydrogenase reaction is readily reversible and interconverts these two ketone bodies, which exist in an equilibrium ratio determined by the $NADH/NAD^+$ ratio of the mitochondrial matrix. Under normal conditions, the ratio of β-hydroxybutyrate to acetoacetate in the blood is approximately 1:1.

An alternate fate of acetoacetate is spontaneous decarboxylation, a nonenzymatic reaction that cleaves acetoacetate into CO2 and acetone (see Fig. 23.18). Because acetone is volatile, it is expired by the lungs. A small amount of acetone may be further metabolized in the body.

B. Oxidation of Ketone Bodies as Fuels

Acetoacetate and β-hydroxybutyrate can be oxidized as fuels in most tissues, including skeletal muscle, brain, certain cells of the kidney, and cells of the intestinal mucosa. Cells transport both acetoacetate and β-hydroxybutyrate from the circulating blood into the cytosol, and into the mitochondrial matrix. Here β-hydroxybutyrate is oxidized back to acetoacetate by β-hydroxybutyrate dehydrogenase. This reaction produces NADH. Subsequent steps convert acetoacetate to acetyl CoA (Fig. 23.19).

In mitochondria, acetoacetate is activated to acetoacetyl CoA by succinyl CoA:acetoacetate CoA transferase. As the name suggests, CoA is transferred from succinyl CoA, a TCA cycle intermediate, to acetoacetate. Although the liver produces ketone bodies, it does not use them, because this thiotransferase enzyme is not present in sufficient quantity.

Acetoacetyl CoA is cleaved to two molecules of acetyl CoA by acetoacetyl CoA thiolase, the same enzyme involved in β-oxidation. The principal fate of this acetyl CoA is oxidation in the TCA cycle.

The energy yield from oxidation of acetoacetate is equivalent to the yield for oxidation of 2 acetyl CoA in the TCA cycle (20 ATP) minus the energy for activation of acetoacetate (1 ATP). The energy of activation is calculated at one high-energy phosphate bond, because succinyl CoA is normally converted to succinate in the TCA cycle, with generation of one molecule of GTP (the energy equivalent of ATP). However, when the high-energy thioester bond of succinyl CoA is transferred to acetoacetate, succinate is produced without the generation of this GTP. Oxidation of β-hydroxybutyrate generates one additional NADH. Therefore the net energy yield from one molecule of β-hydroxybutyrate is approximately 21.5 molecules of ATP.

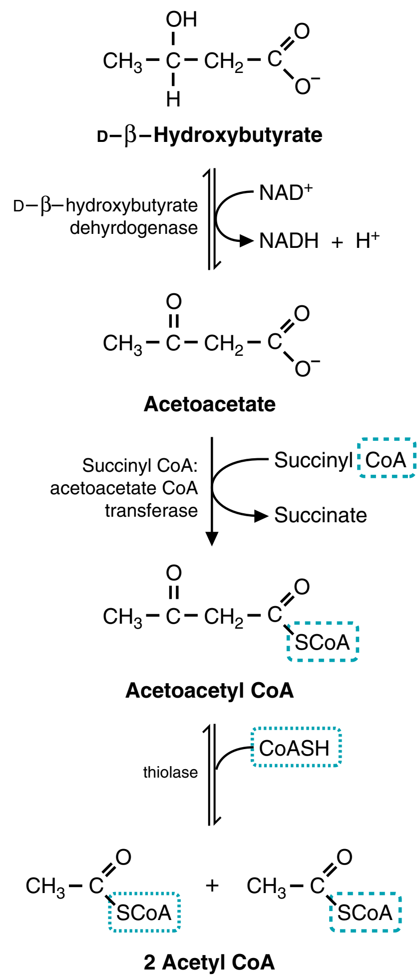


Fig. 23.19. Oxidation of ketone bodies. β-Hydroxybutyrate is oxidized to acetoacetate, which is activated by accepting a CoA group from succinyl CoA. Acetoacetyl CoA is cleaved to two acetyl CoA, which enter the TCA cycle and are oxidized.

C. Alternate Pathways of Ketone Body Metabolism

Although fatty acid oxidation is usually the major source of ketone bodies, they also can be generated from the catabolism of certain amino acids: leucine, isoleucine, lysine, tryptophan, phenylalanine, and tyrosine. These amino acids are called ketogenic amino acids because their carbon skeleton is catabolized to acetyl CoA or acetoacetyl CoA, which may enter the pathway of ketone body synthesis in liver. Leucine and isoleucine also form acetyl CoA and acetoacetyl CoA in other tissues, as well as the liver.

Acetoacetate can be activated to acetoacetyl CoA in the cytosol by an enzyme similar to the acyl CoA synthetases. This acetoacetyl CoA can be used directly in cholesterol synthesis. It also can be cleaved to two molecules of acetyl CoA by a cytosolic thiolase. Cytosolic acetyl CoA is required for processes such as acetylcholine synthesis in neuronal cells.

Ketogenic diets, which are high-fat diets with a 3:1 ratio of lipid to carbohydrate, are being used to reduce the frequency of epileptic seizures in children. The reason for its effectiveness in the treatment of epilepsy is not known. Ketogenic diets are also used to treat children with pyruvate dehydrogenase deficiency. Ketone bodies can be used as a fuel by the brain in the absence of pyruvate dehydrogenase. They also can provide a source of cytosolic acetyl CoA for acetylcholine synthesis. They often contain medium-chain triglycerides, which induce ketosis more effectively than long-chain triglycerides.

IV. THE ROLE OF FATTY ACIDS AND KETONE BODIES IN FUEL HOMEOSTASIS

Fatty acids are used as fuels whenever fatty acid levels are elevated in the blood, that is, during fasting, starvation, as a result of a high-fat, low-carbohydrate diet, or during long-term low- to mild-intensity exercise. Under these conditions, a decrease in insulin and increased levels of glucagon, epinephrine, or other hormones stimulate adipose tissue lipolysis. Fatty acids begin to increase in the blood approximately 3 to 4 hours after a meal and progressively increase with time of fasting up to approximately 2 to 3 days (Fig. 23.20). In the liver, the rate of ketone body synthesis increases as the supply of fatty acids increases. However, the blood level of ketone bodies continues to increase, presumably because their utilization by skeletal muscles decreases.

After 2 to 3 days of starvation, ketone bodies rise to a level in the blood that enables them to enter brain cells, where they are oxidized, thereby reducing the amount of glucose required by the brain. During prolonged fasting, they may supply as much as two thirds of the energy requirements of the brain. The reduction in glucose requirements spares skeletal muscle protein, which is a major source of amino acid precursors for hepatic glucose synthesis from gluconeogenesis.

Children are more prone to ketosis than adults because their body enters the fasting state more rapidly. Their bodies use more energy per unit mass (because their muscle-to-adipose-tissue ratio is higher), and liver glycogen stores are depleted faster (the ratio of their brain mass to liver mass is higher). In children, blood ketone body levels reach 2 mM in 24 hours; in adults, it takes more than 3 days to reach this level. Mild pediatric infections causing anorexia and vomiting are the commonest cause of ketosis in children. Mild ketosis is observed in children after prolonged exercise, perhaps attributable to an abrupt decrease in muscular use of fatty acids liberated during exercise. The liver then oxidizes these fatty acids and produces ketone bodies.

A. Preferential Utilization of Fatty Acids

As fatty acids increase in the blood, they are used by skeletal muscles and certain other tissues in preference to glucose. Fatty acid oxidation generates NADH and FAD(2H) through both β-oxidation and the TCA cycle, resulting in relatively high $NADH/NAD^+$ ratios, acetyl CoA concentration, and ATP/ADP or ATP/AMP levels. In skeletal muscles, AMP-dependent protein kinase (see Section I.E.) adjusts the concentration of malonyl CoA so that CPT1 and β-oxidation operate at a rate that is able to sustain ATP homeostasis. With adequate levels of ATP obtained from fatty acid (or ketone body) oxidation, the rate of glycolysis is decreased. The activity of the regulatory enzymes in glycolysis and the TCA cycle (pyruvate dehydrogenase and PFK-1) are decreased by the changes in concentration of their allosteric regulators (ADP, an activator of PDH,

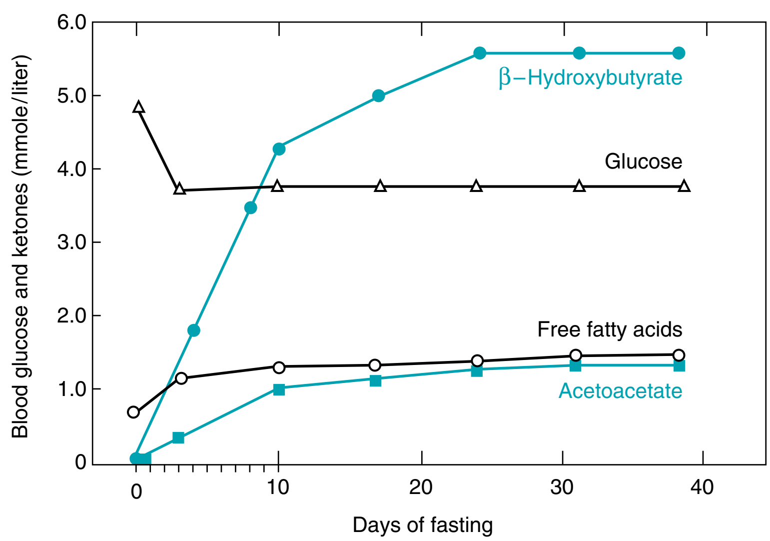


Fig. 23.20. Levels of ketone bodies in the blood at various times during fasting. Glucose levels remain relatively constant, as do levels of fatty acids. Ketone body levels, however, increase markedly, rising to levels at which they can be used by the brain and other nervous tissue. From Cahill GF Jr, Aoki TT. Med Times 1970;98:109.

decreases in concentration; NADH, and acetyl CoA, inhibitors of PDH, are increased in concentration under these conditions; and ATP and citrate, inhibitors of PFK-1, are increased in concentration). As a consequence, glucose-6-P accumulates. Glucose-6-P inhibits hexokinase, thereby decreasing the rate of entry of glucose into glycolysis, and its uptake from the blood. In skeletal muscles, this pattern of fuel metabolism is facilitated by the decrease in insulin concentration (see Chapter 36). Preferential utilization of fatty acids does not, however, restrict the ability of glycolysis to respond to an increase in AMP or ADP levels, such as might occur during exercise or oxygen limitation.

The level of total ketone bodies in **Lofata Burne's** blood greatly exceeds normal fasting levels and the mild ketosis produced during exercise. In a person on a normal mealtime schedule, total blood ketone bodies rarely exceed 0.2 mM. During prolonged fasting, they may rise to 4 to 5 mM. Levels above 7 mM are considered evidence of ketoacidosis, because the acid produced must reach this level to exceed the bicarbonate buffer system in the blood and compensatory respiration (Kussmaul's respiration) (see Chapter 4).

B. Tissues That Use Ketone Bodies

Skeletal muscles, the heart, the liver, and many other tissues use fatty acids as their major fuel during fasting and other conditions that increase fatty acids in the blood. However, a number of other tissues (or cell types), such as the brain, use ketone bodies to a greater extent. For example, cells of the intestinal muscosa, which transport fatty acids from the intestine to the blood, use ketone bodies and amino acids during starvation, rather than fatty acids. Adipocytes, which store fatty acids in triacylglycerols, do not use fatty acids as a fuel during fasting but can use ketone bodies. Ketone bodies cross the placenta, and can be used by the fetus. Almost all tissues and cell types, with the exception of liver and red blood cells, are able to use ketone bodies as fuels.

Q: Why can't red blood cells use ketone bodies for energy?

C. Regulation of Ketone Body Synthesis

A number of events, in addition to the increased supply of fatty acids from adipose triacylglycerols, promote hepatic ketone body synthesis during fasting. The decreased insulin/glucagon ratio results in inhibition of acetyl CoA carboxylase and decreased malonyl CoA levels, which activates CPTI, thereby allowing fatty acyl CoA to enter the pathway of β-oxidation. (Fig. 23.21). When oxidation of fatty acyl CoA to acetyl CoA generates enough NADH and FAD(2H) to supply the ATP needs of the liver, acetyl CoA is diverted from the TCA cycle into ketogenesis and oxaloacetate in the TCA cycle is diverted toward malate and into glucose synthesis (gluconeogenesis). This pattern is regulated by the NADH/NAD^+ ratio, which is relatively high during β-oxidation. As the length of time of fasting continues, increased transcription of the gene for mitochondrial HMG-CoA synthase facilitates high rates of ketone body production. Although the liver has been described as "altruistic" because it provides ketone bodies for other tissues, it is simply getting rid of fuel that it does not need.

CLINICAL COMMENTS

As **Otto Shape** runs, he increases the rate at which his muscles oxidize all fuels. The increased rate of ATP utilization stimulates the electron transport chain, which oxidizes NADH and FAD(2H) much faster, thereby increasing the rate at which fatty acids are oxidized. During exercise, he also uses muscle glycogen stores, which contribute glucose to glycolysis. In some of the fibers, the glucose is used anaerobically, thereby producing lactate. Some of the lactate will be used by his heart, and some will be taken up by the liver to be converted to glucose. As he trains, he increases his mitochondrial capacity, as well as his oxygen delivery, resulting in an increased ability to oxidize fatty acids and ketone bodies. As he runs, he increases fatty acid release from adipose tissue triacylglycerols. In the liver, fatty acids are being converted to ketone bodies, providing his muscles with another fuel. As a consequence, he experiences mild ketosis after his 12-mile run.

A: Red blood cells lack mitochondria, which is the site of ketone body utilization.

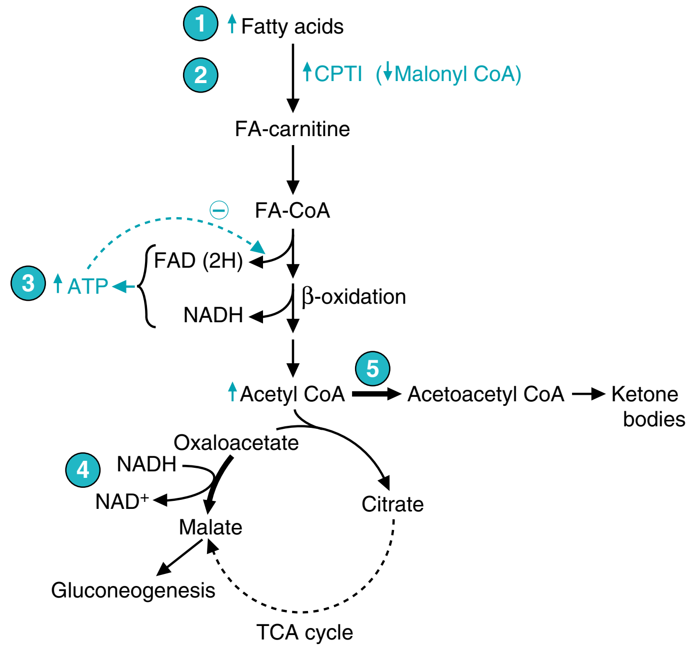


Fig. 23.21. Regulation of ketone body synthesis. (1) The supply of fatty acids is increased. (2) The malonyl CoA inhibition of CPTI is lifted by inactivation of acetyl CoA carboxylase. (3) β-Oxidation supplies NADH and FAD(2H), which are used by the electron transport chain for oxidative phosphorylation. As ATP levels increase, less NADH is oxidized, and the NADH/NAD^+ ratio is increased. (4) Oxaloacetate is converted into malate because of the high NADH levels, and the malate enters the cytoplasm for gluconeogenesis,. (5) Acetyl CoA is diverted from the TCA cycle into ketogenesis, in part because of low oxaloacetate levels, which reduces the rate of the citrate synthase reaction.

More than 25 enzymes and specific transport proteins participate in mitochondrial fatty acid metabolism. At least 15 of these have been implicated in inherited diseases in the human.

Recently, medium-chain acyl-CoA dehydrogenase (MCAD) deficiency, the cause of **Lofata Burne's** problems, has emerged as one of the most common of the inborn errors of metabolism, with a carrier frequency ranging from 1 in 40 in northern European populations to less than 1 in 100 in Asians. Overall, the predicted disease frequency for MCAD deficiency is 1 in 15,000 persons.

MCAD deficiency is an autosomal recessive disorder caused by the substitution of a T for an A at position 985 of the MCAD gene. This mutation causes a lysine to replace a glutamate residue in the protein, resulting in the production of an unstable dehydrogenase.

The most frequent manifestation of MCAD deficiency is intermittent hypoketotic hypoglycemia during fasting (low levels of ketone bodies and low levels of glucose in the blood). Fatty acids normally would be oxidized to CO_2 and H_2O under these conditions. In MCAD deficiency, however, fatty acids are oxidized only until they reach medium-chain length As a result, the body must rely to a greater extent on oxidation of blood glucose to meet its energy needs.

However, hepatic gluconeogenesis appears to be impaired in MCAD. Inhibition of gluconeogenesis may be caused by the lack of hepatic fatty acid oxidation to supply the energy required for gluconeogenesis, or by the accumulation of unoxidized fatty acid metabolites that inhibit gluconeogenic enzymes. As a consequence, liver glycogen stores are depleted more rapidly, and hypoglycemia results. The decrease in hepatic fatty acid oxidation results in less acetyl CoA for ketone body synthesis, and consequently a hypoketotic hypoglycemia develops.

Some of the symptoms once ascribed to hypoglycemia are now believed to be caused by the accumulation of toxic fatty acid intermediates, especially in those patients with only mild reductions in blood glucose levels. **Lofata Burne's** mild elevation in the blood of liver transaminases may reflect an infiltration of her liver cells with unoxidized medium-chain fatty acids.

The management of MCAD-deficient patients includes the intake of a relatively high-carbohydrate diet and the avoidance of prolonged fasting.

Di Abietes, a 26-year-old woman with type 1 diabetes mellitus, was admitted to the hospital in diabetic ketoacidosis. In this complication of diabetes mellitus, an acute deficiency of insulin, coupled with a relative excess of glucagon, results in a rapid mobilization of fuel stores from muscle (amino acids) and adipose tissue (fatty acids). Some of the amino acids are converted to glucose, and fatty acids are converted to ketones (acetoacetate, β-hydroxybutyrate, and acetone). The high glucagon: insulin ratio promotes the hepatic production of ketones. In response to the metabolic "stress," the levels of insulin-antagonistic hormones, such as catecholamines, glucocorticoids, and growth hormone, are increased in the blood. The insulin deficiency further reduces the peripheral utilization of glucose and ketones. As a result of this interrelated dysmetabolism, plasma glucose levels reach 500 mg/dL (27.8 mmol/L) or more (normal fasting levels are 70–100 mg/dL, or 3.9–5.5 mmol/L), and plasma ketones rise to levels of 8 to 15 mmol/L or more (normal is in the range of 0.2–2 mmol/L, depending on the fed state of the individual).

The increased glucose presented to the renal glomeruli induces an osmotic diuresis, which further depletes intravascular volume, further reducing the renal excretion of hydrogen ions and glucose. As a result, the metabolic acidosis worsens, and the hyperosmolarity of the blood increases, at times exceeding 330 mOsm/kg (normal is in the range of 285–295 mOsm/kg). The severity of the hyperosmolar state correlates closely with the degree of central nervous system dysfunction and may end in coma and even death if left untreated.

BIOCHEMICAL COMMENTS

The unripe fruit of the akee tree produces a toxin, hypoglycin, which causes a condition known as Jamaican vomiting sickness. The victims of the toxin are usually unwary children who eat this unripe fruit and develop a severe hypoglycemia, which is often fatal.

Although hypoglycin causes hypoglycemia, it acts by inhibiting an acyl CoA dehydrogenase involved in β-oxidation that has specificity for short- and medium-chain fatty acids. Because more glucose must be oxidized to compensate for the decreased ability of fatty acids to serve as fuel, blood glucose levels may fall to extremely low levels. Fatty acid levels, however, rise because of decreased β-oxidation. As a result of the increased fatty acid levels, α-oxidation increases, and dicarboxylic acids are excreted in the urine. The diminished capacity to oxidize fatty acids in liver mitochondria results in decreased levels of acetyl CoA, the substrate for ketone body synthesis.

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REVIEW QUESTIONS—CHAPTER 23

1. A lack of the enzyme ETF:CoQ oxidoreductase leads to death. This is due to which of the following reasons?

   (A) The energy yield from glucose utilization is dramatically reduced.
   (B) The energy yield from alcohol utilization is dramatically reduced.
   (C) The energy yield from ketone body utilization is dramatically reduced.
   (D) The energy yield from fatty acid utilization is dramatically reduced.
   (E) The energy yield from glycogen utilization is dramatically reduced.

2. The ATP yield from the complete oxidation of 1 mole of a $C_{18:0}$ fatty acid to carbon dioxide and water would be closest to which **ONE** of the following?

   (A) 105
   (B) 115
   (C) 120
   (D) 125
   (E) 130

3. The oxidation of fatty acids is best described by which of the following sets of reactions?

   (A) Oxidation, hydration, oxidation, carbon-carbon bond breaking
   (B) Oxidation, dehydration, oxidation, carbon-carbon bond breaking
   (C) Oxidation, hydration, reduction, carbon-carbon bond breaking
   (D) Oxidation, dehydration, reduction, oxidation, carbon-carbon bond breaking
   (E) Reduction, hydration, oxidation, carbon-carbon bond breaking

4. An individual with a deficiency of an enzyme in the pathway for carnitine synthesis is not eating adequate amounts of carnitine in the diet. Which of the following effects would you expect during fasting as compared with an individual with an adequate intake and synthesis of carnitine?

   (A) Fatty acid oxidation is increased.
   (B) Ketone body synthesis is increased.
   (C) Blood glucose levels are increased.
   (D) The levels of dicarboxylic acids in the blood would be increased.
   (E) The levels of very-long-chain fatty acids in the blood would be increased.

5. At which one of the periods listed below will fatty acids be the major source of fuel for the tissues of the body?

   (A) Immediately after breakfast
   (B) Minutes after a snack
   (C) Immediately after dinner
   (D) While running the first mile of a marathon
   (E) While running the last mile of a marathon

24 Oxygen Toxicity and Free Radical Injury

***O_2* is both essential to human life and *toxic*. We are dependent on O_2 for oxidation reactions in the pathways of adenosine triphosphate (ATP) generation, detoxification, and biosynthesis. However, when O_2 accepts single electrons, it is transformed into highly reactive *oxygen radicals* that damage cellular lipids, proteins, and DNA. Damage by reactive oxygen radicals contributes to cellular death and degeneration in a wide range of diseases (Table 24.1).**

Radicals are compounds that contain a single electron, usually in an outside orbital. Oxygen is a ***biradical***, a molecule that has two unpaired electrons in separate orbitals (Fig. 24.1). Through a number of enzymatic and nonenzymatic processes that routinely occur in cells, O_2 accepts ***single electrons*** to form ***reactive oxygen species (ROS)***. ROS are highly reactive oxygen radicals, or compounds that are readily converted in cells to these reactive radicals. The ROS formed by reduction of O_2 are the radical ***superoxide (O_2^-)***, the nonradical ***hydrogen peroxide (H_2O_2)***, and the ***hydroxyl radical (OH•)***.

ROS may be generated nonenzymatically, or enzymatically as accidental byproducts or major products of reactions. Superoxide may be generated nonenzymatically from CoQ, or from metal-containing enzymes (e.g., ***cytochrome P450, xanthine oxidase,*** *and* ***NADPH oxidase****). The highly toxic hydroxyl radical is formed nonenzymatically from superoxide in the presence of* ***Fe^{3+}*** *or* ***Cu^+*** *by the* ***Fenton reaction,*** *and from hydrogen peroxide in the* ***Haber–Weiss reaction.***

Oxygen radicals and their derivatives can be deadly to cells. The hydroxyl radical causes oxidative damage to proteins and DNA. It also forms ***lipid peroxides*** *and* ***malondialdehyde*** *from membrane lipids containing* ***polyunsaturated fatty acids.*** *In some cases, free radical damage is the direct cause of a disease state (e.g., tissue damage initiated by exposure to ionizing radiation). In* ***neurodegenerative diseases,*** *such as Parkinson's disease, or in ischemia-reperfusion injury, ROS may perpetuate the cellular damage caused by another process.*

Oxygen radicals are joined in their destructive damage by the free radical nitric oxide (NO) and the reactive oxygen species ***hypochlorous acid (HOCl)****. NO*

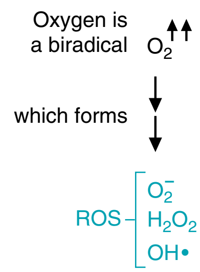

Fig 24.1. O_2 is a biradical. It has two antibonding electrons with parallel spins, denoted by the parallel arrows. It has a tendency to form toxic reactive oxygen species (ROS), such as superoxide (O_2^-), the nonradical hydrogen peroxide (H_2O_2), and the hydroxyl radical (OH•).

Table 24.1. Some Disease States Associated with Free Radical Injury

Atherogenesis	Cerebrovascular disorders
Emphysema bronchitis	Ischemia/reperfusion injury
Duchenne-type muscular dystrophy	Neurodegenerative disorders
Pregnancy/preeclampsia	Amyotrophic lateral sclerosis (Lou Gehrig's disease)
Cervical cancer	Alzheimer's disease
Alcohol-induced liver disease	Down's syndrome
Hemodialysis	Ischemia/reperfusion injury following stroke
Diabetes	Oxphos diseases (Mitochondrial DNA disorders)
Acute renal failure	Multiple sclerosis
Aging	Parkinson's disease
Retrolental fibroplasia	

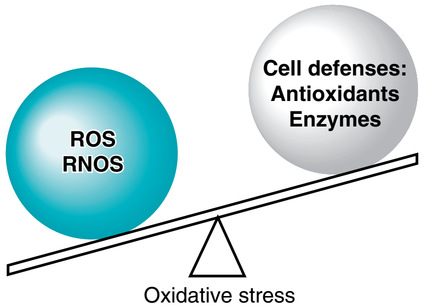

Fig 24.2. Oxidative stress. Oxidative stress occurs when the rate of ROS and RNOS production overbalances the rate of their removal by cellular defense mechanisms. These defense mechanisms include a number of enzymes and antioxidants. Antioxidants usually react nonenzymatically with ROS.

combines with O_2 or superoxide to form reactive ***nitrogen oxygen species (RNOS),*** *such as the nonradical peroxynitrite or the radical* ***nitrogen dioxide.*** *RNOS are present in the environment (e.g., cigarette smoke) and generated in cells. During phagocytosis of invading microorganisms, cells of the immune system produce O_2^-, HOCl, and NO through the actions of* ***NADPH oxidase, myeloperoxidase, and inducible nitric oxide synthase,*** *respectively. In addition to killing phagocytosed invading microorganisms, these toxic metabolites may damage surrounding tissue components.*

Cells ***protect*** *themselves against damage by ROS and other radicals through* ***repair processes, compartmentalization*** *of free radical production,* ***defense enzymes,*** *and* ***endogenous and exogenous antioxidants (free radical scavengers).*** *The defense enzyme* ***superoxide dismutase (SOD)*** *removes the superoxide free radical. Catalase and glutathione peroxidase remove hydrogen peroxide and lipid peroxides.* ***Vitamin E, vitamin C,*** *and* ***plant flavonoids*** *act as* ***antioxidants. Oxidative stress*** *occurs when the rate of ROS generation exceeds the capacity of the cell for their removal (Fig. 24.2).*

THE WAITING ROOM

Two years ago, **Les Dopaman** (less dopamine), a 62-year-old man, noted an increasing tremor of his right hand when sitting quietly (resting tremor). The tremor disappeared if he actively used this hand to do purposeful movement. As this symptom progressed, he also complained of stiffness in his muscles that slowed his movements (bradykinesia). His wife noticed a change in his gait; he had begun taking short, shuffling steps and leaned forward as he walked (postural imbalance). He often appeared to be staring ahead with a rather immobile facial expression. She noted a tremor of his eyelids when he was asleep and, recently, a tremor of his legs when he was at rest. Because of these progressive symptoms and some subtle personality changes (anxiety and emotional lability), she convinced Les to see their family doctor.

The doctor suspected that her patient probably had primary or idiopathic parkinsonism (Parkinson's disease) and referred Mr. Dopaman to a neurologist. In Parkinson's disease, neurons of the substantia nigra pars compacta, containing the pigment melanin and the neurotransmitter dopamine, degenerate.

Cora Nari had done well since the successful lysis of blood clots in her coronary arteries with the use of intravenous recombinant tissue plasminogen activator (TPA)(see Chapters 19 and 21). This therapy had quickly relieved the crushing chest pain (angina) she experienced when she won the lottery. At her first office visit after discharge from the hospital, Cora's cardiologist told her she had developed multiple premature contractions of the ventricular muscle of her heart as the clots were being lysed. This process could have led to a life-threatening arrhythmia known as ventricular fibrillation. However, Cora's arrhythmia responded quickly to pharmacologic suppression and did not recur during the remainder of her hospitalization.

The basal ganglia are part of a neuronal feedback loop that modulates and integrates the flow of information from the cerebral cortex to the motor neurons of the spinal cord. The neostriatum is the major input structure from the cerebral cortex. The substantia nigra pars compacta consists of neurons that provide integrative input to the neostriatum through pigmented neurons that use dopamine as a neurotransmitter (the nigrastriatal pathway). Integrated information feeds back to the basal ganglia and to the cerebral cortex to control voluntary movement. In Parkinson's disease, a decrease in the amount of dopamine reaching the basal ganglia results in the movement disorder.

In ventricular fibrillation, rapid premature beats from an irritative focus in ventricular muscle occur in runs of varying duration. Persistent fibrillation compromises cardiac output, leading to death. This arrythmia can result from severe ischemia (lack of blood flow) in the ventricular muscle of the heart caused by clots forming at the site of a ruptured atherosclerotic plaque. However, **Cora Nari's** rapid beats began during the infusion of TPA as the clot was lysed. Thus, they probably resulted from reperfusing a previously ischemic area of her heart with oxygenated blood. This phenomenon is known as ischemia–reperfusion injury, and it is caused by cytotoxic ROS derived from oxygen in the blood that reperfuses previously hypoxic cells. Ischemic–reperfusion injury also may occur when tissue oxygenation is interrupted during surgery or transplantation.

I. O_2 AND THE GENERATION OF ROS

The generation of reactive oxygen species from O_2 in our cells is a natural everyday occurrence. They are formed as accidental products of nonenzymatic and enzymatic reactions. Occasionally, they are deliberately synthesized in enzyme-catalyzed reactions. Ultraviolet radiation and pollutants in the air can increase formation of toxic oxygen-containing compounds.

The two unpaired electrons in oxygen have the same (parallel) spin and are called antibonding electrons. In contrast, carbon–carbon and carbon–hydrogen bonds each contain two electrons, which have antiparallel spins and form a thermodynamically stable pair. As a consequence, O_2 cannot readily oxidize a covalent bond because one of its electrons would have to flip its spin around to make new pairs. The difficulty in changing spins is called the **spin restriction.** Without the spin restriction, organic life forms could not have developed in the oxygen atmosphere on earth because they would be spontaneously oxidized by O_2. Instead, O_2 is confined to slower one-electron reactions catalyzed by metals (or metalloenzymes).

A. The Radical Nature of O_2

A radical, by definition, is a molecule that has a single unpaired electron in an orbital. A free radical is a radical capable of independent existence. (Radicals formed in an enzyme active site during a reaction, for example, are not considered free radicals unless they can dissociate from the protein to interact with other molecules.) Radicals are highly reactive and initiate chain reactions by extracting an electron from a neighboring molecule to complete their own orbitals. Although the transition metals (e.g., Fe, Cu, and Mo) have single electrons in orbitals, they are not usually considered free radicals because they are relatively stable, do not initiate chain reactions, and are bound to proteins in the cell.

The oxygen atom is a biradical, which means it has two single electrons in different orbitals. These electrons cannot both travel in the same orbital because they have parallel spins (spin in the same direction). Although oxygen is very reactive from a thermodynamic standpoint, its single electrons cannot react rapidly with the paired electrons found in the covalent bonds of organic molecules. As a consequence, O_2 reacts slowly through the acceptance of single electrons in reactions that require a catalyst (such as a metal-containing enzyme).

O_2 is capable of accepting a total of four electrons, which reduces it to water (Fig. 24.3). When O_2 accepts one electron, superoxide is formed. Superoxide is still a radical because it has one unpaired electron remaining. This reaction is not thermodynamically favorable and requires a moderately strong reducing agent that can donate single electrons (e.g., CoQH· in the electron transport chain). When superoxide accepts an electron, it is reduced to hydrogen peroxide, which is not a radical. The hydroxyl radical is formed in the next one-electron reduction step in the reduction sequence. Finally, acceptance of the last electron reduces the hydroxyl radical to H_2O.

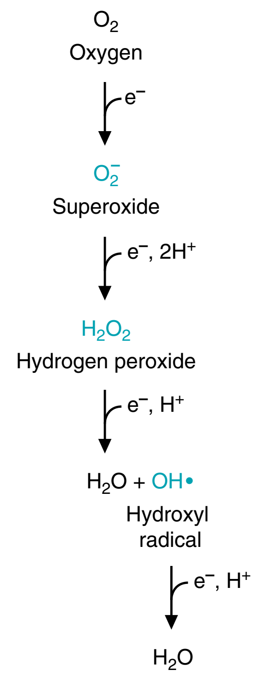

Fig 24.3. Reduction of oxygen by four one-electron steps. The four one-electron reduction steps for O_2 progressively generate superoxide, hydrogen peroxide, and the hydroxyl radical plus water. Superoxide is sometimes written $O_2^{-\cdot}$ to better illustrate its single unpaired electron. H_2O_2, the half-reduced form of O_2, has accepted two electrons and is, therefore, not an oxygen radical.

B. Characteristics of Reactive Oxygen Species

Reactive oxygen species (ROS) are oxygen-containing compounds that are highly reactive free radicals, or compounds readily converted to these oxygen free radicals in the cell. The major oxygen metabolites produced by one-electron reduction of oxygen (superoxide, hydrogen peroxide, and the hydroxyl radical) are classified as ROS (Table 24.2).

Reactive free radicals extract electrons (usually as hydrogen atoms) from other compounds to complete their own orbitals, thereby initiating free radical chain reactions. The hydroxyl radical is probably the most potent of the ROS. It initiates chain reactions that form lipid peroxides and organic radicals and adds directly to compounds. The superoxide anion is also highly reactive, but has limited lipid solubility and cannot diffuse far. However, it can generate the more reactive hydroxyl and hydroperoxy radicals by reacting nonenzymatically with hydrogen peroxide in the Haber–Weiss reaction (Fig 24.4).

Hydrogen peroxide, although not actually a radical, is a weak oxidizing agent that is classified as an ROS because it can generate the hydroxyl radical (OH•). Transition metals, such as Fe^{2+} or Cu^+, catalyze formation of the hydroxyl radical from hydrogen peroxide in the nonenzymatic Fenton reaction (see Fig. 24.4.).

To decrease occurrence of the Fenton reaction, accessibility to transition metals, such as Fe^{2+} and Cu^+, are highly restricted in cells, or in the body as a whole. Events that release iron from cellular storage sites, such as a crushing injury, are associated with increased free radical injury.

Table 24.2. Reactive Oxygen Species (ROS) and Reactive Nitrogen–Oxygen Species (RNOS)

Reactive Species	Properties
O_2^- Superoxide anion	Produced by the electron transport chain and at other sites. Cannot diffuse far from the site of origin. Generates other ROS.
H_2O_2 Hydrogen peroxide	Not a free radical, but can generate free radicals by reaction with a transition metal (e.g., Fe^{2+}). Can diffuse into and through cell membranes.
OH• Hydroxyl radical	The most reactive species in attacking biologic molecules. Produced from H_2O_2 in the Fenton reaction in the presence of Fe^{2+} or Cu^+.
RO•, R•, R-S• Organic radicals	Organic free radicals (R denotes remainder of the compound.) Produced from ROH, RH (e.g., at the carbon of a double bond in a fatty acid) or RSH by OH• attack.
RCOO• Peroxyl radical	An organic peroxyl radical, such as occurs during lipid degradation (also denoted LOO•)
HOCl Hypochlorous acid	Produced in neutrophils during the respiratory burst to destroy invading organisms. Toxicity is through halogenation and oxidation reactions. Attacking species is OCl^-
$O_2^{\downarrow\uparrow}$ Singlet oxygen	Oxygen with antiparallel spins. Produced at high oxygen tensions from absorption of *uv* light. Decays so fast that it is probably not a significant in vivo source of toxicity.
NO Nitric oxide	RNOS. A free radical produced endogenously by nitric oxide synthase. Binds to metal ions. Combines with O_2 or other oxygen-containing radicals to produce additional RNOS.
$ONOO^-$ Peroxynitrite	RNOS. A strong oxidizing agent that is not a free radical. It can generate NO_2 (nitrogen dioxide), which is a radical.

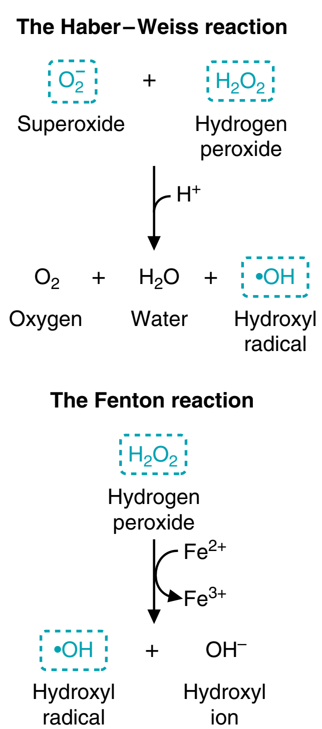

Fig 24.4. Generation of the hydroxyl radical by the nonenzymatic Haber–Weiss and Fenton reactions. In the simplified versions of these reactions shown here, the transfer of single electrons generates the hydroxyl radical. ROS are shown in blue. In addition to Fe^{2+}, Cu^+ and many other metals can also serve as single-electron donors in the Fenton reaction.

Because hydrogen peroxide is lipid soluble, it can diffuse through membranes and generate OH• at localized Fe^{2+}- or Cu^+-containing sites, such as the mitochondria. Hydrogen peroxide is also the precursor of hypochlorous acid (HOCl), a powerful oxidizing agent that is produced endogenously and enzymatically by phagocytic cells.

Organic radicals are generated when superoxide or the hydroxyl radical indiscriminately extract electrons from other molecules. Organic peroxy radicals are intermediates of chain reactions, such as lipid peroxidation. Other organic radicals, such as the ethoxy radical, are intermediates of enzymatic reactions that escape into solution (see Table 24.2).

An additional group of oxygen-containing radicals, termed RNOS, contain nitrogen as well as oxygen. These are derived principally from the free radical nitric oxide (NO), which is produced endogenously by the enzyme nitric oxide synthase. Nitric oxide combines with O_2 or superoxide to produce additional RNOS.

C. Major Sources of Primary Reactive Oxygen Species in the Cell

ROS are constantly being formed in the cell; approximately 3 to 5% of the oxygen we consume is converted to oxygen free radicals. Some are produced as accidental by-products of normal enzymatic reactions that escape from the active site of metal-containing enzymes during oxidation reactions. Others, such as hydrogen peroxide, are physiologic products of oxidases in peroxisomes. Deliberate production of toxic free radicals occurs in the inflammatory response. Drugs, natural radiation, air pollutants, and other chemicals also can increase formation of free radicals in cells.

1. CoQ GENERATES SUPEROXIDE

One of the major sites of superoxide generation is Coenzyme Q (CoQ) in the mitochondrial electron transport chain (Fig. 24.5). The one-electron reduced form of CoQ (CoQH•) is free within the membrane and can accidentally transfer an electron to dissolved O_2, thereby forming superoxide. In contrast, when O_2 binds to cytochrome oxidase and accepts electrons, none of the O_2 radical intermediates are released from the enzyme, and no ROS are generated.

With insufficient oxygen, **Cora Nari's** ischemic heart muscle mitochondria were unable to maintain cellular ATP levels, resulting in high intracellular Na^+ and Ca^{2+} levels. The reduced state of the electron carriers in the absence of oxygen, and loss of mitochondrial ion gradients or membrane integrity, leads to increased superoxide production once oxygen becomes available during reperfusion. The damage can be self-perpetuating, especially if iron bound to components of the electron transport chain becomes available for the Fenton reaction, or the mitochondrial permeability transition is activated.

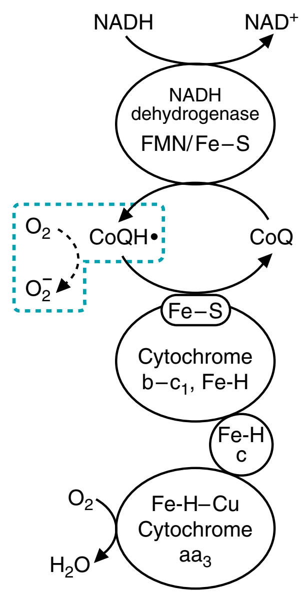

Fig 24.5. Generation of superoxide by CoQ in the electron transport chain. In the process of transporting electrons to O_2, some of the electrons escape when CoQH• accidentally interacts with O_2 to form superoxide. Fe-H represents the Fe-heme center of the cytochromes.

2. OXIDASES, OXYGENASES, AND PEROXIDASES

Most of the oxidases, peroxidases, and oxygenases in the cell bind O_2 and transfer single electrons to it via a metal. Free radical intermediates of these reactions may be accidentally released before the reduction is complete.

Cytochrome P450 enzymes are a major source of free radicals "leaked" from reactions.

Because these enzymes catalyze reactions in which single electrons are transferred to O_2 and an organic substrate, the possibility of accidentally generating and releasing free radical intermediates is high (see Chapters 19 and 25). Induction of P450 enzymes by alcohol, drugs, or chemical toxicants leads to increased cellular injury. When substrates for cytochrome P450 enzymes are not present, its potential for destructive damage is diminished by repression of gene transcription.

Hydrogen peroxide and lipid peroxides are generated enzymatically as major reaction products by a number of oxidases present in peroxisomes, mitochondria, and the endoplasmic reticulum. For example, monoamine oxidase, which oxidatively degrades the neurotransmitter dopamine, generates H_2O_2 at the mitochondrial membrane of certain neurons. Peroxisomal fatty acid oxidase generates H_2O_2 rather than FAD(2H) during the oxidation of very-long-chain fatty acids (see Chapter 23). Xanthine oxidase, an enzyme of purine degradation that can reduce O_2 to O_2^- or H_2O_2 in the cytosol, is thought to be a major contributor to ischemia–reperfusion injury, especially in intestinal mucosal and endothelial cells. Lipid peroxides are also formed enzymatically as intermediates in the pathways for synthesis of many eicosanoids, including leukotrienes and prostaglandins.

3. IONIZING RADIATION

Cosmic rays that continuously bombard the earth, radioactive chemicals, and x-rays are forms of ionizing radiation. Ionizing radiation has a high enough energy level that it can split water into the hydroxyl and hydrogen radicals, thus leading to radiation damage to the skin, mutations, cancer, and cell death (Fig. 24.6). It also may generate organic radicals through direct collision with organic cellular components.

Production of ROS by xanthine oxidase in endothelial cells may be enhanced during ischemia–reperfusion in **Cora Nari's** heart. In undamaged tissues, xanthine oxidase exists as a dehydrogenase that uses NAD^+ rather than O_2 as an electron acceptor in the pathway for degradation of purines (hypoxanthine ↔ xanthine ↔ uric acid (see Chapter 41). When O_2 levels decrease, phosphorylation of ADP to ATP decreases, and degradation of ADP and adenine through xanthine oxidase increases. In the process, xanthine dehydrogenase is converted to an oxidase. As long as O_2 levels are below the high K_m of the enzyme for O_2, little damage is done. However, during reperfusion when O_2 levels return to normal, xanthine oxidase generates H_2O_2 and O_2^- at the site of injury.

Carbon tetrachloride (CCl_4), which is used as a solvent in the dry-cleaning industry, is converted by cytochrome P450 to a highly reactive free radical that has caused hepatocellular necrosis in workers. When the enzyme-bound CCl_4 accepts an electron, it dissociates into CCl_3· and Cl·. The CCl_3· radical, which cannot continue through the P450 reaction sequence, "leaks" from the enzyme active site and initiates chain reactions in the surrounding polyunsaturated lipids of the endoplasmic reticulum. These reactions spread into the plasma membrane and to proteins, eventually resulting in cell swelling, accumulation of lipids, and cell death.

Les Dopaman, who is in the early stages of Parkinson's disease, is treated with a monoamine oxidase B inhibitor. Monoamine oxidase is a copper-containing enzyme that inactivates dopamine in neurons, producing H_2O_2. The drug was originally administered to inhibit dopamine degradation. However, current theory suggests that the effectiveness of the drug is also related to decrease of free radical formation within the cells of the basal ganglia. The dopaminergic neurons involved are particularly susceptible to the cytotoxic effects of ROS and RNOS that may arise from H_2O_2.

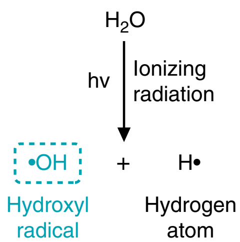

Fig 24.6. Generation of free radicals from radiation.

The appearance of lipofuscin granules in many tissues increases during aging. The pigment lipofuscin (from the Greek "lipos" for lipids and the Latin "fuscus" for dark) consists of a heterogeneous mixture of cross-linked polymerized lipids and protein formed by reactions between amino acid residues and lipid peroxidation products, such as malondialdehyde. These cross-linked products are probably derived from peroxidatively damaged cell organelles that were autophagocytized by lysosomes but could not be digested. When these dark pigments appear on the skin of the hands in aged individuals, they are referred to as "liver spots," a traditional hallmark of aging. In **Les Dopaman** and other patients with Parkinson's disease, lipofuscin appears as Lewy bodies in degenerating neurons.

Evidence of protein damage shows up in many diseases, particularly those associated with aging. In patients with cataracts, proteins in the lens of the eye exhibit free radical damage and contain methionine sulfoxide residues and tryptophan degradation products.

II. OXYGEN RADICAL REACTIONS WITH CELLULAR COMPONENTS

Oxygen radicals produce cellular dysfunction by reacting with lipids, proteins, carbohydrates, and DNA to extract electrons (summarized in Fig. 24.7). Evidence of free radical damage has been described in over 100 disease states. In some of these diseases, free radical damage is the primary cause of the disease; in others, it enhances complications of the disease.

A. Membrane Attack: Formation of Lipid and Lipid Peroxy Radicals

Chain reactions that form lipid free radicals and lipid peroxides in membranes make a major contribution to ROS-induced injury (Fig. 24.8). An initiator (such as a hydroxyl radical produced locally in the Fenton reaction) begins the chain reaction. It extracts a hydrogen atom, preferably from the double bond of a polyunsaturated fatty acid in a membrane lipid. The chain reaction is propagated when O_2 adds to form lipid peroxyl radicals and lipid peroxides. Eventually lipid degradation occurs, forming such products as malondialdehyde (from fatty acids with three or more double bonds), and ethane and pentane (from the ω-terminal carbons of 3 and 6 fatty acids, respectively). Malondialdehyde appears in the blood and urine and is used as an indicator of free radical damage.

Peroxidation of lipid molecules invariably changes or damages lipid molecular structure. In addition to the self-destructive nature of membrane lipid peroxidation, the aldehydes that are formed can cross-link proteins. When the damaged lipids are the constituents of biologic membranes, the cohesive lipid bilayer arrangement and stable structural organization is disrupted (see Fig. 24.7). Disruption of mitochondrial membrane integrity may result in further free radical production.

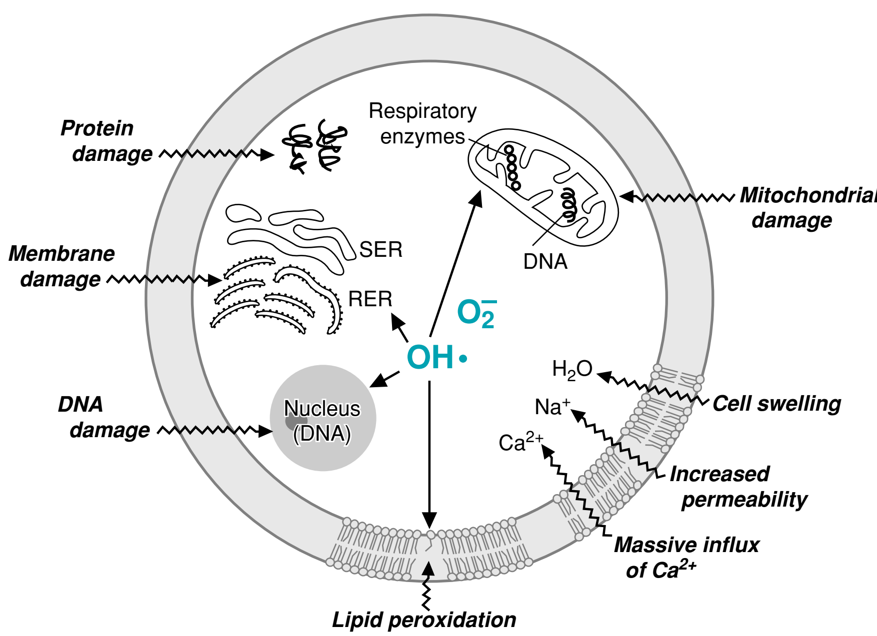

Fig 24.7. Free radical–mediated cellular injury. Superoxide and the hydroxyl radical initiate lipid peroxidation in the cellular, mitochondrial, nuclear, and endoplasmic reticulum membranes. The increase in cellular permeability results in an influx of Ca^{2+}, which causes further mitochondrial damage. The cysteine sulfhydryl groups and other amino acid residues on proteins are oxidized and degraded. Nuclear and mitochondrial DNA can be oxidized, resulting in strand breaks and other types of damage. RNOS (NO, NO_2, and peroxynitrite) have similar effects.

B. Proteins and Peptides

In proteins, the amino acids proline, histidine, arginine, cysteine, and methionine are particularity susceptible to hydroxyl radical attack and oxidative damage. As a consequence of oxidative damage, the protein may fragment or residues cross-link with other residues. Free radical attack on protein cysteine residues can result in cross-linking and formation of aggregates that prevents their degradation. However, oxidative damage increases the susceptibility of other proteins to proteolytic digestion.

Free radical attack and oxidation of the cytsteine sulfhydryl residues of the tripeptide glutathione (γ-glutamyl-cysteinyl-glycine; see section V.A.3.) increases oxidative damage throughout the cell. Glutathione is a major component of cellular defense against free radical injury, and its oxidation reduces its protective effects.

C. DNA

Oxygen-derived free radicals are also a major source of DNA damage. Approximately 20 types of oxidatively altered DNA molecules have been identified. The nonspecific binding of Fe^{2+} to DNA facilitates localized production of the hydroxyl radical, which can cause base alterations in the DNA (Fig. 24.9). It also can attack the deoxyribose backbone and cause strand breaks. This DNA damage can be repaired to some extent by the cell (see Chapter 12), or minimized by apoptosis of the cell.

III. NITRIC OXIDE AND REACTIVE NITROGEN-OXYGEN SPECIES (RNOS)

Nitric oxide (NO) is an oxygen-containing free radical which, like O_2, is both essential to life and toxic. NO has a single electron, and therefore binds to other compounds containing single electrons, such as Fe^{3+}. As a gas, it diffuses through the cytosol and lipid membranes and into cells. At low concentrations, it functions physiologically as a neurotransmitter and a hormone that causes vasodilation. However, at high concentrations, it combines with O_2 or with superoxide to form additional reactive and toxic species containing both nitrogen and oxygen (RNOS). RNOS are involved in neurodegenerative diseases, such as Parkinson's disease, and in chronic inflammatory diseases, such as rheumatoid arthritis.

A. Nitric Oxide Synthase

At low concentrations, nitric oxide serves as a neurotransmitter or a hormone. It is synthesized from arginine by nitric oxide synthases (Fig 24.10). As a gas, it is able to diffuse through water and lipid membranes, and into target cells. In the target cell, it exerts its physiologic effects by high-affinity binding to Fe-heme in the enzyme guanylyl cyclase, thereby activating a signal transduction cascade. However, NO is rapidly inactivated by nonspecific binding to many molecules, and therefore cells that produce NO need to be close to the target cells.

The body has three different tissue-specific isoforms of NO synthase, each encoded by a different gene: neuronal nitric oxide synthase (nNOS, isoform I), inducible nitric oxide synthase (iNOS, isoform II), and endothelial nitric oxide synthase (eNOS, isoform III). nNOS and eNOS are tightly regulated by Ca^{2+} concentration to produce the small amounts of NO required for its role as a neurotransmitter and hormone. In contrast, iNOS is present in many cells of the immune system and cell types with a similar lineage, such as macrophages and

Nitroglycerin, in tablet form, is often given to patients with coronary artery disease who experience ischemia-induced chest pain (angina). The nitroglycerin decomposes in the blood, forming NO, a potent vasodilator, which increases blood flow to the heart and relieves the angina.

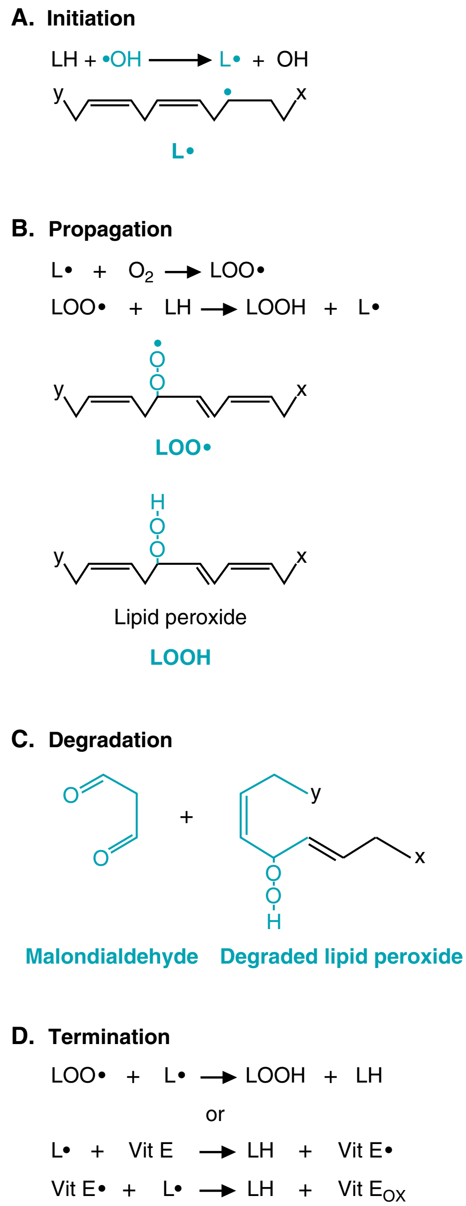

Fig 24.8. Lipid peroxidation: a free radical chain reaction. **A.** Lipid peroxidation is initiated by a hydroxyl or other radical that extracts a hydrogen atom from a polyunsaturated lipid (LH), thereby forming a lipid radical (L•). **B.** The free radical chain reaction is propagated by reaction with O_2, forming the lipid peroxy radical (LOO•) and lipid peroxide (LOOH). **C.** Rearrangements of the single electron result in degradation of the lipid. Malondialdehyde, one of the compounds formed, is soluble and appears in blood. **D.** The chain reaction can be terminated by vitamin E and other lipid-soluble antioxidants that donate single electrons. Two subsequent reduction steps form a stable, oxidized antioxidant.

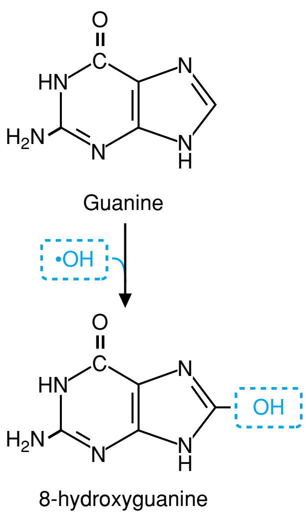

Fig 24.9. Conversion of guanine to 8-hydroxyguanine by the hydroxy radical. The amount of 8-hydroxyguanosine present in cells can be used to estimate the amount of oxidative damage they have sustained. The addition of the hydroxyl group to guanine allows it to mispair with T residues, leading to the creation of a daughter molecule with an A-T base pair in this position.

brain astroglia. This isoenzyme of nitric oxide synthase is regulated principally by induction of gene transcription, and not by changes in Ca^{2+} concentration. It produces high and toxic levels of NO to assist in killing invading microorganisms. It is these very high levels of NO that are associated with generation of RNOS and NO toxicity.

B. NO Toxicity

The toxic actions of NO can be divided into two categories: direct toxic effects resulting from binding to Fe-containing proteins, and indirect effects mediated by compounds formed when NO combines with O_2 or with superoxide to form RNOS.

1. DIRECT TOXIC EFFECTS OF NO

NO, as a radical, exerts direct toxic effects by combining with Fe-containing compounds that also have single electrons. Major destructive sites of attack include Fe-S centers (e.g., electron transport chain complexes I-III, aconitase) and Fe-heme proteins (e.g., hemoglobin and electron transport chain cytochromes). However, there is usually little damage because NO is present in low concentrations and Fe-heme compounds are present in excess capacity. NO can cause serious damage, however, through direct inhibition of respiration in cells that are already compromised through oxidative phosphorylation diseases or ischemia.

2. RNOS TOXICITY

When present in very high concentrations (e.g., during inflammation), NO combines nonenzymatically with superoxide to form peroxynitrite ($ONOO^-$), or with O_2 to form N_2O_3 (Fig. 24.11). Peroxynitrite, although not a free radical, is a strong

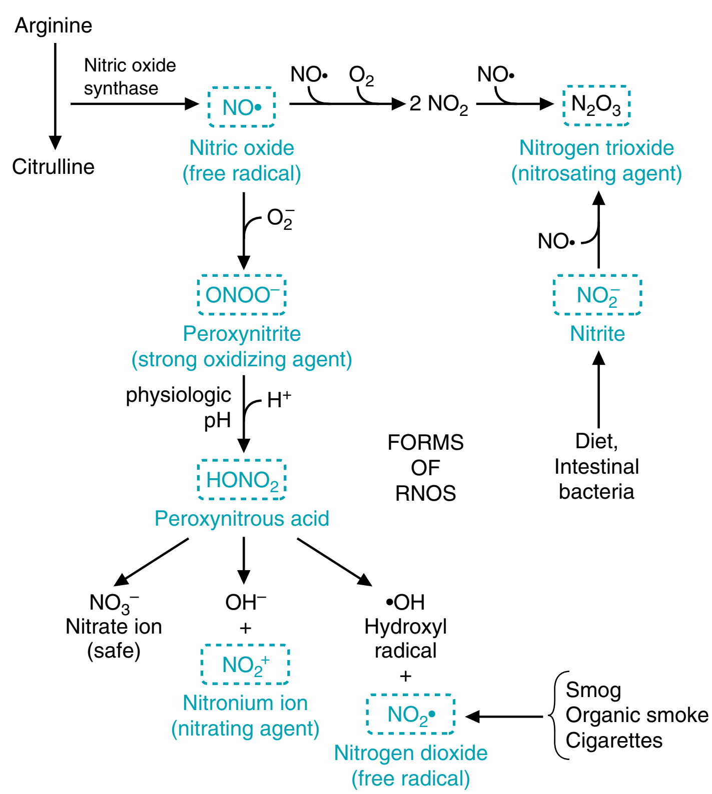

Fig 24.11. Formation of RNOS from nitric oxide. RNOS are shown in blue. The type of damage caused by each RNOS is shown in parentheses. Of all the nitrogen–oxygen-containing compounds shown, only nitrate is relatively nontoxic.

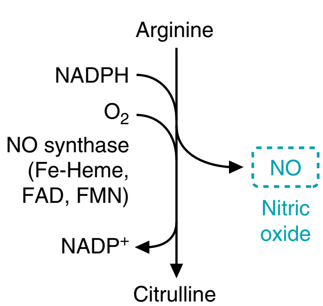

Fig 24.10. Nitric oxide synthase synthesizes the free radical NO. Like cytochrome P450 enzymes, NO synthase uses Fe-heme, FAD, and FMN to transfer single electrons from NADPH to O_2.

oxidizing agent that is stable and directly toxic. It can diffuse through the cell and lipid membranes to interact with a wide range of targets, including protein methionine and -SH groups (e.g., Fe-S centers in the electron transport chain). It also breaks down to form additional RNOS, including the free radical nitrogen dioxide (NO_2), an effective initiator of lipid peroxidation. Peroxynitrite products also nitrate aromatic rings, forming compounds such as nitrotyrosine or nitroguanosine. N_2O_3, which can be derived either from NO_2 or nitrite, is the agent of nitrosative stress, and nitrosylates sulfhydryl and similarily reactive groups in the cell. Nitrosylation will usually interefere with the proper functioning of the protein or lipid that has been modified. Thus, RNOS can do as much oxidative and free radical damage as non–nitrogen-containing ROS, as well as nitrating and nitrosylating compounds. The result is widespread and includes inhibition of a large number of enzymes; mitochondrial lipid peroxidation; inhibition of the electron transport chain and energy depletion; single-stranded or double-stranded breaks in DNA; and modification of bases in DNA.

NO_2 is one of the toxic agents present in smog, automobile exhaust, gas ranges, pilot lights, cigarette smoke, and smoke from forest fires or burning buildings.

IV. FORMATION OF FREE RADICALS DURING PHAGOCYTOSIS AND INFLAMMATION

In response to infectious agents and other stimuli, phagocytic cells of the immune system (neutrophils, eosinophils, and monocytes/macrophages) exhibit a rapid consumption of O_2 called the respiratory burst. The respiratory burst is a major source of superoxide, hydrogen peroxide, the hydroxyl radical, hypochlorous acid (HOCl), and RNOS. The generation of free radicals is part of the human antimicrobial defense system and is intended to destroy invading microorganisms, tumor cells, and other cells targeted for removal.

A. NADPH Oxidase

The respiratory burst results from the activity of NADPH oxidase, which catalyzes the transfer of an electron from NADPH to O_2 to form superoxide (Fig. 24.12). NADPH oxidase is assembled from cytosol and membranous proteins recruited into the phagolysosome membrane as it surrounds an invading microorganism.

Superoxide is released into the intramembranous space of the phagolysosome, where it is generally converted to hydrogen peroxide and other ROS that are effective against bacteria and fungal pathogens. Hydrogen peroxide is formed by superoxide dismutase, which may come from the phagocytic cell or the invading microorganism.

In patients with chronic granulomatous disease, phagocytes have genetic defects in NADPH oxidase. NADPH oxidase has four different subunits (two in the cell membrane and two recruited from the cytosol), and the genetic defect may be in any of the genes that encode these subunits. The membrane catalytic subunit β of NADPH oxidase is a 91-kDa flavocytochrome glycoprotein. It transfers electrons from bound NADPH to FAD, which transfers them to the Fe–heme components. The membranous α-subunit (p22) is required for stabilization. Two additional cytosolic proteins (p47phox and p67phox) are also required for assembly of the complex. Rac, a monomeric GTPase in the Ras subfamily of the Rho superfamily (see Chapter 9), is also required for assembly. The 91-kDa subunit is affected most often in X-linked chronic granulatomous disease, whereas the α-subunit is affected in a rare autosomal recessive form. The cytosolic subunits are affected most often in patients with the autosomal recessive form of granulomatous disease. In addition to their enhanced susceptibility to bacterial and fungal infections, these patients suffer from an apparent dysregulation of normal inflammatory responses.

B. Myeloperoxidase and HOCl

The formation of hypochlorous acid from H_2O_2 is catalyzed by myeloperoxidase, a heme-containing enzyme that is present only in phagocytic cells of the immune system (predominantly neutrophils).

$$H_2O_2 + Cl^- + H^+ \xrightarrow{\text{Myeloperoxidase}} HOCl + H_2O \xrightarrow{\text{Dissociation}} {}^-OCl + H^+ + H_2O$$

Myeloperoxidase contains two Fe heme-like centers, which give it the green color seen in pus. Hypochlorous acid is a powerful toxin that destroys bacteria within seconds through halogenation and oxidation reactions. It oxidizes many Fe and S-containing groups (e.g., sulfhydryl groups, iron-sulfur centers, ferredoxin, heme-proteins, methionine), oxidatively decarboxylates and deaminates proteins, and cleaves peptide bonds. Aerobic bacteria under attack rapidly lose membrane

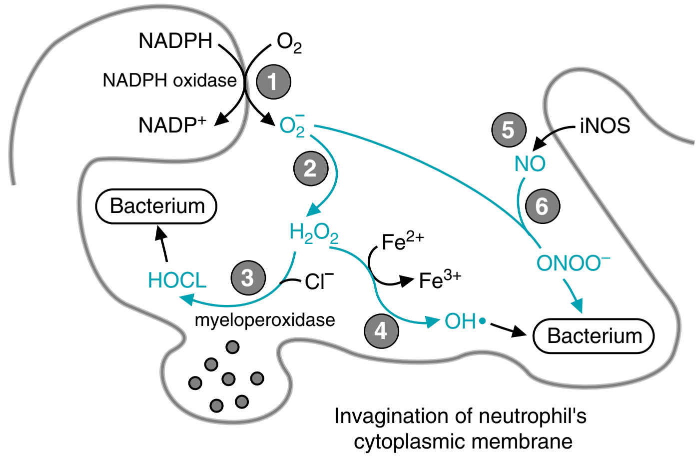

Fig 24.12. Production of reactive oxygen species during the phagocytic respiratory burst by activated neutrophils. (1) Activation of NADPH oxidase on the outer side of the plasma membrane initiates the respiratory burst with the generation of superoxide. During phagocytosis, the plasma membrane invaginates, so superoxide is released into the vacuole space. (2) Superoxide (either spontaneously or enzymatically via superoxide dismutase [SOD]) generates H_2O_2. (3) Granules containing myeloperoxidase are secreted into the phagosome, where myeloperoxidase generates HOCl and other halides. (4) H_2O_2 can also generate the hydroxyl radical from the Fenton reaction. (5) Inducible nitric oxide synthase may be activated and generate NO. (6) Nitric oxide combines with superoxide to form peroxynitrite, which may generate additional RNOS. The result is an attack on the membranes and other components of phagocytosed cells, and eventual lysis. The whole process is referred to as the respiratory burst because it lasts only 30 to 60 minutes and consumes O_2.

transport, possibly because of damage to ATP synthase or electron transport chain components (which reside in the plasma membrane of bacteria).

C. RNOS and Inflammation

When human neutrophils of the immune system are activated to produce NO, NADPH oxidase is also activated. NO reacts rapidly with superoxide to generate peroxynitrite, which forms additional RNOS. NO also may be released into the surrounding medium, to combine with superoxide in target cells.

In a number of disease states, free radical release by neutrophils or macrophages during an inflammation contributes to injury in the surrounding tissues. During stroke or myocardial infarction, phagocytic cells that move into the ischemic area to remove dead cells may increase the area and extent of damage. The self-perpetuating mechanism of radical release by neutrophils during inflammation and immune complex formation may explain some of the features of chronic inflammation in patients with rheumatoid arthritis. As a result of free radical release, the immunoglobulin G (IgG) proteins present in the synovial fluid are partially oxidized, which improves their binding with the rheumatoid factor antibody. This binding, in turn, stimulates the neutrophils to release more free radicals.

During **Cora Nari's** ischemia (decreased blood flow), the ability of her heart to generate ATP from oxidative phosphorylation was compromised. The damage appeared to accelerate when oxygen was first reintroduced (reperfused) into the tissue. During ischemia, CoQ and the other single-electron components of the electron transport chain become saturated with electrons. When oxygen is reintroduced (reperfusion), electron donation to O_2 to form superoxide is increased. The increase of superoxide results in enhanced formation of hydrogen peroxide and the hydroxyl radical. Macrophages in the area to clean up cell debris from ischemic injury produce nitric oxide, which may further damage mitochondria by generating RNOS that attack Fe-S centers and cytochromes in the electron transport chain membrane lipids. Thus, the RNOS may increase the infarct size.

V. CELLULAR DEFENSES AGAINST OXYGEN TOXICITY

Our defenses against oxygen toxicity fall into the categories of antioxidant defense enzymes, dietary and endogenous antioxidants (free radical scavengers), cellular compartmentation, metal sequestration, and repair of damaged cellular components. The antioxidant defense enzymes react with ROS and cellular products of free radical chain reactions to convert them to nontoxic products. Dietary antioxidants, such as vitamin E and flavonoids, and endogenous antioxidants, such as urate, can

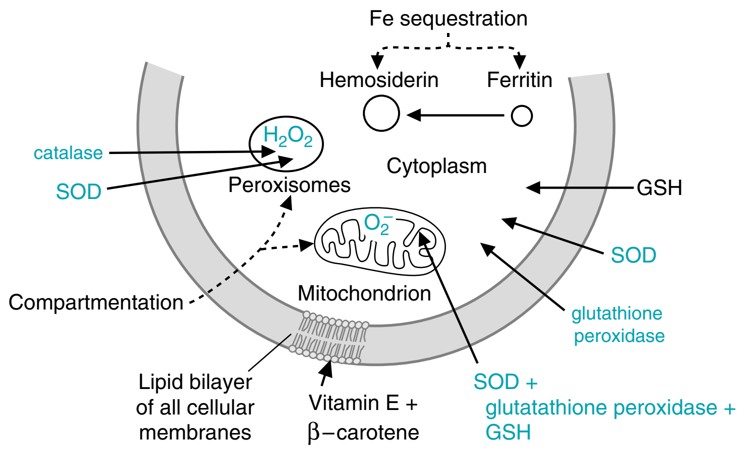

Fig 24.13 Compartmentation of free radical defenses. Various defenses against ROS are found in the different subcellular compartments of the cell. The location of free radical defense enzymes (shown in blue) matches the type and amount of ROS generated in each subcellular compartment. The highest activities of these enzymes are found in the liver, adrenal gland, and kidney, where mitochondrial and peroxisomal contents are high, and cytochrome P450 enzymes are found in abundance in the smooth ER. The enzymes superoxide dismutase (SOD) and glutathione peroxidase are present as isozymes in the different compartments. Another form of compartmentation involves the sequestration of Fe, which is stored as mobilizable Fe in ferritin. Excess Fe is stored in nonmobilizable hemosiderin deposits. Glutathione (GSH) is a nonenzymatic antioxidant.

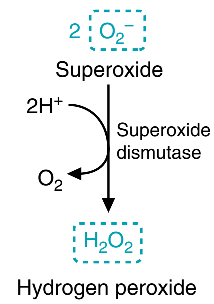

Fig 24.14. Superoxide dismutase converts superoxide to hydrogen peroxide, which is nontoxic unless converted to other ROS.

terminate free radical chain reactions. Defense through compartmentation refers to separation of species and sites involved in ROS generation from the rest of the cell (Fig. 24.13). For example, many of the enzymes that produce hydrogen peroxide are sequestered in peroxisomes with a high content of antioxidant enzymes. Metals are bound to a wide range of proteins within the blood and in cells, preventing their participation in the Fenton reaction. Iron, for example, is tightly bound to its storage protein, ferritin and cannot react with hydrogen peroxide. Repair mechanisms for DNA, and for removal of oxidized fatty acids from membrane lipids, are available to the cell. Oxidized amino acids on proteins are continuously repaired through protein degradation and resynthesis of new proteins.

A. Antioxidant Scavenging Enzymes

The enzymatic defense against ROS includes superoxide dismutase, catalase, and glutathione peroxidase.

1. SUPEROXIDE DISMUTASE (SOD)

Conversion of superoxide anion to hydrogen peroxide and O_2 (dismutation) by superoxide dismutase (SOD) is often called the primary defense against oxidative stress because superoxide is such a strong initiator of chain reactions (Fig 24.14). SOD exists as three isoenzyme forms, a Cu^+-Zn^{2+} form present in the cytosol, a Mn^{2+} form present in mitochondria, and a Cu^+-Zn^{2+} form found extracellularly. The activity of Cu^+-Zn^{2+} SOD is increased by chemicals or conditions (such as hyperbaric oxygen) that increase the production of superoxide.

In the body, iron and other metals are sequestered from interaction with ROS or O_2 by their binding to transport proteins (haptoglobin, hemoglobin, transferrin, ceruloplasmin, and metallothionein) in the blood, and to intracellular storage proteins (ferritin, hemosiderin). Metals also are found bound to many enzymes, particularly those that react with O_2. Usually, these enzymes have reaction mechanisms that minimize nonspecific single-electron transfer from the metal to other compounds.

The intracellular form of the Cu^+–Zn^{2+} superoxide dismutase is encoded by the SOD1 gene. To date, 58 mutations in this gene have been discovered in individuals affected by familial amyotrophic lateral sclerosis (Lou Gehrig's disease). How a mutation in this gene leads to the symptoms of this disease has yet to be understood. It is important to note that only 5 to 10% of the total cases of diagnosed amyotrophic lateral sclerosis are caused by the familial form.

Q: Why does the cell need such a high content of SOD in mitochondria?

Premature infants with low levels of lung surfactant (see Chapter 33) require oxygen therapy. The level of oxygen must be closely monitored to prevent retinopathy and subsequent blindness (the retinopathy of prematurity) and to prevent bronchial pulmonary dysplasia. The tendency for these complications to develop is enhanced by the possibility of low levels of SOD and vitamin E in the premature infant.

A: Mitochondria are major sites for generation of superoxide from the interaction of CoQ and O_2. The Mn^{2+} superoxide dismutase present in mitochondria is not regulated through induction/repression of gene transcription, presumably because the rate of superoxide generation is always high. Mitochondria also have a high content of glutathione and glutathione peroxidase, and can thus convert H_2O_2 to H_2O and prevent lipid peroxidation.

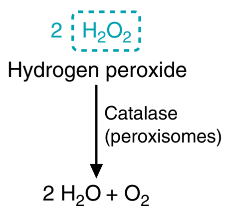

Fig 24.15. Catalase reduces hydrogen peroxide. (ROS is shown in a blue box).

Selenium (Se) is present in human proteins principally as selenocysteine (cysteine with the sulfur group replaced by Se, abbreviated sec). This amino acid functions in catalysis, and has been found in 11 or more human enzymes, including the four enzymes of the glutathione peroxidase family. Selenium is supplied in the diet as selenomethionine from plants (methionine with the Se replacing the sulfur), selenocysteine from animal foods, and inorganic selenium. Se from all of these sources can be converted to selenophosphate. Selenophosphate reacts with a unique tRNA containing bound serine to form a selenocysteine-tRNA, which incorporates selenocystiene into the appropriate protein as it is being synthesized. Se homeostasis in the body is controlled principally through regulation of its secretion as methylated Se. The current dietary requirement is approximately 70 μg/day for adult males and 55 μg for females. Deficiency symptoms reflect diminished antioxidant defenses and include symptoms of vitamin E deficiency.

2. CATALASE

Hydrogen peroxide, once formed, must be reduced to water to prevent it from forming the hydroxyl radical in the Fenton reaction or Haber–Weiss reactions (see Fig. 24.4) One of the enzymes capable of reducing hydrogen peroxide is catalase (Fig.24.15). Catalase is found principally in peroxisomes, and to a lesser extent in the cytosol and microsomal fraction of the cell. The highest activities are found in tissues with a high peroxisomal content (kidney and liver). In cells of the immune system, catalase serves to protect the cell against its own respiratory burst.

3. GLUTATHIONE PEROXIDASE AND GLUTATHIONE REDUCTASE

Glutathione (γ-glutamylcysteinylglycine) is one of the body's principal means of protecting against oxidative damage (see also Chapter 29). Glutathione is a tripeptide composed of glutamate, cysteine, and glycine, with the amino group of cysteine joined in peptide linkage to the γ-carboxyl group of glutamate (Fig. 24.16). In reactions catalyzed by glutathione peroxidases, the reactive sulfhydryl groups reduce hydrogen peroxide to water and lipid peroxides to nontoxic alcohols. In these reactions, two glutathione molecules are oxidized to form a single molecule, glutathione disulfide. The sulfhydryl groups are also oxidized in nonenzymatic chain terminating reactions with organic radicals.

Glutathione peroxidases exist as a family of selenium enzymes with somewhat different properties and tissue locations. Within cells, they are found principally in the cytosol and mitochondria, and are the major means for removing H_2O_2 produced outside of peroxisomes. They contribute to our dietary requirement for selenium and account for the protective effect of selenium in the prevention of free radical injury.

Once oxidized glutathione (GSSG) is formed, it must be reduced back to the sulfhydryl form by glutathione reductase in a redox cycle (Fig. 24.17). Glutathione reductase contains an FAD, and catalyzes transfer of electrons from NADPH to the disulfide bond of GSSG. NADPH is, thus, essential for protection against free radical injury. The major source of NADPH for this reaction is the pentose phosphate pathway (see Chapter 29).

B. Nonenzymatic Antioxidants (Free Radical Scavengers)

Free radical scavengers convert free radicals to a nonradical nontoxic form in nonenzymatic reactions. Most free radical scavengers are antioxidants, compounds

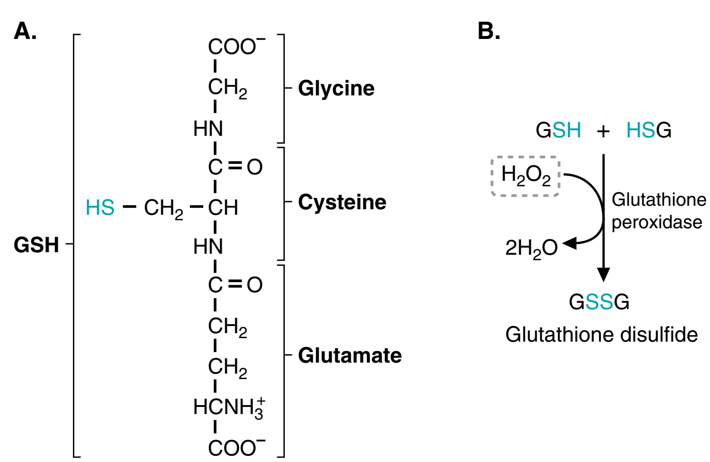

Fig 24.16. Glutathione peroxidase reduces hydrogen peroxide to water. A. The structure of glutathione. The sulfhydryl group of glutathione, which is oxidized to a disulfide, is shown in blue. B. Glutathione peroxidase transfer electrons from glutathione (GSH) to hydrogen peroxide.

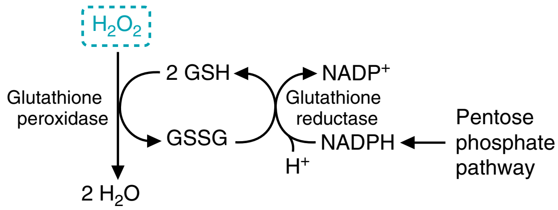

Fig 24.17. Glutathione redox cycle. Glutathione reductase regenerates reduced glutathione. (ROS is shown in the blue box).

that neutralize free radicals by donating a hydrogen atom (with its one electron) to the radical. Antioxidants, therefore, reduce free radicals and are themselves oxidized in the reaction. Dietary free radical scavengers (e.g., vitamin E, ascorbic acid, carotenoids, and flavonoids) as well as endogenously produced free radical scavengers (e.g., urate and melatonin) have a common structural feature, a conjugated double bond system that may be an aromatic ring.

1. VITAMIN E

Vitamin E (α-tocopherol), the most widely distributed antioxidant in nature, is a lipid-soluble antioxidant vitamin that functions principally to protect against lipid peroxidation in membranes (see Fig. 24.13). Vitamin E comprises a number of tocopherols that differ in their methylation pattern. Among these, α-tocopherol is the most potent antioxidant and present in the highest amount in our diet (Fig. 24.18).

Vitamin E is an efficient antioxidant and nonenzymatic terminator of free radical chain reactions, and has little pro-oxidant activity. When Vitamin E donates an electron to a lipid peroxy radical, it is converted to a free radical form that is stabilized by resonance. If this free radical form were to act as a pro-oxidant and abstract an electron from a polyunsaturated lipid, it would be oxidizing that lipid and actually propagate the free radical chain reaction. The chemistry of vitamin E is such that it has a much greater tendency to donate a second electron and go to the fully oxidized form.

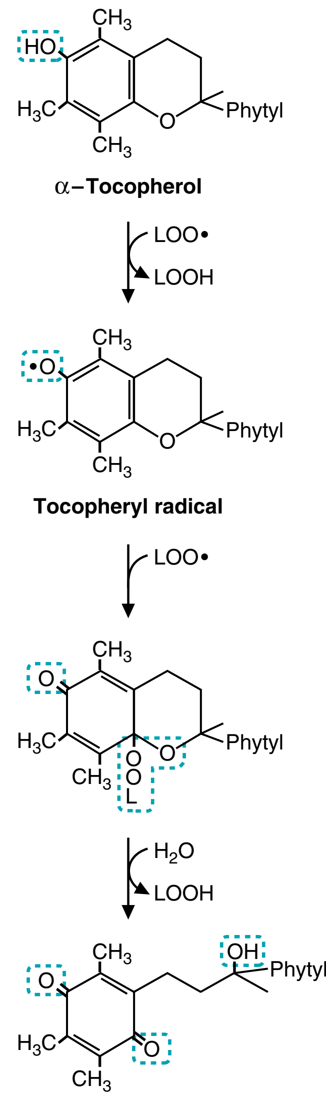

Fig 24.18. Vitamin E (α-tocopherol) terminates free radical lipid peroxidation by donating single electrons to lipid peroxyl radicals (LOO•) to form the more stable lipid peroxide, LOOH. In so doing, the α-tocopherol is converted to the fully oxidized tocopheryl quinone.

2. ASCORBIC ACID

Although ascorbate (vitamin C) is an oxidation-reduction coenzyme that functions in collagen synthesis and other reactions, it also plays a role in free radical defense. Reduced ascorbate can regenerate the reduced form of vitamin E through donating electrons in a redox cycle (Fig. 24.19). It is water-soluble and circulates unbound in blood and extracellular fluid, where it has access to the lipid-soluble vitamin E present in membranes and lipoprotein particles.

3. CAROTENOIDS

Carotenoids is a term applied to β-carotene (the precursor of vitamin A) and similar compounds with functional oxygen-containing substituents on the rings, such as zeaxanthin and lutein (Fig. 24.20). These compounds can exert antioxidant effects, as well as quench singlet O_2 (singlet oxygen is a highly reactive oxygen species in which there are no unpaired electrons in the outer orbitals, but there is one orbital that is completely empty). Epidemiologic studies have shown a correlation between diets high in fruits and vegetables and health benefits, leading to the hypothesis that carotenoids might slow the progression of cancer, atherosclerosis, and other degenerative diseases by acting as chain-breaking antioxidants. However, in clinical

Vitamin E is found in the diet in the lipid fractions of some vegetable oils and in liver, egg yolks, and cereals. It is absorbed together with lipids, and fat malabsorption results in symptomatic deficiencies. Vitamin E circulates in the blood in lipoprotein particles. Its deficiency causes neurologic symptoms, probably because the polyunsaturated lipids in myelin and other membranes of the nervous system are particularly sensitive to free radical injury.

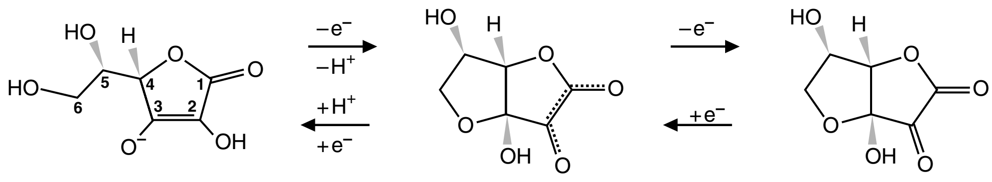

Fig 24.19. L-Ascorbate (the reduced form) donates single electrons to free radicals or disulfides in two steps as it is oxidized to dehydro-L-ascorbic acid. Its principle role in free radical defense is probably regeneration of vitamin E. However, it also may react with superoxide, hydrogen peroxide, hypochlorite, the hydroxyl and peroxyl radicals, and NO_2.

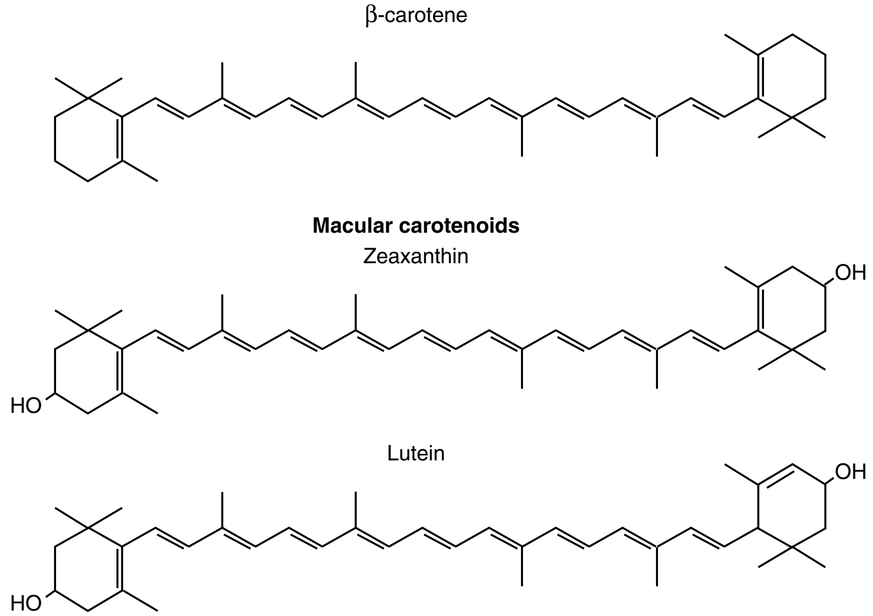

Fig 24.20. Carotenoids are compounds related in structure to β-carotene. Lutein and zeathanthin (the macular carotenoids) are analogs containing hydroxyl groups.

Epidemiologic evidence suggests that individuals with a higher intake of foods containing vitamin E, β-carotene, and vitamin C have a somewhat lower risk of cancer and certain other ROS-related diseases than do individuals on diets deficient in these vitamins. However, studies in which well-nourished populations were given supplements of these antioxidant vitamins found either no effects or harmful effects compared with the beneficial effects from eating foods containing a wide variety of antioxidant compounds. Of the pure chemical supplements tested, there is evidence only for the efficacy of vitamin E. In two clinical trials, β-carotene (or β-carotene + vitamin A) was associated with a higher incidence of lung cancer among smokers and higher mortality rates. In one study, vitamin E intake was associated with a higher incidence of hemorrhagic stroke (possibly because of vitamin K mimicry).

trials, β-carotene supplements had either no effect or an undesirable effect. Its ineffectiveness may be due to the pro-oxidant activity of the free radical form.

In contrast, epidemiologic studies relating the intake of lutein and zeoxanthin with decreased incidence of age-related macular degeneration have received progressive support. These two carotenoids are concentrated in the macula (the central portion of the retina) and are called the macular carotenoids.

Age-related macular degeneration (AMD) is the leading cause of blindness in the United States among persons older than 50 years of age, and it affects 1.7 million people worldwide. In AMD, visual loss is related to oxidative damage to the retinal pigment epithelium (RPE) and the choriocapillaris epithelium. The photoreceptor/retinal pigment complex is exposed to sunlight, it is bathed in near arterial levels of oxygen, and the membranes contain high concentrations of polyunsaturated fatty acids, all of which are conducive to oxidative damage. Lipofuscin granules, which accumulate in the RPE throughout life, may serve as photosensitizers, initiating damage by absorbing blue light and generating singlet oxygen that forms other radicals. Dark sunglasses are protective. Epidemiologic studies showed that the intake of lutein and zeanthin in dark green leafy vegetables (e.g., spinach and collard greens) also may be protective. Lutein and zeaxanthein accumulate in the macula and protect against free radical damage by absorbing blue light and quenching singlet oxygen.

4. OTHER DIETARY ANTIOXIDANTS

Flavonoids are a group of structurally similar compounds containing two spatially separate aromatic rings that are found in red wine, green tea, chocolate, and other plant-derived foods (Fig. 24.21). Flavonoids have been hypothesized to contribute to our free radical defenses in a number of ways. Some flavonoids inhibit enzymes responsible for superoxide anion production, such as xanthine oxidase. Others efficiently chelate Fe and Cu, making it impossible for these metals to participate in the Fenton reaction. They also may act as free radical scavengers by donating electrons to superoxide or lipid peroxy radicals, or stabilize free radicals by complexing with them.

It is difficult to tell how much dietary flavonoids contribute to our free radical defense system; they have a high pro-oxidant activity and are poorly absorbed. Nonetheless, we generally consume large amounts of flavonoids (approximately 800 mg/day), and there is evidence that they can contribute to the maintenance of vitamin E as an antioxidant.

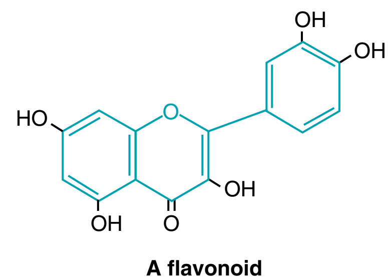

Fig 24.21. The flavonoid quercetin. All flavonoids have the same ring structure, shown in blue. They differ in ring substituents (=O, -OH, and OCH_3). Quercetin is effective in Fe chelation and antioxidant activity. It is widely distributed in fruits (principally in the skins) and in vegetables (e.g., onions).

5. ENDOGENOUS ANTIOXIDANTS

A number of compounds synthesized endogenously for other functions, or as urinary excretion products, also function nonenzymatically as free radical antioxidants. Uric acid is formed from the degradation of purines and is released into extracellular fluids, including blood, saliva, and lung lining fluid (Fig. 24.22). Together with protein thiols, it accounts for the major free radical trapping capacity of plasma. It is particularly important in the upper airways, where there are few other antioxidants. It can directly scavenge hydroxyl radicals, oxyheme oxidants formed between the reaction of hemoglobin and peroxy radicals, and peroxyl radicals themselves. Having acted as a scavenger, uric acid produces a range of oxidation products that are subsequently excreted.

Melatonin, which is a secretory product of the pineal gland, is a neurohormone that functions in regulation of our circadian rhythm, light–dark signal transduction, and sleep induction. In addition to these receptor-mediated functions, it functions as a nonenzymatic free radical scavenger that donates an electron (as hydrogen) to “neutralize” free radicals. It also can react with ROS and RNOS to form addition products, thereby undergoing suicidal transformations. Its effectiveness is related to both its lack of pro-oxidant activity and its joint hydrophilic/hydrophobic nature that allows it to pass through membranes and the blood-brain barrier.

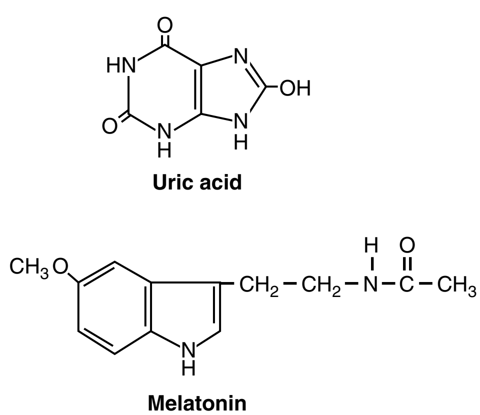

Fig 24.22. Endogenous antioxidants. Uric acid and melatonin both act to successively neutralize several molecules of ROS.

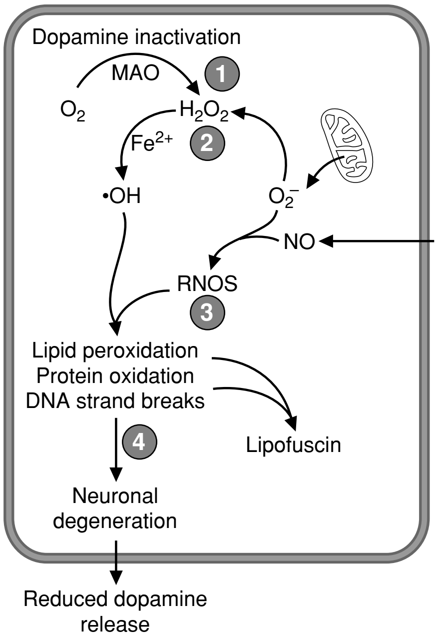

Fig 24.23. A model for the role of ROS and RNOS in neuronal degradation in Parkinson's disease. 1. Dopamine levels are reduced by monoamine oxidase, which generates H_2O_2. 2. Superoxide also can be produced by mitochondria, which SOD will convert to H_2O_2. Iron levels increase, which allows the Fenton reaction to proceed, generating hydroxyl radicals. 3. NO, produced by inducible nitric oxide synthase, reacts with superoxide to form RNOS. 4. The RNOS and hydroxyl radical lead to radical chain reactions that result in lipid peroxidation, protein oxidation, the formation of lipofuscin, and neuronal degeneration. The end result is a reduced production and release of dopamine, which leads to the clinical symptoms observed.

CLINICAL COMMENTS

Les Dopaman has "primary" parkinsonism. The pathogenesis of this disease is not well established and may be multifactorial (Fig. 24.23). The major clinical disturbances in Parkinson's disease are a result of dopamine depletion in the neostriatum, resulting from degeneration of dopaminergic neurons whose cell bodies reside in the substantia nigra pars compacta. The decrease in dopamine production is the result of severe degeneration of these nigrostriatal neurons. Although the agent that initiates the disease is unknown, a variety of studies support a role for free radicals in Parkinson's disease. Within these neurons, dopamine turnover is increased, dopamine levels are lower, glutathione is decreased, and lipofuscin (Lewy bodies) is increased. Iron levels are higher, and ferritin, the storage form of iron, is lower. Furthermore, the disease is mimicked by the compound 1-methyl-4-phenylpyridinium (MPP^+), an inhibitor of NADH dehydrogenase that increases superoxide production in these neurons. Even so, it is not known whether oxidative stress makes a primary or secondary contribution to the disease process.

Drug therapy is based on the severity of the disease. In the early phases of the disease, a monoamine oxidase B-inhibitor is used that inhibits dopamine degradation and decreases hydrogen peroxide formation. In later stages of the disease, patients are treated with levodopa (L-dopa), a precursor of dopamine.

Cora Nari experienced angina caused by severe ischemia in the ventricular muscle of her heart. The ischemia was caused by clots that formed at the site of atherosclerotic plaques within the lumen of the coronary arteries. When TPA was infused to dissolve the clots, the ischemic area of her heart was reperfused with oxygenated blood, resulting in ischemic–reperfusion injury. In her case, the reperfusion injury resulted in ventricular fibrillation.

During ischemia, several events occur simultaneously in cardiomyocytes. A decreased O_2 supply results in decreased ATP generation from mitochondrial oxidative phosphorylation and inhibition of cardiac muscle contraction. As a consequence, cytosolic AMP concentration increases, activating anaerobic glycolysis and lactic acid production. If ATP levels are inadequate to maintain Na^+, K^+ -ATPase activity, intracellular Na^+ increases, resulting in cellular swelling, a further increase in H^+ concentration, and increases of cytosolic and subsequently mitochondrial Ca^{2+} levels. The decrease in ATP and increase in Ca^{2+} may open the mitochondrial permeability transition pore, resulting in permanent inhibition of oxidative phosphorylation. Damage to lipid membranes is further enhanced by Ca^{2+} activation of phospholipases.

Reperfusion with O_2 allows recovery of oxidative phosphorylation, provided that the mitochondrial membrane has maintained some integrity and the mitochondrial transition pore can close. However, it also increases generation of free radicals. The transfer of electrons from CoQ• to O_2 to generate superoxide is increased. Endothelial production of superoxide by xanthine oxidase also may increase. These radicals may go on to form the hydroxyl radical, which can enhance the damage to components of the electron transport chain and mitochondrial lipids, as well as activate the

Currently, an intense study of ischemic insults to a variety of animal organs is underway, in an effort to discover ways of preventing reperfusion injury. These include methods designed to increase endogenous antioxidant activity, to reduce the generation of free radicals, and, finally, to develop exogenous antioxidants that, when administered before reperfusion, would prevent its injurious effects. Each of these approaches has met with some success, but their clinical application awaits further refinement. With the growing number of invasive procedures aimed at restoring arterial blood flow through partially obstructed coronary vessels, such as clot lysis, balloon or laser angioplasty, and coronary artery bypass grafting, development of methods to prevent ischemia–reperfusion injury will become increasingly urgent.

mitochondrial permeability transition. As macrophages move into the area to clean up cellular debris, they may generate NO and superoxide, thus introducing peroxynitrite and other free radicals into the area. Depending on the route and timing involved, the acute results may be cell death through necrosis, with slower cell death through apoptosis in the surrounding tissue.

In **Cora Nari's** case, oxygen was restored before permanent impairment of oxidative phosphorylation had occurred and the stage of irreversible injury was reached. However, reintroduction of oxygen induced ventricular fibrillation, from which she recovered.

BIOCHEMICAL COMMENTS

Protection Against Ozone in Lung Lining Fluid The lung lining fluid, a thin fluid layer extending from the nasal cavity to the most distal lung alveoli, protects the epithelial cells lining our airways from ozone and other pollutants. Although ozone is not a radical species, many of its toxic effects are mediated through generation of the classical ROS, as well as generation of aldehydes and ozonides. Polyunsaturated fatty acids represent the primary target for ozone, and peroxidation of membrane lipids is the most important mechanism of ozone-induced injury. However, ozone also oxidizes proteins.

The lung lining fluid has two phases; a gel-phase that traps microorganisms and large particles, and a sol (soluble) phase containing a variety of ROS defense mechanisms that prevent pollutants from reaching the underlying lung epithelial cells (Fig. 24.24). When the ozone level of inspired air is low, ozone is neutralized principally by uric acid (UA) present in the fluid lining the nasal cavity. In the proximal and distal regions of the respiratory tract, glutathione (GSH) and ascorbic acid (AA), in addition to UA, react directly with ozone. Ozone that escapes this antioxidant screen may react directly with proteins, lipids, and carbohydrates (CHO) to generate secondary oxidants, such as lipid peroxides, that can initiate chain reactions. A second layer of defense protects against these oxidation and peroxidation products: β-tocopherol (vitamin E) and glutathione react directly with lipid radicals; glutathione peroxidase reacts with hydrogen peroxide and lipid peroxides, and

Although most individuals are able to protect against small amounts of ozone in the atmosphere, even slightly elevated ozone concentrations produce respiratory symptoms in 10 to 20% of the healthy population.

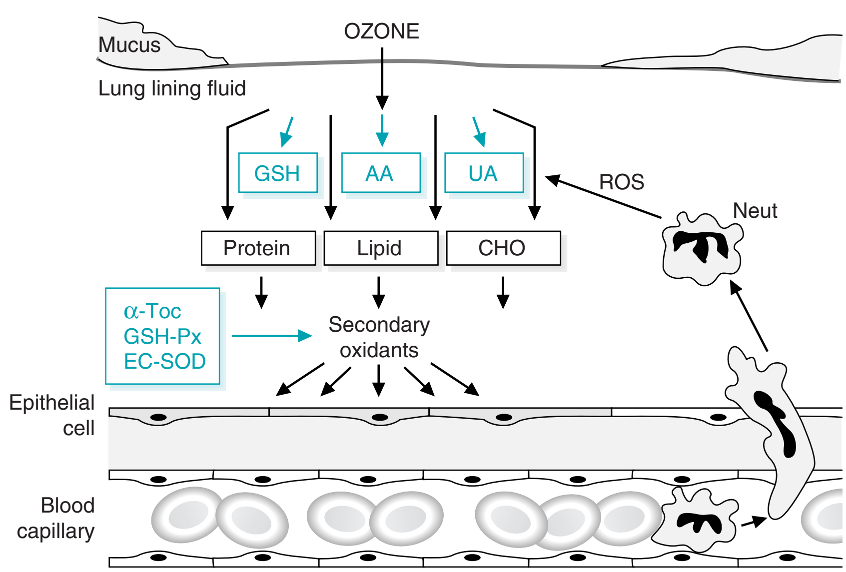

Fig 24.24. Protection against ozone in the lung lining fluid. GSH, glutathione; AA, ascorbic acid (vitamin C); UA, uric acid; CHO, carbohydrate; α-TOC, vitamin E; GSH-Px, glutathione peroxidase; ED-SOD, extracellular superoxide dismutase; *Neut*, neutrophil.

extracellular superoxide dismutase (EC-SOD) converts superoxide to hydrogen peroxide. However, oxidative stress may still overwhelm even this extensive defense network because ozone also promotes neutrophil migration into the lung lining fluid. Once activated, the neutrophils (Neut) produce a second wave of ROS (superoxide, HOCl, and NO).

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REVIEW QUESTIONS—CHAPTER 24

1. Which of the following vitamins or enzymes is unable to protect against free radical damage?

   (A) β-Carotene
   (B) Glutathione peroxidase
   (C) Superoxide dismutase
   (D) Vitamin B6
   (E) Vitamin C
   (F) Vitamin E

2. Superoxide dismutase catalyzes which of the following reactions?

   (A) $O_2^- + e^- + 2H^+$ yields H_2O_2
   (B) $2\ O_2^- + 2H^+$ yields $H_2O_2 + O_2$
   (C) $O_2^- + HO\bullet + H+$ yields $CO_2 + H_2O$
   (D) $H_2O_2 + O_2$ yields $4\ H_2O$
   (E) $O_2^- + H_2O_2 + H^+$ yields $2\ H_2O + O_2$

3. The mechanism of vitamin E as an antioxidant is best described by which of the following?

   (A) Vitamin E binds to free radicals and sequesters them from the contents of the cell.
   (B) Vitamin E participates in the oxidation of the radicals.
   (C) Vitamin E participates in the reduction of the radicals.
   (D) Vitamin E forms a covalent bond with the radicals, thereby stabilizing the radical state.
   (E) Vitamin E inhibits enzymes that produce free radicals.

4. An accumulation of hydrogen peroxide in a cellular compartment can be converted to dangerous radical forms in the presence of which metal?

   (A) Se
   (B) Fe
   (C) Mn
   (D) Mg
   (E) Mb

5. The level of oxidative damage to mitochondrial DNA is 10 times greater than that to nuclear DNA. This could be due, in part, to which of the following?

   (A) Superoxide dismutase is present in the mitochondria.
   (B) The nucleus lacks glutathione.
   (C) The nuclear membrane presents a barrier to reactive oxygen species.
   (D) The mitochondrial membrane is permeable to reactive oxygen species.
   (E) Mitochondrial DNA lacks histones.

25 Metabolism of Ethanol

Ethanol *is a* ***dietary fuel*** *that is metabolized to acetate principally in the liver, with the generation of NADH. The principal route for metabolism of ethanol is through hepatic* ***alcohol dehydrogenases****, which oxidize ethanol to* ***acetaldehyde*** *in the cytosol (Fig. 25.1). Acetaldehyde is further oxidized by* ***acetaldehyde dehydrogenases*** *to* ***acetate,*** *principally* ***in mitochondria****. Acetaldehyde, which is toxic, also may enter the blood.* ***NADH*** *produced by these reactions is used for adenosine triphosphate (ATP) generation through oxidative phosphorylation. Most of the acetate enters the blood and is taken up by skeletal muscles and other tissues, where it is activated to* ***acetyl CoA*** *and is oxidized in the TCA cycle.*

Approximately 10 to 20% of ingested ethanol is oxidized through a microsomal oxidizing system ***(MEOS)****, comprising* ***cytochrome P450*** *enzymes in the endoplasmic reticulum (especially* ***CYP2E1****). CYP2E1 has a high K_m for ethanol and is inducible by ethanol. Therefore, the proportion of ethanol metabolized through this route is greater at high ethanol concentrations, and greater after chronic consumption of ethanol.*

Acute effects *of alcohol ingestion arise principally from the generation of NADH, which greatly increases the* ***NADH/NAD$^+$*** *ratio of the liver. As a consequence,* ***fatty acid oxidation is inhibited****, and* ***ketogenesis*** *may occur. The elevated* ***NADH/NAD$^+$*** *ratio may also cause* ***lactic acidosis*** *and inhibit* ***gluconeogenesis.***

Ethanol metabolism may result in ***alchohol-induced liver disease****, including* ***hepatic steatosis*** *(fatty liver),* ***alcohol-induced hepatitis****, and* ***cirrhosis****. The principal toxic products of ethanol metabolism include* ***acetaldehyde*** *and* ***free radicals.*** *Acetaldehyde forms* ***adducts*** *with proteins and other compounds. The* ***hydroxyethyl radical*** *produced by MEOS and other radicals produced during*

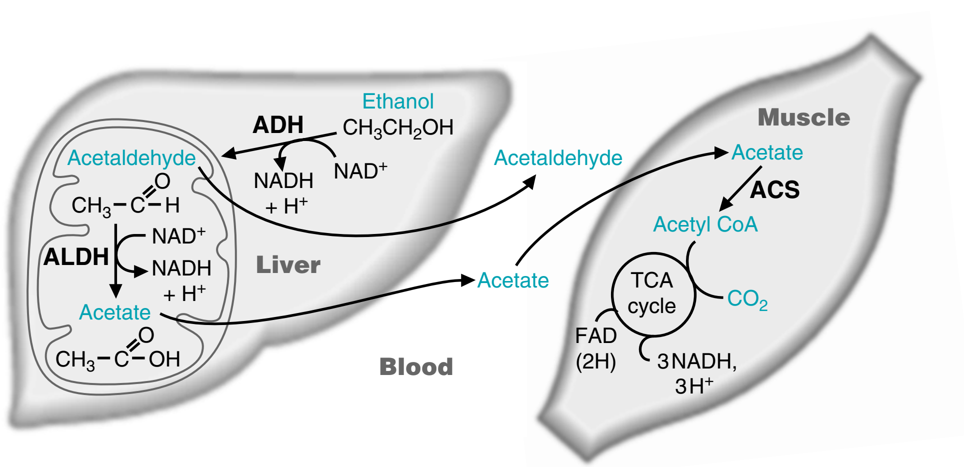

Fig. 25.1. The major route for metabolism of ethanol and use of acetate by the muscle. (ADH, alcohol dehydrogenase; ALDH, acetaldehyde dehydrogenase; ACS, acetyl-CoA synthetase).

inflammation cause irreversible damage to the liver. Many other tissues are adversely affected by ethanol, acetaldehyde, or by the consequences of ***hepatic dysmetabolism*** *and injury.* ***Genetic polymorphisms*** *in the enzymes of ethanol metabolism may be responsible for individual variations in the development of alcoholism or the development of liver cirrhosis.*

THE WAITING ROOM

A dietary history for **Ivan Applebod** showed that he had continued his habit of drinking scotch and soda each evening while watching TV, but he did not add the ethanol calories to his dietary intake. He justifies this calculation on the basis of a comment he heard on a radio program that calories from alcohol ingestion "don't count" because they are empty calories that do not cause weight gain.

Al Martini was found lying semiconscious at the bottom of the stairs by his landlady when she returned from an overnight visit with friends. His face had multiple bruises and his right forearm was grotesquely angulated. Nonbloody dried vomitus stained his clothing. Mr. Martini was rushed by ambulance to the emergency room at the nearest hospital. In addition to multiple bruises and the compound fracture of his right forearm, he had deep and rapid (Kussmaul) respirations and was moderately dehydrated.

Initial laboratory studies showed a relatively large anion gap of 34 mmol/L (reference range = 9–15 mmol/L). An arterial blood gas analysis confirmed the presence of a metabolic acidosis. Mr. Martini's blood alcohol level was only slightly elevated. His serum glucose was 68 mg/dL (low normal).

The anion gap is calculated by subtracting the sum of the value for serum chloride and for the serum HCO_3^- content from the serum sodium concentration. If the gap is greater than normal, it suggests that acids such as the ketone bodies acetoacetate and β-hydroxybutyrate are present in the blood in increased amounts.

Jean Ann Tonich, a 46-year-old commercial artist, recently lost her job because of absenteeism. Her husband of 24 years had left her 10 months earlier. She complains of loss of appetite, fatigue, muscle weakness, and emotional depression. She has had occasional pain in the area of her liver, at times accompanied by nausea and vomiting.

On physical examination she appears disheveled and pale. The physician notes tenderness to light percussion over her liver and detects a small amount of ascites (fluid within the peritoneal cavity around the abdominal organs). The lower edge of her liver is palpable about 2 inches below the lower margin of her right rib cage, suggesting liver enlargement, and feels somewhat more firm and nodular than normal. Jean Ann's spleen is not palpably enlarged. There is a suggestion of mild jaundice. No obvious neurologic or cognitive abnormalities are present.

After detecting a hint of alcohol on Jean Ann's breath, the physician questions her about possible alcohol abuse, which she denies. With more intensive questioning, however, Jean Ann admits that for the last 5 or 6 years she began drinking gin on a daily basis (approximately 4–5 drinks, or 68–85 g ethanol) and eating infrequently. Laboratory tests showed that her serum ethanol level on the initial office visit was 245 mg/dL (0.245%). A serum ethanol level above 150 mg/dL (0.15%) is considered indicative of inebriation.

Jaundice is a yellow discoloration involving the sclerae (the "whites"' of the eyes) and skin. It is caused by the deposition of bilirubin, a yellow degradation product of heme. Bilirubin accumulates in the blood under conditions of liver injury, bile duct obstruction, and excessive degradation of heme.

Jean Ann Tonich's admitted ethanol consumption exceeds the definition of moderate drinking. Moderate drinking is now defined as not more than two drinks per day for men, but only one drink per day for women. A drink is defined as 12 oz of regular beer, 5 oz of wine, or 1.5 oz distilled spirits (80 proof).

I. ETHANOL METABOLISM

Ethanol is a small molecule that is both lipid and water soluble. It is, therefore, readily absorbed from the intestine by passive diffusion. A small percentage of ingested ethanol (0-5%) enters the gastric mucosal cells of the upper GI tract (tongue, mouth,

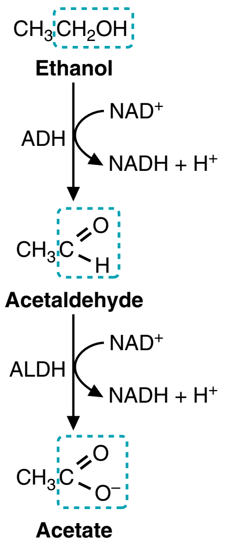

Fig. 25.2. The pathway of ethanol metabolism (ADH, alcohol dehydrogenase; ALDH, acetaldehyde dehydrogenase).

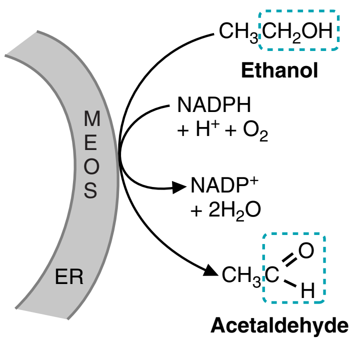

Fig. 25.3. The reaction catalyzed by MEOS (which includes CYP2E1) in the endoplasmic reticulum.

esophagus, and stomach), where it is metabolized. The remainder enters the blood. Of this, 85 to 98% is metabolized in the liver, and only 2 to 10% is excreted through the lungs or kidneys.

The major route of ethanol metabolism in the liver is through liver alcohol dehydrogenase, a cytosolic enzyme that oxidizes ethanol to acetaldehyde with reduction of NAD^+ to NADH (Fig.25.2). If it is not removed by metabolism, acetaldehyde exerts toxic actions in the liver and can enter the blood and exert toxic effects in other tissues.

Approximately 90% of the acetaldehyde that is generated is further metabolized to acetate in the liver. The major enzyme involved is a low K_m mitochondrial acetaldehyde dehydrogenase (ALDH), which oxidizes acetaldehyde to acetate with generation of NADH (see Fig. 25.2). Acetate, which has no toxic effects, may be activated to acetyl CoA in the liver (where it can enter either the TCA cycle or the pathway for fatty acid synthesis). However, most of the acetate that is generated enters the blood and is activated to acetyl CoA in skeletal muscles and other tissues (see Fig. 25.1). Acetate is generally considered nontoxic and is a normal constituent of the diet.

The other principal route of ethanol oxidation in the liver is the microsomal ethanol oxidizing system (MEOS), which also oxidizes ethanol to acetaldehyde (Fig. 25.3). The principal microsomal enzyme involved is a cytochrome P450 mixed-function oxidase isozyme (CYP2E1), which uses NADPH as an additional electron donor and O_2 as an electron acceptor. This route accounts for only 10 to 20% of ethanol oxidation in a moderate drinker.

Each of the enzyme activities involved in ethanol metabolism (alcohol dehydrogenase, acetaldehyde dehydrogenase, and CYP2E1) exist as a family of isoenzymes. Individual variations in the quantity of these isoenzymes influence a number of factors, such as the rate of ethanol clearance from the blood, the degree of inebriation exhibited by an individual, and differences in individual susceptibility to the development of alcohol-induced liver disease.

A. Alcohol Dehydrogenase

Alcohol dehydrogenase (ADH) exists as a family of isoenzymes with varying specificity for chain length of the alcohol substrate (Table 25.1). Ethanol is a small molecule that does not exhibit much in the way of unique structural characteristics and, at high concentrations, is nonspecifically metabolized by many members of the ADH family. The alcohol dehydrogenases that exhibit the highest specificity for ethanol are the class I alcohol dehydrogenases. We have three genes for class I alcohol dehydrogenases, each of which exists as allelic variants (polymorphisms).

Table 25.1. Isozymes of Medium-Chain-Length Alcohol Dehydrogenases

Class	Gene	Sub-Unit	Tissue Distribution	Properties
I	ADH 1 ADH 2 ADH 3	α β γ	Most abundant in liver and adrenal glands. Much lower levels in kidney, lung, colon, small intestine, eye, ovary, blood vessels. None in brain or heart	K_m of 0.05–4 mM for ethanol. Active only with ethanol. High tissue capacity.
II	ADH 4	π	Primarily liver, lower levels in GI tract	K_m of 34 mM for ethanol.
III	ADH 5	χ	Ubiquitously expressed, but at higher levels in liver. The only isozyme present in germinal cells.	Relatively inactive toward ethanol. Active mainly toward long-chain alcohols, and ω-OH fatty acids.
IV	ADH 7	σ	Present in highest levels in upper GI tract, gingiva and mouth, esophagus, down to the stomach. Not present in liver.	K_m of 28 mM. It is the most active of medium-chain alcohol DH toward retinal.
V	ADH 6	-	May be highest in fetal liver.	Some activity toward ethanol

The class I alcohol dehydrogenases are present in high quantities in the liver, representing approximately 3% of all soluble protein. These alcohol dehydrogenases, commonly referred to collectively as liver alcohol dehydrogenase, have low K_ms for ethanol between 0.05 and 4 mM (high affinities). Thus, the liver is the major site of ethanol metabolism and the major site at which the toxic metabolite acetaldehyde is generated.

Although the class IV and class II enzymes make minor contributions to ethanol metabolism, they may contribute to its toxic effects. Ethanol concentrations can be quite high in the upper GI tract (e.g., beer is approximately 0.8 M ethanol), and acetaldehyde generated here by class IV enzymes (gastric ADH) might contribute to the risk for cancer associated with heavy drinking. Class II ADH genes are expressed primarily in the liver and at lower levels in the lower gastrointestinal tract.

B. Acetaldehyde Dehydrogenases

Acetaldehyde is oxidized to acetate, with the generation of NADH, by acetaldehyde dehydrogenases (see Fig. 25.2). More than 80% of acetaldehyde oxidation in the human liver is normally catalyzed by mitochondrial acetaldehyde dehydrogenase (ALDH2), which has a high affinity for acetaldehyde and is highly specific. However, individuals with a common allelic variant of ALDH2 have a greatly decreased capacity for acetaldehyde metabolism

Most of the remainder of acetaldehyde oxidation occurs through a cytosolic acetaldehyde dehydrogenase (ALDH1). Additional aldehyde dehydrogenases act on a variety of organic alcohols, toxins, and pollutants.

C. Fate of Acetate

Metabolism of acetate requires activation to acetyl CoA by acetyl CoA synthetase in a reaction similar to that catalyzed by fatty acyl CoA synthetases (Fig. 25.4). In liver, the principle isoform of acetyl CoA synthetase (ACS I) is a cytosolic enzyme that generates acetyl CoA for the cytosolic pathways of cholesterol and fatty acid synthesis. Acetate entry into these pathways is under regulatory control by mechanisms involving cholesterol or insulin. Thus, most of the acetate generated enters the blood.

Acetate is taken up and oxidized by other tissues, notably heart and skeletal muscle, which have a high concentration of the mitochondrial acetyl CoA synthetase isoform (ACSII). This enzyme is present in the mitochondrial matrix. It therefore generates acetyl CoA that can directly enter the TCA cycle and be oxidized to CO_2.

D. Microsomal Ethanol Oxidizing System

Ethanol is also oxidized to acetaldehyde in the liver by the microsomal ethanol oxidizing system, which comprises members of the cytochrome P450 superfamily of enzymes. Ethanol and NADPH both donate electrons in the reaction, which reduces O_2 to $2H_2O$ (Fig. 25.5). The cytochrome P450 enzymes all have two

The accumulation of acetaldehyde causes nausea and vomiting, and, therefore, inactive acetaldehyde dehydrogenases are associated with a distaste for alcoholic beverages and protection against alcoholism. In one of the common allelic variants of ALDH2 (ALDH2*2), a single substitution increases the K_m for acetaldehyde 260-fold (lowers the affinity) and decreases the V_{max} 10-fold, resulting in a very inactive enzyme. Homozygosity for the ALDH2*2 allele affords absolute protection against alcoholism; no individual with this genotype has been found among alcoholics. Alcoholics are frequently treated with acetaldehyde dehydrogenase inhibitors (e.g., disulfiram) to help them abstain from alcohol intake. Unfortunately, alcoholics who continue to drink while taking this drug are exposed to the toxic effects of elevated acetaldehyde levels.

The human has at least seven, and possibly more, genes that code for specific isoenzymes of medium-chain-length alcohol dehydrogenases, the major enzyme responsible for the oxidation of ethanol to acetaldehyde in the human. These different alcohol dehydrogenases have an approximately 60 to 70% identity and are assumed to have arisen from a common ancestral gene similar to the class III isoenzyme many millions of years ago. The class I alcohol dehydrogenases (ADH 1, ADH 2, and ADH 3) are all present in high concentration in the liver, and have a relatively high affinity and capacity for ethanol at low concentrations. (These properties are quantitatively reflected by their low K_m, a parameter discussed in Chapter 9). They have a 90 to 94% sequence identity and are able to form both homo- and hetero-dimers, among themselves (e.g., $\beta\beta$ or $\beta\gamma$). However, none of the ADHs can form dimers with an ADH from another class. The three genes for class I alcohol dehydrogenases are arranged in tandem, head to tail, on chromosome 4. The genes for the other classes of alcohol dehydrogenase are also on chromosome 4 in nearby locations.

ADH 2 and ADH 3 are present as functional polymorphisms that differ in their properties. Genetic polymorphisms for ADH partially account for the observed differences in ethanol elimination rates among various individuals or populations. Although susceptibility to alcoholism is a complex function of genetics and socioeconomic factors, possession of the ADH 2*2 allele, which encodes a relatively fast ADH (high V_{max}), is associated with a decreased susceptibility to alcoholism—presumably because of nausea and flushing caused by acetaldehyde accumulation (because the aldehyde dehydrogenase gene cannot keep up with the amount of acetaldehyde produced). This particular allele has a relatively high frequency in the East Asian population and a low frequency among white Europeans. In contrast, the ADH 2*1/2*1 genotype (homozygous for allele 1 of the ADH 2 gene) is a risk factor for the development of Wernicke-Korsakoff syndrome, a neuropsychiatric syndrome commonly associated with alcoholism.

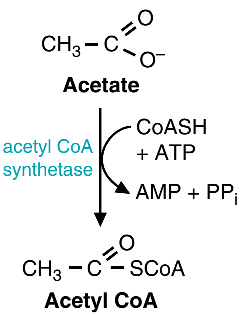

Fig. 25.4. The activation of acetate to acetyl CoA

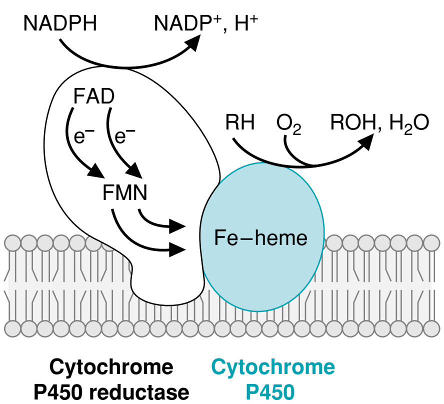

Fig. 25.5. General structure of cytochrome P450 enzymes. O_2 binds to the P450 Fe-heme in the active site and is activated to a reactive form by accepting electrons. The electrons are donated by the cytochrome P450 reductase, which contains an FAD plus an FMN or Fe-S center to facilitate the transfer of single electrons from NADPH to O_2. The P450 enzymes involved in steroidogenesis have a somewhat different structure. For CYP2E1, RH is ethanol (CH_3CH_2OH) and ROH is acetaldehyde (CH_3COH).

CYP represents **cy**tochrome **P**450. P450 is an Fe-heme similar to that found in the cytochromes of the electron transport chain ("P" denotes the heme **p**igment, and 450 is the wavelength of visible light absorbed by the pigment). In CYP2E1, the "2" refers to the gene family, which comprises isoenzymes with greater than 40% amino acid sequence identity. The "E" refers to the subfamily, a grouping of isoenzymes with greater than 55 to 60% sequence identity, and the "1" refers to the individual enzymes within this subfamily.

major catalytic protein components: an electron-donating reductase system that transfers electrons from NADPH (cytochrome P450 reductase) and a cytochrome P450. The cytochrome P450 protein contains the binding sites for O_2 and the substrate (e.g., ethanol) and carries out the reaction. The enzymes are present in the endoplasmic reticulum, which on isolation from disrupted cells forms a membrane fraction after centrifugation that was formerly called "microsomes" by biochemists.

1. CYP2E1

MEOS is part of the superfamily of cytochrome P450 enzymes, all of which catalyze similar oxidative reactions. Within the superfamily, at least 10 distinct gene families are found in mammals. More than 100 different cytochrome P450 isozymes exist within these 10 gene families. Each isoenzyme has a distinct classification according to its structural relationship with other isoenzymes. The isoenzyme that has the highest activity toward ethanol is called CYP2E1. A great deal of overlapping specificity exists among the various P450 isoenzymes, and ethanol is also oxidized by several other P450 isoenzymes. "MEOS" refers to the combined ethanol oxidizing activity of all the P450 enzymes.

CYP2E1 has a much higher K_m for ethanol than the class I alcohol dehydrogenases (11 mM [51 mg/dL] compared with 0.05–4 mM [0.23 to 18.4 mg/dL]). Thus, a greater proportion of ingested ethanol is metabolized through CYP2E1 at high levels of ethanol consumption than at low levels.

2. INDUCTION OF P450 ENZYMES

The P450 enzymes are inducible both by their most specific substrate and by substrates for some of the other cytochrome P450 enzymes. Chronic consumption of ethanol increases hepatic CYP2E1 levels approximately 5- to 10-fold. However, it also causes a twofold to fourfold increase in some of the other P450s from the same subfamily, from different subfamilies, and even from different gene families. The endoplasmic reticulum undergoes proliferation, with a general increase in the content of microsomal enzymes, including those that are not directly involved in ethanol metabolism.

The increase in CYP2E1 with ethanol consumption occurs through transcriptional, post-transcriptional, and post-translational regulation. Increased levels of mRNA, resulting from induction of gene transcription or stabilization of message, are found in actively drinking patients. The protein is also stabilized against degradation. In general, the mechanism for induction of P450 enzymes by their substrates occurs through the binding of the substrate (or related compound) to an intracellular receptor protein, followed by binding of the activated receptor to a response element in the target gene. Whether ethanol induction of CYP2E1 follows this general pattern has not yet been shown.

Overlapping specificity in the catalytic activity of P450 enzymes and in their inducers is responsible for several types of drug interactions. For example, phenobarbital, a barbiturate long used as a sleeping pill or for treatment of epilepsy, is converted to an inactive metabolite by cytochrome P450 monooxygenases CYP2B1 and CYP2B2. After treatment with phenobarbital, CYP2B2 is increased 50- to 100-fold. Individuals who take phenobarbital for prolonged periods develop a drug tolerance as CYP2B2 is induced, and the drug is metabolized to an inactive metabolite more rapidly. Consequently, these individuals use progressively higher doses of phenobarbital.

Ethanol is an inhibitor of the phenobarbital-oxidizing P450 system. When large amounts of ethanol are consumed, the inactivation of phenobarbital is directly or indirectly inhibited. Therefore, when high doses of phenobarbital and ethanol are consumed at the same time, toxic levels of the barbiturate can accumulate in the blood.

Although induction of CYP2E1 increases ethanol clearance from the blood, it has negative consequences. Acetaldehyde may be produced faster than it can be metabolized by acetaldehyde dehydrogenases, thereby increasing the risk of hepatic injury. An increased amount of acetaldehyde can enter the blood and can damage other tissues. In addition, cytochrome P450 enzymes are capable of generating free radicals, which also may lead to increased hepatic injury and cirrhosis (see Chapter 24).

As blood ethanol concentration rises above 18 mM (the legal intoxication limit is now defined as 0.08% in most states of the United States, which is approximately 18 mM), the brain and central nervous system are affected. Induction of CYP2E1 increases the rate of ethanol clearance from the blood, thereby contributing to increased alcohol tolerance. However, the apparent ability of a chronic alcoholic to drink without appearing inebriated is partly a learned behavior.

E. Variations in the Pattern of Ethanol Metabolism

The routes and rates of ethanol oxidation vary from individual to individual. Differences in ethanol metabolism may influence whether an individual becomes a chronic alcoholic, develops alcohol-induced liver disease, or develops other diseases associated with increased alcohol consumption (such as hepatocarcinogenesis, lung cancer, or breast cancer). Factors that determine the rate and route of ethanol oxidation in individuals include:

- Genotype—Polymorphic forms of alcohol dehydrogenases and acetaldehyde dehydrogenases can greatly affect the rate of ethanol oxidation and the accumulation of acetaldehyde. CYP2E1 activity may vary as much as 20-fold between individuals, partly because of differences in the inducibility of different allelic variants.
- Drinking history—The level of gastric alcohol dehydrogenase (ADH) decreases and CYP2E1 increases with the progression from a naïve, to a moderate, and to a heavy and chronic consumer of alcohol.
- Gender—Blood levels of ethanol after consuming a drink are normally higher for women than for men, partly because of lower levels of gastric ADH activity in women. After chronic consumption of ethanol, gastric ADH decreases in both men and women, but the gender differences become even greater. Gender differences in blood alcohol levels also occur because women are normally smaller. Furthermore, in females, alcohol is distributed in a 12% smaller water space because a woman's body composition consists of more fat and less water than that of a man.
- Quantity—The amount of ethanol an individual consumes over a small amount of time determines its metabolic route. Small amounts of ethanol are metabolized most efficiently through the low K_m pathway of class I ADH and class II ALDH. Little accumulation of NADH occurs to inhibit ethanol metabolism via these dehydrogenases. However, when higher amounts of ethanol are consumed in a short period, a disproportionately greater amount is metabolized through MEOS. MEOS, which has a much higher K_m for ethanol, functions principally at high concentrations of ethanol. A higher activity of MEOS would be expected to correlate with tendency to develop alcohol-induced liver disease, because both acetaldehyde and free radical levels would be increased.

F. The Energy Yield of Ethanol Oxidation

The ATP yield from ethanol oxidation to acetate varies with the route of ethanol metabolism. If ethanol is oxidized by the major route of cytosolic ADH and mitochondrial ALDH, one cytosolic and one mitochondrial NADH are generated with a maximum yield of 5 ATP. Oxidation of acetyl CoA in the TCA cycle and electron transport chain leads to the generation of 10 high-energy phosphate bonds. However, activation of acetate to acetyl CoA requires two high-energy phosphate bonds (one in the cleavage of ATP to AMP+ pyrophosphate and one in the cleavage of pyrophosphate to phosphate), which must be subtracted. Thus the maximum total energy yield is 13 moles of ATP per mole of ethanol.

In contrast, oxidation of ethanol to acetaldehyde by CYP2E1 consumes energy in the form of NADPH, which is equivalent to 2.5 ATP. Thus, for every mole of

At **Ivan Applebod's** low level of ethanol consumption, ethanol is oxidized to acetate via ADH and ALDH in the liver and the acetate is activated to acetyl CoA and oxidized to CO_2 in skeletal muscle and other tissues. The overall energy yield of 13 ATP per ethanol molecule accounts for the caloric value of ethanol, approximately 7 Cal/g. However, chronic consumption of substantial amounts of alcohol does not have the effect on body weight expected from the caloric intake. This is partly attributable to induction of MEOS, resulting in a proportionately greater metabolism of ethanol through MEOS with its lower energy yield (only approximately 8 ATP). In general, weight loss diets recommend no, or low, alcohol consumption because ethanol calories are "empty" in the sense that alcoholic beverages are generally low in vitamins, essential amino acids, and other required nutrients, but not empty of calories.

ethanol metabolized by this route, only a maximum of 8.0 moles of ATP can be generated (10 ATP from acetylCoA oxidation through the TCA cycle, minus 2 for acetate activation; the NADH generated by aldehyde dehydrogenase is balanced by the loss of NADPH in the MEOS step).

II. TOXIC EFFECTS OF ETHANOL METABOLISM

Alcohol-induced liver disease, a common and sometimes fatal consequence of chronic ethanol abuse, may manifest itself in three forms: fatty liver, alcohol-induced hepatitis, and cirrhosis. Each may occur alone, or they may be present in any combination in a given patient. Alcohol-induced cirrhosis is discovered in up to 9% of all autopsies performed in the United States, with a peak incidence in patients 40 to 55 years of age.

However, ethanol ingestion also has acute effects on liver metabolism, including inhibition of fatty acid oxidation and stimulation of triacylglycerol synthesis, leading to a fatty liver. It also can result in ketoacidosis or lactic acidosis and cause hypoglycemia or hyperglycemia, depending on the dietary state. These effects are considered reversible.

In contrast, acetaldehyde and free radicals generated from ethanol metabolism can result in alcohol-induced hepatitis, a condition in which the liver is inflamed and cells become necrotic and die. Diffuse damage to hepatocytes results in cirrhosis, characterized by fibrosis (scarring), disturbance of the normal architecture and blood flow, loss of liver function and, ultimately, hepatic failure.

A. Acute Effects of Ethanol Arising from the Increased NADH /NAD^+ Ratio

Many of the acute effects of ethanol ingestion arise from the increased NADH/NAD^+ ratio in the liver (Fig. 25.6). At lower levels of ethanol intake, the rate of ethanol oxidation is regulated by the supply of ethanol (usually determined by how much ethanol we consume) and the rate at which NADH is reoxidized in the electron transport chain. NADH is not a very effective product inhibitor of ADH or ALDH, and there is no other feedback regulation by ATP, ADP, or AMP. As a consequence, NADH generated in the cytosol and mitochondria tends to accumulate, increasing the NADH/NAD^+ ratio to high levels (see Fig. 25.6, circle 1). The increase is even greater as the mitochondria become damaged from acetaldehyde or free radical injury.

1. CHANGES IN FATTY ACID METABOLISM

The high NADH/NAD^+ ratio generated from ethanol oxidation inhibits the oxidation of fatty acids, which accumulate in the liver (see Fig. 25.6, circles 2 and 3) These fatty acids are re-esterified into triacylglycerols by combining with glycerol 3-P. The increased NADH/NAD^+ ratio increases the availability of glycerol 3-P by promoting its synthesis from intermediates of glycolysis. The triacylglycerols are incorporated into VLDL (very-low-density lipoproteins), which accumulate in the liver and enter the blood, resulting in an ethanol-induced hyperlipidemia.

Although just a few drinks may result in hepatic fat accumulation, chronic consumption of alcohol greatly enhances the development of a fatty liver. Re-esterification of fatty acids into triacylglycerols by fatty acyl CoA transferases in the ER is enhanced (see Fig. 25.6). Because the transferases are microsomal enzymes, they are induced by ethanol consumption just as MEOS is induced. The result is a fatty liver (hepatic steatosis).

The source of the fatty acids can be dietary fat, fatty acids synthesized in the liver, or fatty acids released from adipose tissue stores. Adipose tissue lipolysis increases after ethanol consumption, possibly because of a release of epinephrine.

The hyperlipidemia is greatly enhanced if fat is ingested with ethanol. Thus, "happy hour" foods (e.g., pizza, fried potato skins with sour cream, nachos, and deep-fried peppers stuffed with cream cheese and wrapped in bacon) are exactly the wrong things to eat while drinking. Steamed vegetables or salads with your beer would be much better for your liver.

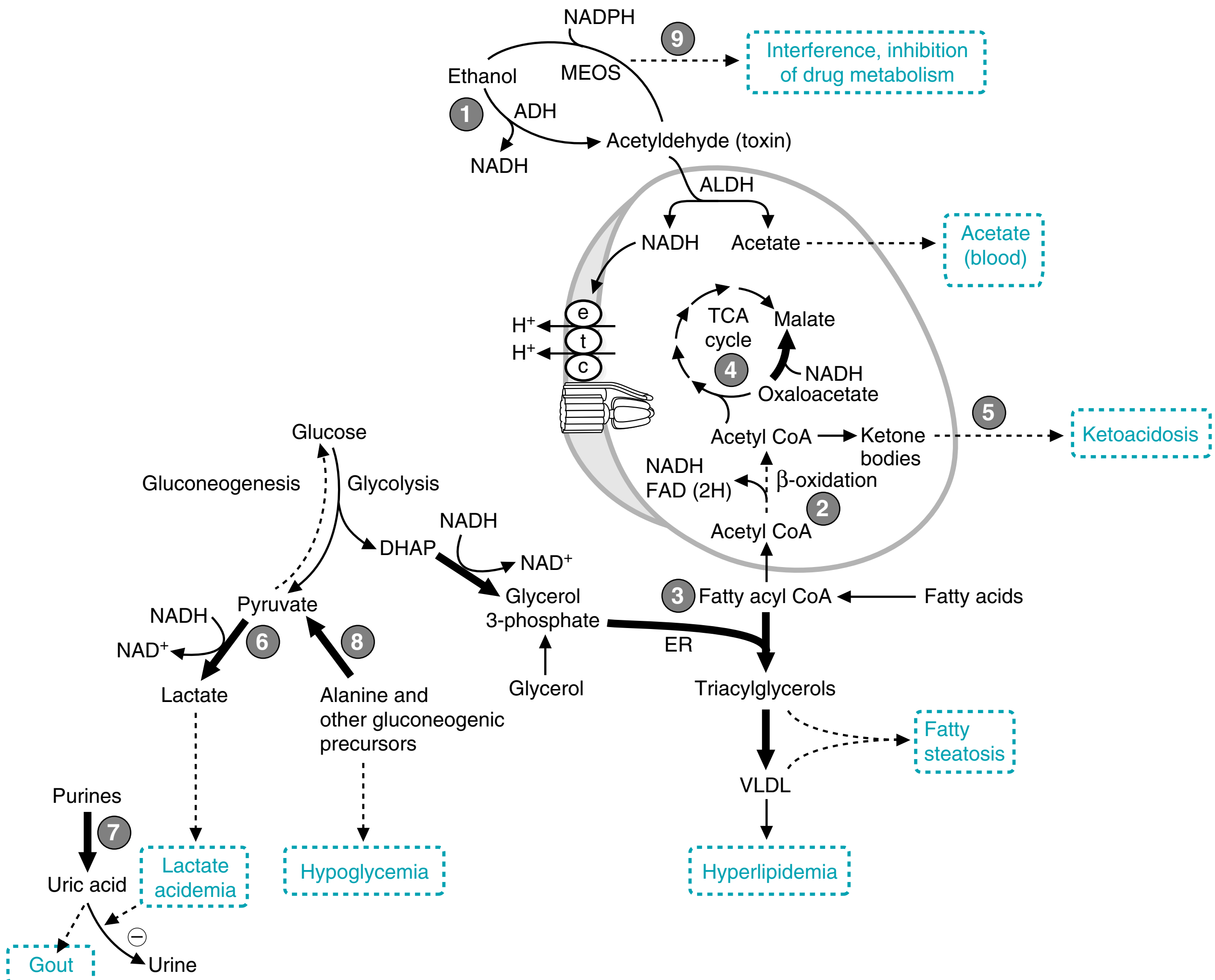

Fig. 25.6. Acute effects of ethanol metabolism on lipid metabolism in the liver. (1) Metabolism of ethanol generates a high $NADH/NAD^+$ ratio. (2) The high $NADH/NAD^+$ ratio inhibits fatty acid oxidation and the TCA cycle, resulting in accumulation of fatty acids. (3) Fatty acids are re-esterified to glycerol 3-P by acyltransferases in the endoplasmic reticulum. Glycerol 3-P levels are increased because a high $NADH/NAD^+$ ratio favors its formation from dihydroxyacetone phosphate (an intermediate of glycolysis). Ethanol-stimulated increases of endoplasmic reticulum enzymes also favors triacylglycerol formation. (4) NADH generated from ethanol oxidation can meet the requirements of the cell for ATP generation from oxidative phosphorylation. Thus, acetyl CoA oxidation in the TCA cycle is inhibited. (5) The high $NADH/NAD^+$ ratio shifts oxaloacetate (OAA) toward malate, and acetyl CoA is directed into ketone body synthesis. Options 6–8 are discussed in the text.

2. ALCOHOL-INDUCED KETOACIDOSIS.

Fatty acids that are oxidized are converted to acetyl CoA and subsequently to ketone bodies (acetoacetate and β-hydroxybutyrate). Enough NADH is generated from oxidation of ethanol and fatty acids that there is no need to oxidize acetyl CoA in the TCA cycle. The very high $NADH/NAD^+$ ratio shifts all of the oxaloacetate in the TCA cycle to malate, leaving the oxaloacetate levels too low for citrate synthase to synhesize citrate (see Fig. 25.6, circle 4). The acetyl CoA enters the pathway for ketone body synthesis instead of the TCA cycle.

Although ketone bodies are being produced at a high rate, their metabolism in other tissues is restricted by the supply of acetate, which is the preferred fuel. Thus, the blood concentration of ketone bodies may be much higher than found under normal fasting conditions.

Al Martini's admitting physician suspected an alcohol-induced ketoacidosis superimposed on a starvation ketoacidosis. Tests showed that his plasma free fatty acid level was elevated, and his plasma β-hydroxybutyrate level was 40 times the upper limit of normal. The increased $NADH/NAD^-$ ratio from ethanol consumption inhibited the TCA cycle and shifted acetyl CoA from fatty acid oxidation into the pathway of ketone body synthesis.

3. LACTIC ACIDOSIS, HYPERURICEMIA, AND HYPOGLYCEMIA

Another consequence of the very high NADH/NAD^+ ratio is that the balance in the lactate dehydrogenase reaction is shifted toward lactate, resulting in a lacticacidosis (see Fig. 25.6, circle 6). The elevation of blood lactate may decrease excretion of uric acid (see Fig. 25.6, circle 7) by the kidney. Consequently patients with gout (which results from precipitated uric acid crystals in the joints) are advised not to drink excessive amounts of ethanol. Increased degradation of purines also may contribute to hyperuricemia.

The increased NADH/NAD^+ ratio also can cause hypoglycemia in a fasting individual who has been drinking and is dependent on gluconeogenesis to maintain blood glucose levels (Fig. 25.6, circles 6 and 8). Alanine and lactate are major gluconeogenic precursors that enter gluconeogenesis as pyruvate. The high NADH/NAD^+ ratio shifts the lactate dehydrogenase equilibrium to lactate, so that pyruvate formed from alanine is converted to lactate and cannot enter gluconeogenesis. The high NADH/NAD^+ ratio also prevents other major gluconeogenic precursors, such as oxaloacetate and glycerol, from entering the gluconeogenic pathway.

In contrast, ethanol consumption with a meal may result in a transient hyperglycemia, possibly because the high NADH/NAD^+ ratio inhibits glycolysis at the glyceraldehyde-3-P dehydrogenase step.

B. Acetaldehyde Toxicity

Many of the toxic effects of chronic ethanol consumption result from accumulation of acetaldehyde, which is produced from ethanol both by alcohol dehydrogenases and MEOS. Acetaldehyde accumulates in the liver and is released into the blood after heavy doses of ethanol (Fig. 25.7). It is highly reactive and binds covalently to amino groups, sulfhydryl groups, nucleotides, and phospholipids to form "adducts."

1. ACETALDEHYDE AND ALCOHOL-INDUCED HEPATITIS

One of the results of acetaldehyde-adduct formation with amino acids is a general decrease in hepatic protein synthesis (see Fig. 25.7, circle 1). Calmodulin, ribonuclease, and tubulin are some of the proteins affected. Proteins in the heart and other tissues also may be affected by acetaldehyde that appears in the blood.

As a consequence of forming acetaldehyde adducts of tubulin, there is a diminished secretion of serum proteins and VLDL lipoproteins from the liver. The liver synthesizes many blood proteins, including serum albumin, blood coagulation factors, and transport proteins for vitamins, steroids, and iron. These proteins accumulate in the liver, together with lipid. The accumulation of proteins results in an influx of water (see Fig. 25.7, circle 6) within the hepatocytes and a swelling of the liver that contributes to portal hypertension and a disruption of hepatic architecture.

The noncaloric effect of heavy and chronic ethanol ingestion that led **Ivan Applebod** to believe ethanol has no calories may be partly attributable to uncoupling of oxidative phosphorylation. The hepatic mitochondria from tissues of chronic alcoholics may be partially uncoupled and unable to maintain the transmembrane proton gradient necessary for normal rates of ATP synthesis. Consequently, a greater proportion of the energy in ethanol would be converted to heat. Metabolic disturbances such as the loss of ketone bodies in urine, or futile cycling of glucose, also might contribute to a dimished energy value for ethanol.

2. ACETALDEHYDE AND FREE RADICAL DAMAGE

Acetaldehyde adduct formation enhances free radical damage. Acetaldehyde binds directly to glutathione and diminishes its ability to protect against H_2O_2 and prevent lipid peroxidation (see Fig. 25.7, circle 2). It also binds to free radical defense enzymes.

Damage to mitochondria from acetaldehyde and free radicals perpetuates a cycle of toxicity (see Fig. 25.7, circles 3 and 4). With chronic consumption of ethanol, mitochondria become damaged, the rate of electron transport is inhibited, and oxidative phosphorylation tends to become uncoupled. Fatty acid

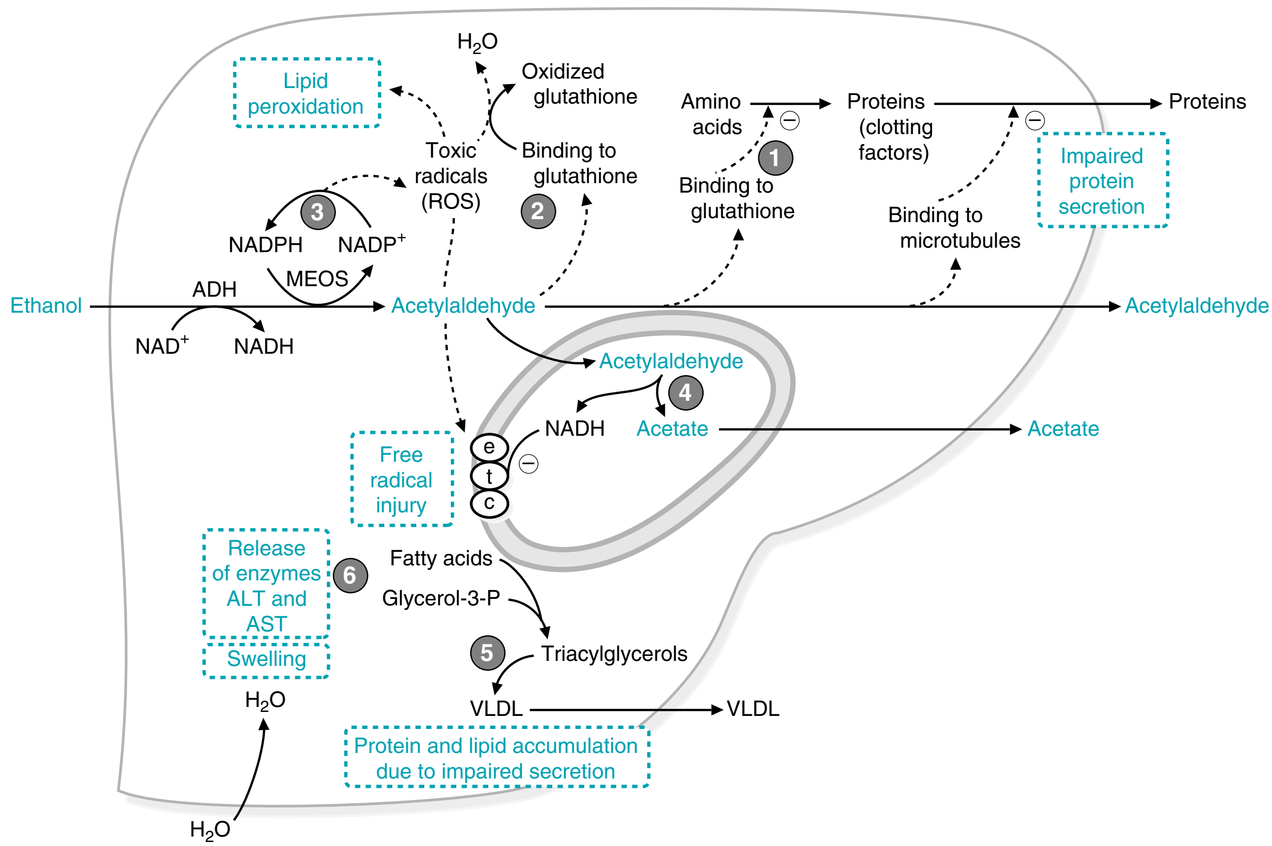

Fig. 25.7. The development of alcohol-induced hepatitis. (1) Acetaldehyde adduct formation decreases protein synthesis and impairs protein secretion. (2) Free radical injury results partly from acetaldehyde adduct formation with glutathione. (3) Induction of MEOS increases formation of free radicals, which leads to lipid peroxidation and cell damage. (4) Mitochondrial damage inhibits the electron transport chain, which decreases acetaldehyde oxidation. (5) Microtubule damage increases VLDL and protein accumulation. (6) Cell damage leads to release of the hepatic enzymes alanine aminotransferase (ALT) and aspartate aminotransferase (AST).

oxidation is decreased even further, thereby enhancing lipid accumulation (see Fig. 25.7, circle 5). The mitochondrial changes further impair mitochondrial acetaldehyde oxidation, thereby initiating a cycle of progressively increasing acetaldehyde damage.

C. Ethanol and Free Radical Formation

Increased oxidative stress in the liver during chronic ethanol intoxication arises from increased production of free radicals, principally by CYP2E1. FAD and FMN in the reductase and heme in the cytochrome P450 system transfer single electrons, thus operating through a mechanism that can generate free radicals. The hydroxyethyl radical ($CH_3CH_2O\cdot$) is produced during ethanol metabolism and can be released as a free radical. Induction of CYP2E1, as well as other cytochrome P450 enzymes, can increase the generation of free radicals from drug metabolism and from the activation of toxins and carcinogens (see Fig. 25.7, circle 3). These effects are enhanced by acetaldehyde-adduct damage.

Phospholipids, the major lipid in cellular membranes, are a primary target of peroxidation caused by free radical release. Peroxidation of lipids in the inner mitochondrial membrane may contribute to the inhibition of electron transport and uncoupling of mitochondria, leading to inflammation and cellular necrosis. Induction of CYP2E1 and other P450 cytochromes also increases formation of other radicals and the activation of hepatocarcinogens.

D. Hepatic Cirrhosis and Loss of Liver Function

Liver injury is irreversible at the stage that hepatic cirrhosis develops. Initially the liver may be enlarged, full of fat, crossed with collagen fibers (fibrosis), and have nodules of regenerating hepatocytes ballooning between the fibers. As liver function is lost, the liver becomes shrunken (Laennec's cirrhosis). During the development of cirrhosis, many of the normal metabolic functions of the liver are lost, including biosynthetic and detoxification pathways. Synthesis of blood proteins, including blood coagulation factors and serum albumin, is decreased. The capacity to incorporate amino groups into urea is decreased, resulting in the accumulation of toxic levels of ammonia in the blood. Conjugation and excretion of the yellow pigment bilirubin (a product of heme degradation) is diminished, and bilirubin accumulates in the blood. It is deposited in many tissues, including the skin and sclerae of the eyes, causing the patient to become visibly yellow. Such a patient is said to be jaundiced.

Because of the possibility of mild alcoholic hepatitis and perhaps chronic alcohol-induced cirrhosis, the physician ordered liver function studies on **Jean Ann Tonich**. The tests indicated an alanine aminotransferase (ALT) level of 46 units/L (reference range = 5–30) and an aspartate aminotransferase (AST) level of 98 units/L (reference range = 10–30). The concentration of these enzymes is high in hepatocytes. When hepatocellular membranes are damaged in any way, these enzymes are released into the blood. Jean Ann Tonich's serum alkaline phosphatase level was 151 units/L (reference range = 56–155 for an adult female). The serum total bilirubin level was 2.4 mg/dL (reference range = 0.2–1.0). These tests show impaired capacity for normal liver function. Her blood hemoglobin and hematocrit levels were slightly below the normal range, consistent with a toxic effect of ethanol on red blood cell production by bone marrow. Serum folate, vitamin B12 and iron levels were also slightly suppressed. Folate is dependent on the liver for its activation and recovery from the enterohepatic circulation. Vitamin B12 and iron are dependent on the liver for synthesis of their blood carrier proteins. Thus, **Jean Ann Tonich** shows many of the consequences of hepatic damage.

In liver fibrosis, disruption of the normal liver architecture, including sinusoids, impairs blood from the portal vein. Increased portal vein pressure (portal hypertension) causes capillaries to anastomose (to meet and unite or run into each other) and form thin-walled dilated esophageal venous conduits known as esophageal varices. When these burst, there is hemorrhaging into the gastrointestinal tract. The bleeding can be very profuse because of the high venous pressure within these varices in addition to the adverse effect of impaired hepatic function on the production of blood clotting proteins.

CLINICAL COMMENTS

Ivan Applebod. When ethanol consumption is low (less than 15% of the calories in the diet), it is efficiently used to produce ATP, thereby contributing to **Ivan Applebod's** weight gain. However, in individuals with chronic consumption of large amounts of ethanol, the caloric content of ethanol is not converted to ATP as effectively. Some of the factors that may contribute to this decreased efficiency include mitochondrial damage (inhibition of oxidative phosphorylation and uncoupling) resulting in the loss of calories as heat, increased recycling of metabolites such as ketone bodies, and inhibition of the normal pathways of fatty acid and glucose oxidation. In addition, heavier drinkers metabolize an increased amount of alcohol through MEOS, which generates less ATP.

Al Martini. **Al Martini** was suffering from acute effects of high ethanol ingestion in the absence of food intake. Both heavy ethanol consumption and low caloric intake increase adipose tissue lipolysis and elevate blood fatty acids. As a consequence of his elevated hepatic NADH/NAD^+ ratio, acetyl CoA produced from fatty acid oxidation was diverted from the TCA cycle into the pathway of ketone body synthesis. Because his skeletal muscles were using acetate as a fuel, ketone body utilization was diminished, resulting in ketoacidosis. **Al Martini's** moderately low blood glucose level also suggests that his high hepatic NADH level prevented pyruvate and glycerol from entering the gluconeogenic pathway. Pyruvate is diverted to lactate, which may have contributed to his metabolic acidosis and anion gap.

Rehydration with intravenous fluids containing glucose and potassium was initiated. His initial potassium was low, possibly secondary to vomiting. An orthopedic surgeon was consulted regarding the compound fracture of his right forearm.

Jean Ann Tonich. Jean Ann Tonich's signs and symptoms, as well as her laboratory profile, were consistent with the presence of mild reversible alcohol-induced hepatocellular inflammation (alcohol-induced hepatitis) superimposed on a degree of irreversible scarring of liver tissues known as chronic alcoholic (Laennec's) cirrhosis of the liver. The chronic inflammatory process associated with long-term ethanol abuse in patients such as **Jean Ann Tonich** is accompanied by increases in the levels of serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST). Her elevated bilirubin and alkaline phosphatase were consistent with hepatic damage. Her values for ALT and

AST were significantly below those seen in acute viral hepatitis. In addition, the ratio of the absolute values for serum ALT and AST often differ in the two diseases, tending to be greater than 1 in acute viral hepatitis and less than 1 in chronic alcohol-induced cirrhosis. The reason for the difference in ratio of enzyme activities released is not understood, but a lower level of ALT in the serum may be attributable to an alcohol-induced deficiency of pyridoxal phosphate. In addition, serologic tests for viral hepatitis were nonreactive. Her serum folate, vitamin B12, and iron levels were also slightly suppressed, indicating impaired nutritional status.

Jean Ann Tonich was strongly cautioned to abstain from alcohol immediately and to improve her nutritional status. In addition, Jean Ann was referred to the hospital drug and alcohol rehabilitation unit for appropriate psychological therapy and supportive social counseling. The physician also arranged for a follow-up office visit in 2 weeks.

Although the full spectrum of alcohol-induced liver disease may be present in a well-nourished individual, the presence of nutritional deficiencies enhances the progression of the disease. Ethanol creates nutritional deficiencies in a number of different ways. The ingestion of ethanol reduces the gastrointestinal absorption of foods containing essential nutrients, including vitamins, essential fatty acids, and essential amino acids. For example, ethanol interferes with absorption of folate, thiamine, and other nutrients. Secondary malabsorption can occur through gastrointestinal complications, pancreatic insufficiency, and impaired hepatic metabolism or impaired hepatic storage of nutrients, such as vitamin A. Changes in the level of transport proteins produced by the liver also strongly affect nutrient status.

BIOCHEMICAL COMMENTS

Fibrosis in Chronic Alcohol-Induced Liver Disease Fibrosis is the excessive accumulation of connective tissue in parenchymal organs. In the liver, it is a frequent event following a repeated or chronic insult of sufficient intensity (such as chronic ethanol intoxication or infection by a hepatitis virus) to trigger a "wound healing–like" reaction. Regardless of the insult, the events are similar: an overproduction of extracellular matrix components occurs, with the tendency to progress into sclerosis, accompanied by a degenerative alteration in the composition of matrix components. (Table 25.2) Some individuals (fewer than 20% of those who chronically consume alcohol) go on to develop cirrhosis.

The development of hepatic fibrosis after ethanol consumption is related to stimulation of the mitogenic development of stellate (Ito) cells into myofibroblasts, and stimulation of the production of collagen type I and fibronectin by these cells. The stellate cells are perisinusoidal cells lodged in the space of Disse that produce extracellular matrix protein. Normally the space of Disse contains basement membrane–-like collagen (collagen type IV) and laminin. As the stellate cells are activated, they change from a resting cell filled with lipids and vitamin A to one that proliferates, loses its vitamin A content, and secretes large quantities of extracellular matrix components.

One of the initial events in the activation and proliferation of stellate cells is the activation of Kupffer cells, which are macrophages resident in the liver sinusoids

Table 25.2. Hepatic Injury

Stage of Injury	Main Features
Fibrosis: Increase of connective tissue	
	Accumulation of both fibrillar and basement membrane–like collagens
	Increase of laminen and fibronectin
	Thickening of connective tissue septae
	Capillarization of the sinusoids
Sclerosis: Aging of fibrotic tissue	
	Decrease of hyaluronic acid and heparan sulfate proteoglycans
	Increase of chondroitin sulfate proteoglycans
	Progressive fragmentation and disappearance of elastic fibers
	Distortion of sinusoidal architecture and parenchymal damage
Cirrhosis: End-stage process of liver fibrotic degeneration	
	Whole liver heavily distorted by thick bands of collagen surrounding nodules of hepatocytes with regenerative foci

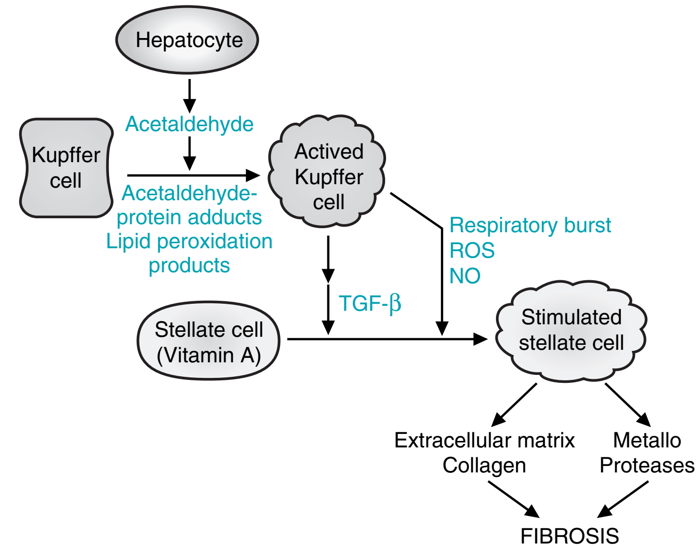

Fig. 25.8. Proposed model for the development of hepatic fibrosis involving hepatocytes, Kupffer cells, and stellate (Ito) cells. ROS, reactive oxygen species; NO, nitric oxide: TGFβ1, transforming growth factor β1.

(Fig.25.8). The Kupffer cells are probably activated by a product of the damaged hepatocytes, such as necrotic debris, iron, ROS, acetaldehyde, or aldehyde products of lipid peroxidation. Kupffer cells also may produce acetaldehyde from ethanol internally through their own MEOS pathway.

Activated Kupffer cells produce a number of products that contribute to activation of stellate cells. They generate additional ROS through NADPH oxidase during the oxidative burst and NOS through inducible NO synthase (see Chapter 24). In addition, they secrete an impressive array of growth factors, such as cytokines, chemokines, prostaglandins, and other reactive molecules. The cytokine transforming growth factor β1 (TGFβ1), produced by both Kupffer cells and sinusoidal endothelial cells, is a major player in the activation of stellate cells. Once activated, the stellate cells produce collagen and proteases, leading to an enhanced fibrotic network within the liver.

Cytokines are proteins produced by inflammatory cells that serve as communicators with other cells. Chemokines are even smaller proteins produced by inflammatory cells that promote migration of other inflammatory cells (e.g., from the blood into the site of injury).

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REVIEW QUESTIONS—CHAPTER 25

1. The fate of acetate, the product of ethanol metabolism, is which of the following?

(A) It is taken up by other tissues and activated to acetyl CoA.
(B) It is toxic to the tissues of the body and can lead to hepatic necrosis.
(C) It is excreted in bile.
(D) It enters the TCA cycle directly to be oxidized.
(E) It is converted into NADH by alcohol dehydrogenase.

2. Which of the following would be expected to occur after acute alcohol ingestion?

   (A) The activation of fatty acid oxidation
   (B) Lactic acidosis
   (C) The inhibition of ketogenesis
   (D) An increase in the NAD^+/NADH ratio
   (E) An increase in gluconeogenesis

3. A chronic alcoholic is in treatment for alcohol abuse. The drug disulfiram is prescribed for the patient. This drug deters the consumption of alcohol by which of the following mechanisms?

   (A) Inhibiting the absorption of ethanol so that an individual cannot become intoxicated, regardless of how much he drinks
   (B) Inhibiting the conversion of ethanol to acetaldehyde, which would cause the excretion of unmetabolized ethanol
   (C) Blocking the conversion of acetaldehyde to acetate, which causes the accumulation of acetaldehyde
   (D) Activating the excessive metabolism of ethanol to acetate, which causes inebriation with consumption of a small amount of alcohol
   (E) Preventing the excretion of acetate, which causes nausea and vomiting

4. Induction of CYP2E1 would result in which of the following?

   (A) A decreased clearance of ethanol from the blood
   (B) A decrease in the rate of acetaldehyde production
   (C) A low possibility of the generation of free radicals
   (D) Protection from hepatic damage
   (E) An increase of one's alcohol tolerance level

5. Which one of the following consequences of chronic alcohol consumption is irreversible?

   (A) Inhibition of fatty acid oxidation
   (B) Activation of triacylglycerol synthesis
   (C) Ketoacidosis
   (D) Lactic acidosis
   (E) Liver cirrhosis

Carbohydrate Metabolism

Glucose is central to all of metabolism. It is the universal fuel for human cells and the source of carbon for the synthesis of most other compounds. Every human cell type uses glucose to obtain energy. The release of insulin and glucagon by the pancreas aids in the body's use and storage of glucose. Other dietary sugars (mainly fructose and galactose) are converted to glucose or to intermediates of glucose metabolism.

Glucose is the precursor for the synthesis of an array of other sugars required for the production of specialized compounds, such as lactose, cell surface antigens, nucleotides, or glycosaminoglycans. Glucose is also the fundamental precursor of noncarbohydrate compounds; it can be converted to lipids (including fatty acids, cholesterol, and steroid hormones), amino acids, and nucleic acids. Only those compounds that are synthesized from vitamins, essential amino acids, and essential fatty acids cannot be synthesized from glucose in humans.

More than 40% of the calories in the typical diet in the United States are obtained from starch, sucrose, and lactose. These dietary carbohydrates are converted to glucose, galactose, and fructose in the digestive tract (Fig. 1). Monosaccharides are absorbed from the intestine, enter the blood, and travel to the tissues where they are metabolized.

After glucose is transported into cells, it is phosphorylated by a hexokinase to form glucose 6-phosphate. Glucose 6-phosphate can then enter a number of metabolic pathways. The three that are common to all cell types are glycolysis, the pentose phosphate pathway, and glycogen synthesis (Fig. 2). In tissues, fructose and galactose are converted to intermediates of glucose metabolism. Thus, the fate of these sugars parallels that of glucose (Fig. 3).

The major fate of glucose 6-phosphate is oxidation via the pathway of glycolysis (see Chapter 22), which provides a source of ATP for all cell types. Cells that lack mitochondria cannot oxidize other fuels. They produce ATP from anaerobic glycolysis (the conversion of glucose to lactic acid). Cells that contain mitochondria

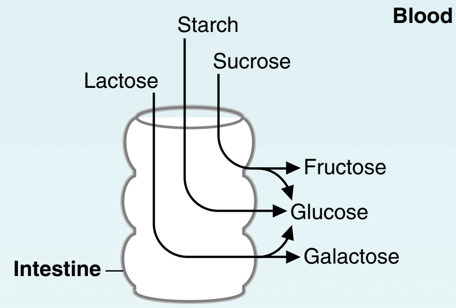


Fig 1. Overview of carbohydrate digestion. The major carbohydrates of the diet (starch, lactose, and sucrose) are digested to produce monosaccharides (glucose, fructose, and galactose), which enter the blood.

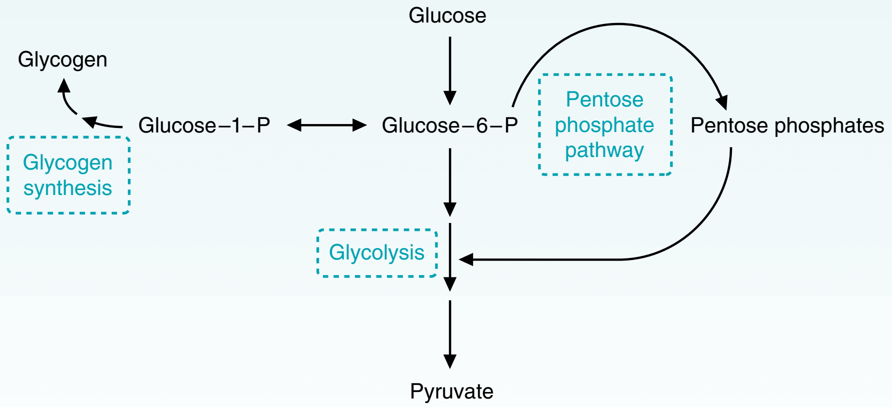


Fig 2. Major pathways of glucose metabolism.

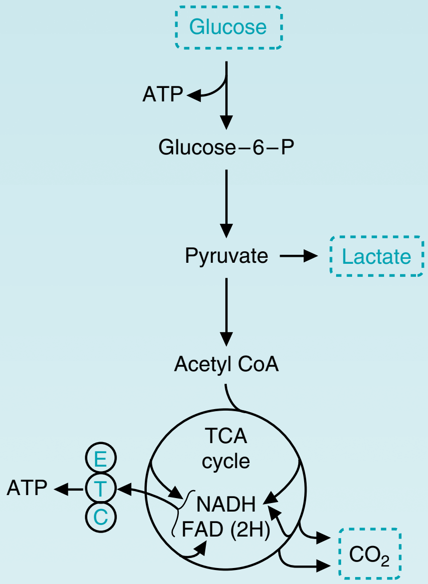


Fig 4. Conversion of glucose to lactate or to CO_2. ETC = electron transport chain.

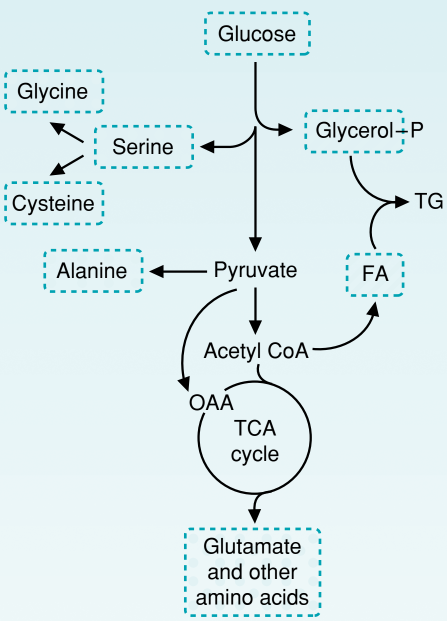


Fig 5. Conversion of glucose to amino acids and to the glycerol and fatty acid (FA) moieties of triacylglycerols (TG). OAA = oxaloacetate.

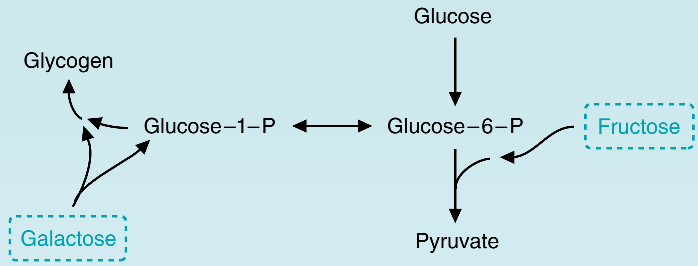


Fig 3. Overview of fructose and galactose metabolism. Fructose and galactose are converted to intermediates of glucose metabolism.

oxidize glucose to CO_2 and H_2O via glycolysis and the TCA cycle (Fig. 4). Some tissues, such as the brain, depend on the oxidation of glucose to CO_2 and H_2O for energy because they have a limited capacity to use other fuels.

Glucose produces the intermediates of glycolysis and the TCA cycle that are used for the synthesis of amino acids and both the glycerol and fatty acid moieties of triacylglycerols (Fig. 5).

Another important fate of glucose 6-phosphate is oxidation via the pentose phosphate pathway, which generates NADPH. The reducing equivalents of NADPH are used for biosynthetic reactions and for the prevention of oxidative damage to cells (see Chapter 24). In this pathway, glucose is oxidatively decarboxylated to 5-carbon sugars (pentoses), which may reenter the glycolytic pathway. They also may be used for nucleotide synthesis (Fig. 6). There are also non-oxidative reactions, which can convert six- and five-carbon sugars.

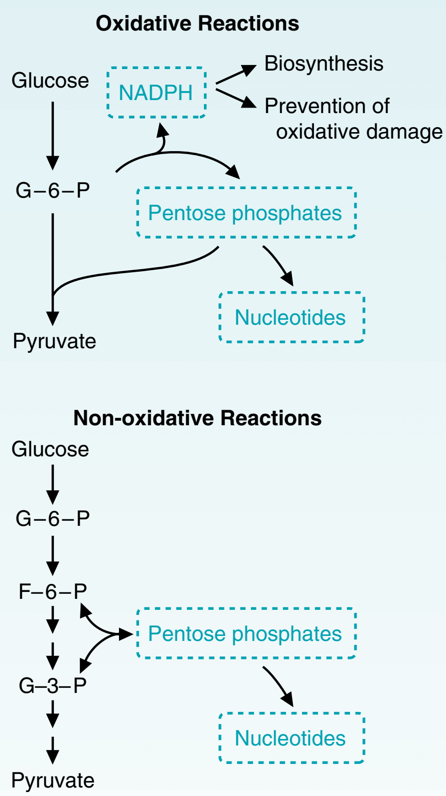


Fig 6. Overview of the pentose phosphate pathway. The oxidative reactions generate both NADPH and pentose phosphates. The non-oxidative reactions only generate pentose phosphates.

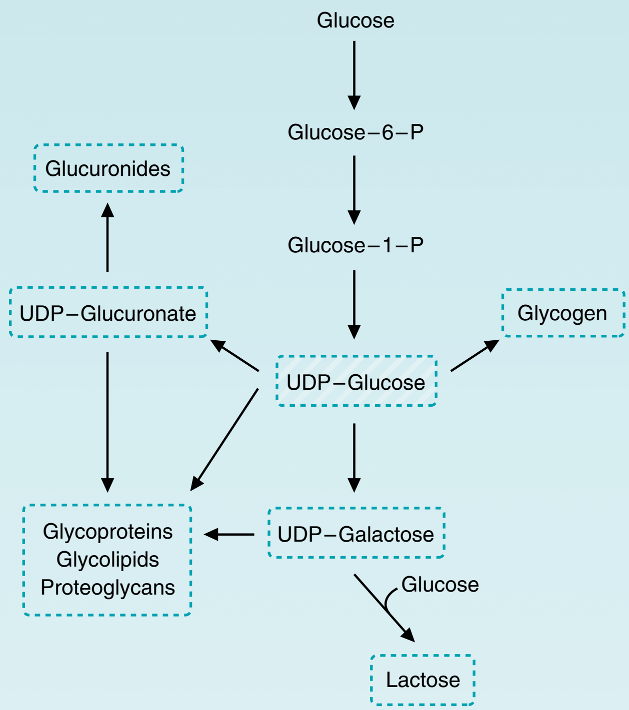


Fig 7. Products derived from UDP-glucose.

Glucose 6-phosphate is also converted to UDP-glucose, which has many functions in the cell (Fig. 7). The major fate of UDP-glucose is the synthesis of glycogen, the storage polymer of glucose. Although most cells have glycogen to provide emergency supplies of glucose, the largest stores are in muscle and liver. Muscle glycogen is used to generate ATP during muscle contraction. Liver glycogen is used to maintain blood glucose during fasting and during exercise or periods of enhanced need. UDP-Glucose is also used for the formation of other sugars, and galactose and glucose are interconverted while attached to UDP. UDP-Galactose is used for lactose synthesis in the mammary gland. In the liver, UDP-glucose is oxidized to UDP-glucuronate, which is used to convert bilirubin and other toxic compounds to glucuronides for excretion (see Fig. 7).

Nucleotide sugars are also used for the synthesis of proteoglycans, glycoproteins, and glycolipids (see Fig. 7). Proteoglycans are major carbohydrate components of the extracellular matrix, cartilage, and extracellular fluids (such as the synovial fluid of joints), and they are discussed in more detail in Chapter 49. Most extracellular proteins are glycoproteins, i.e., they contain covalently attached carbohydrates. For both cell membrane glycoproteins and glycolipids, the carbohydrate portion extends into the extracellular space.

All cells are continuously supplied with glucose under normal circumstances; the body maintains a relatively narrow range of glucose concentration in the blood (approximately 80-100 mg/dL) in spite of the changes in dietary supply and tissue demand as we sleep and exercise. This process is called glucose homeostasis. Low blood glucose levels (hypoglycemia) are prevented by a release of glucose from the large glycogen stores in the liver (glycogenolysis); by synthesis of glucose from lactate, glycerol, and amino acids in liver (gluconeogenesis) (Fig. 8); and to a limited extent by a release of fatty acids from adipose tissue stores (lipolysis) to provide an alternate fuel when glucose is in short supply. High blood glucose levels (hyperglycemia) are prevented both by the conversion of glucose to glycogen and by its conversion to triacylglycerols in liver and adipose tissue. Thus, the pathways for glucose utilization as a fuel cannot be considered as totally separate from pathways involving amino acid and fatty acid metabolism (Fig. 9).

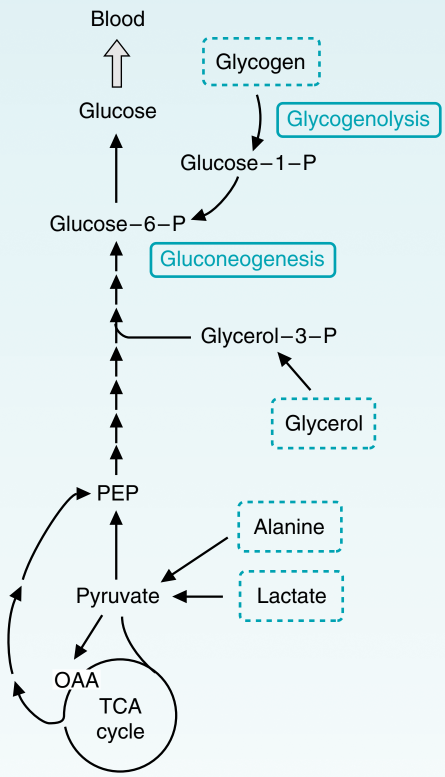


Fig 8. Production of blood glucose from glycogen (by glycogenolysis) and from alanine, lactate, and glycerol (by gluconeogenesis). PEP = phosphoenolpyruvate; OAA = oxaloacetate.

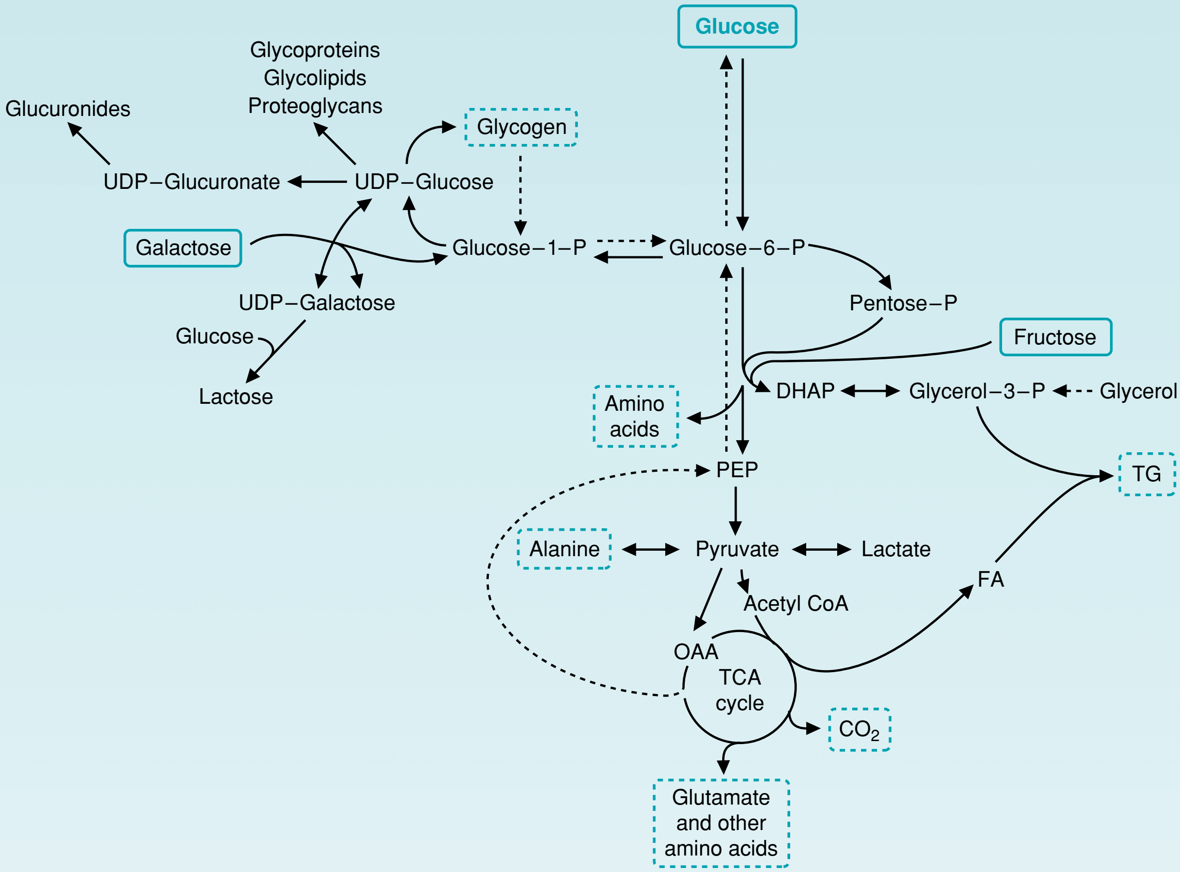


Fig 9. Overview of the major pathways of glucose metabolism. Pathways for production of blood glucose are shown by dashed lines. FA = fatty acids; TG = triacylglycerols; OAA = oxaloacetate; PEP = phosphoenolpyruvate; UDP-G = UDP-glucose; DHAP = dihydroxyacetone phosphate.

Intertissue balance in the utilization and storage of glucose during fasting and feeding is accomplished principally by the actions of the hormones of metabolic homeostasis—insulin and glucagon (Fig. 10). However, cortisol, epinephrine, norepinephrine, and other hormones are also involved in intertissue adjustments of supply and demand in response to changes of physiologic state.

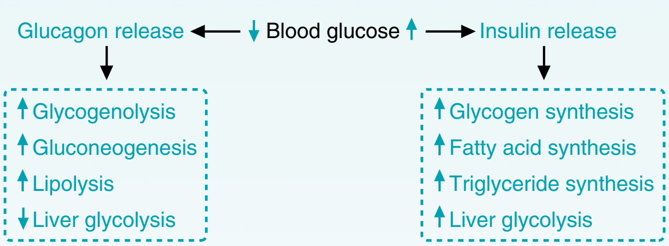


Fig 10. Pathways regulated by the release of glucagon (in response to a lowering of blood glucose levels) and insulin (released in response to an elevation of blood glucose levels). Tissue-specific differences occur in the response to these hormones, as detailed in the subsequent chapters of this section.

26 Basic Concepts in the Regulation of Fuel Metabolism by Insulin, Glucagon, and Other Hormones

All cells continuously use adenosine triphosphate (ATP) and require a constant supply of fuels to provide energy for ATP generation. ***Insulin*** *and* ***glucagon*** *are the two major hormones that regulate fuel mobilization and storage. Their function is to ensure that cells have a constant source of glucose, fatty acids, and amino acids for ATP generation and for cellular maintenance (Fig. 26.1).*

Because most tissues are partially or totally dependent on glucose for ATP generation and for production of precursors of other pathways, insulin and glucagon ***maintain blood glucose*** *levels near 80 to 100 mg/dL (90 mg/dL is the same as 5 mM), despite the fact that carbohydrate intake varies considerably over the course of a day. The maintenance of constant blood glucose levels (****glucose homeostasis****) requires these two hormones to regulate* ***carbohydrate***, ***lipid***, *and* ***amino acid metabolism*** *in accordance with the needs and capacities of individual tissues. Basically, the dietary intake of all fuels in excess of immediate need is stored, and the appropriate fuel is mobilized when a demand occurs. For example, when dietary glucose is not available in sufficient quantities that all tissues can use it, fatty acids are mobilized and made available to skeletal muscle for use as a fuel (see Chapters 2 and 23), and the liver can convert fatty acids to ketone bodies for use by the brain. Fatty acids spare glucose for use by the brain and other glucose-dependent tissues (such as the red blood cell).*

The concentrations of insulin and glucagon in the blood regulate fuel storage and mobilization (Fig. 26.2). Insulin, released in response to carbohydrate ingestion, promotes glucose utilization as a fuel and glucose storage as fat and glycogen. ***Insulin is also the major anabolic hormone of the body****. It increases protein synthesis and cell growth in addition to fuel storage. Blood insulin levels decrease as glucose is taken up by tissues and used. Glucagon, the major* ***counterregulatory hormone*** *of insulin, is decreased in response to a carbohydrate meal and elevated during fasting. Its concentration in the blood signals the absence of dietary glucose, and it promotes glucose production via* ***glycogenolysis*** *(glycogen degradation) and* ***gluconeogenesis*** *(glucose synthesis from amino acids and other noncarbohydrate precursors). Increased levels of glucagon relative to insulin also stimulate the* ***mobilization of fatty acids*** *from adipose tissue.* ***Epinephrine*** *(the fight or flight hormone) and cortisol (a glucocorticoid released from the adrenal cortex in response to fasting and chronic stress) have effects on fuel metabolism that oppose those of insulin. These two hormones are therefore also considered insulin counterregulatory hormones.*

Insulin and glucagon are polypeptide hormones synthesized as ***prohormones*** *in the β and α cells, respectively, in the islets of Langerhans in the pancreas.* ***Proinsulin*** *is cleaved into mature insulin and* ***C-peptide*** *in vesicles and precipitated*

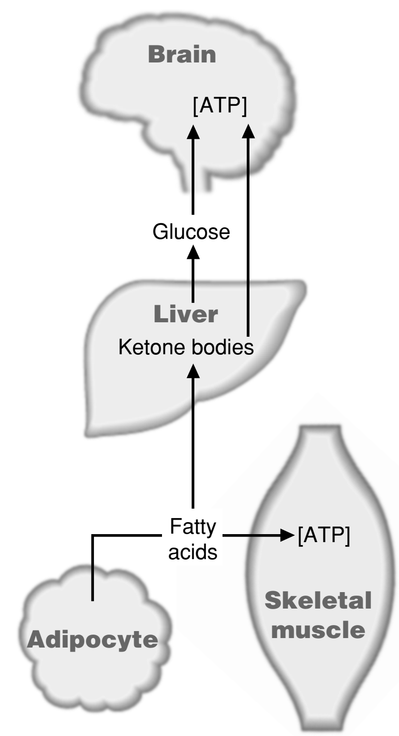


Fig 26.1. Maintenance of fuel supplies to tissues. Glucagon release activates the pathways shown.

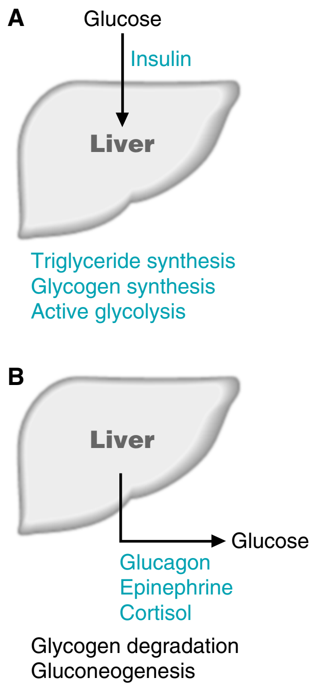


Fig 26.2. Insulin and the insulin counterregulatory hormones. (A) Insulin promotes glucose storage, as triglyceride (TG) or glycogen. (B) Glucagon, epinephrine, and cortisol promote glucose release from the liver, activating glycogenolysis and gluconeogenesis.

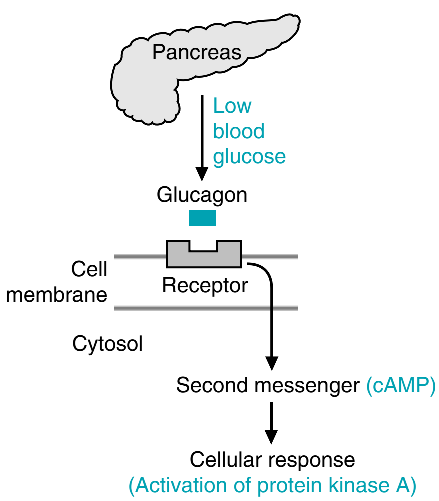


Fig 26.3. Cellular response to glucagon, which is released from the pancreas in response to a decrease in blood glucose levels.

with Zn^{2+}. Insulin secretion is regulated principally by blood glucose levels. Glucagon is also synthesized as a prohormone and cleaved into mature glucagon within storage vesicles. Its release is regulated principally through suppression by glucose and by insulin.

*Glucagon exerts its effects on cells by binding to a receptor on the cell surface, which stimulates the synthesis of the intracellular second messenger, cyclic adenosine monophosphate (**cAMP**) (Fig. 26.3). cAMP activates **protein kinase A**, which phosphorylates key regulatory enzymes, activating some and inhibiting others. Changes of cAMP levels also induce or repress the synthesis of a number of enzymes. Insulin promotes the dephosphorylation of these key enzymes.*

*Insulin binds to a receptor on the cell surface, but the postreceptor events that follow differ from those stimulated by glucagon. Insulin binding activates both autophosphorylation of the receptor and the phosphorylation of other enzymes by the receptor's **tyrosine kinase** domain (see Chapter 11, section III.B.3). The complete routes for **signal transduction** between this point and the final effects of insulin on the regulatory enzymes of fuel metabolism have not yet been fully established.*

THE WAITING ROOM

Ann Sulin returned to her physician for her monthly office visit. She has been seeing her physician for over a year because of obesity and elevated blood glucose levels. She still weighed 198 lb, despite her insistence that she had adhered strictly to her diet. Her blood glucose level at the time of the visit, 2 hours after lunch, was 180 mg/dL (reference range = 80–140).

Bea Selmass is a 46-year-old woman who 6 months earlier began noting episodes of fatigue and confusion as she finished her daily pre-breakfast jog. These episodes were occasionally accompanied by blurred vision and an unusually urgent hunger. The ingestion of food relieved all of her symptoms within 25 to 30 minutes. In the last month, these attacks have occurred more frequently throughout the day, and she has learned to diminish their occurrence by eating between meals. As a result, she has recently gained 8 lb.

A random serum glucose level done at 4:30 PM during her first office visit was subnormal at 46 mg/dL. Her physician, suspecting she had a form of fasting hypoglycemia, ordered a series of fasting serum glucose levels. In addition, he asked Bea to keep a careful daily diary of all of the symptoms that she experienced when her attacks were most severe.

I. METABOLIC HOMEOSTASIS

Living cells require a constant source of fuels from which to derive ATP for the maintenance of normal cell function and growth. Therefore, a balance must be achieved between carbohydrate, fat, and protein intake, their storage when present in excess of immediate need, and their mobilization and synthesis when in demand. The balance between need and availability is referred to as metabolic homeostasis (Fig. 26.4). The intertissue integration required for metabolic homeostasis is achieved in three principal ways:

- The concentration of nutrients or metabolites in the blood affects the rate at which they are used and stored in different tissues;

- Hormones carry messages to individual tissues about the physiologic state of the body and nutrient supply or demand;
- The central nervous system uses neural signals to control tissue metabolism, directly or through the release of hormones.

Insulin and glucagon are the two major hormones that regulate fuel storage and mobilization (see Fig. 26.2). Insulin is the major anabolic hormone of the body. It promotes the storage of fuels and the utilization of fuels for growth. Glucagon is the major hormone of fuel mobilization. Other hormones, such as epinephrine, are released as a response of the central nervous system to hypoglycemia, exercise, or other types of physiologic stress. Epinephrine and other stress hormones also increase the availability of fuels (Fig. 26.5).

The special role of glucose in metabolic homeostasis is dictated by the fact that many tissues (e.g., the brain, red blood cells, the lens of the eye, the kidney medulla, exercising skeletal muscle) are dependent on glycolysis for all or a portion of their energy needs and require uninterrupted access to glucose on a second-to-second basis to meet their rapid rate of ATP utilization. In the adult, a minimum of 190 g glucose is required per day; approximately 150 g for the brain and 40 g for other tissues. Significant decreases of blood glucose below 60 mg/dL limit glucose metabolism in the brain and elicit hypoglycemic symptoms (as experienced by **Bea Selmass**), presumably because the overall process of glucose flux through the blood-brain barrier, into the interstitial fluid, and subsequently into the neuronal cells, is slow at low blood glucose levels because of the K_m values of the glucose transporters required for this to occur (see Chapter 27)

The continuous movement of fuels into and out of storage depots is necessitated by the high amounts of fuel required each day to meet the need for ATP. Disastrous results would occur if even a day's supply of glucose, amino acids, and fatty acids were left circulating in the blood. Glucose and amino acids would be at such high concentrations that the hyperosmolar effect would cause progressively severe neurologic deficits and even coma. The concentration of glucose and amino acids would be above the renal tubular threshold for these substances (the maximal concentration in the blood at which the kidney can completely resorb metabolites), and some of these compounds would be wasted as they spilled over into the urine. Nonenzymatic glycosylation of proteins would increase at higher blood glucose

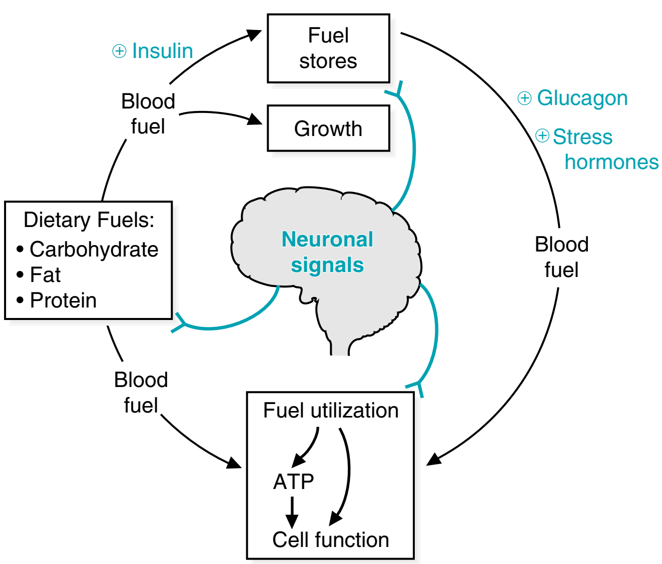


Fig 26.5. Signals that regulate metabolic homeostasis. The major stress hormones are epinephrine and cortisol.

Fatty acids provide an example of the influence that the level of a compound in the blood has on its own rate of metabolism. The concentration of fatty acids in the blood is the major factor determining whether skeletal muscles will use fatty acids or glucose as a fuel (see Chapter 23). In contrast, hormones are (by definition) carriers of messages between tissues. Insulin and glucagon, for example, are two hormonal messengers that participate in the regulation of fuel metabolism by carrying messages that reflect the timing and composition of our dietary intake of fuels. Epinephrine, however, is a flight-or-flight hormone that signals an immediate need for increased fuel availability. Its level is regulated principally through the activation of the sympathetic nervous system.

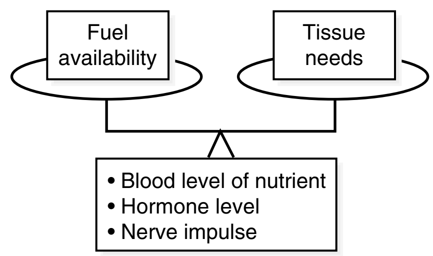


Fig 26.4. Metabolic homeostasis. The balance between fuel availability and the needs of tissues for different fuels is achieved by three types of messages: the level of the fuel or nutrients in the blood, the level of one of the hormones of metabolic homeostasis, or nerve impulses that affect tissue metabolism or the release of hormones.

Hyperglycemia may cause a constellation of symptoms such as polyuria and subsequent polydipsia (increased thirst). The inability to move glucose into cells necessitates the oxidation of lipids as an alternative fuel. As a result adipose stores are used, and the patient with poorly controlled diabetes mellitus loses weight in spite of a good appetite. Extremely high levels of serum glucose can cause nonketotic hyperosmolar coma in patients with type 2 diabetes mellitus. Such patients usually have sufficient insulin responsiveness to block fatty acid release and ketone body formation, but they are unable to significantly stimulate glucose entry into peripheral tissues. The severely elevated levels of glucose in the blood compared with inside the cell leads to an osmotic effect that causes water to leave the cells and enter the blood. Because of the osmotic diuretic effect of hyperglycemia, the kidney produces more urine, leading to dehydration, which in turn may lead to even higher levels of blood glucose. If dehydration becomes severe, further cerebral dysfunction occurs and the patient may become comatose. Chronic hyperglycemia also produces pathologic effects through the nonenzymatic glycosylation of a variety of proteins. Hemoglobin A (HbA), one of the proteins that becomes glycosylated, forms HbA_{1c} (see Chapter 7). **Ann Sulin's** high levels of HbA_{1c} (14% of the total HbA, compared with the reference range of 4.7–6.4%) indicate that her blood glucose has been significantly elevated over the last 12 to 14 weeks, the half-life of hemoglobin in the bloodstream.

All membrane and serum proteins exposed to high levels of glucose in the blood or interstitial fluid are candidates for nonenzymatic glycosylation. This process distorts protein structure and slows protein degradation, which leads to an accumulation of these products in various organs, thereby adversely affecting organ function. These events contribute to the long-term microvascular and macrovascular complications of diabetes mellitus, which include diabetic retinopathy, nephropathy, and neuropathy (microvascular), in addition to coronary artery, cerebral artery, and peripheral artery insufficiency (macrovascular).

levels. Triacylglycerols circulate in cholesterol-containing lipoproteins, and the levels of these lipoproteins would be chronically elevated, increasing the likelihood of atherosclerotic vascular disease. Consequently, glucose and other fuels are continuously moved in and out of storage depots as needed.

II. MAJOR HORMONES OF METABOLIC HOMEOSTASIS

The hormones that contribute to metabolic homeostasis respond to changes in the circulating levels of fuels that, in part, are determined by the timing and composition of our diet. Insulin and glucagon are considered the major hormones of metabolic homeostasis because they continuously fluctuate in response to our daily eating pattern. They provide good examples of the basic concepts of hormonal regulation. Certain features of the release and action of other insulin counterregulatory hormones, such as epinephrine, norepinephrine, and cortisol, will be described and compared with insulin and glucagon.

Insulin is the major anabolic hormone that promotes the storage of nutrients: glucose storage as glycogen in liver and muscle, conversion of glucose to triacylglycerols in liver and their storage in adipose tissue, and amino acid uptake and protein synthesis in skeletal muscle (Fig. 26.6). It also increases the synthesis of albumin and other blood proteins by the liver. Insulin promotes the utilization of glucose as a fuel by stimulating its transport into muscle and adipose tissue. At the same time, insulin acts to inhibit fuel mobilization.

Glucagon acts to maintain fuel availability in the absence of dietary glucose by stimulating the release of glucose from liver glycogen (see Chapter 28), by stimulating gluconeogenesis from lactate, glycerol, and amino acids (see Chapter 31), and, in conjunction with decreased insulin, by mobilizing fatty acids from adipose triacylglycerols to provide an alternate source of fuel (see Chapter 23 and Fig. 26.7). Its sites of action are principally the liver and adipose tissue; it has no influence on skeletal muscle metabolism because muscle cells lack glucagon receptors.

The release of insulin from the beta cells of the pancreas is dictated primarily by the blood glucose level. The highest levels of insulin occur approximately 30 to 45 minutes after a high-carbohydrate meal (Fig. 26.8). They return to basal levels as the blood glucose concentration falls, approximately 120 minutes after the meal. The release of glucagon from the alpha cells of the pancreas, conversely, is controlled principally through suppression by glucose and insulin. Therefore, the lowest levels of glucagon occur after a high-carbohydrate meal. Because all of the effects of glucagon are opposed by insulin, the simultaneous stimulation of insulin release and suppression of glucagon secretion by a high-carbohydrate meal provides an integrated control of carbohydrate, fat, and protein metabolism.

Bea Selmass's studies confirmed that her fasting serum glucose levels were below normal. She continued to experience the fatigue, confusion, and blurred vision she had described on her first office visit. These symptoms are called neuroglycopenic (neurologic symptoms resulting from an inadequate supply of glucose to the brain for the generation of ATP).

Bea also noted the symptoms that are part of the adrenergic response to hypoglycemic stress. Stimulation of the sympathetic nervous system (because of the low levels of glucose reaching the brain) results in the release of epinephrine, a stress hormone, from the adrenal medulla. Elevated epinephrine levels cause tachycardia, palpitations, anxiety, tremulousness, pallor, and sweating.

In addition to the symptoms described by **Bea Selmass**, individuals may experience confusion, lightheadedness, headache, aberrant behavior, blurred vision, loss of consciousness, or seizures.

Ms. Selmass's doctor explained that the general diagnosis of "fasting" hypoglycemia was now established and that a specific cause for this disorder must be found.

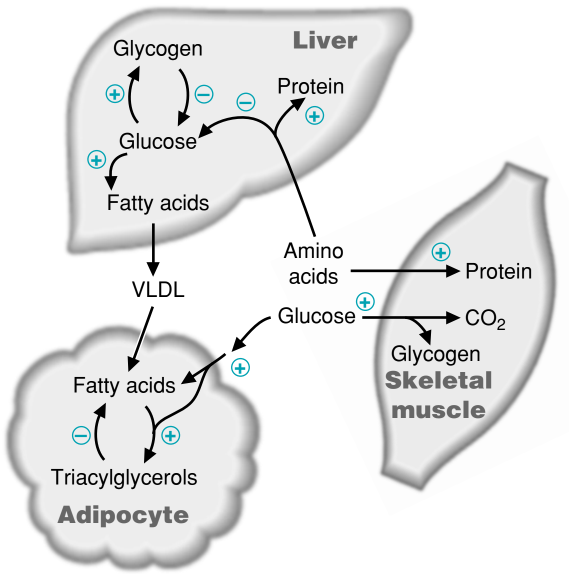


Fig 26.6. Major sites of insulin action on fuel metabolism. + = stimulated by insulin; − = inhibited by insulin.

Insulin and glucagon are not the only regulators of fuel metabolism. The intertissue balance between the utilization and storage of glucose, fat, and protein is also accomplished by the circulating levels of metabolites in the blood, by neuronal signals, and by the other hormones of metabolic homeostasis (epinephrine, norepinephrine, cortisol, and others) (Table 26.1). These hormones oppose the actions of insulin by mobilizing fuels. Like glucagon, they are called insulin counterregulatory hormones (Fig. 26.9). Of all these hormones, only insulin and glucagon are synthe-

The message carried by glucagon is that "glucose is gone"; i.e., the current supply of glucose is inadequate to meet the immediate fuel requirements of the body.

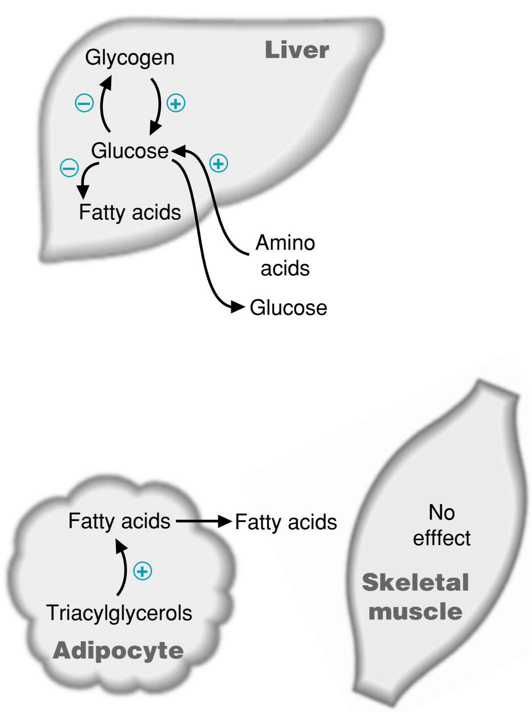


Fig 26.7. Major sites of glucagon action in fuel metabolism. + = pathways stimulated by glucagon; − = pathways inhibited by glucagon.

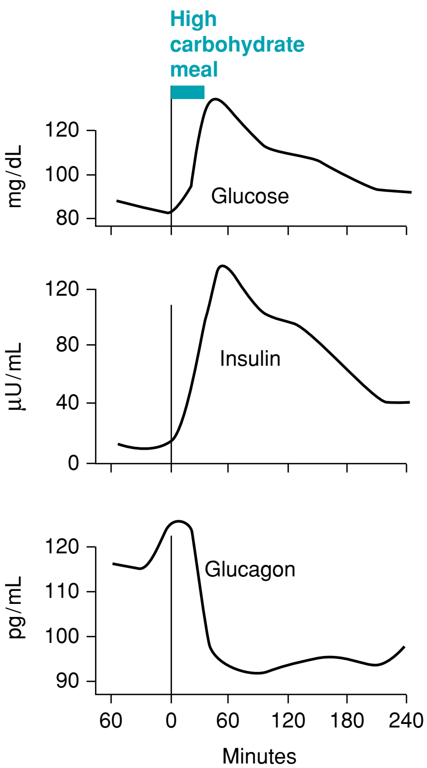


Fig 26.8. Blood glucose, insulin, and glucagon levels after a high-carbohydrate meal.

Table 26.1. Physiologic Actions of Insulin and Insulin Counterregulatory Hormones

Hormone	Function	Major Metabolic Pathways Affected
Insulin	• Promotes fuel storage after a meal • Promotes growth	• Stimulates glucose storage as glycogen (muscle and liver) • Stimulates fatty acid synthesis and storage after a high-carbohydrate meal • Stimulates amino acid uptake and protein synthesis
Glucagon	• Mobilizes fuels • Maintains blood glucose levels during fasting	• Activates gluconeogenesis and glycogenolysis (liver) during fasting • Activates fatty acid release from adipose tissue
Epinephrine	• Mobilizes fuels during acute stress	• Stimulates glucose production from glycogen (muscle and liver) • Stimulates fatty acid release from adipose issue
Cortisol	• Provides for changing requirements over the long-term	• Stimulates amino acid mobilization from muscle protein • Stimulates gluconeogenesis • Stimulates fatty acid release from adipose issue

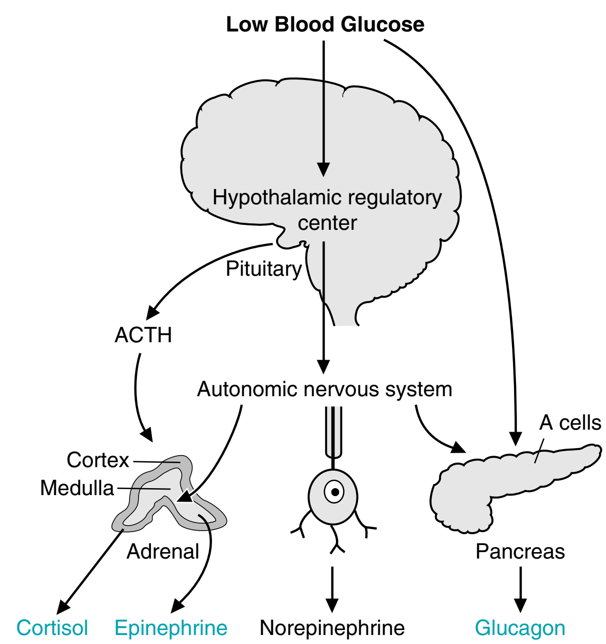


Fig 26.9. Major insulin counterregulatory hormones. The stress of a low blood glucose level mediates the release of the major insulin counterregulatory hormones through neuronal signals. Hypoglycemia is one of the stress signals that stimulates the release of cortisol, epinephrine, and norepinephrine. Adrenocorticotropic hormone (ACTH) is released from the pituitary and stimulates the release of cortisol (a glucocorticoid) from the adrenal cortex. Neuronal signals stimulate the release of epinephrine from the adrenal medulla and norepinephrine from nerve endings. Neuronal signals also play a minor role in the release of glucagon. Although norepinephrine has counterregulatory actions, it is not a major counterregulatory hormone.

sized and released in direct response to changing levels of fuels in the blood. The release of cortisol, epinephrine, and norepinephrine is mediated by neuronal signals. Rising levels of the insulin counterregulatory hormones in the blood, reflect, for the most part, a current increase in the demand for fuel.

III. SYNTHESIS AND RELEASE OF INSULIN AND GLUCAGON

A. Endocrine Pancreas

Insulin and glucagon are synthesized in different cell types of the endocrine pancreas, which consists of microscopic clusters of small glands, the islets of Langerhans, scattered among the cells of the exocrine pancreas. The α cells secrete glucagon, and the β cells secrete insulin into the hepatic portal vein via the pancreatic veins.

B. Synthesis and Secretion of Insulin

Insulin is a polypeptide hormone. The active form of insulin is composed of two polypeptide chains (the A-chain and the B-chain) linked by two interchain disulfide bonds. The A-chain has an additional intrachain disulfide bond (Fig. 26.10).

Insulin, like many other polypeptide hormones, is synthesized as a preprohormone that is converted in the rough endoplasmic reticulum (RER) to proinsulin. The "pre" sequence, a short hydrophobic signal sequence at the N-terminal end, is cleaved as it enters the lumen of the RER. Proinsulin folds into the proper conformation, and disulfide bonds are formed between the cysteine residues. It is then transported in microvesicles to the Golgi complex. It leaves the Golgi complex in storage vesicles, where a protease removes the C-peptide (a fragment with no hormonal activity) and a few small remnants, resulting in the formation of biologically active insulin (see Fig. 26.10). Zinc ions are also transported in these storage vesicles. Cleavage of the C-peptide decreases the solubility of the resulting insulin, which then coprecipitates with zinc. Exocytosis of the insulin storage vesicles from the cytosol of the β cell into the blood is stimulated by rising levels of glucose in the blood bathing the β cells.

Glucose enters the β cell via specific glucose transporter proteins known as GLUT2 (see Chapter 27). Glucose is phosphorylated through the action of glucokinase to form glucose 6-phosphate, which is metabolized through glycolysis, the TCA cycle, and oxidative phosphorylation. These reactions result in an increase in ATP levels within the β cell (circle 1 in Fig. 26.11). As the β cell [ATP]/[ADP] ratio increases, the activity of a membrane-bound, ATP-dependent K^+ channel (K^+_{ATP}) is inhibited (i.e., the channel is closed)(circle 2 in Fig. 26.11). The closing of this channel leads to a membrane depolarization (circle 3, Fig. 26.11), which activates a voltage-gated Ca^{2+} channel that allows Ca^{2+} to enter the β cell such that intracellular Ca^{2+} levels increase significantly (circle 4, Fig. 26.11). The increase in intracellular Ca^{2+} stimulates the fusion of insulin containing exocytotic vesicles with the plasma membrane, resulting in insulin secretion (circle 5, Fig. 26.11). Thus, an increase in glucose levels within the β cells initiates insulin release.

A form of diabetes known as MODY (maturity onset diabetes of the young) results from mutations in either pancreatic glucokinase or specific nuclear transcription factors. MODY type 2 is due to a glucokinase mutation that results in an enzyme with reduced activity, either due to an elevated K_m for glucose or a reduced V_{max} for the reaction. Because insulin release is dependent on normal glucose metabolism within the β cell that yields a critical [ATP]/[ADP] ratio in the β cell, individuals with this glucokinase mutation cannot significantly metabolize glucose unless glucose levels are higher than normal. Thus, although these patients can release insulin, they do so at higher than normal glucose levels, and are therefore almost always in a hyperglycemic state. Interestingly, however, these patients are somewhat resistant to the long-term complications of chronic hyperglycemia.

The message that insulin carries to tissues is that glucose is plentiful and it can be used as an immediate fuel or can be converted to storage forms such as triacylglycerol in adipocytes or glycogen in liver and muscle.

Because insulin stimulates the uptake of glucose into tissues where it may be immediately oxidized or stored for later oxidation, this regulatory hormone lowers blood glucose levels. Therefore, one of the possible causes of **Bea Selmass's** hypoglycemia is an insulinoma, a tumor that produces excessive insulin.

Whenever an endocrine gland continues to release its hormone in spite of the presence of signals that normally would suppress its secretion, this persistent inappropriate release is said to be "autonomous." Secretory neoplasms of endocrine glands generally produce their hormonal product autonomously in a chronic fashion.

Autonomous hypersecretion of insulin from a suspected pancreatic β-cell tumor (an insulinoma) can be demonstrated in several ways. The simplest test is to simultaneously draw blood for the measurement of both glucose and insulin at a time when the patient is spontaneously experiencing the characteristic adrenergic or neuroglycopenic symptoms of hypoglycemia. During such a test, **Bea Selmass's** glucose levels fell to 45 mg/dL (normal = 80–100), and her ratio of insulin to glucose was far higher than normal. The elevated insulin levels markedly increased glucose uptake by the peripheral tissues, resulting in a dramatic lowering of blood glucose levels. In normal individuals, as blood glucose levels drop, insulin levels also drop.

Di Abietes has type 1 diabetes mellitus, formerly known as insulin-dependent diabetes mellitus (IDDM). This metabolic disorder is usually caused by antibody-mediated (autoimmune) destruction of the β cells of the pancreas. Susceptibility to type 1 diabetes mellitus is, in part, conferred by a genetic defect in the human leukocyte antigen (HLA) region of β cells which codes for the major histocompatibility complex II (MHC II). This protein presents an intracellular antigen to the cell surface for "self-recognition" by the cells involved in the immune response. Because of this defective protein, a cell-mediated immune response leads to varying degrees of β cell destruction and eventually to dependence on exogenous insulin administration to control the levels of glucose in the blood.

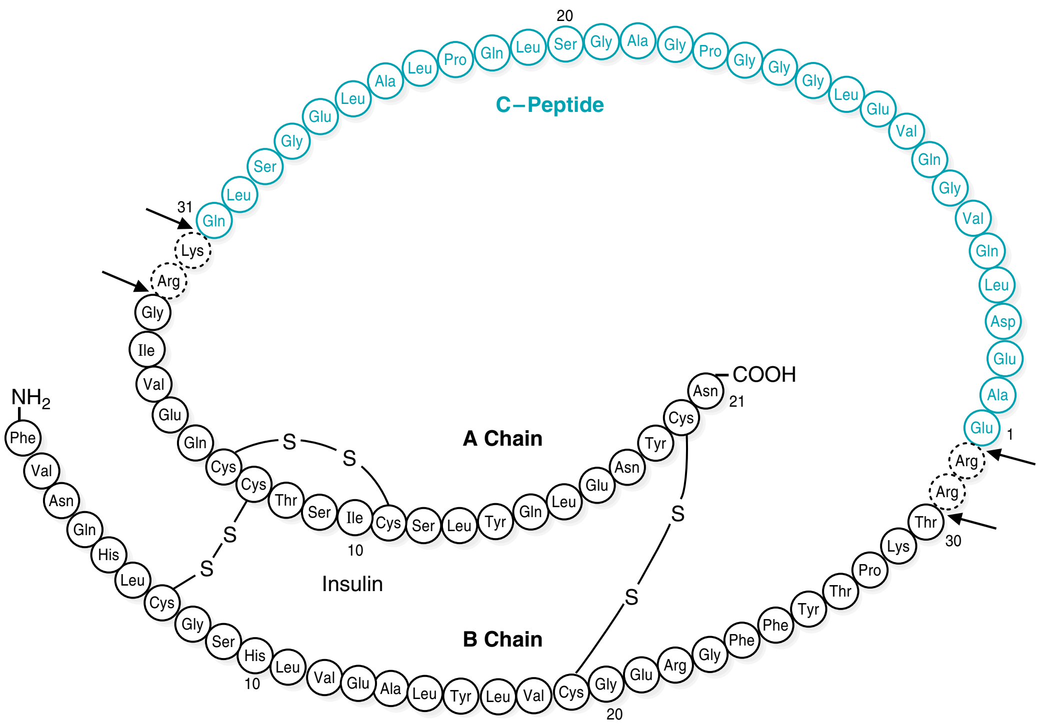


Fig 26.10. Cleavage of proinsulin to insulin. Proinsulin is converted to insulin by proteolytic cleavage, which removes the C-peptide and a few additional amino acid residues. Cleavage occurs at the arrows. From Murray RK, et al. Harper's Biochemistry, 23rd Ed. Stanford, CT: Appleton & Lange, 1993:560.

C. Stimulation and Inhibition of Insulin Release

The release of insulin occurs within minutes after the pancreas is exposed to a high glucose concentration. The threshold for insulin release is approximately 80 mg glucose/dL. Above 80 mg/dL, the rate of insulin release is not an all-or-nothing response but is proportional to the glucose concentration up to approximately 300 mg/dL glucose. As insulin is secreted, the synthesis of new insulin molecules is stimulated, so that secretion is maintained until blood glucose levels fall. Insulin is rapidly removed from the circulation and degraded by the liver (and, to a lesser

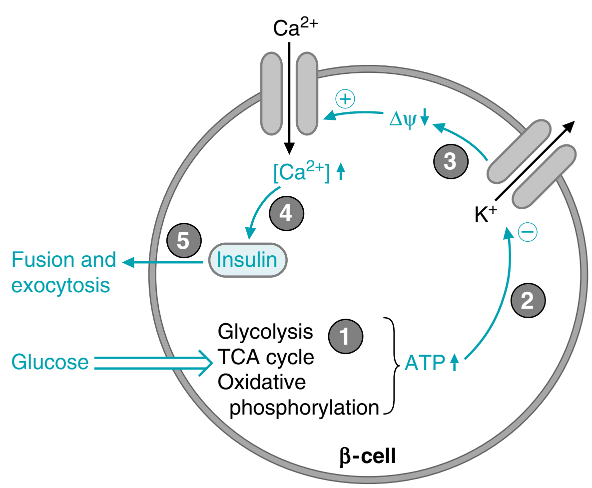


Fig 26.11. Release of insulin by the β-cells. Details are provided in the text.

Table 26.2. Regulators of Insulin Release[a]

Major Regulators	Effect
Glucose	+
Minor regulators	
Amino acids	+
Neural input	+
Gut hormones[b]	+
Epinephrine (adrenergic)	−

[a] + = stimulates
− = inhibits
[b] gut hormones that regulate fuel metabolism are discussed in Chapter 43.

extent, by kidney and skeletal muscle), so that blood insulin levels decrease rapidly once the rate of secretion slows.

A number of factors other than the blood glucose concentration can modulate insulin release (Table 26.2). The pancreatic islets are innervated by the autonomic nervous system, including a branch of the vagus nerve. These neural signals help to coordinate insulin release with the secretory signals initiated by the ingestion of fuels. However, signals from the central nervous system are not required for insulin secretion. Certain amino acids also can stimulate insulin secretion, although the amount of insulin released during a high-protein meal is very much lower than that released by a high-carbohydrate meal. Gastric inhibitory polypeptide (GIP, a gut hormone released after the ingestion of food) also aids in the onset of insulin release. Epinephrine, secreted in response to fasting, stress, trauma, and vigorous exercise, decreases the release of insulin. Epinephrine release signals energy utilization, which indicates that less insulin needs to be secreted, as insulin stimulates energy storage.

Ann Sulin is taking a sulfonylurea compound known as glipizide to treat her diabetes. The sulfonylureas act on the K^+_{ATP} channels on the surface of the pancreatic β cells. The binding of the drug to these channels closes K^+ channels (as do elevated ATP levels), which, in turn, increases Ca^{2+} movement into the interior of the β cell. This influx of calcium modulates the interaction of the insulin storage vesicles with the plasma membrane of the β cell, resulting in the release of insulin into the circulation.

Measurements of proinsulin and the connecting peptide between the α and β chains of insulin (C-peptide) in **Bea Selmass's** blood during her hospital fast provided confirmation that she had an insulinoma. Insulin and C-peptide are secreted in approximately equal proportions from the β cell, but C-peptide is not cleared from the blood as rapidly as insulin. Therefore, it provides a reasonably accurate estimate of the rate of insulin secretion. Plasma C-peptide measurements are also potentially useful in treating patients with diabetes mellitus because they provide a way to estimate the degree of endogenous insulin secretion in patients who are receiving exogenous insulin, which lacks the C-peptide.

D. Synthesis and Secretion of Glucagon

Glucagon, a polypeptide hormone, is synthesized in the α cells of the pancreas by cleavage of the much larger preproglucagon, a 160–amino acid peptide. Like insulin, preproglucagon is produced on the rough endoplasmic reticulum and is converted to proglucagon as it enters the ER lumen. Proteolytic cleavage at various sites produces the mature 29–amino acid glucagon (molecular weight 3,500) and larger glucagon-containing fragments (named glucagon-like peptides 1 and 2). Glucagon is rapidly metabolized, primarily in the liver and kidneys. Its plasma half-life is only about 3 to 5 minutes.

Glucagon secretion is regulated principally by circulating levels of glucose and insulin. Increasing levels of each inhibit glucagon release. Glucose probably has both a direct suppressive effect on secretion of glucagon from the α cell as well as an indirect effect, the latter being mediated by its ability to stimulate the release of insulin. The direction of blood flow in the islets of the pancreas carries insulin from the β cells in the center of the islets to the peripheral α cells, where it suppresses glucagon secretion.

Patients with type 1 diabetes mellitus, such as **Di Abietes,** have almost undetectable levels of insulin in their blood. Patients with type 2 diabetes mellitus, such as **Ann Sulin**, conversely, have normal or even elevated levels of insulin in their blood; however, the level of insulin in their blood is inappropriately low relative to their elevated blood glucose concentration. In type 2 diabetes mellitus, skeletal muscle, liver, and other tissues exhibit a resistance to the actions of insulin. As a result, insulin has a smaller than normal effect on glucose and fat metabolism in such patients. Levels of insulin in the blood must be higher than normal to maintain normal blood glucose levels. In the early stages of type 2 diabetes mellitus, these compensatory adjustments in insulin release may keep the blood glucose levels near the normal range. Over time, as the β cells capacity to secrete high levels of insulin declines, blood glucose levels increase, and exogenous insulin becomes necessary.

Table 26.3 Regulators of Glucagon Release[a]

Major Regulators	Effect
Glucose	−
Insulin	−
Amino acids	+
Minor regulators	
Cortisol	+
Neural (stress)	+
Epinephrine	+
Gut hormones	+

[a] + = stimulates
− = inhibits

Amino acids induce both insulin and glucagon secretion. Although this may seem paradoxical, it actually makes good sense. Insulin release stimulates amino acid uptake by tissues and enhances protein synthesis. However, because glucagon levels also increase in response to a protein meal, gluconeogenesis is enhanced (at the expense of protein synthesis), and the amino acids that are taken up by the tissues serve as a substrate for gluconeogenesis. The synthesis of glycogen and triglycerides is also reduced when glucagon levels rise in the blood.

In fasting subjects, the average level of immunoreactive glucagon in the blood is 75 pg/mL and does not vary as much as insulin during the daily fasting–feeding cycle. However, only 30 to 40% of the measured immunoreactive glucagon is mature pancreatic glucagon. The rest is composed of larger immunoreactive fragments also produced in the pancreas or in the intestinal L cells.

The physiologic importance of insulin's usual action of mediating the suppressive effect of glucose on glucagon secretion is apparent in patients with types 1 and 2 diabetes mellitus. Despite the presence of hyperglycemia, glucagon levels in such patients initially remain elevated (near fasting levels) either because of the absence of insulin's suppressive effect or because of the resistance of the α cells to insulin's suppressive effect even in the face of adequate insulin levels in type 2 patients. Thus, these patients have inappropriately high glucagon levels, leading to the suggestion that diabetes mellitus is actually a "bi-hormonal" disorder.

Certain hormones stimulate glucagon secretion. Among these are the catecholamines (including epinephrine), cortisol, and certain gastrointestinal (gut) hormones (Table 26.3).

Many amino acids also stimulate glucagon release (Fig. 26.12). Thus, the high levels of glucagon that would be expected in the fasting state do not decrease after a high-protein meal. In fact, glucagon levels may increase, stimulating gluconeogenesis in the absence of dietary glucose. The relative amounts of insulin and glucagon in the blood after a mixed meal are dependent on the composition of the meal, because glucose stimulates insulin release, and amino acids stimulate glucagon release.

IV. MECHANISMS OF HORMONE ACTION

For a hormone to affect the flux of substrates through a metabolic pathway, it must be able to change the rate at which that pathway proceeds by increasing or decreasing the rate of the slowest step(s). Either directly or indirectly, hormones affect the

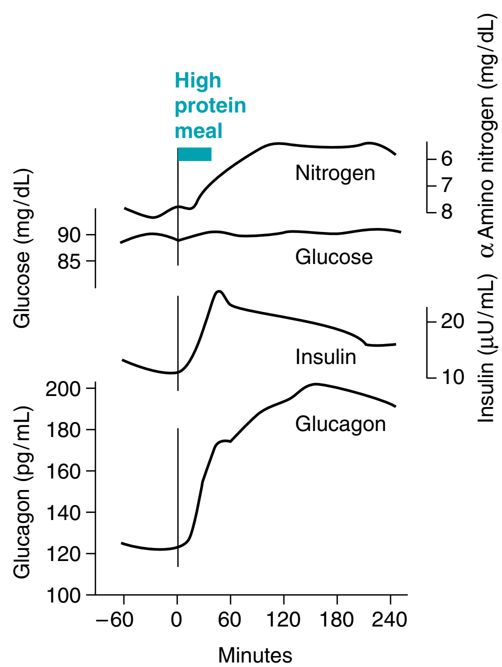


Fig 26.12. Release of insulin and glucagon in response to a high-protein meal. This figure shows the increase in the release of insulin and glucagon into the blood after an overnight fast followed by the ingestion of 100 g protein (equivalent to a slice of roast beef). Insulin levels do not increase nearly as much as they do after a high-carbohydrate meal (see Fig. 26.8). The levels of glucagon, however, significantly increase above those present in the fasting state.

activity of specific enzymes or transport proteins that regulate the flux through a pathway. Thus, ultimately, the hormone must either cause the amount of the substrate for the enzyme to increase (if substrate supply is a rate-limiting factor), change the conformation at the active site by phosphorylating the enzyme, change the concentration of an allosteric effector of the enzyme, or change the amount of the protein by inducing or repressing its synthesis or by changing its turnover rate or location. Insulin, glucagon, and other hormones use all of these regulatory mechanisms to regulate the rate of flux in metabolic pathways. The effects mediated by phosphorylation or changes in the kinetic properties of an enzyme occur rapidly, within minutes. In contrast, it may take hours for induction or repression of enzyme synthesis to change the amount of an enzyme in the cell.

The details of hormone action were previously described in Chapter 11 and are only summarized here.

During the "stress" of hypoglycemia, the autonomic nervous system stimulates the pancreas to secrete glucagon, which tends to restore the serum glucose level to normal. The increased activity of the adrenergic nervous system (through epinephrine) also alerts a patient, such as **Bea Selmass,** to the presence of increasingly severe hypoglycemia. Hopefully, this will induce the patient to ingest simple sugars or other carbohydrates, which, in turn, will also increase glucose levels in the blood. **Bea Selmass** gained 8 lb before resection of her pancreatic insulin-secreting adenoma through this mechanism.

A. Signal Transduction by Hormones That Bind to Plasma Membrane Receptors

Hormones initiate their actions on target cells by binding to specific receptors or binding proteins. In the case of polypeptide hormones (such as insulin and glucagon), and catecholamines (epinephrine and norepinephrine), the action of the hormone is mediated through binding to a specific receptor on the plasma membrane (see Chapter 11, section III). The first message of the hormone is transmitted to intracellular enzymes by the activated receptor and an intracellular second messenger; the hormone does not need to enter the cell to exert its effects. (In contrast, steroid hormones such as cortisol and the thyroid hormone triiodothyronine [T_3] enter the cytosol and eventually move into the cell nucleus to exert their effects.)

The mechanism by which the message carried by the hormone ultimately affects the rate of the regulatory enzyme in the target cell is called signal transduction. The three basic types of signal transduction for hormones binding to receptors on the plasma membrane are (*a*) receptor coupling to adenylate cyclase which produces cAMP, (*b*) receptor kinase activity, and (*c*) receptor coupling to hydrolysis of phosphatidylinositol bisphosphate (PIP_2). The hormones of metabolic homeostasis each use one of these mechanisms to carry out their physiologic effect. In addition, some hormones and neurotransmitters act through receptor coupling to gated ion channels (previously described in Chapter 11).

1. SIGNAL TRANSDUCTION BY INSULIN

Insulin initiates its action by binding to a receptor on the plasma membrane of insulin's many target cells (see Fig. 11.13). The insulin receptor has two types of subunits, the α-subunits to which insulin binds, and the β-subunits, which span the membrane and protrude into the cytosol. The cytosolic portion of the β-subunit has tyrosine kinase activity. On binding of insulin, the tyrosine kinase phosphorylates tyrosine residues on the β-subunit (autophosphorylation) as well as on several other enzymes within the cytosol. A principal substrate for phosphorylation by the receptor, insulin receptor substrate (IRS-1), then recognizes and binds to various signal transduction proteins in regions referred to as SH_2 domains. IRS-1 is involved in many of the physiologic responses to insulin through complex mechanisms that are the subject of intensive investigation. The basic tissue-specific cellular responses to insulin, however, can be grouped into five major categories: (*a*) insulin reverses glucagon-stimulated phosphorylation, (*b*) insulin works through a phosphorylation cascade that stimulates the phosphorylation of several enzymes, (*c*) insulin induces and represses the synthesis of specific enzymes, (*d*) insulin acts as a growth factor and has a general stimulatory effect on protein synthesis, and (*e*) insulin stimulates glucose and amino acid transport into cells (see Fig. 10 of section introduction, p. 484).

A number of mechanisms have been proposed for the action of insulin in reversing glucagon-stimulated phosphorylation of the enzymes of carbohydrate metabolism. From the student's point of view, the ability of insulin to reverse glucagon-stimulated phosphorylation occurs as if it were lowering cAMP and stimulating phosphatases that could remove those phosphates added by protein kinase A. In reality, the mechanism is more complex and still not fully understood.

2. SIGNAL TRANSDUCTION BY GLUCAGON

The pathway for signal transduction by glucagon is one common to a number of hormones; the glucagon receptor is coupled to adenylate cyclase and cAMP production (see Fig. 11.11). Glucagon, through G proteins, activates the membrane-bound adenylate cyclase, increasing the synthesis of the intracellular second messenger 3′,5′-cyclic AMP (cAMP) (see Fig. 9.10). cAMP activates protein kinase A (cAMP-dependent protein kinase), which changes the activity of enzymes by phosphorylating them at specific serine residues. Phosphorylation activates some enzymes and inhibits others.

The G proteins, which couple the glucagon receptor to adenylate cyclase, are proteins in the plasma membrane that bind guanosine triphosphate (GTP) and have dissociable subunits that interact with both the receptor and adenylate cyclase. In the absence of glucagon, the stimulatory G_s protein complex binds guanosine diphosphate (GDP) but cannot bind to the unoccupied receptor or adenylate cyclase (see Fig. 11.17). Once glucagon binds to the receptor, the receptor also binds the G_s complex, which then releases GDP and binds GTP. The α-subunit then dissociates from the $\beta\gamma$-subunits and binds to adenylate cyclase, thereby activating it. As the GTP on the α-subunit is hydrolyzed to GDP, the subunit dissociates and recomplexes with the β- and γ-subunits. Only continued occupancy of the glucagon receptor can keep adenylate cyclase active.

Although glucagon works by activating adenylate cyclase, a few hormones inhibit adenylate cyclase. In this case, the inhibitory G protein complex is called a G_i complex.

cAMP is very rapidly degraded to AMP by a membrane-bound phosphodiesterase. The concentration of cAMP is thus very low in the cell so that changes in its concentration can occur rapidly in response to changes in the rate of synthesis. The amount of cAMP present at any time is a direct reflection of hormone binding and the activity of adenylate cyclase. It is not affected by ATP, ADP, or AMP levels in the cell.

cAMP transmits the hormone signal to the cell by activating protein kinase A (cAMP-dependent protein kinase). As cAMP binds to the regulatory subunits of protein kinase A, these subunits dissociate from the catalytic subunits, which are thereby activated (see Figs. 9.9 and 9.11). Activated protein kinase A phosphorylates serine residues of key regulatory enzymes in the pathways of carbohydrate and fat metabolism. Some enzymes are activated and others are inhibited by this change in phosphorylation state. The message of the hormone is terminated by the action of semispecific protein phosphatases that remove phosphate groups from the enzymes. The activity of the protein phosphatases is also controlled through hormonal regulation.

Changes in the phosphorylation state of proteins that bind to cAMP response elements (CREs) in the promoter region of genes contribute to the regulation of gene transcription by a number of cAMP-coupled hormones (see Chapter 16). For instance, cAMP response element binding protein (CREB) is directly phosphorylated by protein kinase A, a step essential for the initiation of transcription. Phosphorylation at other sites on CREB, by a variety of kinases, also may play a role in regulating transcription.

cAMP is the intracellular second messenger for a number of hormones that regulate fuel metabolism. The specificity of the physiologic response to each hormone results from the presence of specific receptors for that hormone in target tissues. For example, glucagon activates glucose production from glycogen in liver, but not in skeletal muscle because glucagon receptors are present in liver but absent in skeletal muscle. However, skeletal muscle has adenylate cyclase, cAMP, and protein kinase A, which can be activated by epinephrine binding to the β_2 receptors in the membrane of muscle cells. Liver cells also have epinephrine receptors.

Q: Phosphodiesterase is inhibited by methylxanthines, a class of compounds that includes caffeine. Would the effect of a methylxanthine on fuel metabolism be similar to fasting or to a high-carbohydrate meal?

The mechanism for signal transduction by glucagon illustrates some of the important principles of hormonal signaling mechanisms. The first principle is that specificity of action in tissues is conferred by the receptor on a target cell for glucagon. In general, the major actions of glucagon occur in liver, adipose tissue, and certain cells of the kidney that contain glucagon receptors. The second principle is that signal transduction involves amplification of the first message. Glucagon and other hormones are present in the blood in very low concentrations. However, these minute concentrations of hormone are adequate to initiate a cellular response because the binding of one molecule of glucagon to one receptor ultimately activates many protein kinase A molecules, each of which phosphorylates hundreds of downstream enzymes. The third principle involves integration of metabolic responses. For instance, the glucagon-stimulated phosphorylation of enzymes simultaneously activates glycogen degradation, inhibits glycogen synthesis, and inhibits glycolysis in the liver (see Fig. 10 in section introduction, p. 484.). The fourth principle involves augmentation and antagonism of signals. An example of augmentation involves the actions of glucagon and epinephrine (which is released during exercise). Although these hormones bind to different receptors, each can increase cAMP and stimulate glycogen degradation. A fifth principle is that of rapid signal termination. In the case of glucagon, both the termination of the G_s protein activation and the rapid degradation of cAMP contribute to signal termination.

A: Inhibition of phosphodiesterase by methylxanthine would increase cAMP and have the same effects on fuel metabolism as would an increase of glucagon and epinephrine, in the fasted state. Increased fuel mobilization would occur through glycogenolysis (the release of glucose from glycogen), and through lipolysis (the release of fatty acids from triacylglycerols).

Protein kinases are a class of enzymes that change the activity of other enzymes by phosphorylating them at serine or tyrosine residues and are referred to as serine or tyrosine kinases, respectively. Serine kinases also phosphorylate some enzymes at threonine residues. Protein kinase A (a serine kinase) phosphorylates many of the enzymes in the pathways of fuel metabolism.

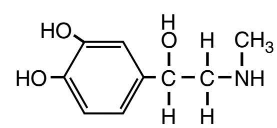
B. Signal Transduction by Cortisol and Other Hormones That Interact with Intracellular Receptors

Signal transduction by the glucocorticoid cortisol and other steroids and by thyroid hormone involves hormone binding to intracellular (cytosolic) receptors or binding proteins, after which this hormone–binding protein complex moves into the nucleus, where it interacts with chromatin. This interaction changes the rate of gene transcription in the target cells (see Chapter 16). The cellular responses to these hormones will continue as long as the target cell is exposed to the specific hormones. Thus, disorders that cause a chronic excess in their secretion will result in an equally persistent influence on fuel metabolism. For example, chronic stress such as that seen in prolonged sepsis may lead to varying degrees of glucose intolerance if high levels of epinephrine and cortisol persist.

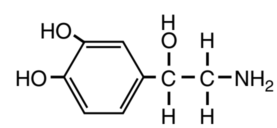
The effects of cortisol on gene transcription are usually synergistic to those of certain other hormones. For instance, the rates of gene transcription for some of the enzymes in the pathway for glucose synthesis from amino acids (gluconeogenesis) are induced by glucagon as well as by cortisol.

C. Signal Transduction by Epinephrine and Norepinephrine

Epinephrine and norepinephrine are catecholamines (Fig. 26.13). They can act as neurotransmitters or as hormones. A neurotransmitter allows a neural signal to be transmitted across the juncture or synapse between the nerve terminal of a proximal nerve axon and the cell body of a distal neuron. A hormone, conversely, is released into the blood and travels in the circulation to interact with specific receptors on the plasma membrane or cytosol of the target organ. The general effect of these catecholamines is to prepare us for fight or flight. Under these acutely stressful circumstances, these "stress" hormones increase fuel mobilization, cardiac output, blood flow, etc., which enable us to meet these stresses. The catecholamines bind to adrenergic receptors (the term *adrenergic* refers to nerve cells or fibers that are part of the involuntary or autonomic nervous system, a system that employs norepinephrine as a neurotransmitter).



Epinephrine



Norepinephrine

Fig 26.13. Structure of epinephrine and norepinephrine. Epinephrine and norepinephrine are synthesized from tyrosine and act as both hormones and neurotransmitters. They are catecholamines, the term *catechol* referring to a ring structure containing two hydroxyl groups.

There are nine different types of adrenergic receptors: α_{1A}, α_{1B}, α_{1D}, α_{2A}, α_{2B}, α_{2C}, β_1, β_2, and β_3. Only the three β and α_1 receptors are discussed here. The three β receptors work through the adenylate cyclase–cAMP system, activating a G_s protein, which activates adenylate cyclase, and eventually protein kinase A. The β_1 receptor is the major adrenergic receptor in the human heart and is primarily stimulated by norepinephrine. On activation, the β_1 receptor increases the rate of muscle contraction, in part because of PKA-mediated phosphorylation of phospholamban (see Chapter 47). The β_2 receptor is present in liver, skeletal muscle, and other tissues and is involved in the mobilization of fuels (such as the release of glucose through glycogenolysis). It also mediates vascular, bronchial, and uterine smooth muscle contraction. Epinephrine is a much more potent agonist for this receptor than norepinephrine, whose major action is neurotransmission. The β_3 receptor is found predominantly in adipose tissue and to a lesser extent in skeletal muscle. Activation of this receptor stimulates fatty acid oxidation and thermogenesis, and agonists for this receptor may prove to be beneficial weight loss agents. The α_1 receptors, which are postsynaptic receptors, mediate vascular and smooth muscle contraction. They work through the phosphatidylinositol bisphosphate system (see Chapter 11, section III.B.2) through activation of a G_q protein, and phospholipase C-β. This receptor also mediates glycogenolysis in liver.

Ann O'Rexia, to stay thin, frequently fasts for prolonged periods, but jogs every morning (see Chapter 2). The release of epinephrine and norepinephrine and the increase of glucagon and fall of insulin during her exercise provide coordinated and augmented signals that stimulate the release of fuels above the fasting levels. Fuel mobilization will occur, of course, only as long as she has fuel stored as triacylglycerols.

Ann Sulin, a patient with type 2 diabetes mellitus, is experiencing insulin resistance. Her levels of circulating insulin are normal to high, although inappropriately low for her elevated level of blood glucose. However, her insulin target cells, such as muscle and fat, do not respond as those of a nondiabetic subject would to this level of insulin. For most type 2 patients, the site of insulin resistance is subsequent to binding of insulin to its receptor; i.e., the number of receptors and their affinity for insulin is near normal. However, the binding of insulin at these receptors does not elicit most of the normal intracellular effects of insulin discussed earlier. Consequently, there is little stimulation of glucose metabolism and storage after a high-carbohydrate meal and little inhibition of hepatic gluconeogenesis.

CLINICAL COMMENTS

Di Abietes has type 1 diabetes mellitus (formally designated insulin-dependent diabetes mellitus, IDDM) whereas **Ann Sulin** has type 2 diabetes mellitus (formally called non–insulin-dependent diabetes mellitus). Although the pathogenesis differs for these major forms of diabetes mellitus, both cause varying degrees of hyperglycemia. In type 1, the pancreatic β cells are gradually destroyed by antibodies directed at a variety of proteins within the β cells. As insulin secretory capacity by the β cells gradually diminishes below a critical level, the symptoms of chronic hyperglycemia develop rapidly. In type 2 diabetes mellitus, these symptoms develop more subtly and gradually over the course of months or years. Eighty-five percent or more of type 2 patients are obese and, like **Ivan Applebod,** have a high waist–hip ratio with regard to adipose tissue disposition. This abnormal distribution of fat in the visceral (peri-intestinal) adipocytes is associated with reduced sensitivity of fat cells, muscle cells, and liver cells to the actions of insulin outlined above. This insulin resistance can be diminished through weight loss, specifically in the visceral depots.

Bea Selmass underwent a high-resolution ultrasonographic (ultrasound) study of her upper abdomen, which showed a 2.6-cm mass in the midportion of her pancreas. With this finding, her physicians decided that further noninvasive studies would not be necessary before surgical exploration of her upper abdomen was performed. At the time of surgery, a yellow-white 2.8-cm mass consisting primarily of insulin-rich β cells was resected from her pancreas. No cytologic changes of malignancy were seen on cytologic examination of the surgical specimen, and no gross evidence of malignant behavior by the tumor (such as local metastases) was found. Bea had an uneventful postoperative recovery and no longer experienced the signs and symptoms of insulin-induced hypoglycemia.

BIOCHEMICAL COMMENTS

One of the important cellular responses to insulin is the reversal of glucagon-stimulated phosphorylation of enzymes. Mechanisms proposed for this action include the inhibition of adenylate cyclase, a reduction of cAMP levels, the stimulation of phosphodiesterase, the production of a specific protein (insulin factor), the release of a second messenger from a bound glycosylated phosphatidylinositol, and the phosphorylation of enzymes at a site that antagonizes protein kinase A phosphorylation. Not all of these physiologic actions of insulin occur in each of the insulin-sensitive organs of the body.

Insulin is also able to antagonize the actions of glucagon at the level of specific induction or repression of key regulatory enzymes of carbohydrate metabolism. For instance, the rate of synthesis of mRNA for phosphoenolpyruvate carboxykinase, a key enzyme of the gluconeogenic pathway, is increased severalfold by glucagon (via cAMP) and decreased by insulin. Thus, all of the effects of glucagon, even the induction of certain enzymes, can be reversed by insulin. This antagonism is exerted through an insulin-sensitive hormone response element (IRE) in the promoter region of the genes. Insulin causes repression of the synthesis of enzymes that are induced by glucagon.

The general stimulation of protein synthesis by insulin (its mitogenic or growth-promoting effect) appears to occur through both a general increase in rates of mRNA translation for a broad spectrum of structural proteins. These actions result from a phosphorylation cascade initiated by autophosphorylation of the insulin receptor and ending in the phosphorylation of subunits of proteins that bind to and inhibit eukaryotic protein synthesis initiation factors (eIFs). When phosphorylated, the inhibitory proteins are released from the eIFs, allowing translation of mRNA to be stimulated. In this respect, the actions of insulin are similar to those of hormones that act as growth factors and also have receptors with tyrosine kinase activity.

In addition to signal transduction, activation of the insulin receptor mediates the internalization of receptor-bound insulin molecules increasing their subsequent degradation. Although unoccupied receptors can be internalized and eventually recycled to the plasma membrane, the receptor can be irreversibly degraded after prolonged occupation by insulin. The result of this process, referred to as receptor downregulation, is an attenuation of the insulin signal. The physiologic importance of receptor internalization on insulin sensitivity is poorly understood but could eventually lead to chronic hyperglycemia.

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REVIEW QUESTIONS—CHAPTER 26

1. A patient with type I diabetes mellitus takes an insulin injection before eating dinner but then gets distracted and does not eat. Approximately 3 hours later, the patient becomes shaky, sweaty, and confused. These symptoms have occurred because of which of the following?

   (A) Increased glucagon release from the pancreas
   (B) Decreased glucagon release from the pancreas
   (C) High blood glucose levels
   (D) Low blood glucose levels
   (E) Elevated ketone body levels

2. Caffeine is a potent inhibitor of the enzyme cAMP phosphodiesterase. Which of the following consequences would you expect to occur in the liver after drinking two cups of strong expresso coffee?

   (A) A prolonged response to insulin
   (B) A prolonged response to glucagon
   (C) An inhibition of protein kinase A
   (D) An enhancement of glycolytic activity
   (E) A reduced rate of glucose export to the circulation

3. Assume that an increase in blood glucose concentration from 5 to 10 mM would result in insulin release by the pancreas. A mutation in pancreatic glucokinase can lead to MODY because of which of the following within the pancreatic β-cell?

   (A) An inability to raise cAMP levels
   (B) An inability to raise ATP levels
   (C) An inability to stimulate gene transcription
   (D) An inability to activate glycogen degradation
   (E) An inability to raise intracellular lactate levels

4. Which one of the following organs has the highest demand for glucose as a fuel?

   (A) Brain
   (B) Muscle (skeletal)
   (C) Heart
   (D) Liver
   (E) Pancreas

5. Glucagon release does not alter muscle metabolism because of which of the following?

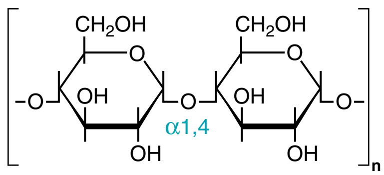
   (A) Muscle cells lack adenylate cyclase.
   (B) Muscle cells lack protein kinase A.
   (C) Muscle cells lack G proteins.
   (D) Muscle cells lack GTP.
   (E) Muscle cells lack the glucagon receptor.

27 Digestion, Absorption, and Transport of Carbohydrates

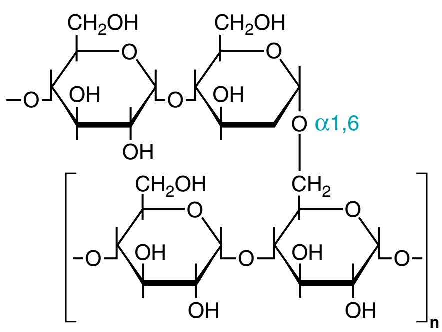
Carbohydrates are the largest source of dietary calories for most of the world's population. The major carbohydrates in the American diet are starch, lactose, and sucrose. The ***starches amylose*** *and* ***amylopectin*** *are polysaccharides composed of hundreds to millions of glucosyl units linked together through α-1,4 and α-1,6 glycosidic bonds (Fig. 27.1).* ***Lactose*** *is a disaccharide composed of glucose and galactose, linked together through a β-1,4 glycosidic bond.* ***Sucrose*** *is a disaccharide composed of glucose and fructose, linked through an α-1,2 glycosidic bond. The digestive processes convert all of these dietary carbohydrates to their constituent monosaccharides by* ***hydrolyzing glycosidic bonds*** *between the sugars.*

The digestion of starch begins in the mouth (Fig. 27.2). The ***salivary*** *gland releases* ***a-amylase****, which converts starch to smaller polysaccharides called* ***α-dextrins.*** *Salivary α-amylase is inactivated by the acidity of the stomach (HCl).* ***Pancreatic α-amylase*** *and bicarbonate are secreted by the exocrine pancreas into the lumen of the small intestine, where bicarbonate neutralizes the gastric secretions. Pancreatic α-amylase continues the digestion of α-dextrins, converting them to disaccharides (****maltose****), trisaccharides (****maltotriose****), and oligosaccharides called* ***limit dextrins****. Limit dextrins usually contain four to nine glucosyl residues and an* ***isomaltose*** *branch (two glucosyl residues attached through an α-1,6 glycosidic bond).*

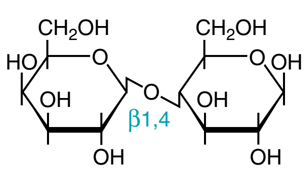
The digestion of the disaccharides lactose and sucrose, as well as further digestion of maltose, maltotriose and limit dextrins, occurs through ***disaccharidases*** *attached to the membrane surface of the* ***brush border (microvilli)*** *of intestinal epithelial cells.* ***Glucoamylase*** *hydrolyzes the α-1,4 bonds of dextrins. The* ***sucrase–isomaltase complex*** *hydrolyzes sucrose, most of maltose, and almost all of the isomaltose formed by glucoamylase from limit dextrins.* ***Lactase-glycosylceramidase*** *(β-glycosidase) hydrolyzes the β-glycosidic bonds in* ***lactose*** *and* ***glycolipids****. A fourth disaccharidase complex,* ***trehalase****, hydrolyzes the bond (an α-1,1 glycosidic bond) between two glucosyl units in the sugar trehalose. The monosaccharides produced by these hydrolases (glucose, fructose, and galactose) are then transported into the intestinal epithelial cells.*



Amylose



Amylopectin

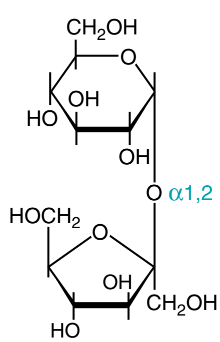


Galactose **Glucose**

Lactose

Glucose

Fructose



Sucrose

Fig. 27.1. The structures of common dietary carbohydrates. For disaccharides and greater, the sugars are linked through glycosidic bonds between the anomeric carbon of one sugar and a hydroxyl group on another sugar. The glycosidic bond may be either α or β, depending on its position above or below the plane of the sugar containing the anomeric carbon. (see Chapter 5, Section II.A, to review terms used in the description of sugars). The starch amylose is a polysaccharide of glucose residues linked with α-1,4 glycosidic bonds. Amylopectin is amylase with the addition of α-1,6 glycosidic branchpoints. Dietary sugars may be monosaccharides (single sugar residues), disaccharides (two sugar residues), oligosaccharides (several sugar residues) or polysaccharides (hundreds of sugar residues).

A common malabsorption syndrome, **lactose intolerance,** is characterized by nausea, diarrhea, and flatulence after ingesting dairy products or other foods containing lactose. One of the causes of lactose intolerance is a low level of lactase, which decreases after infancy in most of the world's population **(nonpersistant lactase** or **adult hypolactasia).** However, lactase activity remains high in some populations (**persistent lactase),** including Northwestern Europeans and their descendants.

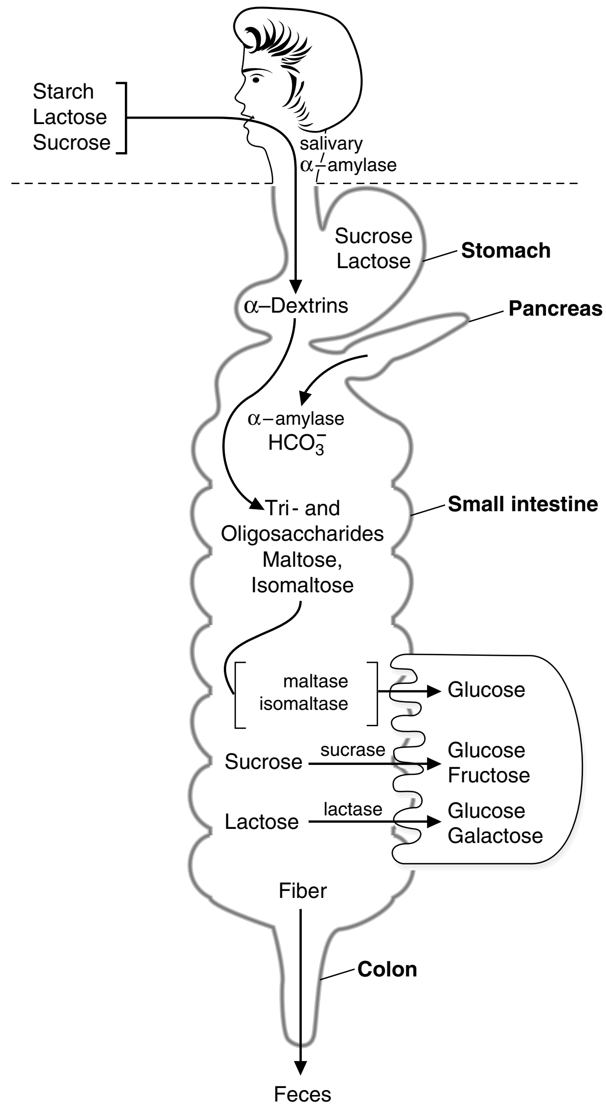


Fig. 27.2. Overview of carbohydrate digestion. **Digestion** of the carbohydrates occurs first, followed by **absorption** of monosaccharides. Subsequent **metabolic** reactions occur after the sugars are absorbed.

Dietary fiber*, composed principally of polysaccharides, cannot be digested by human enzymes in the intestinal tract. In the colon, dietary fiber and other nondigested carbohydrates may be converted to gases (H_2, CO_2, and methane) and short-chain fatty acids (principally acetic acid, propionic acid, and butyric acid) by bacteria in the colon.*

Glucose*,* ***galactose****, and* ***fructose*** *formed by the digestive enzymes are transported into the absorptive epithelial cells of the small intestine by protein-mediated* ***Na^+-dependent active transport*** *and* ***facilitative diffusion.*** *Monosaccharides are transported from these cells into the blood and circulate to the liver and peripheral tissues, where they are taken up by facilitative transporters. Facilitative transport of glucose across epithelial cells and other cell membranes is mediated by a family of* ***tissue-specific glucose transport proteins (GLUT I–V)****. The type of transporter found in each cell reflects the role of glucose metabolism in that cell.*

THE WAITING ROOM

Deria Voider is a 20-year-old exchange student from Nigeria who has noted gastrointestinal bloating, abdominal cramps, and intermittent diarrhea ever since arriving in the United States 6 months earlier. A careful history shows that these symptoms occur most commonly about 45 minutes to 1 hour after eating breakfast but may occur after other meals as well. Dairy products, not a part of Deria's diet in Nigeria, were identified as the probable offending agent because her gastrointestinal symptoms disappeared when milk and milk products were eliminated from her diet.

Ann Sulin's fasting and postprandial blood glucose levels are frequently above the normal range in spite of good compliance with insulin therapy. Her physician has referred her to a dietician skilled in training diabetic patients in the successful application of an appropriate American Diabetes Association diet. As part of the program, Ms. Sulin is asked to incorporate foods containing fiber into her diet, such as whole grains (e.g., wheat, oats, corn), legumes (e.g., peas, beans, lentils), tubers (e.g., potatoes, peanuts), and fruits.

Nona Melos (no sweets) is a 7-month-old baby girl, the second child born to unrelated parents. Her mother had a healthy, full-term pregnancy, and Nona's birth weight was normal. She did not respond well to breastfeeding and was changed entirely to a formula based on cow's milk at 4 weeks. Between 7 and 12 weeks of age, she was admitted to the hospital twice with a history of screaming after feeding but was discharged after observation without a specific diagnosis. Elimination of cow's milk from her diet did not relieve her symptoms; Nona's mother reported that the screaming bouts were worse after Nona drank juice and that Nona frequently had gas and a distended abdomen. At 7 months she was still thriving (weight above 97th percentile) with no abnormal findings on physical examination. A stool sample was taken.

The dietary sugar in fruit juice and other sweets is sucrose, a disaccharide composed of glucose and fructose joined through their anomeric carbons. **Nona Melos'** symptoms of pain and abdominal distension are caused by an inability to digest sucrose or absorb fructose, which are converted to gas by colonic bacteria. "Melos" is Latin for sweets, and her name means "no sweets." Nona Melos's stool sample had a pH of 5 and gave a positive test for sugar. The possibility of carbohydrate malabsorption was considered, and a hydrogen breath test was recommended.

I. DIETARY CARBOHYDRATES

Carbohydrates are the largest source of calories in the average American diet and usually constitute 40 to 45% of our caloric intake. The plant starches amylopectin and amylose, which are present in grains, tubers, and vegetables, constitute approximately 50 to 60% of the carbohydrate calories consumed. These starches are polysaccharides, containing 10,000 to 1 million glucosyl units. In amylose, the glucosyl residues form a straight chain linked via α-1,4 glycosidic bonds; in amylopectin, the α-1,4 chains contain branches connected via α-1,6 glycosidic bonds (see Fig. 27.1). The other major sugar found in fruits and vegetables is sucrose, a disaccharide of glucose and fructose (see Fig. 27.1). Sucrose and small amounts of the monosaccharides glucose and fructose are the major natural sweeteners found in fruit, honey, and vegetables. Dietary fiber, that portion of the diet that cannot be digested by human enzymes of the intestinal tract, is also composed principally of plant polysaccharides and a polymer called lignan.

Most foods derived from animals, such as meat or fish, contain very little carbohydrate except for small amounts of glycogen (which has a structure similar to amylopectin) and glycolipids. The major dietary carbohydrate of animal origin is lactose, a disaccharide composed of glucose and galactose found exclusively in milk and milk products (see Fig. 27.1).

Sweeteners, in the form of sucrose and high-fructose corn syrup (starch, partly hydrolyzed and isomerized to fructose), also appear in the diet as additives to processed foods. On average, a person in the United States consumes 65 lb added sucrose and 40 lb high-fructose corn syrup solids per year.

Although all cells require glucose for metabolic functions, neither glucose nor other sugars are specifically required in the diet. Glucose can be synthesized from many amino acids found in dietary protein. Fructose, galactose, xylose, and all the other sugars required for metabolic processes in the human can be synthesized from glucose.

Starch blockers had been marketed many years ago as a means of losing weight without having to exercise or reduce your daily caloric intake. Starch blockers were based on a protein found in beans, which blocked the action of amylase. Thus, as the advertisements proclaimed, one could eat a large amount of starch during a meal, and as long as you took the starch blocker, the starch would pass through the digestive track without being metabolized. Unfortunately, this was too good to be true, and starch blockers were never shown to be effective in aiding weight loss. This was probably because of a combination of factors, such as inactivation of the inhibitor by the low pH in the stomach, and an excess of amylase activity as compared with the amount of starch blocker ingested. Recently this issue has been revisited, as a starch blocker from wheat has been developed that may work as advertised, although much more work is required to determine whether this amylase inhibitor will be safe and effective in humans.

II. DIGESTION OF DIETARY CARBOHYDRATES

In the digestive tract, dietary polysaccharides and disaccharides are converted to monosaccharides by glycosidases, enzymes that hydrolyze the glycosidic bonds between the sugars. All of these enzymes exhibit some specificity for the sugar, the glycosidic bond (α or β), and the number of saccharide units in the chain. The monosaccharides formed by glycosidases are transported across the intestinal mucosal cells into the interstitial fluid and subsequently enter the bloodstream. Undigested carbohydrates enter the colon, where they may be fermented by bacteria.

A. Salivary and Pancreatic α-Amylase

The digestion of starch (amylopectin and amylose) begins in the mouth, where chewing mixes the food with saliva. The salivary glands secrete approximately 1 liter of liquid per day into the mouth, containing salivary α-amylase and other components. α-Amylase is an endoglucosidase, which means that it hydrolyzes internal α-1,4 bonds between glucosyl residues at random intervals in the polysaccharide chains (Fig. 27.3). The shortened polysaccharide chains that are formed are called α-dextrins. Salivary α-amylase may be largely inactivated by the acidity of the stomach contents, which contain HCl secreted by the peptic cells.

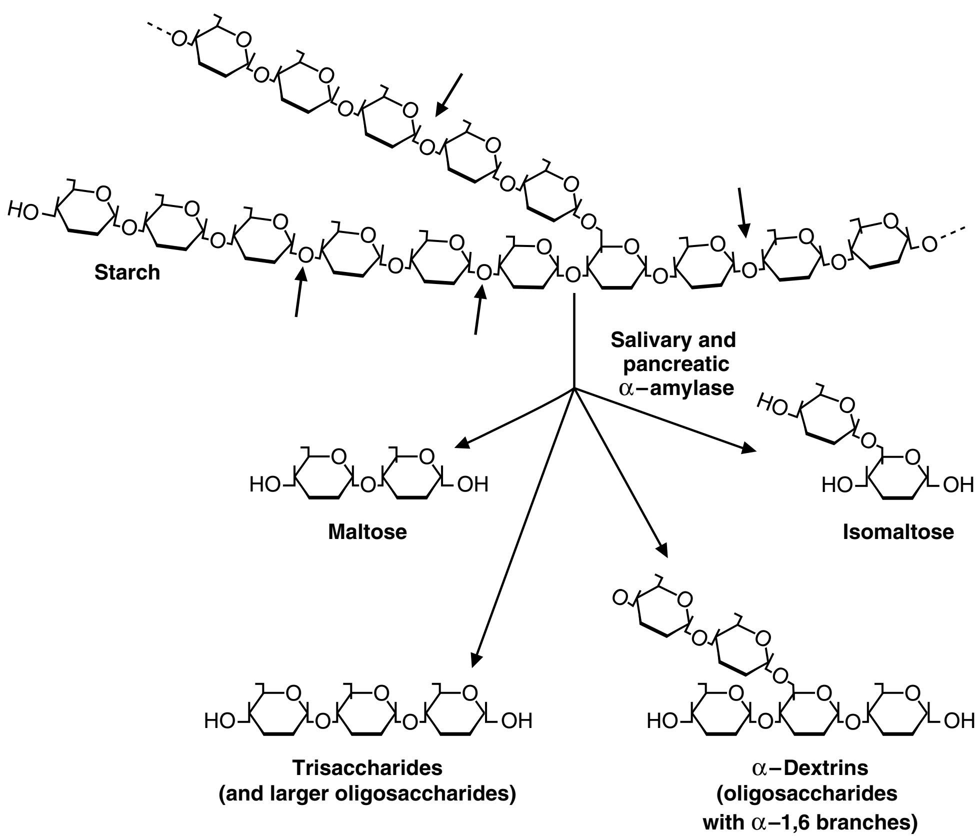


Fig. 27.3. Action of pancreatic and α-amylase.

The acidic gastric juice enters the duodenum, the upper part of the small intestine, where digestion continues. Secretions from the exocrine pancreas (approximately 1.5 liters/day) flow down the pancreatic duct and also enter the duodenum. These secretions contain bicarbonate (HCO_3^-), which neutralizes the acidic pH of stomach contents, and digestive enzymes, including pancreatic α-amylase.

Pancreatic α-amylase continues to hydrolyze the starches and glycogen, forming the disaccharide maltose, the trisaccharide maltotriose, and oligosaccharides. These oligosaccharides, called limit dextrins, are usually four to nine glucosyl units long and contain one or more α-1,6 branches. The two glucosyl residues that contain the α-1,6 glycosidic bond will eventually become the disaccharide isomaltose, but α-amylase does not cleave these branched oligosaccharides all the way down to isomaltose.

α-Amylase has no activity toward sugar containing polymers other than glucose linked by α-1,4 bonds. α-Amylase displays no activity toward the α-1,6- bond at branchpoints and has little activity for the α-1,4 bond at the nonreducing end of a chain.

B. Disaccharidases of the Intestinal Brush-Border Membrane

The dietary disaccharides lactose and sucrose, as well as the products of starch digestion, are converted to monosaccharides by glycosidases attached to the membrane in the brush-border of absorptive cells (Fig. 27.4). The different glycosidase activities are found in four glycoproteins: glucoamylase, the sucrase–maltase complex, the smaller glycoprotein trehalase, and lactase-glucosylceramidase (Table 27.1). These glycosidases are collectively called the small intestinal disaccharidases, although glucoamylase is really an oligosaccharidase.

1. GLUCOAMYLASE

Glucoamylase and the sucrase–isomaltase complex have similar structures and exhibit a great deal of sequence homogeneity (Fig. 27.5). A membrane-spanning domain near the N-terminal attaches the protein to the luminal membrane. The long polypeptide chain forms two globular domains, each with a catalytic site. In glucoamylase, the two catalytic sites have similar activities, with only small differences in substrate specificity. The protein is heavily glycosylated with oligosaccharides that protect it from digestive proteases.

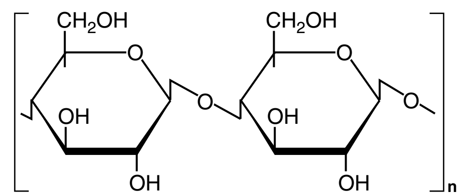
Glucoamylase is an exoglucosidase that is specific for the α–1,4 bonds between glucosyl residues (Fig. 27.6). It begins at the nonreducing end of a polysaccharide or limit dextrin, and sequentially hydrolyzes the bonds to release glucose monosaccharides. It will digest a limit dextrin down to isomaltose, the glucosyl disaccharide with an α–1,6-branch, that is subsequently hydrolyzed principally by the isomaltase activity in the sucrase–isomaltase complex.

2. SUCRASE–ISOMALTASE COMPLEX

The structure of the sucrase–isomaltase complex is very similar to that of glucoamylase, and these two proteins have a high degree of sequence homology. However, after the single polypeptide chain of sucrase–isomaltase is inserted through the membrane and the protein protrudes into the intestinal lumen, an intestinal protease clips it into two separate subunits that remain attached to each other. Each subunit has a catalytic site that differs in substrate specificity from the other through noncovalent interactions. The sucrase–maltase site accounts for approximately 100% of the intestine's ability to hydrolyze sucrose in addition to maltase activity; the isomaltase–maltase site accounts for almost all of the intestine's ability to hydrolyze α-1,6-bonds (Fig. 27.7), in addition to maltase activity. Together, these sites account for approximately 80% of the maltase activity of the small intestine. The remainder of the maltase activity is found in the glucoamylase complex.

Amylase activity in the gut is abundant and is not normally rate limiting for the process of digestion. Alcohol-induced pancreatitis or surgical removal of part of the pancreas can decrease pancreatic secretion. Pancreatic exocrine secretion into the intestine also can be decreased through cystic fibrosis, in which mucus blocks the pancreatic duct, which eventually degenerates. However, pancreatic exocrine secretion can be decreased to 10% of normal and still not affect the rate of starch digestion, because amylases are secreted in the saliva and pancreatic fluid in excessive amounts. In contrast, protein and fat digestion is more strongly affected in cystic fibrosis.

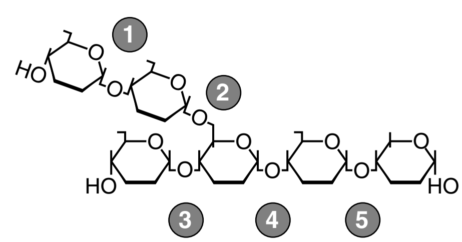
Q:



Can the glycosidic bonds of the structure shown above be hydrolyzed by α-amylose?

Individuals with genetic deficiencies of the sucrase-isomaltase complex show symptoms of sucrose intolerance but are able to digest normal amounts of starch in a meal, without problems. The maltase activity in the glucoamylase complex, and residual activity in the sucrase-isomaltase complex (which is normally present in excess of need) is apparently sufficient to digest normal amounts of dietary starch.

Q:



Which of the bonds in the structure above are hydrolyzed by the sucrase–isomaltase complex? Which by glucoamylase?

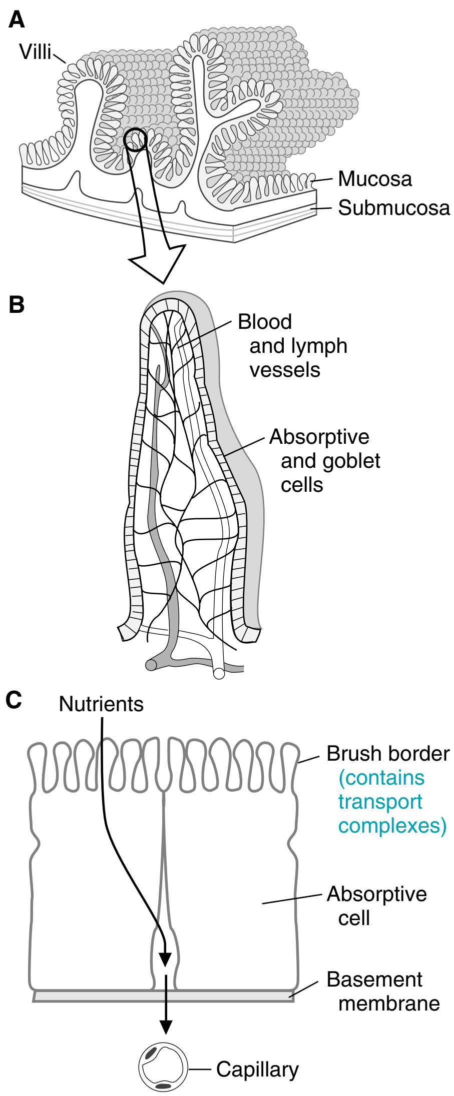


Fig. 27.4. Location of disaccharide complexes in intestinal villi.

Table 27.1. The Different Forms of the Brush Border Glycosidases

Complex	Catalytic Sites	Principal Activities
β-Glucoamylase	α-Glucosidase	Split α-1,4 glycosidic bonds between glucosyl units, beginning sequentially with the residue at the tail end (nonreducing end) of the chain. This is an exoglycosidase. Substrates include amylase, amylopectin, glycogen and maltose.
	α-Glucosidase	Same as above, but with slightly different specificity and affinities for the substrates
Sucrase–Isomaltase	Sucrase–maltase	Splits sucrose, maltose, and maltotriose
	Isomaltase–maltase	Splits α-1, 6 bonds in a number of limit dextrins, as well as the α-1,4 bonds in maltose and maltotriose.
β-Glycosidase	Glucosyl–ceramidase (Phlorizin hydrolase)	Splits β-glycosidic bonds between glucose or galactose and hydrophobic residues, such as the glycolipids glucosylceramide and galactosylceramide
	Lactase	Splits the β-1,4 bond between glucose and galactose. To a lesser extent also splits the β-1,4 bond between some cellulose disaccharides.
Trehalase	Trehalase	Splits bond in trehalose, which is 2 glucosyl units linked α-1,1 through their anomeric carbons.

3. TREHALASE

Trehalase is only half as long as the other disaccharidases and has only one catalytic site. It hydrolyzes the glycosidic bond in trehalose, a disaccharide composed of two glucosyl units linked by an α-bond between their anomeric carbons (Fig. 27.8). Trehalose, which is found in insects, algae, mushrooms, and other fungi, is not currently a major dietary component in the United States. However, unwitting consumption of trehalose can cause nausea, vomiting, and other symptoms of severe gastrointestinal distress if consumed by an individual deficient in the enzyme. Trehalase deficiency was discovered when a woman became very sick after eating mushrooms and was initially thought to have α-amanitin poisoning.

4. β-GLYCOSIDASE COMPLEX (LACTASE-GLUCOSYLCERAMIDASE)

The β-glycosidase complex is another large glycoprotein found in the brush border that has two catalytic sites extending in the lumen of the intestine. However, its primary structure is very different from the other enzymes, and it is attached to the membrane through its carboxyl end by a phosphatidylglycan anchor (see Fig.10.7). The lactase catalytic site hydrolyzes the β-bond connecting glucose and galactose in lactose (a β-galactosidase activity; Fig. 27.9). The major activity of the other catalytic site in humans is the β-bond between glucose or galactose and ceramide in glycolipids (this catalytic site is sometimes called phlorizin hydrolase, named for its ability to hydrolyze an artificial substrate).

5. LOCATION WITHIN THE INTESTINE

The production of maltose, maltotriose, and limit dextrins by pancreatic α-amylase occurs in the duodenum, the most proximal portion of the small intestine. Sucrase–isomaltase activity is highest in the jejunum, where the enzymes can hydrolyze sucrose and the products of starch digestion. β-Glycosidase activity is also highest in the jejenum. Glucoamylase activity progressively increases along the length of the small intestine, and its activity is highest in the ileum. Thus, it

A: No. This polysaccharide is cellulose, which contains β-1,4 glycosidic bonds. Pancreatic and salivary α-amylase cleave only α-1,4 bonds between glucosyl units.

A: Bonds (1) and (3) would first be hydrolyzed by glucoamylase. Bond (2) would require isomaltase. Bonds (4) and (5) could then be hydrolyzed by the sucrase–isomaltase complex, or by the glucoamylase complex, all of which can convert maltotriose and maltose to glucose.

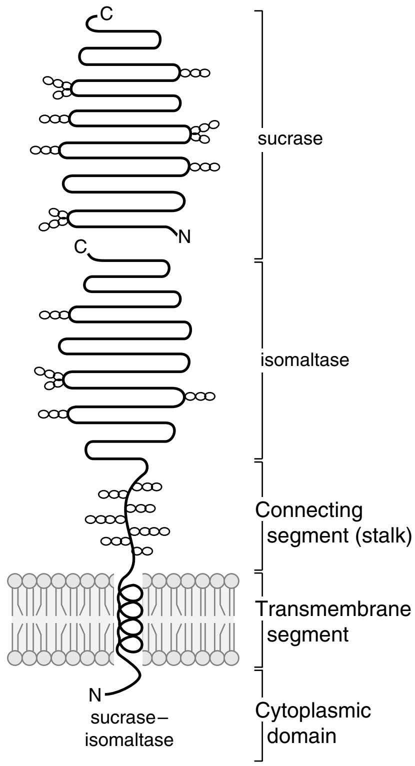


Fig. 27.5. The major portion of the sucrase-isomaltase complex, containing the catalytic sites, protrudes from the absorptive cells into the lumen of the intestine. Other domains of the protein form a connecting segment (stalk), and an anchoring segment that extends through the membrane into the cell. The complex is synthesized as a single polypeptide chain that is split into its two enzyme subunits extracellularly. Each subunit is a domain with a catalytic site (sucrase-maltase) and isomaltase-maltase sites. In spite of their maltase activity, these catalytic sites are often called just sucrase and isomaltase.

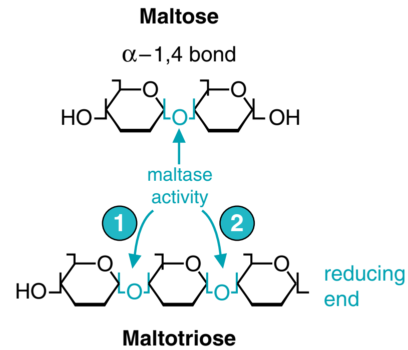


Fig. 27.6. Glucoamylase activity. Glucoamylase is an α-1,4 exoglycosidase, which initiates cleavage at the nonreducing end of the sugar. Thus, for malotriose, the bond labeled 1 will be hydrolyzed first, which frees up the bond at position 2 to be the next one hydrolyzed.

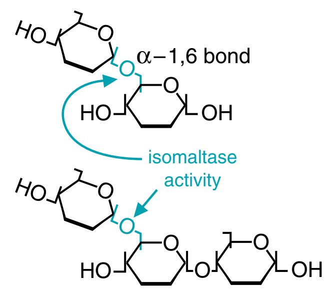


Fig. 27.7. Isomaltase activity. Arrows indicate the α-1,6 bonds that are cleaved.

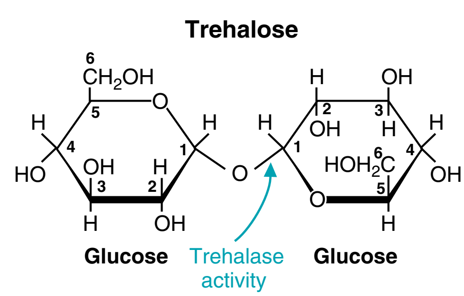


Fig. 27.8. Trehalose. This disaccharide contains two glucose moieties linked by an unusual bond that joins their anomeric carbons. It is cleaved by trehalase.

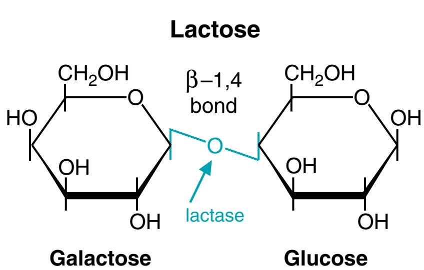


Fig. 27.9. Lactase activity. Lactase is a β-galactosidase. It cleaves the β-galactoside lactose, the major sugar in milk, forming galactose and glucose.

presents a final opportunity for digestion of starch oligomers that have escaped amylase and disaccharidase activities at the more proximal regions of the intestine.

C. Metabolism of Sugars by Colonic Bacteria

Not all of the starch ingested as part of foods is normally digested in the small intestine (Fig. 27.10). Starches high in amylose, or less well hydrated (e.g., starch in dried beans), are resistant to digestion and enter the colon. Dietary fiber and undigested sugars also enter the colon. Here colonic bacteria rapidly metabolize the saccharides, forming gases, short-chain fatty acids, and lactate. The major short-chain fatty acids formed are acetic acid (two carbon), propionic acid (three carbon), and butyric acid (four carbon). The short-chain fatty acids are absorbed by the colonic mucosal cells and can provide a substantial source of energy for these cells. The major gases formed are hydrogen gas (H_2), carbon dioxide (CO_2), and methane (CH_4). These gases are released through the colon, resulting in flatulence, or in the breath. Incomplete products of digestion in the intestines increase the retention of water in the colon, resulting in diarrhea.

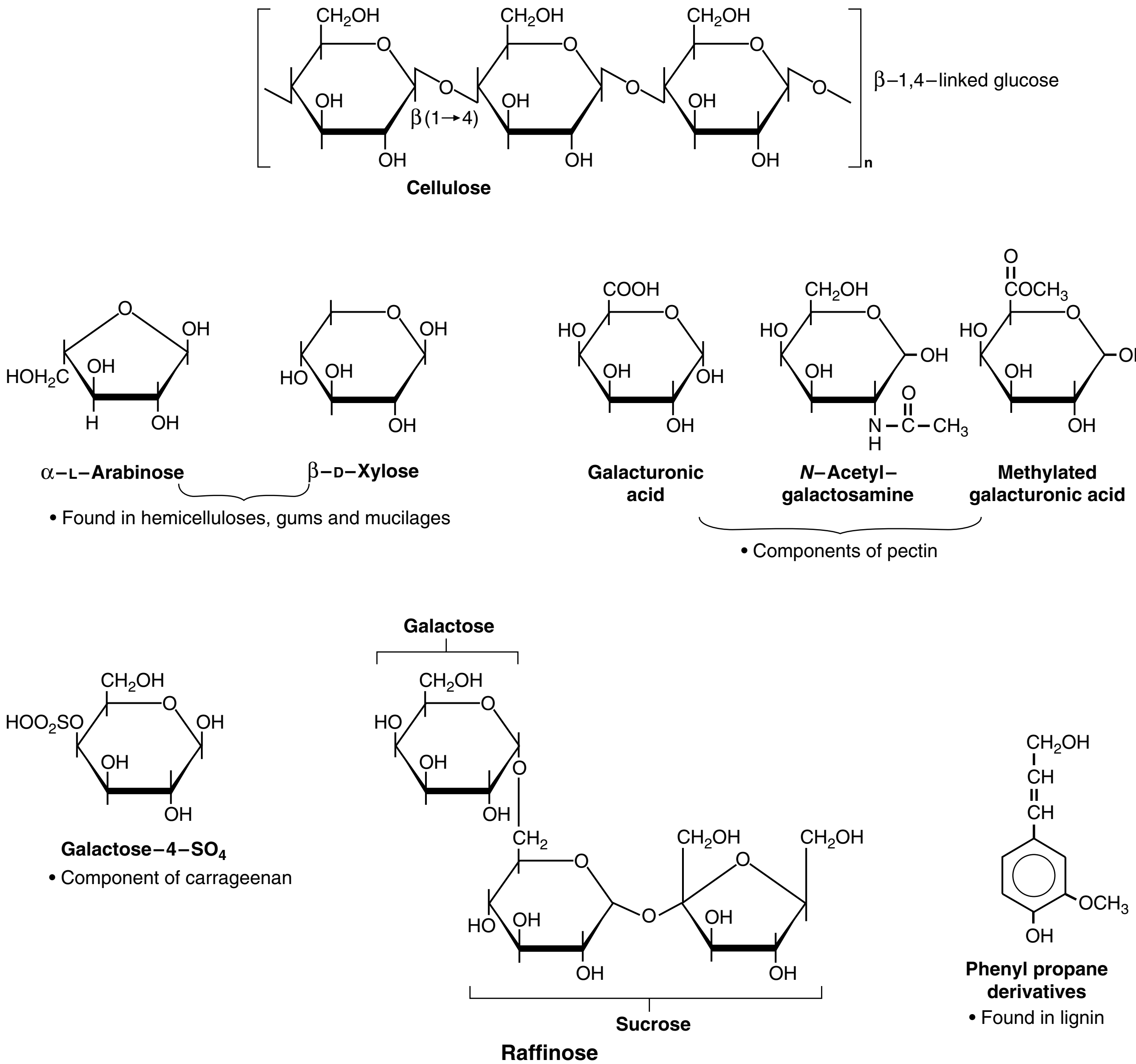


Fig. 27.10. Some indigestible carbohydrates. These compounds are components of dietary fiber.

Nona Melos was given a hydrogen breath test, a test measuring the amount of hydrogen gas released after consuming a test dose of sugar. The association of **Nona Melos's** symptoms with her ingestion of fruit juices suggests that she might have a problem resulting from a low sucrase activity or an inability to absorb fructose. Her ability to thrive and her adequate weight gain suggest that any deficiencies of the sucrase–isomaltase complex must be partial and do not result in a functionally important reduction in maltase activity (maltase activity is also present in the glucoamylase complex). Her urine tested negative for sugar, suggesting the problem is in digestion or absorption, because only sugars that are absorbed and enter the blood can be found in urine. The basis of the hydrogen breath test is that if a sugar is not absorbed, it is metabolized in the intestinal lumen by bacteria that produce various gases, including hydrogen. The test is often accompanied by measurements of the amount of sugar appearing in the blood or feces, and acidity of the feces.

Q: Beans, peas, soybeans, and other leguminous plants contain oligosaccharides with (1,6)-linked galactose residues that cannot be hydrolyzed for absorption, including sucrose with 1, 2, or 3 galactose residues attached (see Fig. 27.10). What is the fate of these polysaccharides in the intestine?

D. Lactose Intolerance

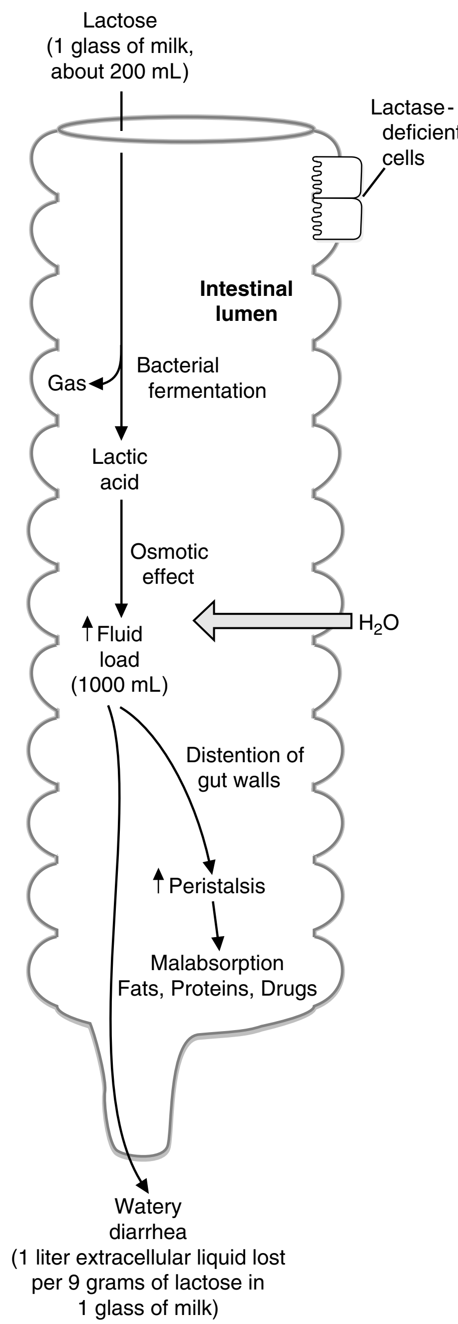
Lactose intolerance refers to a condition of pain, nausea, and flatulence after the ingestion of foods containing lactose, most notably dairy products. Although it is often caused by low levels of lactase, it also can be caused by intestinal injury (defined below).

1. NONPERSISTENT AND PERSISTANT LACTASE

Lactase activity increases in the human from about 6 to 8 weeks of gestation, and it rises during the late gestational period (27–32 weeks) through full term. It remains high for about 1 month after birth and then begins to decline. For most of the world's population, lactase activity decreases to adult levels at approximately 5 to 7 years of age. Adult levels are less than 10% of that present in infants. These populations have adult hypolactasia (formerly called adult lactase deficiency) and exhibit the lactase nonpersistence phenotype. In people who are derived mainly from western Northern Europeans, and milk-dependent Nomadic tribes of Saharan Africa, the levels of lactase remain at, or only slightly below, infant levels throughout adulthood (lactase persistence phenotype). Thus, adult hypolactasia is the normal condition for most of the world's population. (Table 27.2).

2. INTESTINAL INJURY

Intestinal diseases that injure the absorptive cells of the intestinal villi diminish lactase activity along the intestine, producing a condition known as secondary lactase deficiency. Kwashiorkor (protein malnutrition), colitis, gastroenteritis, tropical and



Lactose intolerance can either be the result of a primary deficiency of lactase production in the small bowel (as is the case for **Deria Voider**) or it can be secondary to an injury to the intestinal mucosa, where lactase is normally produced. The lactose that is not absorbed is converted by colonic bacteria to lactic acid, methane gas (CH_4), and H_2 gas (see figure on left). The osmotic effect of the lactose and lactic acid in the bowel lumen is responsible for the diarrhea often seen as part of this syndrome. Similar symptoms can result from sensitivity to milk proteins (milk intolerance) or from the malabsorption of other dietary sugars.

In adults suspected of having a lactase deficiency, the diagnosis is usually made inferentially when avoidance of all dairy products results in relief of symptoms and a rechallenge with these foods reproduces the characteristic syndrome. If the results of these measures are equivocal, however, the malabsorption of lactose can be more specifically determined by measuring the H_2 content of the patient's breath after a test dose of lactose has been consumed.

Deria Voider's symptoms did not appear if she took tablets containing lactase when she ate dairy products.

A: These sugars are not digested well by the human intestine but form good sources of energy for the bacteria of the gut. These bacteria convert the sugars to H_2, lactic acid and short-chain fatty acids. The amount of gas released after a meal containing beans is especially notorious.

Table 27.2. Prevalence of Late-Onset Lactase Deficiency

Group	Prevalence (%)
U.S. population	
Asians	100
American Indians (Oklahoma)	95
Black Americans	81
Mexican Americans	56
White Americans	24
Other Populations	
Ibo, Yoruba (Nigeria)	89
Italians	71
Aborigines (Australia)	67
Greeks	53
Danes	3
Dutch	0

nontropical sprue, and excessive alcohol consumption fall into this category. These diseases also affect other disaccharidases, but sucrase, maltase, isomaltase, and glucoamylase activities are usually present at such excessive levels that there are no pathologic effects. Lactase is usually the first activity lost and the last to recover.

III. DIETARY FIBER

Dietary fiber is the portion of the diet resistant to digestion by human digestive enzymes. It consists principally of plant materials that are polysaccharide derivatives and lignan (see Fig.27.10). The components of fiber are often divided into the categories of soluble and insoluble fiber, according to their ability to dissolve in water. Insoluble fiber consists of three major categories; cellulose, hemicellulose, and lignins. Soluble fiber categories include pectins, mucilages, and gums (Table 27.3). Although human enzymes cannot digest fiber, the bacterial flora in the normal human gut may metabolize the more soluble dietary fibers to gases and short-chain fatty acids, much as they do undigested starch and sugars. Some of these

Table 27.3 Types of Fiber in the Diet

Classical Nomenclature	Classes of compounds	Dietary Sources
Insoluble Fiber		
Cellulose	Polysaccharide composed of glucosyl residues linked β-1,4.	Whole wheat flour, unprocessed bran, cabbage, peas, green beans, wax beans, broccoli, brussel sprouts, cucumber with skin, green peppers, apples, carrots
Hemicelluloses	Polymers of arabinoxylans or galactomannans	Bran cereals, whole grains, brussel sprouts, mustard beans, beet root
Lignin	Noncarbohydrate, polymeric derivatives of phenylpropane	Bran cereals, unprocessed bran, strawberries, eggplant, peas, green beans, radishes
Water Soluble Fiber (or dispersable)		
Pectic Substances	Galactouranans, arabinoglactans, β-glucans, arabinoxylans	Squash, apples, citrus fruits
Gums	Galactomannans, arabinogalactans	Oatmeal, dried beans, cauliflower, green beans, cabbage, carrots, dried peas, potatoes, strawberries
Mucilages	Wide range of branched and substituted galactans	Flax seed, psyllium, mustard seed

fatty acids may be absorbed and used by the colonic epithelial cells of the gut, and some may travel to the liver through the hepatic portal vein. We may obtain as much as 10% of our total calories from compounds produced by bacterial digestion of substances in our digestive tract.

In 2002, the Committee on Dietary Reference Intakes issued new guidelines for fiber ingestion; anywhere from 19 to 38 g/day, depending on age and sex of the individual. No distinction was made between soluble and insoluble fibers. Adult males between the ages of 14 and 50 years require 38 grams of fiber per day. Females from ages 4 to 8 years require 25 g/day; from ages 9 to 16 years, 26 g/day; and from ages 19 to 30, 25 g/day. These numbers are increased during pregnancy and lactation. One beneficial effect of fiber is seen in diverticular disease, in which sacs or pouches may develop in the colon because of a weakening of the muscle and submucosal structures. Fiber is thought to "soften" the stool, thereby reducing pressure on the colonic wall and enhancing expulsion of feces.

Certain types of soluble fiber have been associated with disease prevention. For example, pectins may lower blood cholesterol levels by binding bile acids. β-glucan (obtained from oats) has also been shown, in some studies, to reduce cholesterol levels through a reduction in bile acid resorption in the intestine (see Chapter 34). Pectins also may have a beneficial effect in the diet of individuals with diabetes mellitus by slowing the rate of absorption of simple sugars and preventing high blood glucose levels after meals. However, each of the beneficial effects which have been related to "fiber" are relatively specific for the type of fiber, and the physical form of food which contains the fiber. This factor, along with many others, has made it difficult to obtain conclusive results from studies of the effects of fiber on human health.

Pectins are found in fruits, such as apples. Could this be the basis for the saying "An apple a day keeps the doctor away"?

Carrageenan is a type of fiber derived from seaweed. It is composed of sulfated galactose and galacturonic acid derivatives (see Fig. 27.10). The negatively charged sulfate groups form hydrogen bonds with water and convert the polysaccharide into a gel-like substance. It is added to many foods, such as ice cream and McDonald's McLean burger.

IV. ABSORPTION OF SUGARS

Once the carbohydrates have been split into monosaccharides, the sugars are transported across the intestinal epithelial cells and into the blood for distribution to all tissues. Not all complex carbohydrates are digested at the same rate within the intestine, and some carbohydrate sources lead to a near-immediate rise in blood glucose levels after ingestion, whereas others slowly raise blood glucose levels over an extended period after ingestion. The glycemic index of a food is an indication of how rapidly blood glucose levels rise after consumption. Glucose and maltose have the highest glycemic indices (142, with white bread defined as an index of 100). Table 27.4 indicates the glycemic index for a variety of food types. Although there is no need to memorize this table, note that cornflakes and potatoes have high glycemic indices, whereas yogurt and skim milk have particularly low glycemic indices.

The glycemic response to ingested foods depends not only on the glycemic index of the foods, but also on the fiber and fat content of the food, as well as its method of preparation. Highly glycemic carbohydrates can be consumed before and after exercise, as their metabolism results in a rapid entry of glucose into the blood, where it is then immediately available for muscle use. Low glycemic carbohydrates enter the circulation slowly and can be used to best advantage if consumed before exercise, such that as exercise progresses glucose is slowly being absorbed from the intestine into the circulation, where it can be used to maintain blood glucose levels during the exercise period.

A. Absorption by the Intestinal Epithelium

Glucose is transported through the absorptive cells of the intestine by facilitated diffusion and by Na^+-dependent facilitated transport. (See Chapter 10 for a description of transport mechanisms.) Glucose, therefore, enters the absorptive cells by binding

The dietician explained to **Ann Sulin** the rationale for a person with diabetes to watch their diet. It is important for Ann to add a variety of fibers to her diet. The gel-forming, water-retaining pectins and gums delay gastric emptying and retard the rate of absorption of disaccharides and monosaccharides, thus reducing the rate at which blood glucose levels rise. The glycemic index of foods also needs to be considered for appropriate maintenance of blood glucose levels in diabetic patients. Consumption of a low glycemic index diet results in a lower rise in blood glucose levels after eating, which can be more easily controlled by exogenous insulin. For example, Ms. Sulin is advised to eat pasta and rice (glycemic index of 67 and 65, respectively) instead of potatoes (glycemic index of 80–120, depending on the method of preparation), and to incorporate breakfast cereals composed of wheat bran, barley, and oats into her morning routine.

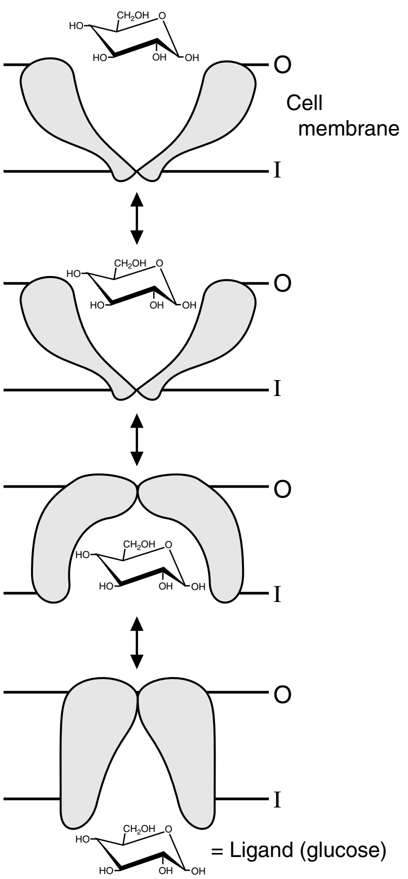


Fig. 27.11. Facilitative transport. Transport of glucose occurs without rotation of the glucose molecule. Multiple groups on the protein bind the hydroxyl groups of glucose and close behind it as it is released into the cell (i.e., the transporter acts like a "gated pore"). O = outside; I = inside.

The glucose molecule is extremely polar and cannot diffuse through the hydrophobic phospholipid bilayer of the cell membrane. Each hydroxyl group of the glucose molecule forms at least two hydrogen bonds with water molecules, and random movement would require energy to dislodge the polar hydroxyl groups from their hydrogen bonds and to disrupt the Van der Waals' forces between the hydrocarbon tails of the fatty acids in the membrane phospholipid.

The epithelial cells of the kidney, which reabsorb glucose into the blood, have Na^+-dependent glucose transporters similar to those of intestinal epithelial cells. They are thus also able to transport glucose against its concentration gradient. Other types of cells use mainly facilitative glucose transporters that carry glucose down its concentration gradient.

Table 27.4 Glycemic Index of Selected Foods, with Values Adjusted to White Bread of 100

Breads		Legumes	
Whole wheat	100	Baked beans (canned)	70
Pumpernickel (whole grain rye)	88	Butter beans	46
Pasta		Garden peas (frozen)	85
Spaghetti, white, boiled	67		
		Kidney beans (dried)	43
Cereal grains		Kidney beans (canned)	74
Barley (pearled)	36	Peanuts	15
Rice (instant, boiled 1 min)	65	Fruit	
Rice, polished (boiled 10–25 min)	81	Apple	52
Sweet corn	80	Apple juice	45
Breakfast cereals		Orange	59
All bran	74	Raisins	93
Cornflakes	121	Sugars	
Muesli	96	Fructose	27
Cookies		Glucose	142
Oatmeal	78	Lactose	57
Plain water crackers	100	Sucrose	83
Root vegetables		Dairy Products	
Potatoes (instant)	120	Ice cream	69
Potato (new,white, boiled)	80	Whole milk	44
Potato chips	77	Skim milk	46
Yam	74	Yogurt	52

to transport proteins, membrane-spanning proteins that bind the glucose molecule on one side of the membrane and release it on the opposite side (Fig. 27.11). Two types of glucose transport proteins are present in the intestinal absorptive cells: the Na^+-dependent glucose transporters and the facilitative glucose transporters (Fig. 27.12).

1. NA$^+$-DEPENDENT TRANSPORTERS

Na^+-dependent glucose transporters, which are located on the luminal side of the absorptive cells, enable these cells to concentrate glucose from the intestinal lumen. A low intracellular Na^+ concentration is maintained by a Na^+,K^+-ATPase on the serosal (blood) side of the cell that uses the energy from ATP cleavage to pump Na^+ out of the cell into the blood. Thus, the transport of glucose from a low concentration in the lumen to a high concentration in the cell is promoted by the cotransport of Na^+ from a high concentration in the lumen to a low concentration in the cell (secondary active transport).

2. FACILITATIVE GLUCOSE TRANSPORTERS

Facilitative glucose transporters, which do not bind Na^+, are located on the serosal side of the cells. Glucose moves via the facilitative transporters from the high concentration inside the cell to the lower concentration in the blood without the expenditure of energy. In addition to the Na^+-dependent glucose transporters, facilitative transporters for glucose also exist on the luminal side of the absorptive cells. The various types of facilitative glucose transporters found in the plasma membranes of cells (referred to as GLUT 1 to GLUT 5), are described in Table 27.5. One common structural theme to these proteins is that they all contain 12 membrane-spanning domains. Note that the sodium-linked transporter on the luminal side of the intestinal epithelial cell is not a member of the GLUT family.

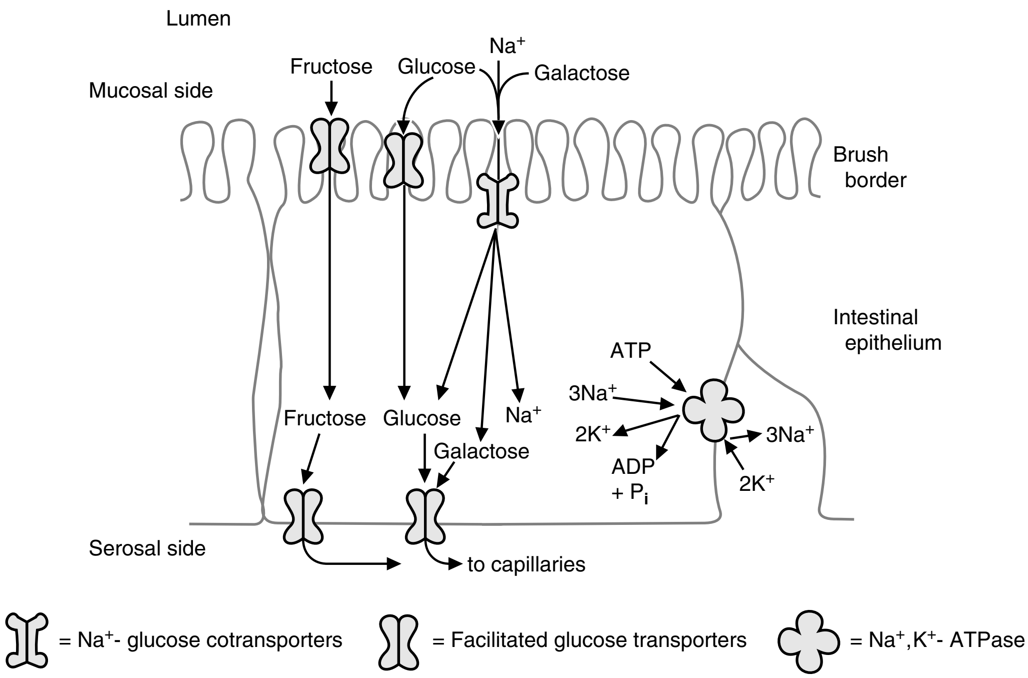


Fig. 27.12. Na^+-dependent and facilitative transporters in the intestinal epithelial cells. Both glucose and fructose are transported by the facilitated glucose transporters on the luminal and serosal sides of the absorptive cells. Glucose and galactose are transported by the Na^+-glucose cotransporters on the luminal (mucosal) side of the absorptive cells.

Table 27.5. Properties of the GLUT 1-GLUT 5 Isoforms of the Glucose Transport Proteins

Transporter	Tissue Distribution	Comments
GLUT 1	Human erythrocyte Blood-brain barrier Blood-retinal barrier Blood-placental barrier Blood-testis barrier	Expressed in cell types with barrier functions; a high-affinity glucose transport system
GLUT 2	Liver Kidney Pancreatic β-cell Serosal surface of Intestinal mucosa cells	A high capacity, low affinity transporter. May be used as the glucose sensor in the pancreas.
GLUT 3	Brain (neurons)	Major transporter in the central nervous system. A high-affinity system.
GLUT 4	Adipose tissue Skeletal muscle Heart muscle	Insulin-sensitive transporter. In the presence of insulin the number of GLUT 4 transporters increases on the cell surface. A high-affinity system
GLUT 5	Intestinal epithelium Spermatozoa	This is actually a fructose transporter.

Genetic techniques have identified additional GLUT transporters (GLUT 7-12), but the role of these transporters has not yet been fully described.

3. GALACTOSE AND FRUCTOSE ABSORPTION THROUGH GLUCOSE TRANSPORTERS

Galactose is absorbed through the same mechanisms as glucose. It enters the absorptive cells on the luminal side via the Na^+-dependent glucose transporters and facilitative glucose transporters and is transported through the serosal side on the facilitative glucose transporters.

Fructose both enters and leaves absorptive epithelial cells by facilitated diffusion, apparently via transport proteins that are part of the GLUT family. The transporter on the luminal side has been identified as GLUT 5. Although this transporter can transport glucose, it has a much higher activity with fructose (see Fig. 27.12). Other fructose transport proteins also may be present. For reasons as yet unknown, fructose is absorbed at a much more rapid rate when it is ingested as sucrose than when it is ingested as a monosaccharide.

B. Transport of Monosaccharides into Tissues

The erythrocyte (red blood cell) is an example of a tissue in which glucose transport is not rate-limiting. Although the glucose transporter (GLUT 1) has a K_m of 1 to 7 mM, it is present in extremely high concentrations, constituting approximately 5% of all membrane proteins. Consequently, as the blood glucose levels fall from a postprandial level of 140 mg/dL (7.5 mM) to the normal fasting level of 80 mg/dL (4.5 mM), or even the hypoglycemic level of 40 mg/dL (2.2 mM), the supply of glucose is still adequate for the rates at which glycolysis and the pentose phosphate pathway operate.

The properties of the GLUT transport proteins differ between tissues, reflecting the function of glucose metabolism in each tissue. In most cell types, the rate of glucose transport across the cell membrane is not rate-limiting for glucose metabolism. This is because the isoform of transporter present in these cell types has a relatively low K_m for glucose (that is, a low concentration of glucose will result in half the maximal rate of glucose transport) or is present in relatively high concentration in the cell membrane so that the intracellular glucose concentration reflects that in the blood. Because the hexokinase isozyme present in these cells has an even lower K_m for glucose (0.05–0.10 mM), variations in blood glucose levels do not affect the intracellular rate of glucose phosphorylation. However, in several tissues, the rate of transport becomes rate limiting when the serum level of glucose is low or when low levels of insulin signal the absence of dietary glucose.

In the liver, the K_m for the glucose transporter (GLUT 2) is relatively high compared with that of other tissues, probably 15 mM or above. This is in keeping with the liver's role as the organ that maintains blood glucose levels. As such, the liver will only convert glucose into other energy storage molecules when the blood glucose levels are high, such as the time immediately after ingestion of a meal. In muscle and adipose tissue, the transport of glucose is greatly stimulated by insulin. The mechanism involves the recruitment of glucose transporters (specifically, GLUT 4) from intracellular vesicles into the plasma membrane (Fig. 27.13). In adipose tissue, the stimulation of glucose transport across the plasma membrane by insulin increases its availability for the synthesis of fatty acids and glycerol from the glycolytic pathway. In skeletal muscle, the stimulation of glucose transport by insulin increases its availability for glycolysis and glycogen synthesis.

V. GLUCOSE TRANSPORT THROUGH THE BLOOD-BRAIN BARRIER AND INTO NEURONS

A hypoglycemic response is elicited by a decrease of blood glucose concentration to some point between 18 and 54 mg/dL (1 and 3 mM). The hypoglycemic response is a result of a decreased supply of glucose to the brain and starts with light-headedness and dizziness and may progress to coma. The slow rate of transport of glucose through the blood-brain barrier (from the blood into the cerebrospinal fluid) at low levels of glucose is thought to be responsible for this neuroglycopenic response. Glucose transport from the cerebrospinal fluid across the plasma membranes of neurons is rapid and is not rate limiting for ATP generation from glycolysis.

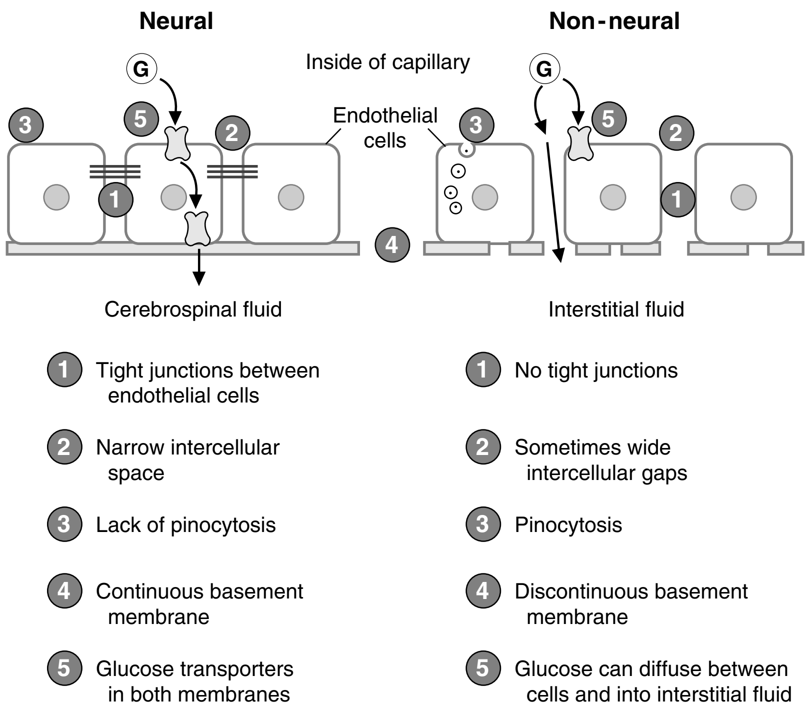


Fig. 27.14. Glucose transport through the capillary endothelium in neural and nonneural tissues. Characteristics of transport in each type of tissue are listed by numbers that refer to the numbers in the drawing. G = glucose.

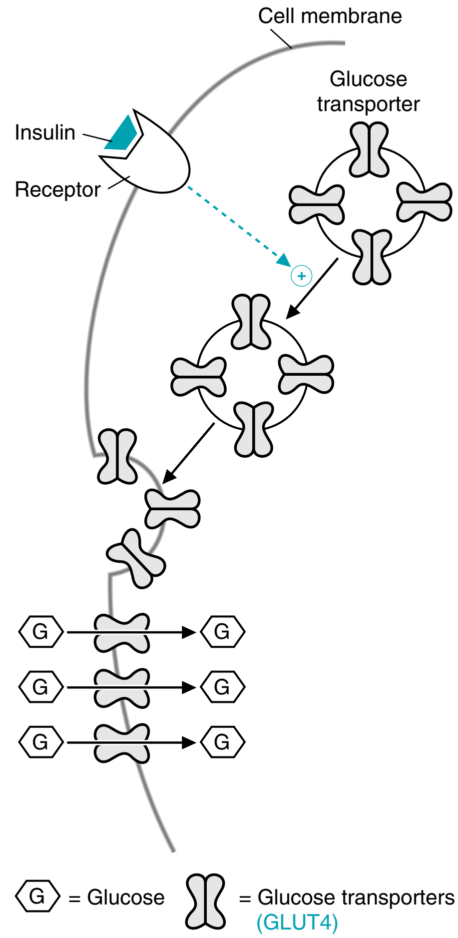


Fig. 27.13. Stimulation by insulin of glucose transport into muscle and adipose cells. Binding of insulin to its cell membrane receptor causes vesicles containing glucose transport proteins to move from inside the cell to the cell membrane.

In the brain, the endothelial cells of the capillaries have extremely tight junctions, and glucose must pass from the blood into the extracellular cerebrospinal fluid by GLUT 1 transporters in the endothelial cell membranes (Fig. 27.14), and then through the basement membrane. Measurements of the overall process of glucose transport from the blood into the brain (mediated by GLUT 3 on neural cells) show a $K_{m,app}$ of 7 to 11 mM, and a maximal velocity not much greater than the rate of glucose utilization by the brain. Thus, decreases of blood glucose below the fasting level of 80 to 90 mg/dL (approximately 5 mM) are likely to significantly affect the rate of glucose metabolism in the brain, because of reduced glucose transport into the brain.

CLINICAL COMMENTS

One of five Americans experiences some form of gastrointestinal discomfort from 30 minutes to 12 hours after ingesting lactose-rich foods. Most become symptomatic when they consume more than 25 g lactose at one time (e.g., 1 pint of milk or its equivalent). **Deria Voider's** symptoms were caused by her "new" diet in this country, which included a glass of milk in addition to the milk she used on her cereal with breakfast each morning.

Management of lactose intolerance includes a reduction or avoidance of lactose-containing foods depending on the severity of the deficiency of intestinal lactase. Hard cheeses (cheddar, Swiss, Jarlsberg) are low in lactose and may be tolerated by patients with only moderate lactase deficiency. Yogurt with "live and active cultures" printed on the package contain bacteria that release free lactases when the bacteria are lysed by gastric acid and proteolytic enzymes. The free lactases then digest the lactose. Commercially available milk products that have been hydrolyzed with a lactase enzyme provide a 70% reduction in total lactose content, which may be adequate to prevent digestive symptoms in mildly affected patients. Tablets and capsules containing lactase are also available and should be taken one-half hour before meals.

Many adults who have a lactase deficiency develop the ability to ingest small amounts of lactose in dairy products without experiencing symptoms. This adaptation probably involves an increase in the population of colonic bacteria that can cleave lactose and not a recovery or induction of human lactase synthesis. For many individuals, dairy products are the major dietary source of calcium, and their complete elimination from the diet can lead to osteoporosis.

Lactose, however, is used as a "filler" or carrying agent in more than 1,000 prescription and over-the-counter drugs in this country. People with lactose intolerance often unwittingly ingest lactose with their medications.

Poorly controlled diabetic patients such as **Ann Sulin** frequently have elevations in serum glucose levels (hyperglycemia). This is often attributable to a lack of circulating, active insulin, which will stimulate glucose uptake (through the recruitment of GLUT 4 transporters from the endoplasmic reticulum to the plasma membrane) by the peripheral tissues (heart, muscle, and adipose tissue). Without uptake by these tissues, glucose tends to accumulate within the bloodstream, leading to hyperglycemia.

The large amount of H_2 produced on fructose ingestion suggested that **Nona Melos's** problem was one of a deficiency in fructose transport into the absorptive cells of the intestinal villi. If fructose were being absorbed properly, the fructose would not have traveled to the colonic bacteria, which metabolized the fructose to generate the hydrogen gas. To confirm the diagnosis, a jejunal biopsy was taken; lactase, sucrase, maltase, and trehalase activities were normal in the jejunal cells. The tissue was also tested for the enzymes of fructose metabolism; these were in the normal range as well. Although Nona had no sugar in her urine, malabsorption of disaccharides can result in their appearance in the urine if damage to the intestinal mucosal cells allows their passage into the interstitial fluid. When Nona was placed on a diet free of fruit juices and other foods containing fructose, she did well and could tolerate small amounts of pure sucrose.

More than 50% of the adult population are estimated to be unable to absorb fructose in high doses (50 g), and more than 10% cannot completely absorb 25 g fructose. These individuals, like those with other disorders of fructose metabolism, must avoid fruits and other foods containing high concentrations of fructose.

BIOCHEMICAL COMMENTS

Cholera is an acute watery diarrheal disorder caused by the water-borne, Gram-negative bacterium *Vibrio cholerae*. It is a disease of antiquity; descriptions of epidemics of the disease date to before 500 BC. During epidemics, the infection is spread by large numbers of vibrio that enter water sources from the voluminous liquid stools and contaminate the environment, particularly in areas of extreme poverty where plumbing and modern waste-disposal systems are primitive or nonexistent.

After being ingested, the *V. cholerae* organisms attach to the brush border of the intestinal epithelium and secrete an exotoxin that binds irreversibly to a specific chemical receptor (G_{M1} ganglioside) on the cell surface. This exotoxin catalyzes an ADP-ribosylation reaction that increases adenylate cyclase activity and thus cAMP levels in the enterocyte. As a result, the normal absorption of sodium, anions, and water from the gut lumen into the intestinal cell is markedly diminished. The exotoxin also stimulates the crypt cells to secrete chloride, accompanied by cations and water, from the bloodstream into the lumen of the gut. The resulting loss of solute-rich diarrheal fluid may, in severe cases, exceed 1 liter/hour, leading to rapid dehydration and even death.

The therapeutic approach to cholera takes advantage of the fact that the Na^+-dependent transporters for glucose and amino acids are not affected by the cholera exotoxin. As a result, coadministration of glucose and Na^+ by mouth results in the uptake of glucose and Na^+, accompanied by chloride and water, thereby partially correcting the ion deficits and fluid loss. Amino acids and small peptides are also adsorbed by Na^+-dependent cotransport involving transport proteins distinct from the Na^+-dependent glucose transporters. Therefore, addition of protein to the glucose–sodium replacement solution enhances its effectiveness and markedly decreases the severity of the diarrhea. Adjunctive antibiotic therapy also shortens the diarrheal phase of cholera but does not decrease the need for the oral replacement therapy outlined earlier.

Suggested Readings

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REVIEW QUESTIONS—CHAPTER 27

1. The facilitative transporter most responsible for transporting fructose from the blood into cells is which of the following?

   (A) GLUT 1
   (B) GLUT 2
   (C) GLUT 3
   (D) GLUT 4
   (E) GLUT 5

2. An alcoholic patient developed a pancreatitis that affected his exocrine pancreatic function. He exhibited discomfort after eating a high-carbohydrate meal. The patient most likely had a reduced ability to digest which of the following?

   (A) Starch
   (B) Lactose
   (C) Fiber
   (D) Sucrose
   (E) Maltose

3. A type I diabetic neglects to take his insulin injections while on a weekend vacation. Cells of which tissue would be most greatly affected by this mistake?

   (A) Brain
   (B) Liver
   (C) Muscle
   (D) Red blood cells
   (E) Pancreas

4. After digestion of a piece of cake that contains flour, milk, and sucrose as its primary ingredients, the major carbohydrate products entering the blood are which of the following?

   (A) Glucose
   (B) Fructose and galactose
   (C) Galactose and glucose
   (D) Fructose and glucose
   (E) Glucose, galactose and fructose

5. A patient has a genetic defect that causes intestinal epithelial cells to produce disaccharidases of much lower activity than normal. Compared with a normal person, after eating a bowl of milk and oatmeal sweetened with table sugar, this patient will exhibit higher levels of which of the following?

   (A) Maltose, sucrose, and lactose in the stool
   (B) Starch in the stool
   (C) Galactose and fructose in the blood
   (D) Glycogen in the muscles
   (E) Insulin in the blood

28 Formation and Degradation of Glycogen

*Glycogen is the storage form of glucose found in most types of cells. It is composed of glucosyl units linked by **α-1,4 glycosidic bonds**, with **α-1,6 branches** occurring roughly every 8 to 10 glucosyl units (Fig. 28.1). The liver and skeletal muscle contain the largest glycogen stores.*

*The formation of glycogen from glucose is an energy-requiring pathway that begins, like most of glucose metabolism, with the phosphorylation of glucose to **glucose 6-phosphate**. Glycogen synthesis from glucose 6-phosphate involves the formation of uridine diphosphate glucose **(UDP-glucose)** and the transfer of glucosyl units from UDP-glucose to the ends of the glycogen chains by the enzyme **glycogen synthase**. Once the chains reach approximately 11 glucosyl units, a **branching enzyme** moves six to eight units to form an α (1,6) branch.*

***Glycogenolysis**, the pathway for glycogen degradation, is not the reverse of the biosynthetic pathway. The degradative enzyme **glycogen phosphorylase** removes glucosyl units one at a time from the ends of the glycogen chains, converting them to **glucose 1-phosphate** without resynthesizing UDP-glucose or UTP. A **debranching enzyme** removes the glucosyl residues near each branchpoint.*

*Liver glycogen serves as a source of **blood glucose**. To generate glucose, the glucose 1-phosphate produced from glycogen degradation is converted to*

Glycogen degradation is a phosphorolysis reaction (breaking of a bond using a phosphate ion as a nucleophile). Enzymes that catalyze phosphorolysis reactions are named phosphorylase. Because more than one type of phosphorylase exists, the substrate usually is included in the name of the enzyme, such as glycogen phosphorylase or purine nucleoside phosphorylase.

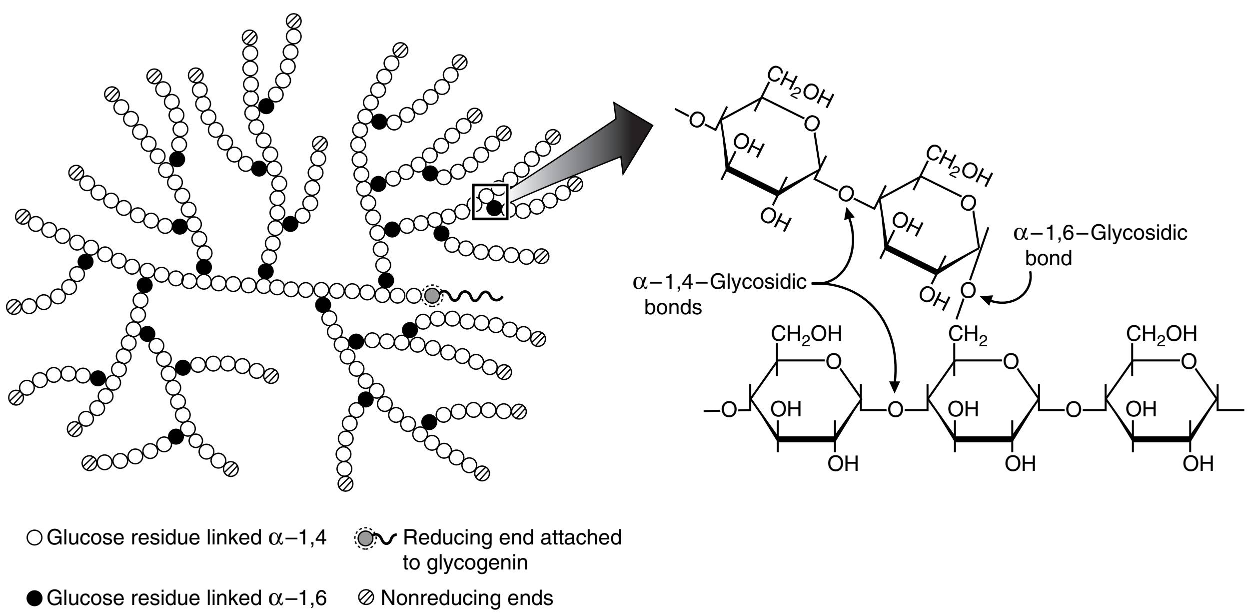


Fig. 28.1. Glycogen structure. Glycogen is composed of glucosyl units linked by α-1,4-glycosidic bonds and α-1,6-glycosidic bonds. The branches occur more frequently in the center of the molecule, and less frequently in the periphery. The anomeric carbon that is not attached to another glucosyl residue (the reducing end) is attached to the protein glycogenin by a glycosidic bond.

glucose 6-phosphate. ***Glucose 6-phosphatase****, an enzyme found only in liver and kidney, converts glucose 6-phosphate to free glucose, which then enters the blood.*

Glycogen synthesis and degradation are regulated in liver by ***hormonal changes*** *that signal the need for blood glucose (see Chapter 26). The body maintains fasting blood glucose levels at approximately 80 mg/dL to ensure that the brain and other tissues that are dependent on glucose for the generation of adenosine triphosphate (ATP) have a continuous supply. The lack of dietary glucose, signaled by a decrease of the* ***insulin/glucagon ratio****, activates liver glycogenolysis and inhibits glycogen synthesis.* ***Epinephrine****, which signals an increased utilization of blood glucose and other fuels for exercise or emergency situations, also activates liver glycogenolysis. The hormones that regulate liver glycogen metabolism work principally through changes in the* ***phosphorylation*** *state of glycogen synthase in the biosynthetic pathway and glycogen phosphorylase in the degradative pathway.*

In skeletal muscle, glycogen supplies glucose 6-phosphate for ATP synthesis in the glycolytic pathway. Muscle glycogen phosphorylase is stimulated during exercise by the increase of adenosine monophosphate ***(AMP)****, an* ***allosteric activator*** *of the enzyme, and also by phosphorylation. The phosphorylation is stimulated by* ***calcium*** *released during contraction, and by the "fight-or-flight" hormone epinephrine. Glycogen synthesis is activated in resting muscles by the elevation of insulin after carbohydrate ingestion.*

The neonate must rapidly adapt to an intermittent fuel supply after birth. Once the umbilical cord is clamped, the supply of glucose from the maternal circulation is interrupted. The combined effect of epinephrine and glucagon on the liver glycogen stores of the neonate rapidly restore glucose levels to normal.

THE WAITING ROOM

A newborn baby girl, **Getta Carbo,** was born after a 38-week gestation. Her mother, a 36-year-old woman, had moderate hypertension during the last trimester of pregnancy related to a recurrent urinary tract infection that resulted in a severe loss of appetite and recurrent vomiting in the month preceding delivery. Fetal bradycardia (slower than normal fetal heart rate) was detected with each uterine contraction of labor, a sign of possible fetal distress.

At birth Getta was cyanotic (a bluish discoloration caused by a lack of adequate oxygenation of tissues) and limp. She responded to several minutes of assisted ventilation. Her Apgar score of 3 was low at 1 minute after birth, but improved to a score of 7 at 5 minutes.

The Apgar score is an objective estimate of the overall condition of the newborn, determined at both 1 and 5 minutes after birth. The best score is 10 (normal in all respects).

Physical examination in the nursery at 10 minutes showed a thin, malnourished female newborn. Her body temperature was slightly low, her heart rate was rapid, and her respiratory rate of 35 breaths/minute was elevated. Getta's birth weight was only 2,100 g, compared with a normal value of 3,300 g. Her length was 47 cm, and her head circumference was 33 cm (low normal). The laboratory reported that Getta's serum glucose level when she was unresponsive was 14 mg/dL. A glucose value below 40 mg/dL (2.5 mM) is considered to be abnormal in newborn infants.

At 5 hours of age, she was apneic (not breathing) and unresponsive. Ventilatory resuscitation was initiated and a cannula placed in the umbilical vein. Blood for a glucose level was drawn through this cannula, and 5 mL of a 20% glucose solution was injected. Getta slowly responded to this therapy.

Jim Bodie, a 19-year-old body builder, was rushed to the hospital emergency room in a coma. One-half hour earlier, his mother had heard a loud crashing sound in the basement where Jim had been lifting weights and completing his daily workout on the treadmill. She found her son on the floor having severe jerking movements of all muscles (a grand mal seizure).

In the emergency room, the doctors learned that despite the objections of his family and friends, Jim regularly used androgens and other anabolic steroids in an effort to bulk up his muscle mass.

On initial physical examination, he was comatose with occasional involuntary jerking movements of his extremities. Foamy saliva dripped from his mouth. He had bitten his tongue and had lost bowel and bladder control at the height of the seizure.

The laboratory reported a serum glucose level of 18 mg/dL (extremely low). The intravenous infusion of 5% glucose (5 g of glucose per 100 mL of solution), which had been started earlier, was increased to 10%. In addition, 50 g glucose was given over 30 seconds through the intravenous tubing.

Jim Bodie's treadmill exercise and most other types of moderate exercise involving whole body movement (running, skiing, dancing, tennis) increase the utilization of blood glucose and other fuels by skeletal muscles. The blood glucose is normally supplied by the stimulation of liver glycogenolysis and gluconeogenesis.

I. STRUCTURE OF GLYCOGEN

Glycogen, the storage form of glucose, is a branched glucose polysaccharide composed of chains of glucosyl units linked by α-1,4 bonds with α-1,6 branches every 8 to 10 residues (see Fig. 28.1). In a molecule of this highly branched structure, only one glucosyl residue has an anomeric carbon that is not linked to another glucose residue. This anomeric carbon at the beginning of the chain is attached to the protein glycogenin. The other ends of the chains are called nonreducing ends (see Chapter 5). The branched structure permits rapid degradation and rapid synthesis of glycogen because enzymes can work on several chains simultaneously from the multiple nonreducing ends.

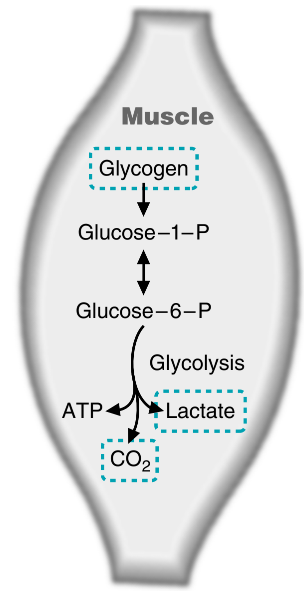
Glycogen is present in tissues as polymers of very high molecular weight (10^7–10^8) collected together in glycogen particles. The enzymes involved in glycogen synthesis and degradation, and some of the regulatory enzymes, are bound to the surface of the glycogen particles.

II. FUNCTION OF GLYCOGEN IN SKELETAL MUSCLE AND LIVER

Glycogen is found in most cell types, where it serves as a reservoir of glucosyl units for ATP generation from glycolysis.

Glycogen is degraded mainly to glucose 1-phosphate, which is converted to glucose 6-phosphate. In skeletal muscle and other cell types, the glucose 6-phosphate enters the glycolytic pathway (Fig. 28.2). Glycogen is an extremely important fuel source for skeletal muscle when ATP demands are high and when glucose 6-phosphate is used rapidly in anaerobic glycolysis. In many other cell types, the small glycogen reservoir serves a similar purpose; it is an emergency fuel source that supplies glucose for the generation of ATP in the absence of oxygen or during restricted blood flow. In general, glycogenolysis and glycolysis are activated together in these cells.

Glycogen serves a very different purpose in liver than in skeletal muscle and other tissues (see Fig. 28.2). Liver glycogen is the first and immediate source of glucose for the maintenance of blood glucose levels. In the liver, the glucose 6-phosphate that is generated from glycogen degradation is hydrolyzed to glucose by glucose 6-phosphatase, an enzyme present only in the liver and kidneys. Glycogen degradation thus provides a readily mobilized source of blood glucose as dietary glucose decreases, or as exercise increases the utilization of blood glucose by muscles.



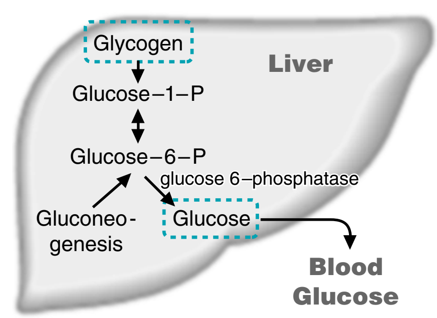


Fig. 28.2. Glycogenolysis in skeletal muscle and liver. Glycogen stores serve different functions in muscle cells and liver. In the muscle and most other cell types, glycogen stores serve as a fuel source for the generation of ATP. In the liver, glycogen stores serve as a source of blood glucose.

Regulation of glycogen synthesis serves to prevent futile cycling and waste of ATP. Futile cycling refers to a situation in which a substrate is converted to a product through one pathway, and the product converted back to the substrate through another pathway. Because the biosynthetic pathway is energy-requiring, futile cycling results in a waste of high-energy phosphate bonds. Thus, glycogen synthesis is activated when glycogen degradation is inhibited, and vice versa.

The pathways of glycogenolysis and gluconeogenesis in the liver both supply blood glucose, and, consequently, these two pathways are activated together by glucagon. Gluconeogenesis, the synthesis of glucose from amino acids and other gluconeogenic precursors (discussed in detail in Chapter 31), also forms glucose 6-phosphate, so that glucose 6-phosphatase serves as a "gateway" to the blood for both pathways (see Fig. 28.2).

III. SYNTHESIS AND DEGRADATION OF GLYCOGEN

Glycogen synthesis, like almost all the pathways of glucose metabolism, begins with the phosphorylation of glucose to glucose 6-phosphate by hexokinase or, in the liver, glucokinase (Fig. 28.3). Glucose 6-phosphate is the precursor of glycolysis, the pentose phosphate pathway, and of pathways for the synthesis of other sugars. In the pathway for glycogen synthesis, glucose 6-phosphate is converted to glucose 1-phosphate by phosphoglucomutase, a reversible reaction.

Glycogen is both formed from and degraded to glucose 1-phosphate, but the biosynthetic and degradative pathways are separate and involve different enzymes (see Fig. 28.3). The biosynthetic pathway is an energy-requiring pathway; high-energy phosphate from UTP is used to activate the glucosyl residues to UDP-glucose (Fig. 28.4). In the degradative pathway, the glycosidic bonds between the glucosy1 residues in glycogen are simply cleaved by the addition of phosphate to produce glucose 1-phosphate (or water to produce free glucose), and UDP-glucose is not resynthesized. The existence of separate pathways for the formation and degradation of important compounds is a common theme in metabolism. Because

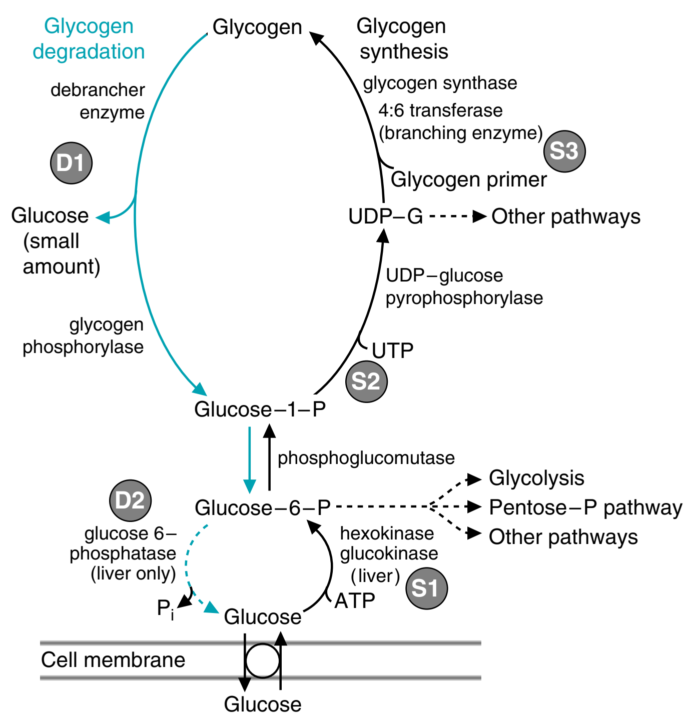


Fig. 28.3. Scheme of glycogen synthesis and degradation. **S1.** Glucose 6-phosphate is formed from glucose by hexokinase in most cells, and glucokinase in the liver. It is a metabolic branchpoint for the pathways of glycolysis, the pentose phosphate pathway, and glycogen synthesis. **S2.** UDP-Glucose (UDP-G) is synthesized from glucose 1-phosphate. UDP-Glucose is the branchpoint for glycogen synthesis and other pathways requiring the addition of carbohydrate units. **S3.** Glycogen synthesis is catalyzed by glycogen synthase and the branching enzyme. **D1.** Glycogen degradation is catalyzed by glycogen phosphorylase and a debrancher enzyme. **D2.** Glucose 6-phosphatase in the liver generates free glucose from glucose 6-phosphate.

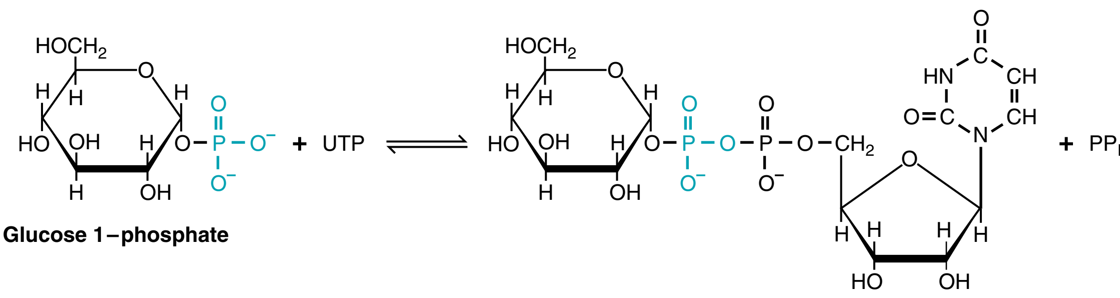


Fig. 28.4. Formation of UDP-glucose. The high-energy phosphate bond of UTP provides the energy for the formation of a high-energy bond in UDP-glucose. Pyrophosphate (PP_i), released by the reaction, is cleaved to 2 P_i.

the synthesis and degradation pathways use different enzymes, one can be activated while the other is inhibited.

A. Glycogen Synthesis

Glycogen synthesis requires the formation of α-1,4-glycosidic bonds to link glucosy1 residues in long chains and the formation of an α-1,6 branch every 8 to 10 residues (Fig. 28.5). Most of glycogen synthesis occurs through the lengthening of the polysaccharide chains of a preexisting glycogen molecule (a glycogen primer) in which the reducing end of the glycogen is attached to the protein glycogenin. To lengthen the glycogen chains, glucosyl residues are added from UDP-glucose to the nonreducing ends of the chain by glycogen synthase. The anomeric carbon of each glucosyl residue is attached in an α-1,4 bond to the hydroxyl on carbon 4 of the terminal glucosyl residue. When the chain reaches 11 residues in length, a 6- to 8-residue piece is cleaved by amylo-4:6-transferase and reattached to a glucosyl unit by an α-1,6 bond. Both chains continue to lengthen until they are long enough to produce two new branches. This process continues, producing highly branched molecules. Glycogen synthase, the enzyme that attaches the glucosyl residues in 1,4-bonds, is the regulated step in the pathway.

The synthesis of new glycogen primer molecules also occurs. Glycogenin, the protein to which glycogen is attached, glycosylates itself (autoglycosylation) by attaching the glucosyl residue of UDP-glucose to the hydroxyl side chain of a serine residue in the protein. The protein then extends the carbohydrate chain (using UDP-glucose as the substrate) until the glucosyl chain is long enough to serve as a substrate for glycogen synthase.

B. Degradation of Glycogen

Glycogen is degraded by two enzymes, glycogen phosphorylase and the debrancher enzyme (Fig. 28.6). The enzyme glycogen phosphorylase starts at the end of a chain and successively cleaves glucosyl residues by adding phosphate to the terminal glycosidic bond, thereby releasing glucose 1-phosphate. However, glycogen phosphorylase cannot act on the glycosidic bonds of the four glucosyl residues closest to a branchpoint because the branching chain sterically hinders a proper fit into the catalytic site of the enzyme. The debrancher enzyme, which catalyzes the removal of the four residues closest to the branchpoint, has two catalytic activities: it acts as a transferase and as an α 1,6-glucosidase. As a transferase, the debrancher first removes a unit containing three glucose residues, and adds it to the end of a longer chain by an α-1,4 bond. The one glucosyl residue remaining at the 1,6-branch is hydrolyzed by the amylo-1,6-glucosidase activity of the debrancher, resulting in the release of free glucose. Thus, one glucose and approximately 7 to 9 glucose 1-phosphate residues are released for every branchpoint.

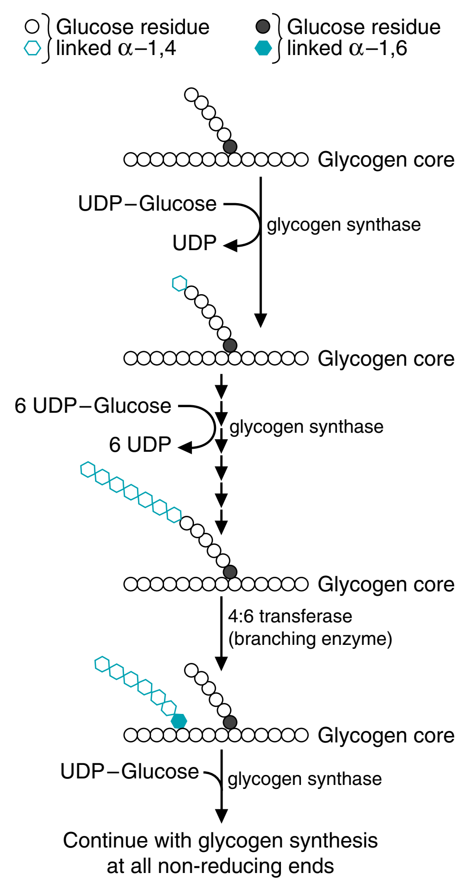


Fig. 28.5. Glycogen synthesis. See text for details.

Branching of glycogen serves two major roles; increased sites for synthesis and degradation, and enhancing the solubility of the molecule.

A genetic defect of lysosomal glucosidase, called type II glycogen storage disease, leads to the accumulation of glycogen particles in large, membrane-enclosed residual bodies, which disrupt the function of liver and muscle cells. Children with this disease usually die of heart failure at a few months of age.

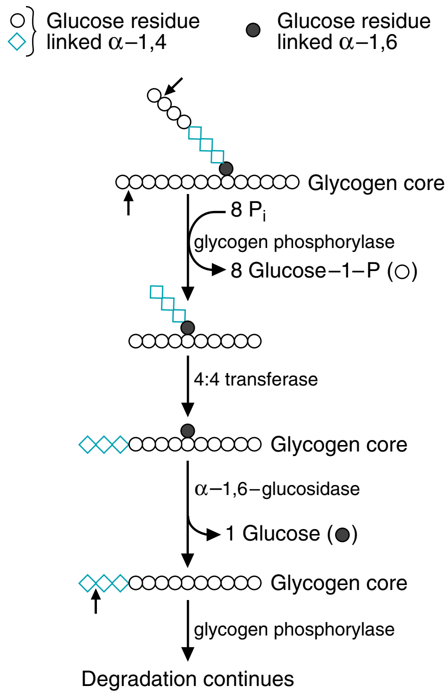


Fig. 28.6. Glycogen degradation. See text for details.

Q: A series of inborn errors of metabolism, the glycogen storage diseases, result from deficiencies in the enzymes of glycogenolysis (see Table 28.1). Muscle glycogen phosphorylase, the key regulatory enzyme of glycogen degradation, is genetically different from liver glycogen phosphorylase, and thus a person may have a defect in one and not the other. Why do you think that a genetic deficiency in muscle glycogen phosphorylase (McArdle's disease) is a mere inconvenience, whereas a deficiency of liver glycogen phosphorylase (Hers' disease) can be lethal?

Some degradation of glycogen also occurs within lysosomes when glycogen particles become surrounded by membranes that then fuse with the lysosomal membranes. A lysosomal glucosidase hydrolyzes this glycogen to glucose.

IV. REGULATION OF GLYCOGEN SYNTHESIS AND DEGRADATION

The regulation of glycogen synthesis in different tissues matches the function of glycogen in each tissue. Liver glycogen serves principally for the support of blood glucose during fasting or during extreme need (e.g., exercise), and the degradative and biosynthetic pathways are regulated principally by changes in

Table 28.1. Glycogen Storage Diseases

Type	Enzyme Affected	Primary Organ Involved	Manifestations[a]
O	Glycogen synthase	Liver	Hypoglycemia, hyperketonemia, FTT[b] early death
I[c]	Glucose 6-phosphatase (Von Gierke's disease)	Liver	Enlarged liver and kidney, growth failure, fasting hypoglycemia, acidosis, lipemia, thrombocyte dysfunction. Hypoglycemia is the most severe.
II	Lysosomal α-glucosidase	All organs with lysosomes	Infantile form: early-onset progressive muscle hypotonia, cardiac failure,death before 2 years; juvenile form: later-onset myopathy with variable cardiac involvement, adultform: limb-girdle muscular dystrophy-like features.Glycogen deposits accumulate in lysosomes.
III	Amylo-1,6-glucosidase (debrancher)	Liver, skeletal muscle, heart	Fasting hypoglycemia; hepatomegaly in infancy in some. myopathic features. Glycogen deposits have short outer branches.
IV	Amylo-4,6-glucosidase (branching enzyme)	Liver	Hepatosplenomegaly; symptoms may arise from a hepatic reaction to the presence of a foreign body (glycogen with long outer branches). Usually fatal.
V	Muscle glycogen phosphorylase (McArdle's disease)	Skeletal muscle	Exercise-induced muscular pain, cramps, and progressive weakness, sometimes with myoglobinuria
VI	Liver glycogen phosphorylase	Liver	Hepatomegaly, mild hypoglycemia, good prognosis
VII	Phosphofructokinase-I	Muscle, red blood cells	As in type V, in addition, enzymopathic hemolysis
IX[d]	Phosphorylase kinase	Liver	As in VI. Hepatomegaly.
X	cAMP-dependant Protein kinase A	Liver	Hepatomegaly.

[a] All of these diseases but type O are characterized by increased glycogen deposits.
[b] FTT = failure to thrive
[c] Glucose 6-phosphatase is composed of several subunits that also transport glucose, glucose 6-phosphate, phosphate, and pyrophosphate across the endoplasmic reticulum membranes. Therefore, there are several subtypes of this disease, corresponding to defects in the different subunits.
[d] There are several subtypes of this disease, corresponding to different mutations and patterns of inheritance.

Table 28.2. Regulation of Liver and Muscle Glycogen Stores[a]

State	Regulators	Response of Tissue
	Liver	
Fasting	Blood: Glucagon ↑ Insulin ↓ Tissue: cAMP ↑	Glycogen degradation ↑ Glycogen synthesis ↓
Carbohydrate meal	Blood: Glucagon ↓ Insulin ↑ Glucose ↑ Tissue: cAMP ↓ Glucose ↑	Glycogen degradation ↓ Glycogen synthesis ↑
Exercise and stress	Blood: Epinephrine ↑ Tissue: cAMP ↑ Ca^{2+}-calmodulin ↑	Glycogen degradation ↑ Glycogen synthesis ↓
	Muscle	
Fasting (rest)	Blood: Insulin ↓	Glycogen synthesis ↓ Glucose transport ↓
Carbohydrate meal (rest)	Blood: Insulin ↑	Glycogen synthesis ↑ Glucose transport ↑
Exercise	Blood: Epinephrine ↑ Tissue: AMP ↑ Ca^{2+}-calmodulin ↑ cAMP ↑	Glycogen synthesis ↓ Glycogen degradation ↑ Glycolysis ↑

[a]↑ = increased compared with other physiologic states; ↓ = decreased compared with other physiologic states.

the insulin/glucagon ratio and by blood glucose levels, which reflect the availability of dietary glucose (Table 28.2). Degradation of liver glycogen is also activated by epinephrine, which is released in response to exercise, hypoglycemia, or other stress situations in which there is an immediate demand for blood glucose. In contrast, in skeletal muscles, glycogen is a reservoir of glucosyl units for the generation of ATP from glycolysis and glucose oxidation. As a consequence, muscle glycogenolysis is regulated principally by AMP, which signals a lack of ATP, and by Ca^{2+} released during contraction. Epinephrine, which is released in response to exercise and other stress situations, also activates skeletal muscle glycogenolysis. The glycogen stores of resting muscle decrease very little during fasting.

A. Regulation of Glycogen Metabolism in Liver

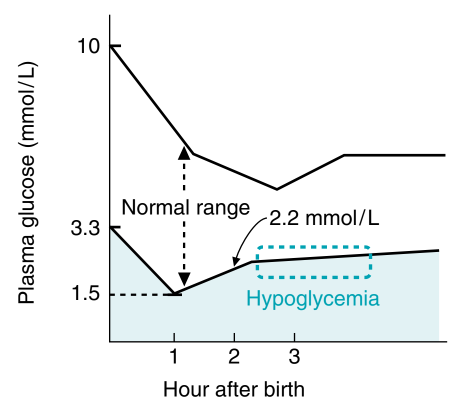
Liver glycogen is synthesized after a carbohydrate meal when blood glucose levels are elevated, and degraded as blood glucose levels decrease. When an individual eats a carbohydrate-containing meal, blood glucose levels immediately increase, insulin levels increase, and glucagon levels decrease (see Fig. 26.8). The increase of blood glucose levels and the rise of the insulin/glucagon ratio inhibit glycogen degradation and stimulate glycogen synthesis. The immediate increased transport of glucose into peripheral tissues, and storage of blood glucose as glycogen, helps to bring circulating blood glucose levels back to the normal 80- to 100-mg/dL range of the fasted state. As the length of time after a carbohydrate-containing meal increases, insulin levels decrease, and glucagon levels increase. The fall of the insulin/glucagon ratio results in inhibition of the biosynthetic pathway and

A: Muscle glycogen is used within the muscle to support exercise. Thus, an individual with McArdle's disease (type V glycogen storage disease) experiences no other symptoms but unusual fatigue and muscle cramps during exercise. These symptoms may be accompanied by myoglobinuria and release of muscle creatine kinase into the blood.

Liver glycogen is the first reservoir for the support of blood glucose levels, and a deficiency in glycogen phosphorylase or any of the other enzymes of liver glycogen degradation can result in fasting hypoglycemia. The hypoglycemia is usually mild because patients can still synthesize glucose from gluconeogenesis (see Table 28.1).

Maternal blood glucose readily crosses the placenta to enter the fetal circulation. During the last 9 or 10 weeks of gestation, glycogen formed from maternal glucose is deposited in the fetal liver under the influence of the insulin-dominated hormonal milieu of that period. At birth, maternal glucose supplies cease, causing a temporary physiologic drop in glucose levels in the newborn's blood, even in normal healthy infants. This drop serves as one of the signals for glucagon release from the newborn's pancreas, which, in turn, stimulates glycogenolysis. As a result, the glucose levels in the newborn return to normal.

Healthy full-term babies have adequate stores of liver glycogen to survive short (12 hours) periods of caloric deprivation provided other aspects of fuel metabolism are normal. Because **Getta Carbo's** mother was markedly anorexic during the critical period when the fetal liver is normally synthesizing glycogen from glucose supplied in the maternal blood, Getta's liver glycogen stores were below normal. Thus, because fetal glycogen is the major source of fuel for the newborn in the early hours of life, Getta became profoundly hypoglycemic within 5 hours of birth because of her low levels of stored carbohydrate.



Plasma glucose levels in the neonate. The normal range of blood glucose levels in the neonate lies between the two black lines. The stippled blue area represents the range of hypoglycemia in the neonate that should be treated. Treatment of neonates with blood glucose levels that fall within the dashed blue box, the zone of clinical uncertainty, is controversial. The units of plasma glucose are given in millimoles/L. Both milligrams/dL (milligrams/100 mL) and millimoles/L are used clinically for the values of blood glucose: 80 mg/dL glucose is equivalent to 5 mmol/L (5 mM). From Mehta A. Arch Dis Child 1994;70:F54.

Q: A patient was diagnosed as an infant with type III glycogen storage disease, a deficiency of debrancher enzyme (see Table 28.1). The patient had hepatomegaly (an enlarged liver) and experienced bouts of mild hypoglycemia. To diagnose the disease, glycogen was obtained from the patient's liver by biopsy after the patient had fasted overnight and compared with normal glycogen. The glycogen samples were treated with a preparation of commercial glycogen phosphorylase and commerical debrancher enzyme. The amounts of glucose 1-phosphate and glucose produced in the assay were then measured. The ratio of glucose 1-phosphate to glucose for the normal glycogen sample was 9:1, and the ratio for the patient was 3:1. Can you explain these results?

Table 28.3. Effect of Fasting on Liver Glycogen Content in the Human

Length of Fast (hours)	Glycogen Content (μmol/g liver)	Rate of Glycogenolysis (μmol/kg-min)
0	300	—
2	260	4.3
4	216	4.3
24	42	1.7
64	16	0.3

activation of the degradative pathway. As a result, liver glycogen is rapidly degraded to glucose, which is released into the blood.

Although glycogenolysis and gluconeogenesis are activated together by the same regulatory mechanisms, glycogenolysis responds more rapidly, with a greater outpouring of glucose. A substantial proportion of liver glycogen is degraded within the first few hours after eating (Table 28.3). The rate of glycogenolysis is fairly constant for the first 22 hours, but in a prolonged fast the rate decreases significantly as the liver glycogen supplies dwindle. Liver glycogen stores are, therefore, a rapidly rebuilt and degraded store of glucose, ever responsive to small and rapid changes of blood glucose levels.

1. NOMENCLATURE CONCERNS WITH ENZYMES METABOLIZING GLYCOGEN

Both glycogen phosphorylase and glycogen synthase will be covalently modified to regulate their activity (Fig. 28.7). When activated by covalent modification, glycogen phosphorylase is referred to as glycogen phosphorylase **a** (remember **a** for active); when the covalent modification is removed, and the enzyme is inactive, it is referred to as glycogen phosphorylase **b**. Glycogen synthase, when not covalently modified is active, and can be designated glycogen synthase **a** or glycogen synthase **I** (the I stands for *independent* of modifiers for activity). When glycogen synthase is covalently modified, it is inactive, in the form of glycogen synthase **b** or glycogen synthase **D** (for *dependent* on a modifier for activity).

2. REGULATION OF LIVER GLYCOGEN METABOLISM BY INSULIN AND GLUCAGON

Insulin and glucagon regulate liver glycogen metabolism by changing the phosphorylation state of glycogen phosphorylase in the degradative pathway and glycogen synthase in the biosynthetic pathway. An increase of glucagon and decrease of insulin during the fasting state initiates a cAMP-directed phosphorylation cascade, which results in the phosphorylation of glycogen phosphorylase to an active enzyme, and the

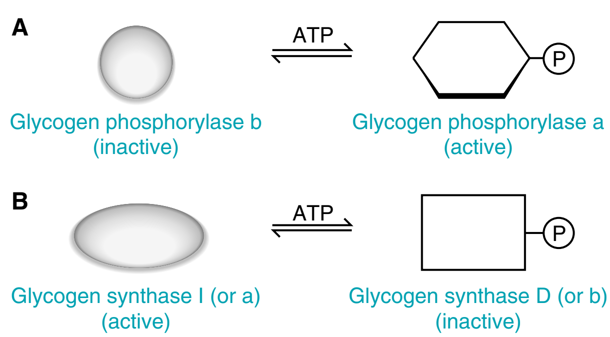


Fig. 28.7. The conversion of active and inactive forms of glycogen phosphorylase (**A**) and glycogen synthase (**B**). Note how the nomenclature changes depending on the phosphorylation and activity state of the enzyme.

phosphorylation of glycogen synthase to an inactive enzyme (Fig. 28.8). As a consequence, glycogen degradation is stimulated, and glycogen synthesis is inhibited.

3. GLUCAGON ACTIVATES A PHOSPHORYLATION CASCADE THAT CONVERTS GLYCOGEN PHOSPHORYLASE b TO GLYCOGEN PHOSPHORYLASE a

Glucagon regulates glycogen metabolism through its intracellular second messenger cAMP and protein kinase A (see Chapter 26). Glucagon, by binding to its cell membrane receptor, transmits a signal through G proteins that activates adenylate cyclase, causing cAMP levels to increase (see Fig. 28.8). cAMP binds to the regulatory subunits of protein kinase A, which dissociate from the catalytic subunits. The catalytic subunits of protein kinase A are activated by the dissociation and phosphorylate the enzyme phosphorylase kinase, activating it. Phosphorylase kinase is the protein kinase that converts the inactive liver glycogen phosphorylase b conformer to the active glycogen phosphorylase a conformer by transferring a phosphate from ATP to a specific serine residue on the

A: With a deficiency of debrancher enzyme, but normal levels of glycogen phosphorylase, the glycogen chains of the patient could be degraded in vivo only to within 4 residues of the branchpoint. When the glycogen samples were treated with the commercial preparation containing normal enzymes, one glucose residue was released for each α-1,6 branch. However, in the patient's glycogen sample, with the short outer branches, three glucose 1-phosphates and one glucose residue were obtained for each α-1,6 branch. Normal glycogen has 8-10 glucosyl residues per branch, and thus gives a ratio of approximately 9 moles of glucose 1-phosphate to 1 mole of glucose.

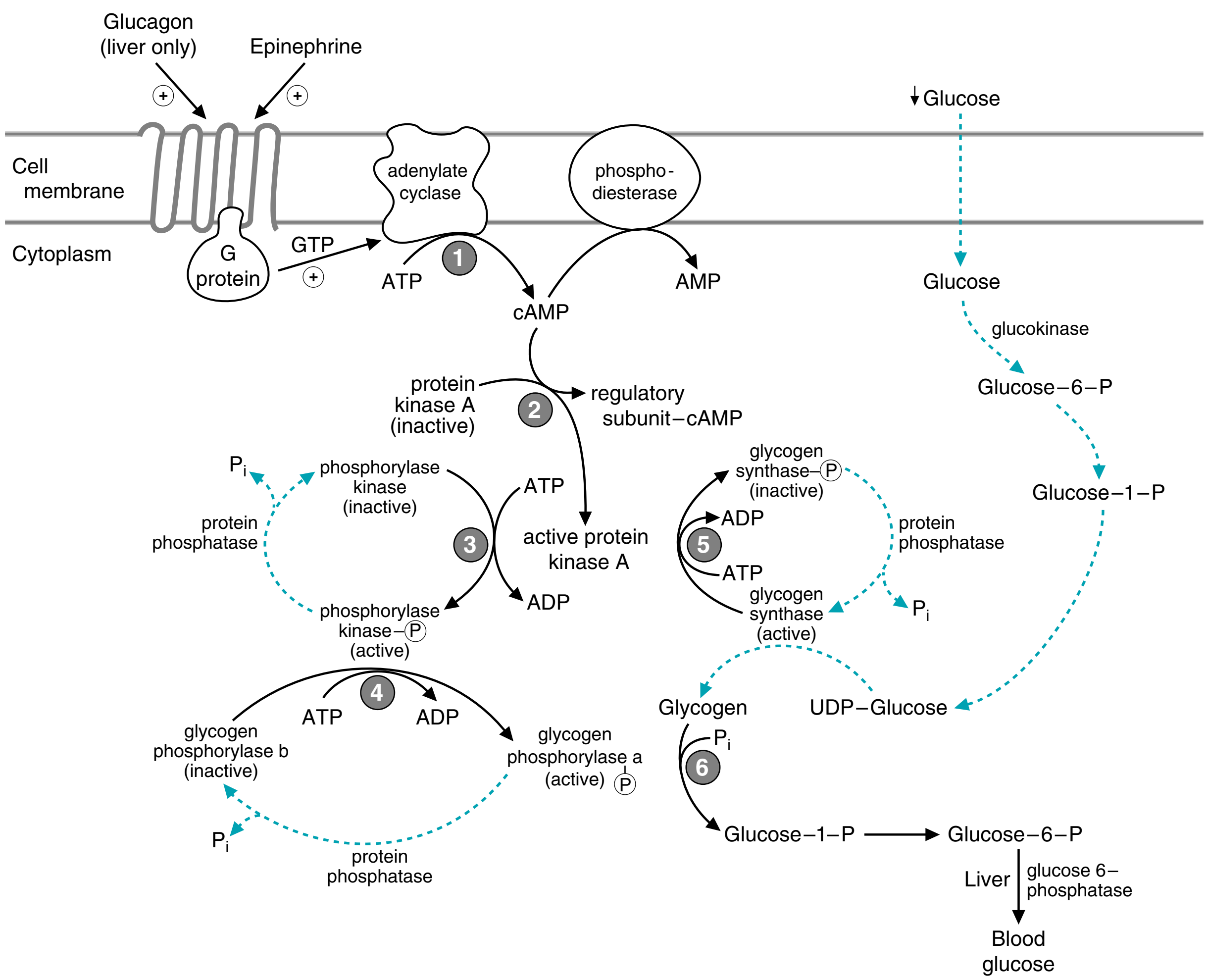


Fig. 28.8. Regulation of glycogen synthesis and degradation in the liver. 1. Glucagon binding to the glucagon receptor or epinephrine binding to a β receptor in the liver activates adenylate cyclase, via G proteins, which synthesizes cAMP from ATP. 2. cAMP binds to protein kinase A (cAMP-dependent protein kinase), thereby activating the catalytic subunits. 3. Protein kinase A activates phosphorylase kinase by phosphorylation. 4. Phosphorylase kinase adds a phosphate to specific serine residues on glycogen phosphorylase b, thereby converting it to the active glycogen phosphorylase a. 5. Protein kinase A also phosphorylates glycogen synthase, thereby decreasing its activity. 6. As a result of the inhibition of glycogen synthase and the activation of glycogen phosphorylase, glycogen is degraded to glucose 1-phosphate. The blue dashed lines denote reactions that are decreased in the livers of fasting individuals.

phosphorylase subunits. As a result of the activation of glycogen phosphorylase, glycogenolysis is stimulated.

To remember whether a particular enzyme has been activated or inhibited by cAMP-dependent phosphorylation, consider whether it makes sense for the enzyme to be active or inhibited under fasting conditions (In a **PHast, PHosphorylate**).

4. INHIBITION OF GLYCOGEN SYNTHASE BY GLUCAGON-DIRECTED PHOSPHORYLATION

When glycogen degradation is activated by the cAMP-stimulated phosphorylation cascade, glycogen synthesis is simultaneously inhibited. The enzyme glycogen synthase is also phosphorylated by protein kinase A, but this phosphorylation results in a less active form, glycogen synthase b.

The phosphorylation of glycogen synthase is far more complex than glycogen phosphorylase. Glycogen synthase has multiple phosphorylation sites and is acted on by up to 10 different protein kinases. Phosphorylation by protein kinase A does not, by itself, inactivate glycogen synthase. Instead, phosphorylation by protein kinase A facilitates the subsequent addition of phosphate groups by other kinases, and these inactivate the enzyme. A term that has been applied to changes of activity resulting from multiple phosphorylation is hierarchical or synergistic phosphorylation-the phosphorylation of one site makes another site more reactive and easier to phosphorylate by a different protein kinase

Most of the enzymes that are regulated by phosphorylation have multiple phosphorylation sites. Glycogen phosphorylase, which has only one serine per subunit, and can be phosphorylated only by phosphorylase kinase, is the exception. For some enzymes, the phosphorylation sites are antagonistic, and phosphorylation initiated by one hormone counteracts the effects of other hormones. For other enzymes, the phosphorylation sites are synergistic, and phosphorylation at one site stimulated by one hormone can act synergistically with phosphorylation at another site.

5. REGULATION OF PROTEIN PHOSPHATASES

At the same time that protein kinase A and phosphorylase kinase are adding phosphate groups to enzymes, the protein phosphatases that remove this phosphate are inhibited. Protein phosphatases remove the phosphate groups, bound to serine or other residues of enzymes, by hydrolysis. Hepatic protein phosphatase-1 (hepatic PP-1), one of the major protein phosphatases involved in glycogen metabolism, removes phosphate groups from phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. During fasting, hepatic PP-1 is inactivated by a number of mechanisms. One is dissociation from the glycogen particle, such that the substrates are no longer available to the phosphatase. A second is the binding of inhibitor proteins, such as the protein called inhibitor-1, which, when phosphorylated by a glucagon (or epinephrine)-directed mechanism, binds to and inhibits phosphatase action. Insulin indirectly activates hepatic PP-1 through its own signal transduction cascade initiated at the insulin receptor tyrosine kinase.

Most of the enzymes that are regulated by phosphorylation also can be converted to the active conformation by allosteric effectors. Glycogen synthase b, the less active form of glycogen synthase, can be activated by the accumulation of glucose 6-phosphate above physiologic levels. The activation of glycogen synthase by glucose 6-phosphate may be important in individuals with glucose 6-phosphatase deficiency, a disorder known as type I or von Gierke's glycogen storage disease. When glucose 6-phosphate produced from gluconeogenesis accumulates in the liver, it activates glycogen synthesis even though the individual may be hypoglycemic and have low insulin levels. Glucose 1-phosphate is also elevated, resulting in the inhibition of glycogen phosphorylase. As a consequence, large glycogen deposits accumulate, and hepatomegaly occurs.

6. INSULIN IN LIVER GLYCOGEN METABOLISM

Insulin is antagonistic to glucagon in the degradation and synthesis of glycogen. The glucose level in the blood is the signal controlling the secretion of insulin and glucagon. Glucose stimulates insulin release and suppresses glucagon release; one increases while the other decreases after a high carbohydrate meal. However, insulin levels in the blood change to a greater degree with the fasting-feeding cycle than the glucagon levels, and thus insulin is considered the principal regulator of glycogen synthesis and degradation. The role of insulin in glycogen metabolism is often overlooked because the mechanisms by which insulin reverses all of the effects of glucagon on individual metabolic enzymes is still under investigation. In addition to the activation of hepatic PP-1 through the insulin receptor tyrosine kinase phosphorylation cascade, insulin may activate the phosphodiesterase that converts cAMP to AMP, thereby decreasing cAMP levels and inactivating protein kinase A. Regardless of the mechanisms involved, insulin is able to reverse all of the effects of glucagon and is the most important hormonal regulator of blood glucose levels.

7. BLOOD GLUCOSE LEVELS AND GLYCOGEN SYNTHESIS AND DEGRADATION

When an individual eats a high-carbohydrate meal, glycogen degradation immediately stops. Although the changes in insulin and glucagon levels are relatively rapid (10–15 minutes), the direct inhibitory effect of rising glucose levels on glycogen degradation is even more rapid. Glucose, as an allosteric effector, inhibits liver glycogen phosphorylase a by stimulating dephosphorylation of this enzyme. As insulin levels rise and glucagon levels fall, cAMP levels decrease and protein kinase A reassociates with its inhibitory subunits and becomes inactive. The protein phosphatases are activated, and phosphorylase a and glycogen synthase b are dephosphorylated. The collective result of these effects is rapid inhibition of glycogen degradation, and rapid activation of glycogen synthesis.

8. EPINEPHRINE AND CALCIUM IN THE REGULATION OF LIVER GLYCOGEN

Epinephrine, the "fight-or-flight" hormone, is released from the adrenal medulla in response to neural signals reflecting an increased demand for glucose. To flee from a dangerous situation, skeletal muscles use increased amounts of blood glucose to generate ATP. As a result, liver glycogenolysis must be stimulated. In the liver, epinephrine stimulates glycogenolysis through two different types of receptors, the α- and β-agonist receptors.

a. EPINEPHRINE ACTING AT THE β-RECEPTORS

Epinephrine, acting at the β-receptors, transmits a signal through G proteins to adenylate cyclase, which increases cAMP and activates protein kinase A. Hence, regulation of glycogen degradation and synthesis in liver by epinephrine and glucagon are similar (see Fig. 28.8).

b. EPINEPHRINE ACTING AT α-RECEPTORS

Epinephrine also binds to α-receptors in the liver. This binding activates glycogenolysis and inhibits glycogen synthesis principally by increasing the Ca^{2+} levels in the liver. The effects of epinephrine at the α-agonist receptor are mediated by the phosphatidylinositol bisphosphate (PIP_2)-Ca^{2+} signal transduction system, one of the principal intracellular second messenger systems employed by many hormones (Fig. 28.9) (see Chapter 11).

In the PIP_2-Ca^{2+} signal transduction system, the signal is transferred from the epinephrine receptor to membrane-bound phospholipase C by G proteins. Phospholipase C hydrolyzes PIP_2 to form diacylglycerol (DAG) and inositol trisphosphate (IP_3). IP_3 stimulates the release of Ca^{2+} from the endoplasmic reticulum. Ca^{2+} and DAG activate protein kinase C. The amount of calcium bound to one of the calcium-binding proteins, calmodulin, is also increased.

Calcium/calmodulin associates as a subunit with a number of enzymes and modifies their activities. It binds to inactive phosphorylase kinase, thereby partially activating this enzyme. (The fully activated enzyme is both bound to the calcium/calmodulin subunit and phosphorylated.) Phosphorylase kinase then phosphorylates glycogen phosphorylase b, thereby activating glycogen degradation. Calcium/calmodulin is also a modifier protein that activates one of the glycogen synthase kinases (calcium/calmodulin synthase kinase). Protein kinase C, calcium/calmodulin synthase kinase, and phosphorylase kinase all phosphorylate glycogen synthase at different serine residues on the enzyme, thereby inhibiting glycogen synthase and thus glycogen synthesis.

An inability of liver and muscle to store glucose as glycogen contributes to the hyperglycemia in patients, such as **Di Abietes,** with type 1 diabetes mellitus and in patients, such as **Ann Sulin,** with type 2 diabetes mellitus. The absence of insulin in type 1 diabetes mellitus patients and the high levels of glucagon result in decreased activity of glycogen synthase. Glycogen synthesis in skeletal muscles of type 1 patients is also limited by the lack of insulin-stimulated glucose transport. Insulin resistance in type 2 patients has the same effect.

An injection of insulin suppresses glucagon release and alters the insulin/glucagon ratio. The result is rapid uptake of glucose into skeletal muscle and rapid conversion of glucose to glycogen in skeletal muscle and liver.

In the neonate, the release of epinephrine during labor and birth normally contributes to restoring blood glucose levels. Unfortunately, **Getta Carbo** did not have adequate liver glycogen stores to support a rise in her blood glucose levels.

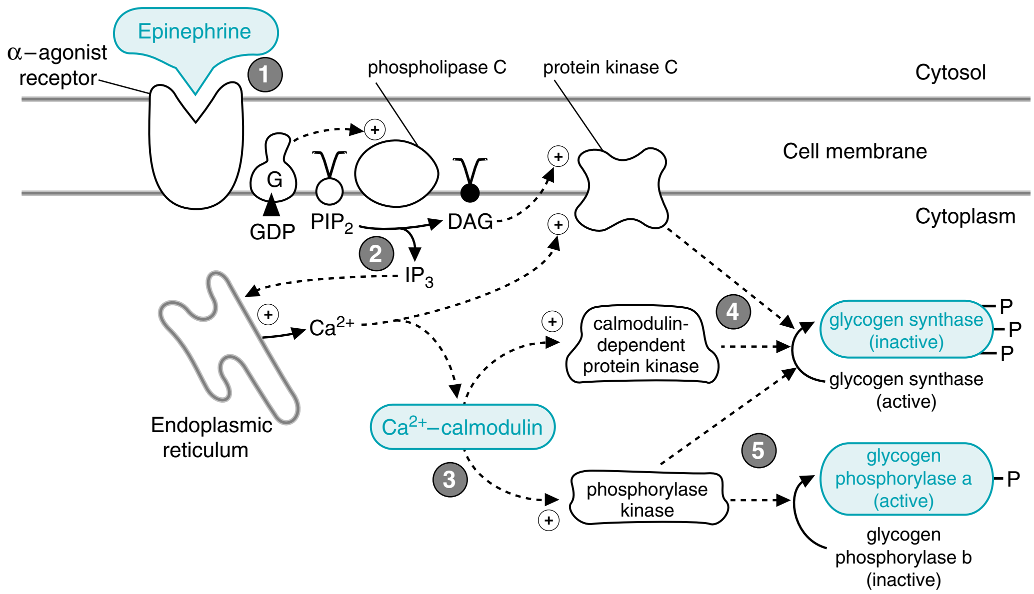


Fig. 28.9. Regulation of glycogen synthesis and degradation by epinephrine and Ca^{2+}. 1. The effect of epinephrine binding to α-agonist receptors in liver transmits a signal via G proteins to phospholipase C, which hydrolyzes PIP_2 to DAG and IP_3. 2. IP_3 stimulates the release of Ca^{2+} from the endoplasmic reticulum. 3. Ca^{2+} binds to the modifier protein calmodulin, which activates calmodulin-dependent protein kinase and phosphorylase kinase. Both Ca^{2+} and DAG activate protein kinase C. 4. These three kinases phosphorylate glycogen synthase at different sites and decrease its activity. 5. Phosphorylase kinase phosphorylates glycogen phosphorylase b to the active form. It therefore activates glycogenolysis as well as inhibiting glycogen synthesis.

The effect of epinephrine in the liver, therefore, enhances or is synergistic with the effects of glucagon. Epinephrine release during bouts of hypoglycemia or during exercise can stimulate hepatic glycogenolysis and inhibit glycogen synthesis very rapidly.

B. Regulation of Glycogen Synthesis and Degradation in Skeletal Muscle

The regulation of glycogenolysis in skeletal muscle is related to the availability of ATP for muscular contraction. Skeletal muscle glycogen produces glucose

Jim Bodie gradually regained consciousness with continued infusions of high-concentration glucose titrated to keep his serum glucose level between 120 and 160 mg/dL. Although he remained somnolent and moderately confused over the next 12 hours, he was eventually able to tell his physicians that he had self-injected approximately 80 units of regular (short-acting) insulin every 6 hours while eating a high-carbohydrate diet for the last 2 days preceding his seizure. Normal subjects under basal conditions secrete an average of 40 units of insulin daily. He had last injected insulin just before exercising. An article in a body-building magazine that he had recently read cited the anabolic effects of insulin on increasing muscle mass. He had purchased the insulin and necessary syringes from the same underground drug source from whom he regularly bought his anabolic steroids.

Normally, muscle glycogenolysis supplies the glucose required for the kinds of high-intensity exercise that require anaerobic glycolysis, such as weight-lifting. Jim's treadmill exercise also uses blood glucose, which is supplied by liver glycogenolysis. The high serum insulin levels, resulting from the injection he gave himself just before his workout, activated both glucose transport into skeletal muscle and glycogen synthesis, while inhibiting glycogen degradation. His exercise, which would continue to use blood glucose, could normally be supported by breakdown of liver glycogen. However, glycogen synthesis in his liver was activated, and glycogen degradation was inhibited by the insulin injection.

1-phosphate and a small amount of free glucose. Glucose 1-phosphate is converted to glucose 6-phosphate, which is committed to the glycolytic pathway; the absence of glucose 6-phosphatase in skeletal muscle prevents conversion of the glucosyl units from glycogen to blood glucose. Skeletal muscle glycogen is therefore degraded only when the demand for ATP generation from glycolysis is high. The highest demands occur during anaerobic glycolysis, which requires more moles of glucose for each ATP produced than oxidation of glucose to CO_2 (see Chapter 22). Anaerobic glycolysis occurs in tissues that have fewer mitochondria, a higher content of glycolytic enzymes, and higher levels of glycogen, or fast-twitch glycolytic fibers. It occurs most frequently at the onset of exercise–before vasodilation occurs to bring in blood-borne fuels. The regulation of skeletal muscle glycogen degradation therefore must respond very rapidly to the need for ATP, indicated by the increase in AMP.

The regulation of skeletal muscle glycogen synthesis and degradation differs from that in liver in several important respects: (*a*) glucagon has no effect on muscle, and thus glycogen levels in muscle do not vary with the fasting/feeding state; (*b*) AMP is an allosteric activator of the muscle isozyme of glycogen phosphorylase, but not liver glycogen phosphorylase (Fig. 28.10); (*c*) the effects of Ca^{2+} in muscle result principally from the release of Ca^{2+} from the sarcoplasmic reticulum after neural stimulation, and not from epinephrine-stimulated uptake; (*d*) glucose is not a physiologic inhibitor of glycogen phosphorylase a in muscle; (*e*) glycogen is a stronger feedback inhibitor of muscle glycogen synthase than of liver glycogen synthase, resulting in a smaller amount of stored glycogen per gram weight of muscle tissue. However, the effects of epinephrine-stimulated phosphorylation by protein kinase A on skeletal muscle glycogen degradation and glycogen synthesis are similar to those occurring in liver (see Fig. 28.8).

Muscle glycogen phosphorylase is a genetically distinct isoenzyme of liver glycogen phosphorylase and contains an amino acid sequence that has a purine nucleotide

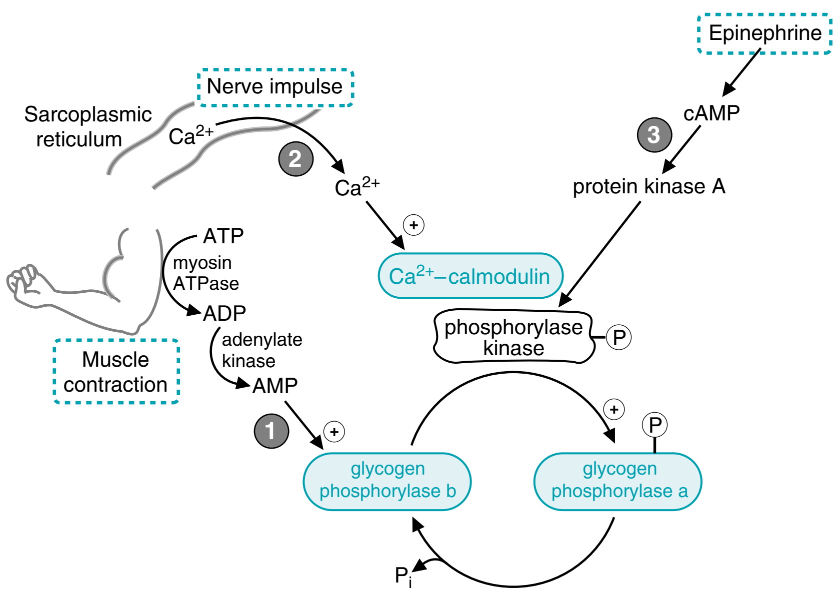


Fig. 28.10. Activation of muscle glycogen phosphorylase during exercise. Glycogenolysis in skeletal muscle is initiated by muscle contraction, neural impulses, and epinephrine. 1. AMP produced from the degradation of ATP during muscular contraction allosterically activates glycogen phosphorylase b. 2. The neural impulses that initiate contraction release Ca^{2+} from the sarcoplasmic reticulum. The Ca^{2+} binds to calmodulin, which is a modifier protein that activates phosphorylase kinase. 3. Phosphorylase kinase is also activated through phosphorylation by protein kinase A. The formation of cAMP and the resultant activation of protein kinase A are initiated by the binding of epinephrine to plasma membrane receptors.

binding site. When AMP binds to this site, it changes the conformation at the catalytic site to a structure very similar to that in the phosphorylated enzyme (see Fig. 9.9). Thus, hydrolysis of ATP to ADP and the consequent increase of AMP generated by adenylate kinase during muscular contraction can directly stimulate glycogenolysis to provide fuel for the glycolytic pathway. AMP also stimulates glycolysis by activating phosphofructokinase-1, so this one effector activates both glycogenolysis and glycolysis. The activation of the calcium/calmodulin subunit of phosphorylase kinase by the Ca^{2+} released from the sarcoplasmic reticulum during muscle contraction also provides a direct and rapid means of stimulating glycogen degradation.

CLINICAL COMMENTS

Getta Carbo's hypoglycemia illustrates the importance of glycogen stores in the neonate. At birth, the fetus must make two major adjustments in the way fuels are used: it must adapt to using a greater variety of fuels than were available in utero, and it must adjust to intermittent feeding. In utero, the fetus receives a relatively constant supply of glucose from the maternal circulation through the placenta, producing a level of glucose in the fetus that approximates 75% of maternal blood levels. With regard to the hormonal regulation of fuel utilization in utero, fetal tissues function in an environment dominated by insulin, which promotes growth. During the last 10 weeks of gestation, this hormonal milieu leads to glycogen formation and storage. At birth, the infant's diet changes to one containing greater amounts of fat and lactose (galactose and glucose in equal ratio), presented at intervals rather than in a constant fashion. At the same time, the neonate's need for glucose will be relatively larger than that of the adult because the newborn's ratio of brain to liver weight is greater. Thus, the infant has even greater difficulty in maintaining glucose homeostasis than the adult.

At the moment that the umbilical cord is clamped, the normal neonate is faced with a metabolic problem: the high insulin levels of late fetal existence must be quickly reversed to prevent hypoglycemia. This reversal is accomplished through the secretion of the counterregulatory hormones epinephrine and glucagon. Glucagon release is triggered by the normal decline of blood glucose after birth. The neural response that stimulates the release of both glucagon and epinephrine is activated by the anoxia, cord clamping, and tactile stimulation that are part of a normal delivery. These responses have been referred to as the "normal sensor function" of the neonate.

Within 3 to 4 hours of birth, these counterregulatory hormones reestablish normal serum glucose levels in the newborn's blood through their glycogenolytic and gluconeogenic actions. The failure of Getta's normal "sensor function" was partly the result of maternal malnutrition, which resulted in an inadequate deposition of glycogen in Getta's liver before birth. The consequence was a serious degree of postnatal hypoglycemia.

The ability to maintain glucose homeostasis during the first few days of life also depends on the activation of gluconeogenesis and the mobilization of fatty acids. Fatty acid oxidation in the liver not only promotes gluconeogenesis (see Chapter 31) but generates ketone bodies. The neonatal brain has an enhanced capacity to use ketone bodies relative to that of infants (fourfold) and adults (40-fold). This ability is consistent with the relatively high fat content of breast milk.

Jim Bodie attempted to build up his muscle mass with androgens and with insulin. The anabolic (nitrogen-retaining) effects of androgens on skeletal muscle cells enhance muscle mass by increasing amino acid flux into muscle and by stimulating protein synthesis. Exogenous insulin has the potential to increase muscle mass by similar actions and also by increasing the content of muscle glycogen.

The most serious side effect of exogenous insulin administration is the development of severe hypoglycemia, such as occurred in **Jim Bodie**. The immediate adverse effect relates to an inadequate flow of fuel (glucose) to the metabolizing brain. When hypoglycemia is extreme, the patient may suffer a seizure and, if the hypoglycemia worsens, may lapse into a coma and die. If untreated, irreversible brain damage occurs in those who survive.

BIOCHEMICAL COMMENTS

The regulatory effect of insulin is frequently described as one of activating protein phosphatases. The effects of insulin on the regulation of hepatic and skeletal PP-1 are complex and not yet fully understood.

PP-1 is targeted to glycogen particles by four tissue-specific targeting factors: G_M is found in striated muscle; G_L is found in liver; PTG (protein targeting to glycogen) is found in almost all tissues; and R6 is also found in most tissues. The targeting factors bind to PP-1 and glycogen and localize the PP-1 to the glycogen particles, where the enzyme will be physically close to the regulated enzymes of glycogen metabolism, phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. Regulation of the phosphatase will involve complex interactions between the target enzymes, the targeting subunit, the phosphatase, and protein inhibitor I. The interactions are also tissue specific in the case of G_M and G_L.

A simplistic view of hepatic PP-1 regulation is as follows. PP-1 is bound to G_L and the glycogen particle. Glycogen phosphorylase a binds to the complex, and in so doing alters the conformation of PP-1, rendering it inactive. When glucose levels rise in the blood (for example, after eating a meal), the glucose is transported into the liver cells via GLUT 2 transporters, and the intracellular glucose level increases. Glucose can bind to glycogen phosphorylase a, which relieves the inhibition of PP-1, and glycogen phosphorylase a will be converted to glycogen phosphorylase b by active PP-1. Additionally, as the intracellular glucose is converted to glucose 6-phosphate by glucokinase, the increase in glucose-6-P levels activates PP-1 to dephosphorylate glycogen synthase, thereby activating the glycogen synthesizing enzyme. The complicated view of hepatic PP-1 regulation also must take into account the PTG-PP-1 interactions (PTG is also expressed in the liver) and the kinases that are activated by either insulin or glucagon/epinephrine, which lead to alterations in glycogen metabolizing enzyme activities.

In contrast to hepatic regulation, muscle regulation of PP-1 activity via G_M is directly responsive to phosphorylation by kinases. A phosphorylation event that appears to be critical is that of ser-67 in G_M. Phosphorylation of ser-67 by the cAMP-dependent protein kinase leads to a dissociation of PP-1 from G_M, and, therefore, the phosphatase is removed from its substrates and cannot reverse the phosphorylation of the target enzymes. If ser-67 is altered to a threonine, the phosphorylation at that site is blocked, and PP-1 does not dissociate from G_M. This indicates the importance of the phosphorylation event in regulating PP-1 action in the muscle.

Future work will be needed before a complete understanding of how insulin reverses glucagon/epinephrine stimulation of glycogenolysis is obtained.

Suggested Readings

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REVIEW QUESTIONS—CHAPTER 28

1. The degradation of glycogen normally produces which of the following?

   (A) More glucose than glucose 1-phosphate
   (B) More glucose 1-phosphate than glucose
   (C) Equal amounts of glucose and glucose 1-phosphate
   (D) Neither glucose or glucose 1-phosphate
   (E) Only glucose 1-phosphate

2. A patient has large deposits of liver glycogen, which, after an overnight fast, had shorter than normal branches. This abnormality could be caused by a defective form of which of the following proteins or activities?

   (A) Glycogen phosphorylase
   (B) Glucagon receptor
   (C) Glycogenin
   (D) Amylo 1,6 glucosidase
   (E) Amylo 4,6 transferase

3. An adolescent patient with a deficiency of muscle phosphorylase was examined while exercising her forearm by squeezing a rubber ball. Compared with a normal person performing the same exercise, this patient would exhibit which of the following?

   (A) Exercise for a longer time without fatigue
   (B) Have increased glucose levels in blood drawn from her forearm
   (C) Have decreased lactate levels in blood drawn from her forearm
   (D) Have lower levels of glycogen in biopsy specimens from her forearm muscle
   (E) Hyperglycemia

4. In a glucose tolerance test, an individual in the basal metabolic state ingests a large amount of glucose. If the individual is normal, this ingestion should result in which of the following?

   (A) An enhanced glycogen synthase activity in the liver
   (B) An increased ratio of glycogen phosphorylase a to glycogen phosphorylase b in the liver
   (C) An increased rate of lactate formation by red blood cells
   (D) An inhibition of protein phosphatase I activity in the liver
   (E) An increase of cAMP levels in the liver

5. Consider a type 1 diabetic who has neglected to take insulin for the past 72 hours and has not eaten much as well. Which of the following best describes the activity level of hepatic enzymes involved in glycogen metabolism under these conditions?

	Glycogen Synthase	Phosphorylase Kinase	Glycogen Phosphorylase
(A)	Active	Active	Active
(B)	Active	Active	Inactive
(C)	Active	Inactive	Inactive
(D)	Inactive	Inactive	Inactive
(E)	Inactive	Active	Inactive
(F)	Inactive	Active	Active

29 Pathways of Sugar Metabolism: Pentose Phosphate Pathway, Fructose, and Galactose Metabolism

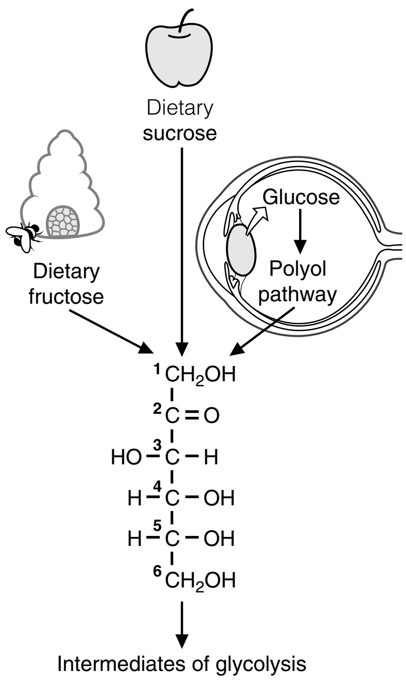

Fig. 29.1. Fructose. The sugar fructose is found in the diet as the free sugar in foods such as honey or as a component of the disaccharide sucrose in fruits and sweets. It also can be synthesized from glucose via the polyol pathway. In the lens of the eye, the polyol pathway contributes to the formation of cataracts. Fructose is metabolized by conversion to intermediates of glycolysis.

Enzymes can generally use either NADPH or NADH, but not both. Reactions requiring the input of electrons as hydride ions are usually catalyzed by enzymes specific for NADPH.

Glucose is at the center of carbohydrate metabolism and is the major dietary sugar. Other sugars in the diet are converted to intermediates of glucose metabolism, and their fates parallel that of glucose. When carbohydrates other than glucose are required for the synthesis of diverse compounds such as lactose, glycoproteins, or glycolipids, they are synthesized from glucose.

***Fructose**, the second most common sugar in the adult diet, is ingested principally as the monosaccharide or as part of **sucrose** (Fig. 29.1). It is metabolized principally in the liver (and to a lesser extent in the small intestine and kidney) by phosphorylation at the 1-position to form **fructose 1-P**, followed by conversion to intermediates of the glycolytic pathway. The major products of its metabolism in liver are, therefore, the same as glucose (including lactate, blood glucose, and glycogen). **Essential fructosuria** (**fructokinase deficiency**) and **hereditary fructose intolerance** (a deficiency of the fructose 1-phosphate cleavage by **aldolase B**) are inherited disorders of fructose metabolism.*

*Fructose synthesis from glucose in the **polyol pathway** occurs in seminal vesicles and other tissues. **Aldose reductase** converts glucose to the sugar alcohol sorbitol (a polyol), which is then oxidized to fructose. In the lens of the eye, elevated levels of **sorbitol** in diabetes mellitus may contribute to **cataract** formation.*

*Galactose is ingested principally as **lactose,** which is converted to galactose and glucose in the intestine. **Galactose** is converted to glucose principally in the liver. It is phosphorylated to galactose 1-phosphate by **galactokinase** and activated to a UDP-sugar by **galactosyl uridylyltransferase.** The metabolic pathway subsequently generates glucose 1-phosphate. **Classical galactosemia**, a deficiency of galactosyl uridylyltransferase, results in the accumulation of galactose 1-phosphate in the liver and the inhibition of hepatic glycogen metabolism and other pathways that require UDP sugars. Cataracts can occur from accumulation of galactose in the blood, which is converted to **galactitol** (the sugar alcohol of galactose) in the lens of the eye.*

*The **pentose phosphate pathway** consists of both **oxidative** and **nonoxidative** components (Fig. 29.2). In the oxidative pathway, glucose 6-phosphate is oxidized to **ribulose 5-phosphate**, CO_2, and **NADPH**. Ribulose 5-phosphate, a pentose, can be converted to **ribose 5-phosphate** for nucleotide biosynthesis. The NADPH is used for reductive pathways, such as **fatty acid biosynthesis**, **detoxification** of drugs by monooxygenases, and the **glutathione defense system** against injury by reactive oxygen species (ROS).*

In the nonoxidative phase of the pathway, ribulose 5-phosphate is converted to ribose 5-phosphate and to intermediates of the glycolytic pathway. This portion of

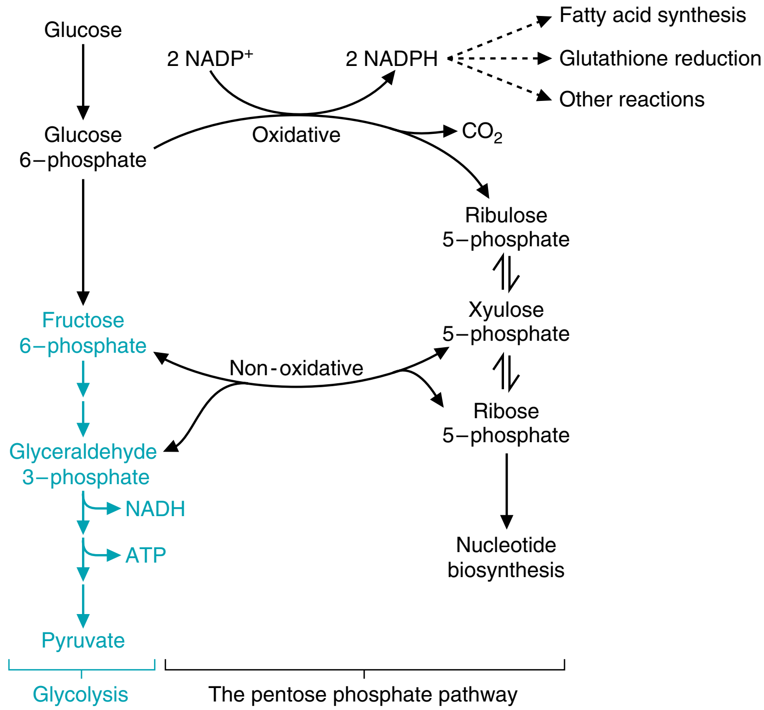

Fig. 29.2. Overview of the pentose phosphate pathway. The pentose phosphate pathway generates NADPH for reactions requiring reducing equivalents (electrons) or ribose 5-phosphate for nucleotide biosynthesis. Glucose 6-phosphate is a substrate for both the pentose phosphate pathway and glycolysis. The 5-carbon sugar intermediates of the pentose phosphate pathway are reversibly interconverted to intermediates of glycolysis. The portion of glycolysis that is not part of the pentose phosphate pathway is shown in blue.

the pathway is reversible; therefore, ribose 5-phosphate can also be formed from intermediates of glycolysis. One of the enzymes involved in these sugar interconversions, ***transketolase****, uses* ***thiamine pyrophosphate*** *as a coenzyme.*

The sugars produced by the pentose phosphate pathway enter glycolysis as fructose 6-phosphate and glyceraldehyde 3-phosphate, and their further metabolism in the glycolytic pathway generates NADH, adenosine triphsphate (ATP), and pyruvate. The overall equation for the conversion of glucose 6-phosphate to fructose 6-phosphate and glyceraldehyde 3-phosphate through both the oxidative and nonoxidative reactions of the pentose phosphate pathway is:

$$3\ glucose\text{-}6\text{-}P + 6\ NADP^+ \rightarrow 3\ CO_2 + 6\ NADPH + 6\ H^+ + 2\ fructose\text{-}6\text{-}P + glyceraldehyde\text{-}3\text{-}P.$$

The pentose phosphate pathway is also called the hexose monophosphate shunt (HMP shunt). It shunts hexoses from glycolysis, forming pentoses, which may be reconverted to glycolytic intermediates.

THE WAITING ROOM

Candice Sucher is an 18-year-old girl who presented to her physician for a precollege physical examination. While taking her medical history, the doctor learned that she carefully avoided eating all fruits and any foods that contained table sugar. She related that, from a very early age, she had learned that these foods caused severe weakness and symptoms suggestive of low blood sugar, such as tremulousness and sweating. Her medical history also indicated that her mother had described her as having been a very irritable baby who often cried incessantly, especially after meals, and vomited frequently. At these times, Candice's abdomen had become distended, and she became drowsy and apathetic. Her mother had intuitively eliminated certain foods from Candice's diet, after which the severity and frequency of these symptoms diminished.

Erin Galway is a 3-week-old female infant who began vomiting 3 days after birth, usually within 30 minutes after breastfeeding. Her abdomen became distended at these times, and she became irritable and cried frequently. When her mother noted that the whites of Erin's eyes were yellow, she took her to a pediatrician. The doctor agreed that Erin was slightly jaundiced. He also noted an enlargement of her liver and questioned the possibility of early cataract formation in the lenses of Erin's eyes. He ordered liver and kidney function tests and did two separate dipstick urine tests in his office, one designed to measure only glucose in the urine and the other capable of detecting any of the reducing sugars.

Al Martini developed a fever of 101.5°F on the second day of his hospitalization for acute alcoholism. He had a cough productive of gray sputum. A chest x-ray showed right lower lobe pneumonia. A stain of his sputum showed many small pleomorphic Gram-negative bacilli. Sputum was sent for culture and a determination of which antibiotics would be effective in treating the causative organism (sensitivity testing). Because his landlady stated that he had an allergy to penicillin, he was started on a course of the antibiotic combination of trimethoprim and sulfamethoxazole (TMP/sulfa). To his landlady's knowledge, he had never been treated with a sulfa drug previously.

On the third day of therapy with TMP/sulfa for his pneumonia, **Al Martini** was slightly jaundiced. His hemoglobin level had fallen by 3.5 g/dL from the value on admission, and his urine was red-brown because of the presence of free hemoglobin. Mr. Martini had apparently suffered acute hemolysis (lysis or destruction of some of his red blood cells) induced by his infection and exposure to the sulfa drug.

I. FRUCTOSE

Fructose is found in the diet as a component of sucrose in fruit, as a free sugar in honey, and in high-fructose corn syrup (see Fig. 29.1). Fructose enters epithelial cells and other types of cells by facilitated diffusion on the GLUT V transporter. It is metabolized to intermediates of glycolysis. Problems with fructose absorption and metabolism are relatively more common than with other sugars.

A. Fructose Metabolism

Fructose is metabolized by conversion to glyceraldehyde-3-P and dihydroxyacetone phosphate, which are intermediates of glycolysis (Fig. 29.3). The steps parallel those of glycolysis. The first step in the metabolism of fructose, as with glucose, is phosphorylation. Fructokinase, the major kinase involved, phosphorylates fructose in the 1-position. Fructokinase has a high V_{max}, and rapidly phosphorylates fructose as it enters the cell. The fructose 1-phosphate formed is not an intermediate of glycolysis but rather is cleaved by aldolase B to dihydroxyacetone phosphate (an intermediate of glycolysis) and glyceraldehyde. Glyceraldehyde is then phosphorylated to glyceraldehyde-3-P by triose kinase. Dihydroxyacetone phosphate and glyceraldehyde 3-phosphate are intermediates of the glycolytic pathway and can proceed through it to pyruvate, the TCA cycle, and fatty acid synthesis. Alternately, these intermediates can also be converted to glucose by gluconeogenesis. In other words, the fate of fructose parallels that of glucose.

When individuals with defects of aldolase B ingest fructose, the extremely high levels of fructose 1-phosphate that accumulate in the liver and kidney cause a number of adverse effects. Hypoglycemia results from inhibition of glycogenolysis and gluconeogenesis. Glycogen phosphorylase (and possibly phosphoglucomutase and other enzymes of glycogen metabolism) are inhibited by the accumulated fructose 1-phosphate. Aldolase B is required for glucose synthesis from glyceraldehyde 3-phosphate and dihydroxyacetone phosphate, and its low activity in aldolase B–deficient individuals is further decreased by the accumulated fructose 1-phosphate. The inhibition of gluconeogenesis results in lactic acidosis.

The accumulation of fructose 1-phosphate also substantially depletes the phosphate pools. The fructokinase reaction uses ATP at a rapid rate such that the mitochondria regenerate ATP rapidly, which leads to a drop in free phosphate levels. The low levels of phosphate release inhibition of AMP deaminase, which converts AMP to inosine monophosphate (IMP). The nitrogenous base of IMP (hypoxanthine) is degraded to uric acid. The lack of phosphate and depletion of adenine nucleotides lead to a loss of ATP, further contributing to the inhibition of biosynthetic pathways, including gluconeogenesis.

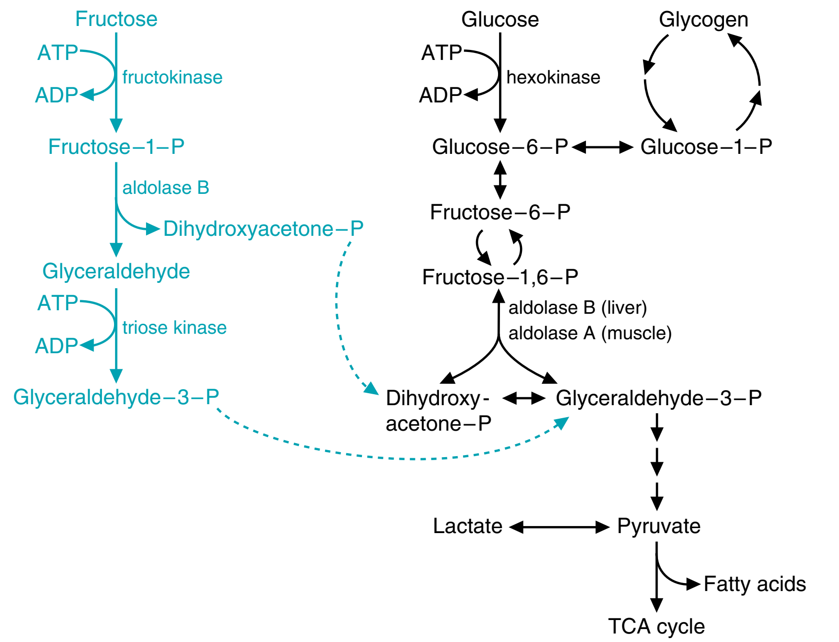

Fig. 29.3. Fructose metabolism. The pathway for the conversion of fructose to dihydroxyacetone phosphate and glyceraldehyde 3-phosphate is shown in blue. These two compounds are intermediates of glycolysis and are converted in the liver principally to glucose, glycogen, or fatty acids. In the liver, aldolase B cleaves both fructose 1-phosphate in the pathway for fructose metabolism, and fructose 1,6-bisphosphate in the pathway for glycolysis.

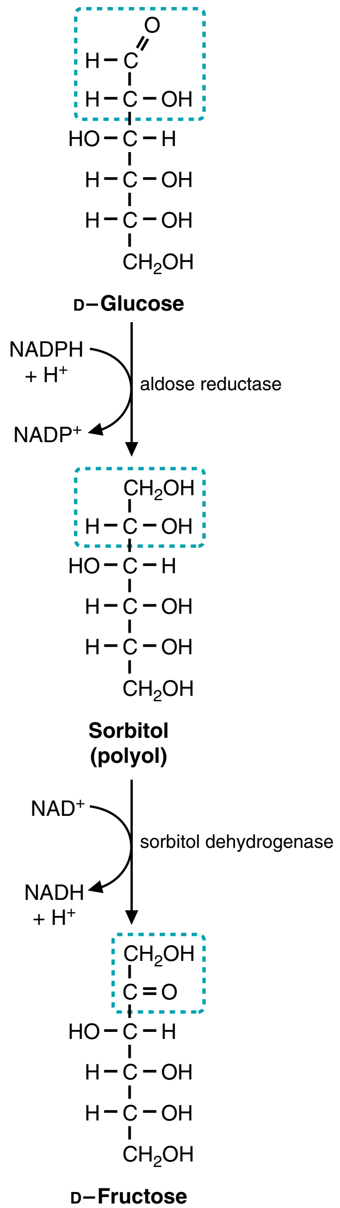

Fig. 29.4. The polyol pathway converts glucose to fructose.

Q: Essential fructosuria is a rare and benign genetic disorder caused by a deficiency of the enzyme fructokinase. Why is this disease benign, when a deficiency of aldolase B (hereditary fructose intolerance) can be fatal? Could **Candice Sucher** have essential fructosuria?

The metabolism of fructose occurs principally in the liver and to a lesser extent in the small intestinal mucosa and proximal epithelium of the renal tubule, because these tissues have both fructokinase and aldolase B. Aldolase exists as several isoforms: aldolases A, B, C, and fetal aldolase. Although all of these aldolase isoforms can cleave fructose 1,6-bisphosphate, the intermediate of glycolysis, only aldolase B can also cleave fructose 1-phosphate. Aldolase A, present in muscle and most other tissues, and aldolase C, present in brain, have almost no ability to cleave fructose 1-phosphate. Fetal aldolase, present in the liver before birth, is similar to aldolase C.

Aldolase B is the rate-limiting enzyme of fructose metabolism, although it is not a rate-limiting enzyme of glycolysis. It has a much lower affinity for fructose l-phosphate than fructose 1,6-bisphosphate (although the k_{cat} is the same) and is very slow at physiologic levels of fructose 1-phosphate. As a consequence, after ingesting a high dose of fructose, normal individuals accumulate fructose 1-phosphate in the liver while it is slowly converted to glycolytic intermediates. Individuals with hereditary fructose intolerance (a deficiency of aldolase B) accumulate much higher amounts of fructose 1-phosphate in their livers.

Other tissues also have the capacity to metabolize fructose but do so much more slowly. The hexokinase isoforms present in muscle, adipose tissue, and other tissues can convert fructose to fructose 6-phosphate, but react so much more efficiently with glucose. As a result, fructose phosphorylation is very slow in the presence of physiologic levels of intracellular glucose and glucose 6-phosphate.

B. Synthesis of Fructose in the Polyol Pathway

Fructose can be synthesized from glucose in the polyol pathway. The polyol pathway is named for the first step of the pathway in which sugars are reduced to the sugar alcohol by the enzyme aldose reductase (Fig. 29.4) Glucose is reduced to the sugar alcohol sorbitol, and sorbitol is then oxidized to fructose.

This pathway is present in seminal vesicles, which synthesize fructose for the seminal fluid. Spermatozoa use fructose as a major fuel source while in the seminal fluid and then switch to glucose once in the female reproductive tract. Utilization of fructose is thought to prevent acrosomal breakdown of the plasma membrane (and consequent activation) while the spermatozoa are still in the seminal fluid.

The polyol pathway is present in many tissues, but its function in all tissues is not understood. Aldose reductase is relatively nonspecific, and its major function may be the metabolism of an aldehyde sugar other than glucose. The activity of this enzyme can lead to major problems in the lens of the eye, where it is responsible for the production of sorbitol from glucose and galactitol from galactose. When the concentration of glucose or galactose is elevated in the blood, their respective sugar alcohols are synthesized in the lens more rapidly than they are removed, resulting in increased osmotic pressure within the lens.

II. GALACTOSE METABOLISM—METABOLISM TO GLUCOSE-1-P

Dietary galactose is metabolized principally by phosphorylation to galactose 1-phosphate, and then conversion to UDP-galactose and glucose 1-phosphate (Fig. 29.5). The phosphorylation of galactose, again an important first step in the pathway, is carried out by a specific kinase, galactokinase. The formation of UDP-galactose is accomplished by attack of the phosphate oxygen on galactose 1-phosphate on the α phosphate of UDP-glucose, releasing glucose 1-phosphate while forming UDP-galactose. The enzyme that catalyzes this reaction is galactose l-phosphate uridylyltransferase. The UDP-galactose is then converted to UDP-glucose by the reversible UDP-glucose epimerase (the configuration of the hydroxyl group on carbon four is reversed in this reaction). The net result of this sequence of reactions is that galactose is converted to glucose 1-phosphate, at the expense of 1 high-energy bond of ATP. The sum of these reactions is indicated in the equations that follow:

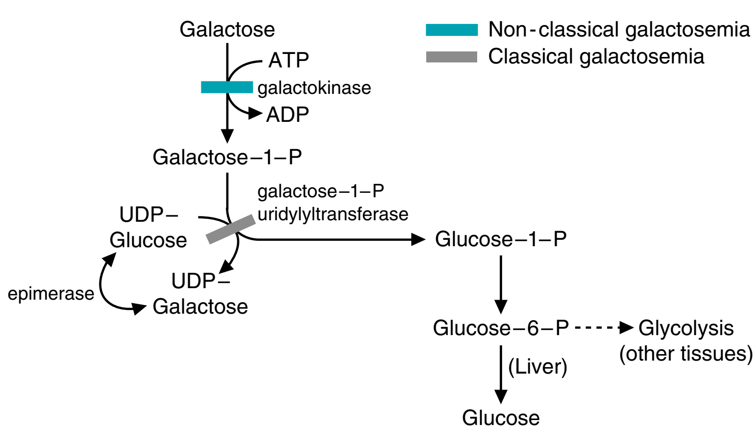

Fig. 29.5. Metabolism of galactose. Galactose is phosphorylated to galactose 1-phosphate by galactokinase. Galactose 1-phosphate reacts with UDP-glucose to release glucose 1-phosphate. Galactose thus can be converted to blood glucose, enter glycolysis, or enter any of the metabolic routes of glucose. In classical galactosemia, a deficiency of galactose 1-phosphate uridylyltransferase (shown in grey) results in the accumulation of galactose 1-phosphate in tissues and the appearance of galactose in the blood and urine. In nonclassical galactosemia, a deficiency of galactokinase results in the accumulation of galactose.

The accumulation of sorbitol in muscle and nerve tissues may contribute to the peripheral neuropathy characteristic of patients with poorly controlled diabetes mellitus. This is one of the reasons it is so important for **Di Abietes** (who has type 1 diabetes mellitus) and **Ann Sulin** (who has type 2 diabetes mellitus) to achieve good glycemic control.

The accumulation of sugars and sugar alcohols in the lens of patients with hyperglycemia (e.g., diabetes mellitus) results in the formation of cataracts. Glucose levels are elevated and increase the synthesis of sorbitol and fructose. As a consequence, a high osmotic pressure is created in the lens. The high glucose and fructose levels also result in nonenzymatic glycosylation of lens proteins. The result of the increased osmotic pressure and the glycosylation of the lens protein is an opaque cloudiness of the lens known as a cataract. **Erin Galway** seemed to have an early cataract, probably caused by the accumulation of galactose and its sugar alcohol galactitol.

A: In essential fructosuria, fructose cannot be converted to fructose 1-phosphate. This condition is benign because no toxic metabolites of fructose accumulate in the liver, and the patient remains nearly asymptomatic. Some of the ingested fructose is slowly phosphorylated by hexokinase in nonhepatic tissues and metabolized by glycolysis, and some appears in the urine. There is no renal threshold for fructose; the appearance of fructose in the urine (fructosuria) does not require a high fructose concentration in the blood.

Hereditary fructose intolerance, conversely, results in the accumulation of fructose 1-phosphate and fructose. By inhibiting glycogenolysis and gluconeogenesis, the high levels of fructose 1-phosphate caused the hypoglycemia that **Candice Sucher** experienced as an infant when she became apathetic and drowsy, and as an adult when she experienced sweating and tremulousness.

Erin Galway's urine was negative for glucose when measured with the glucose oxidase strip but was positive for the presence of a reducing sugar. The reducing sugar was identified as galactose. Her liver function tests showed an increase in serum bilirubin and in several liver enzymes. Albumin was present in her urine. These findings and the clinical history increased her physician's suspicion that Erin had classical galactosemia.

Classical galactosemia is caused by a deficiency of galactose 1-phosphate uridylyltransferase. In this disease, galactose 1-phosphate accumulates in tissues, and galactose is elevated in the blood and urine. This condition differs from the rarer deficiency of galactokinase (nonclassical galactosemia), in which galactosemia and galactosuria occur but galactose 1-phosphate is not formed. Both enzyme defects result in cataracts from galactitol formation by aldose reductase in the polyol pathway. Aldose reductase has a relatively high K_m for galactose, approximately 12 to 20 mM, so that galactitol is formed only in galactosemic patients who have eaten galactose. Galactitol is not further metabolized and diffuses out of the lens very slowly. Thus, hypergalactosemia is even more likely to cause cataracts than hyperglycemia. **Erin Galway**, although only 3 weeks old, appeared to have early cataracts forming in the lens of her eyes.

One of the most serious problems of classical galactosemia is an irreversible mental retardation. Realizing this problem, **Erin Galway's** physician wanted to begin immediate dietary therapy. A test that measures galactose 1-phosphate uridylyltransferase in erythrocytes was ordered. The enzyme activity was virtually absent, confirming the diagnosis of classical galactosemia.

(1) $\text{Galactose} + \text{ATP} \xrightarrow{\textit{galactokinase}} \text{Galactose-1-P} + \text{ADP}$

(2) $\text{Galactose-1-P} + \text{UDP-glucose} \xrightarrow[\textit{uridylyltransferase}]{\textit{galactose-1-P}} \text{UDP-galactose} + \text{glucose-1-P}$

(3) $\text{UDP-galactose} \xrightarrow{\textit{UDP-glucose epimerase}} \text{UDP-glucose}$

Net Equation: $\text{Galactose} + \text{ATP} \longrightarrow \text{Glucose-1-P} + \text{ADP}$

The enzymes for galactose conversion to glucose 1-phosphate are present in many tissues, including the adult erythrocyte, fibroblasts, and fetal tissues. The liver has a high activity of these enzymes, and can convert dietary galactose to blood glucose and glycogen. The fate of dietary galactose, like that of fructose, therefore, parallels that of glucose. The ability to metabolize galactose is even higher in infants than in adults. Newborn infants ingest up to 1 g galactose per kg per feeding (as lactose). Yet the rate of metabolism is so high that the blood level in the systemic circulation is less than 3 mg/dL, and none of the galactose is lost in the urine.

III. THE PENTOSE PHOSPHATE PATHWAY

The pentose phosphate pathway is essentially a scenic bypass route around the first stage of glycolysis that generates NADPH and ribose-5-P (as well as other pentose sugars). Glucose 6-phosphate is the common precursor for both pathways. The oxidative first stage of the pentose phosphate pathway generates two moles of NADPH per glucose 6-phosphate oxidized. The second stage of the pentose phosphate pathway generates ribose-5-P and converts unused intermediates to fructose-6-P and glyceraldehyde-3-P in the glycolytic pathway (see Fig. 29.2). All cells require NADPH for reductive detoxification, and most cells require ribose-5-P for nucleotide synthesis. Consequently, the pathway is present in all cells. The enzymes reside in the cytosol, as do the enzymes of glycolysis.

A. Oxidative Phase of the Pentose Phosphate Pathway

1. NADPH PRODUCTION

In the oxidative first phase of the pentose phosphate pathway, glucose 6-phosphate is oxidatively decarboxylated to a pentose sugar, ribulose 5-phosphate (Fig. 29.6). The first enzyme of this pathway, glucose 6-phosphate dehydrogenase, oxidizes the aldehyde at C1 and reduces $NADP^+$ to NADPH. The gluconolactone that is formed is rapidly hydrolyzed to 6-phosphogluconate, a sugar acid with a carboxylic acid group at C1. The next oxidation step releases this carboxyl group as CO_2, with the electrons being transferred to $NADP^+$. This reaction is mechanistically very similar to the one catalyzed by isocitrate dehydrogenase in the TCA cycle. Thus, two moles of NADPH per mole of glucose 6-phosphate are formed from this portion of the pathway.

NADPH, rather than NADH, is generally used in the cell for pathways that require the input of electrons for reductive reactions because the ratio of $NADPH/NADP^+$ is much greater than the $NADH/NAD^+$ ratio. The NADH generated from fuel oxidation is rapidly oxidized back to NAD^+ by NADH dehydrogenase in the electron transport chain, so the level of NADH is very low in the cell.

NADPH can be generated from a number of reactions in the liver and other tissues, but not the red blood cell. For example, in tissues with mitochondria, an energy-requiring transhydrogenase located near the complexes of the electron transport chain can transfer reducing equivalents from NADH to $NADP^+$ to generate NADPH.

NADPH, however, cannot be directly oxidized by the electron transport chain, and the ratio of NADPH to $NADP^+$ in cells is greater than one. The reduction potential of NADPH therefore can contribute to the energy needed for biosynthetic processes and provide a constant source of reducing power for detoxification reactions.

2. RIBOSE 5-PHOSPHATE FROM THE OXIDATIVE ARM OF THE PATHWAY

To generate ribose 5-phosphate from the oxidative pathway, the ribulose 5-phosphate formed from the action of the two oxidative steps is isomerized to produce ribose 5-phosphate (a ketose-to-aldose conversion, similar to fructose 6-phosphate being isomerized to glucose 6-phosphate; see section III.B.1 below). The ribose 5-phosphate can then enter the pathway for nucleotide synthesis, if needed, or can be converted to glycolytic intermediates, as described below for the nonoxidative phase of the pentose phosphate pathway. The pathway through which the ribose 5-phosphate travels is determined by the needs of the cell at the time of its synthesis.

B. The Nonoxidative Phase of the Pentose Phosphate Pathway

The nonoxidative reactions of this pathway are reversible reactions that allow intermediates of glycolysis (specifically glyceraldehyde-3-P and fructose-6-P) to be converted to five-carbon sugars (such as ribose-5-P), and vice versa. The needs of the cell will determine in which direction this pathway proceeds. If the cell has produced ribose-5-P, but does not need to synthesize nucleotides, then the ribose-5-P will be converted to glycolytic intermediates. If the cell still requires NADPH, the ribose-5-P will be converted back into glucose-6-P using nonoxidative reactions (see below). And finally, if the cell already has a high level of NADPH, but needs to produce nucleotides, the oxidative reactions of the pentose phosphate pathway will be inhibited, and the glycolytic intermediates fructose-6-P and glyceraldehyde-3-P will be used to produce the five carbon sugars using exclusively the nonoxidative phase of the pentose phosphate pathway.

1. THE CONVERSION OF RIBOSE 5-PHOSPHATE TO GLYCOLYTIC INTERMEDIATES

The nonoxidative portion of the pentose phosphate pathway consists of a series of rearrangement and transfer reactions that first convert ribulose 5-phosphate to ribose 5-phosphate and xylulose 5-phosphate, and then the ribose 5-phosphate and xyulose 5-phosphate are converted to intermediates of the glycolytic pathway. The enzymes involved are an epimerase, an isomerase, transketolase, and transaldolase.

The epimerase and isomerase convert ribulose 5-phosphate to two other 5-carbon sugars (Fig. 29.7). The isomerase converts ribulose 5-phosphate to ribose 5-phosphate. The epimerase changes the stereochemical position of one hydroxyl group (at carbon 3), converting ribulose 5-phosphate to xylulose 5-phosphate.

Transketolase transfers 2-carbon fragments of keto sugars (sugars with a keto group at C2) to other sugars. Transketolase picks up a 2-carbon fragment from xylulose 5-phosphate by cleaving the carbon–carbon bond between the keto group and the adjacent carbon, thereby releasing glyceraldehyde 3-phosphate (Fig. 29.8). The 2-carbon fragment is covalently bound to thiamine pyrophosphate, which transfers

Xylulose 5-phosphate has recently been identified as an activator of protein phosphatase 2A (PP2A). PP2A removes phosphates from PFK-2 and from a transcription factor that binds to carbohydrate response elements in promoters of genes such as pyruvate kinase. The hydrolysis of the phosphates activates both proteins, such that xylulose 5-phosphate can regulate pathways relating to both carbohydrate and fat metabolism.

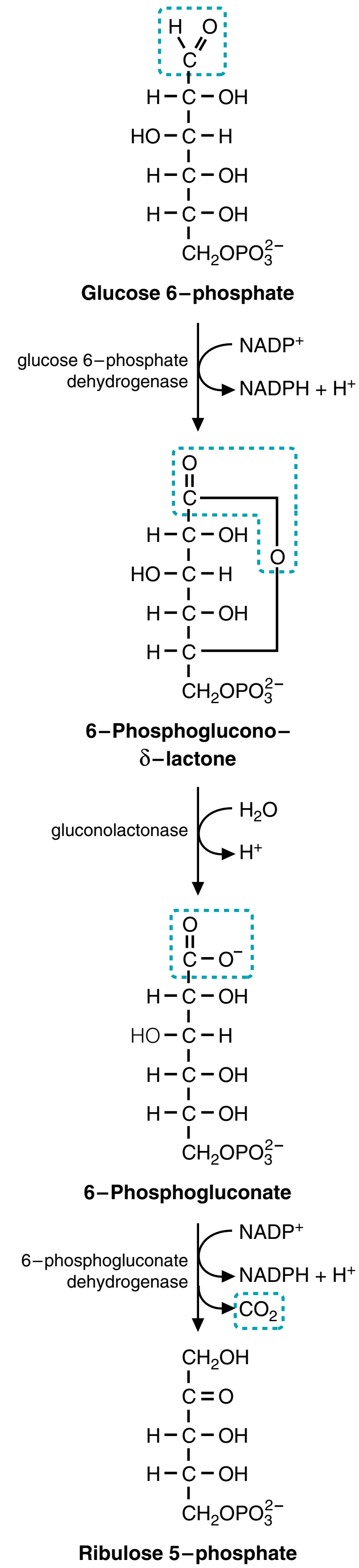

Fig. 29.6. Oxidative portion of the pentose phosphate pathway. Carbon 1 of glucose 6-phosphate is oxidized to an acid and then released as CO_2 in an oxidative decarboxylation reaction. Each oxidation step generates an NADPH.

Doctors suspected that the underlying factor in the destruction of **Al Martini's** red blood cells was an X-linked defect in the gene that codes for glucose 6-phosphate dehydrogenase. The red blood cell is dependent on this enzyme for a source of NADPH to maintain reduced levels of glutathione, one of its major defenses against oxidative stress (see Chapter 24). Glucose 6-phosphate dehydrogenase deficiency is the most common known enzymopathy, and affects approximately 7% of the world's population and about 2% of the U.S. population. Most glucose 6-phosphate dehydrogenase–deficient individuals are asymptomatic but can undergo an episode of hemolytic anemia if exposed to certain drugs, to certain types of infections, or if they ingest fava beans. When questioned, Al Martini replied that he did not know what a fava bean was and had no idea whether he was sensitive to them.

it to the aldehyde carbon of another sugar, forming a new ketose. The role of thiamine-pyrophosphate here is thus very similar to its role in the oxidative decarboxylation of pyruvate and α-ketoglutarate (see Chapter 20, section I.B). Two reactions in the pentose phosphate pathway use transketolase; in the first, the 2-carbon keto fragment from xylulose 5-phosphate is transferred to ribose 5-phosphate to form sedoheptulose 7-phosphate, and in the other, a 2-carbon keto fragment (usually derived from xylulose 5-phosphate) is transferred to erythrose 4-phosphate to form fructose 6-phosphate.

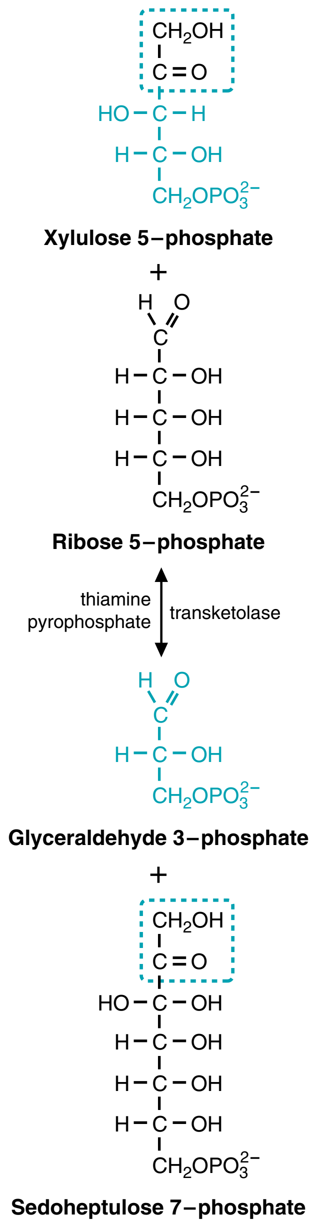

Fig. 29.8. Two-carbon unit transferred by transketolase. Transketolase cleaves the bond next to the keto group and transfers the 2-carbon keto fragment to an aldehyde. Thiamine pyrophosphate carries the 2-carbon fragment, forming a covalent bond with the carbon of the keto group.

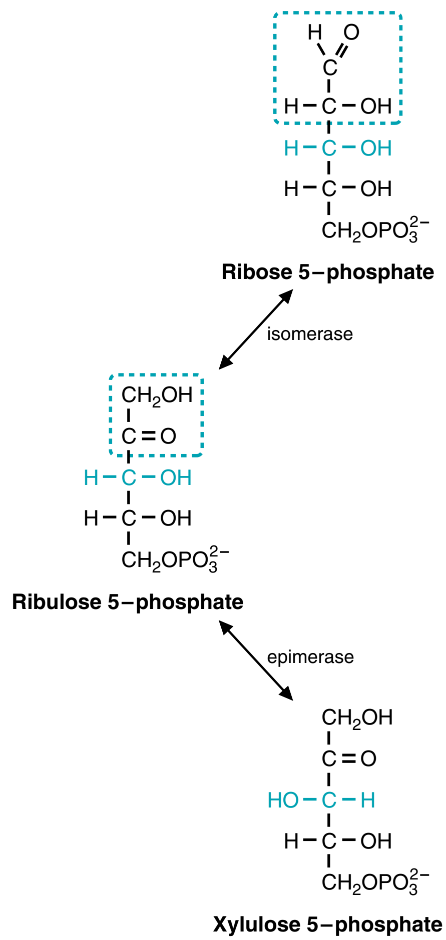

Fig. 29.7. Ribulose 5-phosphate is epimerized (to xylulose 5-phosphate) and isomerized (to ribose 5-phosphate).

Transaldolase transfers a 3-carbon keto fragment from sedoheptulose 7-phosphate to glyceraldehyde 3-phosphate to form erythrose 4-phosphate and fructose 6-phosphate (Fig. 29.9). The aldol cleavage occurs between the two hydroxyl carbons adjacent to the keto group (on carbons 3 and 4 of the sugar). This reaction is similar to the aldolase reaction in glycolysis, and the enzyme uses an active amino group, from the side chain of lysine, to catalyze the reaction.

The net result of the metabolism of 3 moles of ribulose 5-phosphate in the pentose phosphate pathway is the formation of 2 moles of fructose 6-phosphate and 1 mole of glyceraldehyde 3-phosphate, which then continue through the glycolytic pathway with the production of NADH, ATP, and pyruvate. Because the pentose phosphate pathway begins with glucose 6-phosphate, and feeds back into the

The transketolase activity of red blood cells is used to measure thiamine nutritional status and diagnose the presence of thiamine deficiency. The activity of transketolase is measured in the presence and absence of added thiamine pyrophosphate. If the thiamine intake of a patient is adequate, the addition of thiamine pyrophosphate does not increase the activity of transketolase because it already contains bound thiamine pyrophosphate. If the patient is thiamine deficient, transketolase activity will be low, and adding thiamine pyrophosphate will greatly stimulate the reaction. **Al Martini** was diagnosed in Chapter 19 as having beriberi heart disease resulting from thiamine deficiency. The diagnosis was based on laboratory tests confirming the thiamine deficiency.

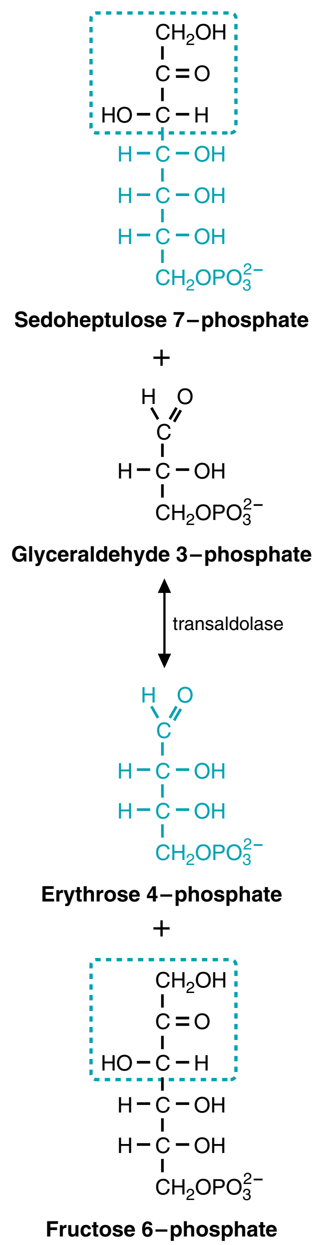

Fig. 29.9. Transaldolase transfers a 3-carbon fragment that contains an alcohol group next to a keto group.

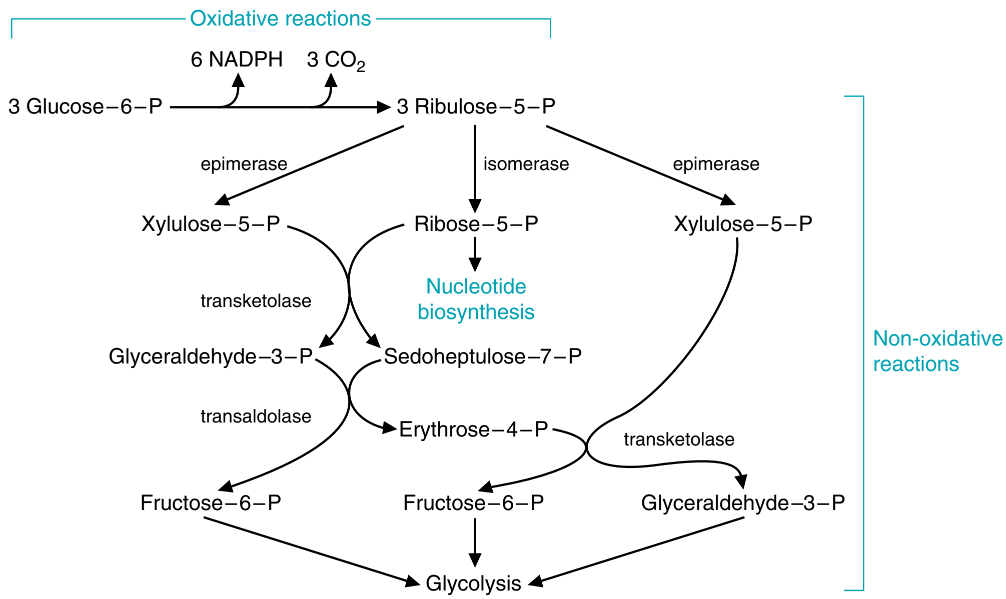

Fig. 29.10. A balanced sequence of reactions in the pentose phosphate pathway. The interconversion of sugars in the pentose phosphate pathway results in conversion of 3 glucose 6-phosphate to 6 NADPH, 3 CO_2, 2 fructose 6-phosphate, and one glyceraldehyde 3-phosphate.

glycolytic pathway, it is sometimes called the hexose monophosphate shunt (a shunt or a pathway for glucose 6-phosphate). The reaction sequence starting from glucose-6-P, and involving both the oxidative and nonoxidative phases of the pathway, is shown in Figure 29.10.

2. GENERATION OF RIBOSE 5-PHOSPHATE FROM INTERMEDIATES OF GLYCOLYSIS

The reactions catalyzed by the epimerase, isomerase, transketolase, and transaldolase are all reversible reactions under physiologic conditions. Thus, ribose 5-phosphate required for purine and pyrimidine synthesis can be generated from intermediates of the glycolytic pathway, as well as from the oxidative phase of the pentose phosphate pathway. The sequence of reactions that generate ribose 5-phosphate from intermediates of glycolysis is indicated below.

(1) Fructose-6-P + glyceraldehyde-3-P $\xleftrightarrow{\textit{Transketolase}}$ Erythrose-4-P + Xylulose-5-P

(2) Erythrose-4-P + Fructose-6-P $\xleftrightarrow{\textit{Transaldolase}}$ Sedoheptulose-7-P + Glyceraldehyde-3-P

(3) Sedoheptulose-7-P+ Glyceraldehyde-3-P $\xleftrightarrow{\textit{Transketolase}}$ Ribose-5-P + Xylulose-5-P

(4) 2 Xylulose-5-P $\xleftrightarrow{\textit{Epimerase}}$ 2 Ribulose-5-P

(5) 2 Ribulose-5-P $\xleftrightarrow{\textit{Isomerase}}$ 2 Ribose-5-P

Net Equation : 2 Fructose-6-P + Glyceraldehyde-3-P $\longleftrightarrow$ *3 Ribose-5-P*

C. Role of the Pentose Phosphate Pathway in the Generation of NADPH

In general, the oxidative phase of the pentose phosphate pathway is the major source of NADPH in cells. NADPH provides the reducing equivalents for biosynthetic reactions and for oxidation–reduction reactions involved in protection against the toxicity of ROS (see Chapter 24). The glutathione-mediated defense against oxidative stress is common to all cell types (including the red blood cell), and the requirement for NADPH to maintain levels of reduced glutathione probably accounts for the universal distribution of the pentose phosphate pathway among different types of cells. Fig. 29.11 illustrates the importance of this pathway in maintaining the membrane integrity of the red blood cell. NADPH is also used for anabolic pathways, such as fatty acid synthesis, cholesterol synthesis, and fatty acid chain elongation (Table 29.1). It is the source of reducing equivalents for cytochrome P450 hydroxylation of aromatic compounds, steroids, alcohols, and drugs. The highest concentrations of glucose 6-phosphate dehydrogenase are found in phagocytic cells, where NADPH oxidase uses NADPH to form superoxide from molecular oxygen. The superoxide then generates hydrogen peroxide, which kills the microorganisms taken up by the phagocytic cells (see Chapter 24).

The entry of glucose 6-phosphate into the pentose phosphate pathway is controlled by the cellular concentration of NADPH. NADPH is a strong product inhibitor of glucose 6-phosphate dehydrogenase, the first enzyme of the pathway. As NADPH is oxidized in other pathways, the product inhibition of glucose 6-phosphate dehydrogenase is relieved, and the rate of the enzyme is accelerated to produce more NADPH.

Table 29.1. Pathways That Require NADPH

Detoxification
• Reduction of oxidized glutathione
• Cytochrome P450 monooxygenases
Reductive synthesis
• Fatty acid synthesis
• Fatty acid chain elongation
• Cholesterol synthesis
• Neurotransmitter synthesis
• Nucleotide synthesis
• Superoxide synthesis

Q: How does the net energy yield from the metabolism of 3 moles of glucose 6-phosphate through the pentose phosphate pathway to pyruvate compare with the yield of 3 moles of glucose 6-phosphate through glycolysis?

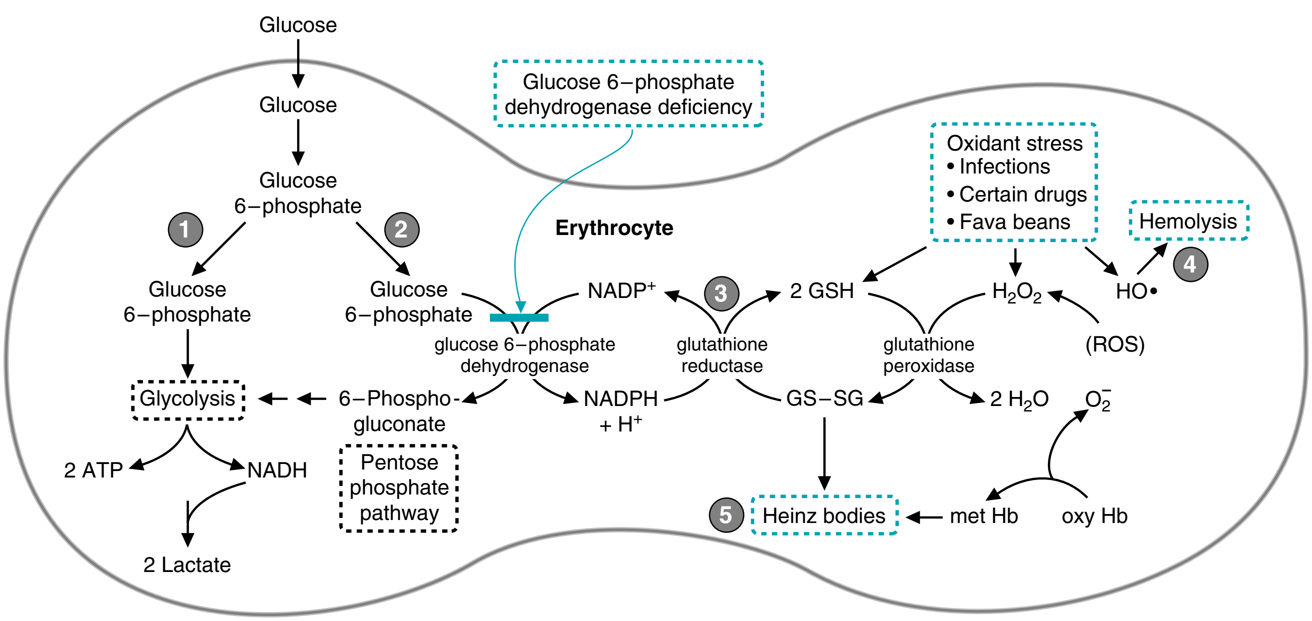

Fig. 29.11. Hemolysis caused by reactive oxygen species. **1.** Maintenance of the integrity of the erythrocyte membrane depends on its ability to generate ATP and NADH from glycolysis. **2.** NADPH is generated by the pentose phosphate pathway. **3.** NADPH is used for the reduction of oxidized glutathione to reduced glutathione. Glutathione is necessary for the removal of H_2O_2 and lipid peroxides generated by reactive oxygen species (ROS). **4.** In the erythrocytes of healthy individuals, the continuous generation of superoxide ion from the nonenzymatic oxidation of hemoglobin provides a source of reactive oxygen species. The glutathione defense system is compromised by glucose 6-phosphate dehydrogenase deficiency, infections, certain drugs, and the purine glycosides of fava beans. **5.** As a consequence, Heinz bodies, aggregates of cross-linked hemoglobin, form on the cell membranes and subject the cell to mechanical stress as it tries to go through small capillaries. The action of the ROS on the cell membrane as well as mechanical stress from the lack of deformability result in hemolysis.

A: The net energy yield from 3 moles of glucose 6-phosphate metabolized through the pentose phosphate pathway and then the last portion of the glycolytic pathway is 6 moles of NADPH, 3 moles of CO_2, 5 moles of NADH, 8 moles of ATP, and 5 moles of pyruvate. In contrast, the metabolism of 3 moles of glucose 6-phosphate through glycolysis is 6 moles of NADH, 9 moles of ATP, and 6 moles of pyruvate.

Table 29.2. Cellular Needs Dictate the Direction of the Pentose Phosphate Pathway Reactions

Cellular Need	Direction of Pathway
NADPH only	Oxidative reactions produce NADPH; nonoxidative reactions convert ribulose-5-P to glucose-6-P to produce more NADPH
NADPH + ribose-5-P	Oxidative reactions produce NADPH and ribulose-5-P; the isomerase converts ribulose-5-P to ribose-5-P.
Ribose-5-P only	Only the nonoxidative reactions. High NADPH inhibits glucose- 6-P dehydrogenase, so transketolase and transaldolase will be used to convert fructose-6-P and glyceraldehyde-3-P to ribose-5-P.
NADPH and pyruvate	Both the oxidative and nonoxidative reactions are used. The oxidative reactions generate NADPH and ribulose-5-P. The nonoxidative reactions convert the ribulose-5-P to fructose-6-P and glyceraldehyde-3-P, and glycolysis will convert these intermediates to pyruvate.

In the liver, the synthesis of fatty acids from glucose is a major route of NADPH reoxidation. The synthesis of liver glucose 6-phosphate dehydrogenase, like the key enzymes of glycolysis and fatty acid synthesis, is induced by the increased insulin/glucagon ratio after a high-carbohydrate meal. A summary of the possible routes glucose-6-P may follow using the pentose phosphate pathway is presented in Table 29.2.

CLINICAL COMMENTS

Hereditary fructose intolerance (HFI) is caused by a low level of fructose 1-phosphate aldolase activity in aldolase B, an isozyme of fructose 1,6-bisphosphate aldolase that is also capable of cleaving fructose 1-phosphate. In patients of European descent, the most common defect is a single missense mutation in exon 5 (G → C), resulting in an amino acid substitution (Ala → Pro). As a result of this substitution, a catalytically impaired aldolase B is synthesized in abundance. The exact prevalence of HFI in the United States is not established but is approximately 1 per 15,000 to 25,000 population. The disease is transmitted by an autosomal recessive inheritance pattern.

When affected patients such as **Candice Sucher** ingest fructose, fructose is converted to fructose 1-phosphate. Because of the deficiency of aldolase B, fructose 1-phosphate cannot be further metabolized to dihydroxyacetone phosphate and glyceraldehyde and accumulates in those tissues that have fructokinase (liver, kidney, and small intestine). Fructose is detected in the urine with the reducing sugar test (see Chapter 5). A DNA screening test (based on the generation of a new restriction site by the mutation) now provides a safe method to confirm a diagnosis of hereditary fructose intolerance.

In the infant and small child, the major symptoms include poor feeding, vomiting, intestinal discomfort, and failure to thrive. The greater the ingestion of dietary fructose, the more severe the clinical reaction. The result of prolonged ingestion of fructose is ultrastructural changes in the liver and kidney resulting in hepatic and renal failure. Hereditary fructose intolerance is usually a disease of infancy, because adults with fructose intolerance who have survived avoid the ingestion of fruits, table sugar, and other sweets.

Erin Galway has galactosemia, which is caused by a deficiency of galactose 1-phosphate uridylyltransferase; it is one of the most common genetic diseases. Galactosemia is an autosomal recessive disorder of galactose metabolism that occurs in about 1 in 60,000 newborns. Approximately two thirds of the states in the United States screen newborns for this disease because failure to begin immediate treatment results in mental retardation. Failure to thrive is the most common initial clinical symptom. Vomiting or diarrhea is found in most patients, usually starting within a few days of milk ingestion. Signs of deranged liver function, either jaundice or hepatomegaly, are present almost as frequently after the first week of life. The jaundice of intrinsic liver disease may be accentuated by the severe hemolysis in some patients. Cataracts have been observed within a few days of birth.

Management of patients requires elimination of galactose from the diet. Failure to eliminate this sugar results in progressive liver failure and death. In infants, artificial milk made from casein or soybean hydrolysate is used.

Al Martini's sputum culture sent on the second day of his admission for acute alcoholism and pneumonia grew out *Haemophilus influenzae*. This organism is sensitive to a variety of antibiotics, including TMP/sulfa. Unfortunately, it appeared that Mr. Martini had suffered an acute hemolysis (lysis or destruction of some of his red blood cells), probably induced by exposure to the sulfa drug and his infection with *H. influenzae*. The hemoglobin that escaped from the lysed red blood cells was filtered by his kidneys and appeared in his urine.

By mechanisms that are not fully delineated, certain drugs (such as sulfa drugs and antimalarials), a variety of infectious agents, and exposure to fava beans can cause red blood cell destruction in individuals with a genetic deficiency of glucose 6-phosphate dehydrogenase. Presumably, these patients cannot generate enough reduced NADPH to defend against the ROS. Although erythrocytes lack most of the other enzymatic sources of NADPH for the glutathione antioxidant system, they do have the defense mechanisms provided by the antioxidant vitamins E and C and catalase. Thus, individuals who are not totally deficient in glucose 6-phosphate dehydrogenase remain asymptomatic unless an additional oxidative stress, such as an infection, generates additional oxygen radicals.

Some drugs, such as the antimalarial primaquine and the sulfonamide which Al Martini is taking, affect the ability of red blood cells to defend against oxidative stress. Fava beans, which look like fat string beans and are sometimes called broad beans, contain the purine glycosides vicine and isouramil. These compounds react with glutathione. It has been suggested that cellular levels of reduced glutathione (GSH) decrease to such an extent that critical sulfhydryl groups in some key proteins cannot be maintained in reduced form.

The highest prevalence rates for glucose 6-phosphate dehydrogenase deficiency are found in tropical Africa and Asia, in some areas of the Middle East and the Mediterranean, and in Papua New Guinea. The geographic distribution of this deficiency is similar to that of sickle cell trait, and is probably also related to the relative resistance it confers against the malaria parasite.

Because the individuals with this deficiency are asymptomatic unless exposed to an "oxidant challenge," the clinical course of the hemolytic anemia is usually self-limited if the causative agent is removed. However, genetic polymorphism accounts for a substantial variability in the severity of the disease. Severely affected patients may have a chronic hemolytic anemia and other sequelae even without known exposure to drugs, infection, and other causative factors. In such patients, neonatal jaundice is also common and can be severe enough to cause death.

BIOCHEMICAL COMMENTS

Before the metabolic toxicity of fructose was appreciated, substitution of fructose for glucose in intravenous solutions, and of fructose for sucrose in enteral tube feeding or diabetic diets, was frequently recommended.

(Enteral tube feeding refers to tubes placed into the gut; parenteral tube feeding refers to tubes placed into the vein, feeding intravenously.) Administration of intravenous fructose to patients with diabetes mellitus or other forms of insulin resistance avoided the hyperglycemia found with intravenous glucose, possibly because fructose metabolism in the liver bypasses the insulin-regulated step at phosphofructokinase-1. However, because of the unregulated flow of fructose through glycolysis, intravenous fructose feeding frequently resulted in lactic acidosis (see Fig. 29.3). In addition, the fructokinase reaction is very rapid, and tissues would become depleted of ATP and phosphate when large quantities of fructose were metabolized over a short period. This would lead to cell death. Fructose is less toxic in the diet or in enteral feeding because of the relatively slow rate of fructose absorption.

Suggested References

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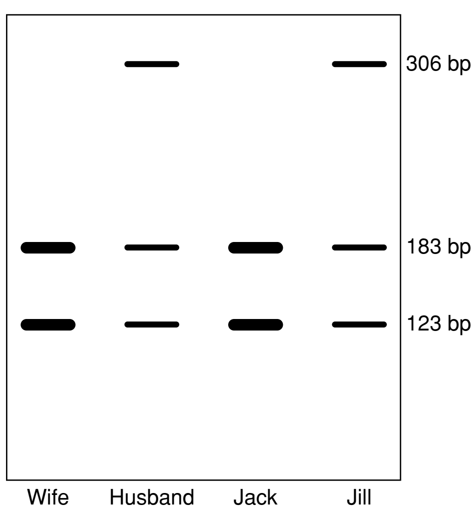
REVIEW QUESTIONS—CHAPTER 29

1. Hereditary fructose intolerance is a rare recessive genetic disease that is most commonly caused by a mutation in exon 5 of the aldolase B gene. The mutation fortuitously creates a new AhaII recognition sequence. To test for the mutation, DNA was extracted from a wife, husband, and their two children, Jack and Jill. The DNA for exon 5 of the aldolase B gene was amplified by polymerase chain reaction (PCR), treated with AhaII, subjected to electrophoresis on an agarose gel, and stained with a dye that binds to DNA.



Which of the following conclusions can be made from the data presented?
(A) Both of the children have the disease.
(B) Neither of the children has the disease.
(C) Jill has the disease, Jack does not.
(D) Jack has the disease, Jill does not.
(E) There is not enough information to make a determination

2. On examining the gel himself, the husband became concerned that he might not be the biologic father of one or both of the children. From the pattern on the gel, you can reasonably conclude which of the following?

(A) He is probably not Jill's father.
(B) He is probably not Jack's father.
(C) He could be the father of both children.
(D) He is probably not the father of either child.
(E) There is not enough information to make a determination

3. An alcoholic is brought to the Emergency Room for a hypoglycemic coma. Because alcoholics are frequently malnourished, which of the following enzymes can be used to test for a thiamine deficiency?

(A) Aldolase
(B) Transaldolase
(C) Transketolase
(D) Glucose 6-phosphate dehydrogenase
(E) UDP-galactose epimerase

4. Intravenous fructose feeding can lead to lactic acidosis caused by which of the following?

(A) Bypassing the regulated pyruvate kinase step
(B) Bypassing the regulated PFK-1 step
(C) Allosterically activating aldolase B
(D) Allosterically activating lactate dehydrogenase
(E) Increasing the [ATP]/[ADP] ratio in liver

5. The polyol pathway of sorbitol production and the HMP shunt pathway are linked by which of the following?

(A) The HMP shunt produces 6-phosphogluconate, an intermediate in the polyol pathway.
(B) The HMP shunt produces NADPH, which is required for the polyol pathway.
(C) The HMP shunt produces ribitol, an intermediate of the polyol pathway.
(D) Both pathways use glucose as the starting material.
(E) Both pathways use fructose as the starting material.

30 Synthesis of Glycosides, Lactose, Glycoproteins and Glycolipids

Many of the pathways for interconversion of sugars or the formation of sugar derivatives use activated sugars attached to nucleotides. Both ***UDP-glucose*** *and* ***UDP-galactose*** *are used for* ***glycosyltransferase*** *reactions in many systems. Lactose, for example, is synthesized from UDP-galactose and glucose in the mammary gland. UDP-glucose also can be oxidized to form UDP-glucuronate, which is used to form* ***glucuronide derivatives*** *of bilirubin and xenobiotic compounds. Glucuronide derivatives are generally more readily excreted in urine or bile than the parent compound.*

In addition to serving as fuel, carbohydrates are often found in ***glycoproteins*** *(carbohydrate chains attached to proteins) and* ***glycolipids*** *(carbohydrate chains attached to lipids).* ***Nucleotide sugars*** *are used to donate sugar residues for the formation of the glycosidic bonds in both glycoproteins and glycolipids. These carbohydrate groups have many different types of functions.*

Glycoproteins contain short chains of carbohydrates (oligosaccharides) that are usually branched. These oligosaccharides are generally composed of glucose, galactose, and their amino derivatives. In addition, mannose, L-fucose, and N-acetylneuraminic acid (NANA) are frequently present. The carbohydrate chains grow by the sequential addition of sugars to a ***serine*** *or* ***threonine*** *residue of the protein. Nucleotide-sugars are the precursors. Branched carbohydrate chains also may be attached to the amide nitrogen of* ***asparagine*** *in the protein. In this case, the chains are synthesized on* ***dolichol phosphate*** *and subsequently transferred to the protein. Glycoproteins are found in mucus, in the blood, in compartments within the cell (such as lysosomes), in the extracellular matrix, and embedded in the cell membrane with the carbohydrate portion extending into the extracellular space.*

Glycolipids belong to the class of ***sphingolipids****. They are synthesized from nuceotide-sugars that add monosaccharides sequentially to the hydroxymethyl group of the lipid ceramide (related to sphingosine). They often contain branches of N-acetylneuraminic acid produced from CMP-NANA. They are found in the cell membrane with the carbohydrate portion extruding from the cell surface. These carbohydrates, as well as some of the carbohydrates of glycoproteins, serve as* ***cell recognition factors****.*

THE WAITING ROOM

To help support herself through medical school, **Erna Nemdy** works evenings in a hospital blood bank. She is responsible for assuring that compatible donor blood is available to patients needing blood transfusions.

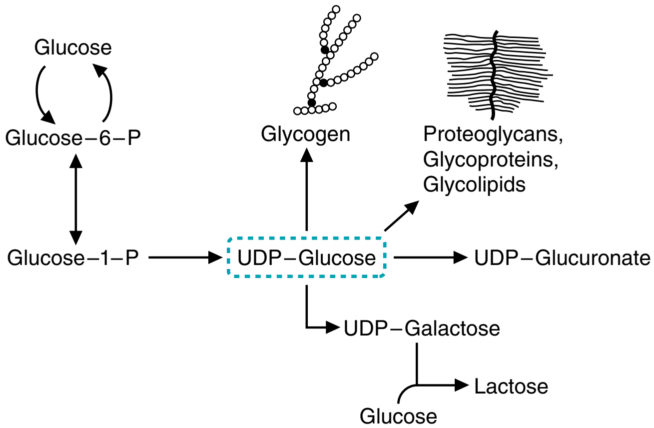


Fig. 30.1. Metabolism of UDP-glucose. The activated glucose moiety of UDP-glucose can be attached by a glycosidic bond to other sugars, as in glycogen or the sugar oligosaccharide and polysaccharide side chains of proteoglycans, glycoproteins, and glycolipids. UDP-glucose also can be oxidized to UDP-glucuronate, or epimerized to UDP-galactose, a precursor of lactose.

As part of her training, Erna has learned that the external surfaces of all blood cells contain large numbers of antigenic determinants. These determinants are often glycoproteins or glycolipids that differ from one individual to another. As a result, all blood transfusions expose the recipient to many foreign immunogens. Most of these, fortunately, do not induce antibodies, or they induce antibodies that elicit little or no immunologic response. For routine blood transfusions, therefore, tests are performed only for the presence of antigens that determine whether the patient's blood type is A, B, AB, or O.

Jay Sakz's psychomotor development has become progressively more abnormal. At 2 years of age, he is obviously mentally retarded and nearly blind. His muscle weakness has progressed to the point that he cannot sit up or even crawl. As the result of a weak cough reflex, he is unable to clear his normal respiratory secretions and has had recurrent respiratory infections.

I. INTERCONVERSIONS INVOLVING NUCLEOTIDE-SUGARS

Activated sugars attached to nucleotides are converted to other sugars, oxidized to sugar acids, and joined to proteins, lipids, or other sugars through glycosidic bonds.

A. Reactions of UDP-Glucose

UDP-glucose is an activated sugar nucleotide that is a precursor of glycogen and lactose, UDP-glucuronate and glucuronides, and the carbohydrate chains in proteoglycans, glycoproteins, and glycolipids (Fig. 30.1). Both proteoglycans and glycosaminoglycans are discussed further in Chapter 49. In the synthesis of many of the carbohydrate portions of these compounds, a sugar is transferred from the nucleotide sugar to an alcohol or other nucleophilic group to form a glycosidic bond (Fig. 30.2). The use of UDP as a leaving group in this reaction provides the energy for formation of the new bond. The enzymes that form glycosidic bonds are sugar transferases (for example, glycogen synthase is a glucosyltransferase). Transferases are also involved in the formation of the glycosidic bonds in bilirubin glucuronides, proteoglycans, and lactose.

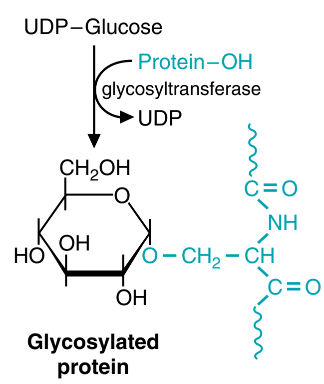


Fig. 30.2. Glycosyltransferases. These enzymes transfer sugars from nucleotide sugars to nucleophilic amino acid residues on proteins, such as the hydroxyl group of serine or the amide group of asparagine. Other transferases transfer specific sugars from a nucleotide sugar to a hydroxyl group of other sugars. The bond formed between the anomeric carbon of the sugar and the nucleophilic group of another compound is a glycosidic bond.

How does glucuronic acid differ from gluconic acid?

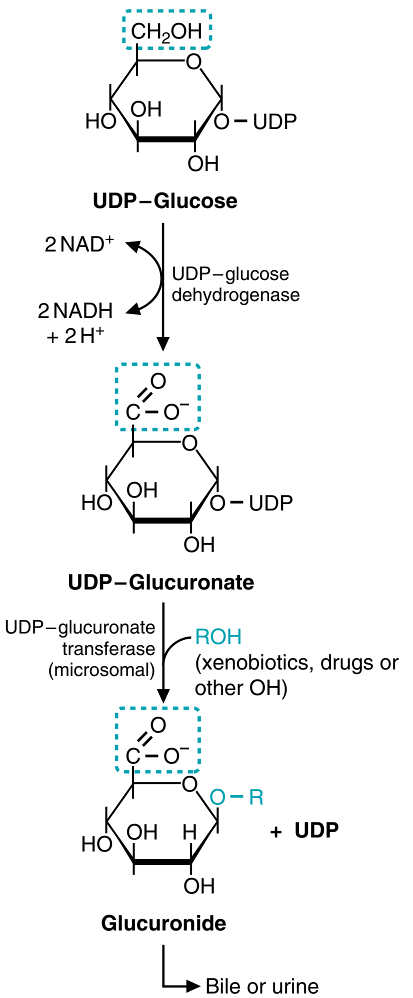


Fig. 30.4. Formation of glucuronate and glucuronides. A glycosidic bond is formed between the anomeric hydroxyl of glucuronate and the hydroxyl group of a nonpolar compound. The negatively charged carboxyl group of the glucuronate increases the water solubility and allows otherwise nonpolar compounds to be excreted in the urine or bile.

High concentrations of galactose 1-phosphate inhibit phosphoglucomutase, the enzyme that converts glucose 6-phosphate to glucose 1-phosphate. How can this inhibition account for the hypoglycemia and jaundice that accompany galactose 1-phosphate uridylyltransferase deficiency?

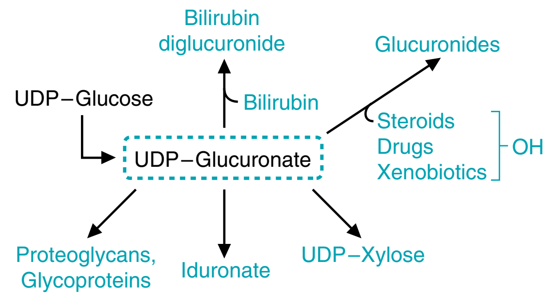


Fig. 30.3. Metabolic routes of UDP-glucuronate. UDP-glucuronate is formed from UDP-glucose (shown in black). Glucuronate from UDP-glucuronate is incorporated into glycosaminoglycans, where certain of the glucuronate residues are converted to iduronate (see Chapter 49). UDP-glucuronate is a precursor of UDP-xylose, another sugar residue incorporated into glycosaminoglycans. Glucuronate is also transferred to the carboxyl groups of bilirubin or the alcohol groups of steroids, drugs, and xenobiotics to form glucuronides. The "ide" in the name glucuronide denotes that these compounds are glycosides. Xenobiotics are pharmacologically, endocrinologically, or toxicologically active substances not endogenously produced and therefore foreign to an organism. Drugs are an example of a xenobiotic.

B. UDP-Glucuronate: A Source of Negative Charges

One of the major routes of UDP-glucose metabolism is the formation of UDP-glucuronate, which serves as a precursor of other sugars and of glucuronides (Fig. 30.3). Glucuronate is formed by the oxidation of the alcohol on C6 of glucose to an acid (through two oxidation states) by an NAD^+-dependent dehydrogenase (Fig. 30.4). Glucuronate is also present in the diet and can be formed from the degradation of inositol (the sugar alcohol that forms inositol trisphosphate (IP3), an intracellular second messenger for many hormones).

C. Formation of Glucuronides

The function of glucuronate in the excretion of bilirubin, drugs, xenobiotics, and other compounds containing a hydroxyl group is to add negative charges and increase their solubility. Bilirubin is a degradation product of heme that is formed in the reticuloendothelial system and is only slightly soluble in plasma. It is transported to the liver bound to albumin. In the liver, glucuronate residues are transferred from UDP-glucuronate to two carboxyl groups on bilirubin, sequentially forming bilirubin monoglucuronide and bilirubin diglucuronide, the "conjugated" forms of bilirubin (Fig. 30.5). The more soluble bilirubin diglucuronide (as compared with unconjugated bilirubin) is then actively transported into the bile for excretion.

Many xenobiotics, drugs, steroids, and other compounds with hydroxyl groups and a low solubility in water are converted to glucuronides in a similar fashion by glucuronyltransferases present in the endoplasmic reticulum and cytoplasm of the liver and kidney (Table 30.1). This is one of the major conjugation pathways for excretion of these compounds.

A failure of the liver to transport, store, or conjugate bilirubin results in the accumulation of unconjugated bilirubin in the blood. Jaundice, the yellowish tinge to the skin and the whites of the eyes (sclera) experienced by **Erin Galway**, occurs when plasma becomes supersaturated with bilirubin (>2–2.5 mg/dL), and the excess diffuses into tissues.

When bilirubin levels are measured in the blood, one can measure either indirect bilirubin (this is the nonconjugated form of bilirubin, which is bound to albumin), direct bilirubin (the conjugated, water-soluble form), or total bilirubin (the sum of the direct and indirect levels). If total bilirubin levels are high, then a determination of direct and indirect bilirubin is needed to appropriately determine a cause for the elevation of total bilirubin.

Glucuronate, once formed, can reenter the pathways of glucose metabolism through reactions that eventually convert it to D-xylulose 5-phosphate, an intermediate of the pentose phosphate pathway. In most mammals other than humans, an intermediate of this pathway is the precursor of ascorbic acid (vitamin C). Humans, however, are deficient in this pathway and cannot synthesize vitamin C.

D. Synthesis of UDP-Galactose and Lactose from Glucose

Lactose is synthesized from UDP-galactose and glucose (Fig. 30.6). However, galactose is not required in the diet for lactose synthesis because galactose can be synthesized from glucose.

1. CONVERSION OF GLUCOSE TO GALACTOSE

Galactose and glucose are epimers; they differ only in the stereochemical position of one hydroxyl group. Thus, the formation of UDP-galactose from UDP-glucose is an epimerization (Fig. 30.7). The epimerase does not actually transfer the hydroxyl group; it oxidizes the hydroxyl to a ketone by transferring electrons to NAD^+, and then donates electrons back to re-form the alcohol group on the other side of the carbon.

2. LACTOSE SYNTHESIS

Lactose is unique in that it is synthesized only in the mammary gland of the adult for short periods during lactation. Lactose synthase, an enzyme present in the endoplasmic reticulum of the lactating mammary gland, catalyzes the last step in lactose biosynthesis, the transfer of galactose from UDP-galactose to glucose (see Fig. 30.6). Lactose synthase has two protein subunits, a galactosyltransferase and α-lactalbumin. α-Lactalbumin is a modifier protein synthesized after parturition (childbirth) in response to the hormone prolactin. This enzyme subunit lowers the K_m of the galactosyltransferase for glucose from 1,200 to 1 mM, thereby increasing the rate of lactose synthesis. In the absence of α-lactalbumin, galactosyltransferase transfers galactosyl units to glycoproteins.

Many (60%) full-term newborns develop jaundice, termed neonatal jaundice. This is usually caused by an increased destruction of red blood cells after birth (the fetus has an unusually large number of red blood cells) and an immature bilirubin conjugating system in the liver. This leads to elevated levels of nonconjugated bilirubin, which is deposited in hydrophobic (fat) environments. If bilirubin levels reach a certain threshold at the age of 48 hours, the newborn is a candidate for phototherapy, in which the child is placed under lamps that emit light between the wavelengths of 425 and 475 nm. Bilirubin absorbs this light, undergoes chemical changes, and becomes more water soluble. Usually, within a week of birth, the newborn's liver can handle the load generated from red blood cell turnover.

A: The inhibition of phosphoglucomutase results in hypoglycemia by interfering with both the formation of UDP-glucose (the glycogen precursor) and the degradation of glycogen back to glucose 6-phosphate. Ninety percent of glycogen degradation leads to glucose 1-phosphate, which can only be converted to glucose 6-phosphate by phosphoglucomutase. When phosphoglucomutase activity is inhibited, less glucose-6-P production occurs, and hence, less glucose is available for export. Thus, the stored glycogen is only approximately 10% efficient in raising blood glucose levels, and hypoglycemia results. UDP-glucose levels are reduced because glucose-1-P is required to synthesize UDP-glucose, and in the absence of phosphoglucomutase activity glucose-6-P cannot be converted to glucose-1-P. This prevents the formation of UDP-glucuronate, which is necessary to convert bilirubin to the diglucuronide form for transport into the bile. Bilirubin accumulates in tissues, giving them a yellow color (jaundice).

Table 30.1. Some Compounds Degraded and Excreted as Urinary Glucuronides

Estrogen (female sex hormone)
Progesterone (steroid hormone)
Triiodothyronine (thyroid hormone)
Acetylaminofluorene (xenobiotic carcinogen)
Meprobamate (drug for sleep)
Morphine (painkiller)

A: 6-Phosphogluconate is produced by the first oxidative reaction in the pentose phosphate pathway, in which carbon 1 of glucose is oxidized to a carboxylate. In contrast, glucuronic acid is oxidized at carbon 6 to the carboxylate form.

Q: A pregnant woman who was extremely lactose intolerant asked her physician if she would still be able to breastfeed her infant since she could not drink milk or dairy products. What advice should she be given?

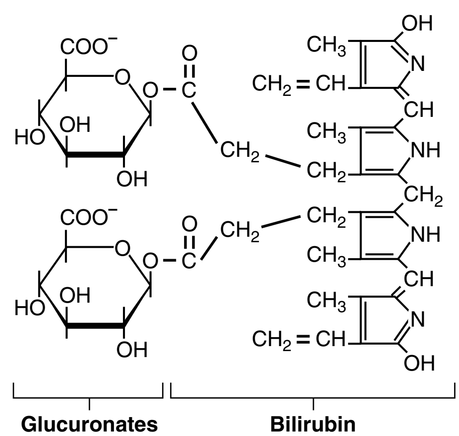


Fig. 30.5. Formation of bilirubin diglucuronide. A glycosidic bond is formed between the anomeric hydroxyl of glucuronate and the carboxylate groups of bilirubin. The addition of the hydrophilic carbohydrate group, and the negatively charged carboxyl group of the glucuronate, increases the water solubility of the conjugated bilirubin and allows the otherwise insoluble bilirubin to be excreted in the urine or bile.

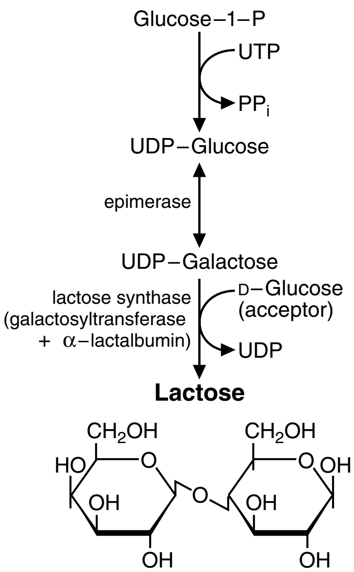


Fig. 30.6. Lactose synthesis. Lactose is a disaccharide composed of galactose and glucose. UDP-galactose for the synthesis of lactose in the mammary gland is usually formed from the epimerization of UDP-glucose. Lactose synthase attaches the anomeric carbon of the galactose to the C4 alcohol group of glucose to form a glycosidic bond. Lactose synthase is composed of a galactosyltransferase and α-lactalbumin, which is a regulatory subunit.

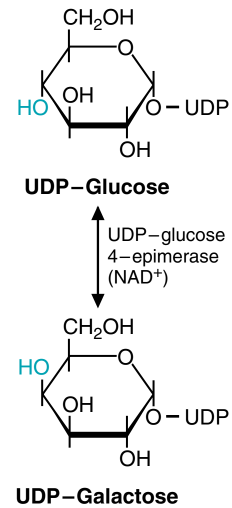


Fig. 30.7. Epimerization of UDP-glucose to UDP-galactose. The epimerization of glucose to galactose occurs on UDP-sugars. The epimerase uses NAD$^+$ to oxidize the alcohol to a ketone, and then reduce the ketone back to an alcohol. The reaction is reversible; glucose being converted to galactose forms galactose for lactose synthesis, and galactose being converted to glucose is part of the pathway for the metabolism of dietary galactose.

Table 30.2. Some Sugar Nucleotides That Are Precursors for Transferase Reactions

UDP-glucose
UDP-galactose
UDP-glucuronic acid
UDP-xylose
UDP-N-acetylglucosamine
UDP-N-acetylgalactosamine
CMP-N-acetylneuraminic acid
GDP-fucose
GDP-mannose

A: Although the lactose in dairy products is a major source of galactose, the ingestion of lactose is not required for lactation. UDP-galactose in the mammary gland is derived principally from the epimerization of glucose. Dairy products are, however, a major dietary source of Ca^{2+}, so breastfeeding mothers need increased Ca^{2+} from another source.

E. Formation of Sugars for Glycolipid and Glycoprotein Synthesis

The transferases that produce the oligosaccharide and polysaccharide side chains of glycolipids and attach sugar residues to proteins are specific for the sugar moiety and for the donating nucleotide (e.g., UDP, CMP, or GDP). Some of the sugar-nucleotides used for glycoprotein, proteoglycan (see Chapter 49) and glycolipid formation are listed in Table 30.2. They include the derivatives of glucose and galactose that we have already discussed, as well as acetylated amino sugars and derivatives of mannose. The reason for the large variety of sugars attached to proteins and lipids is that they have relatively specific and different functions, such as targeting a protein toward a membrane, providing recognition sites on the cell surface for other cells, hormones, or viruses, or acting as lubricants or molecular sieves (see Chapter 42).

The pathways for utilization and formation of many of these sugars are summarized in Figure 30.8. Note that many of the steps are reversible, so that glucose and other dietary sugars enter a common pool from which the diverse sugars can be formed.

The amino sugars are all derived from glucosamine 6-phosphate. To synthesize glucosamine 6-phosphate, an amino group is transferred from the amide of glutamine to fructose 6-phosphate (Fig. 30.9). Amino sugars, such as glucosamine, can then be N-acetylated by an acetyltransferase.

Mannose is found in the diet in small amounts. Like galactose, it is an epimer of glucose, and mannose and glucose are interconverted by epimerization reactions. The interconversion can take place either at the level of fructose 6-phosphate to mannose 6-phosphate, or at the level of the derivatized sugars (see Fig. 30.8).

N-Acetyltransferases are present in the endoplasmic reticulum and cytosol and provide another means of chemically modifying sugars, metabolites, drugs, and xenobiotic compounds. Individuals may vary greatly in their capacity for acetylation reactions.

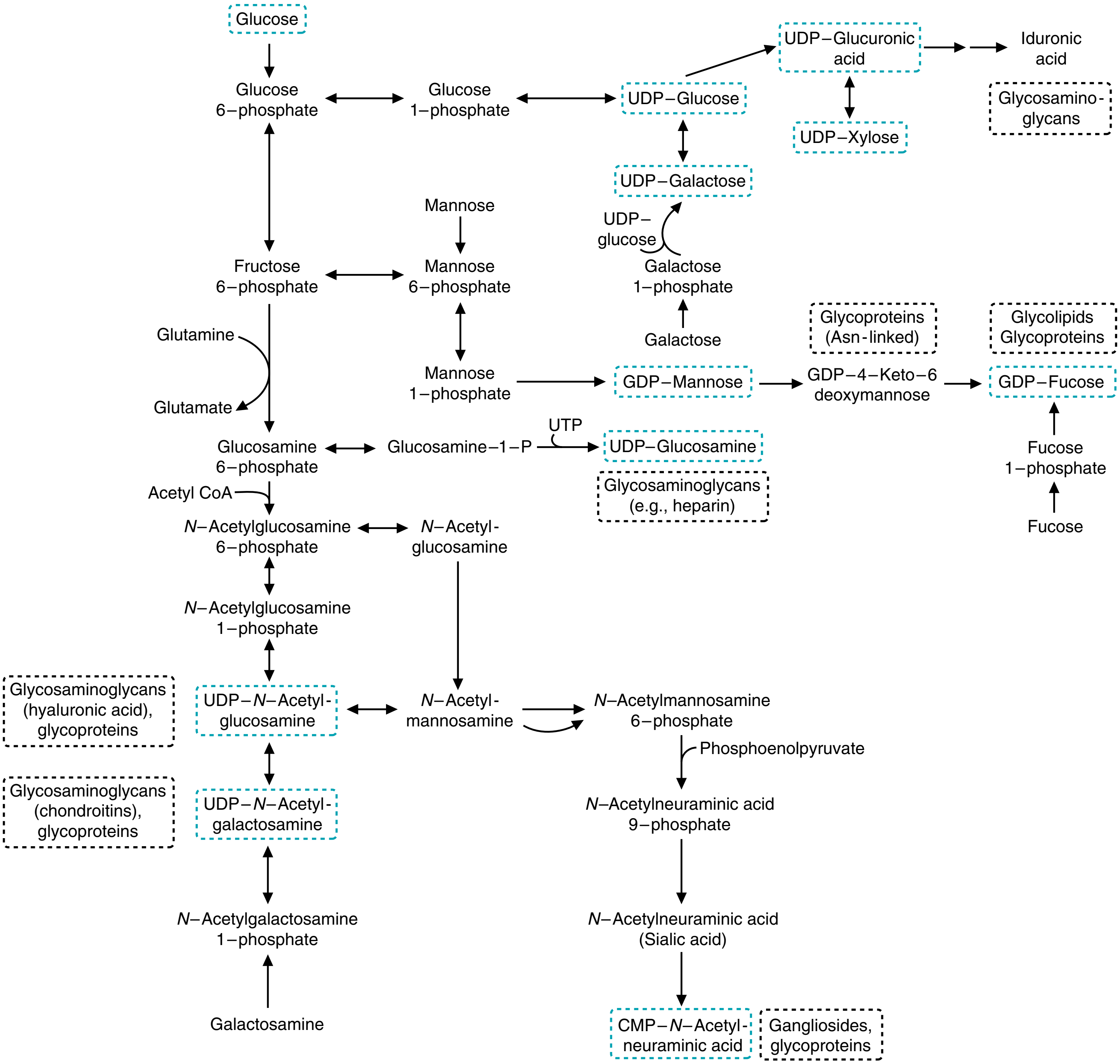


Fig. 30.8. Pathways for the interconversion of sugars. All of the different sugars found in glycosaminoglycans, gangliosides, and other compounds in the body can be synthesized from glucose. Dietary glucose, fructose, galactose, mannose, and other sugars enter a common pool from which other sugars are derived. The activated sugar is transferred from the nucleotide sugar, shown in blue boxes, to form a glycosidic bond with another sugar or amino acid residue. The box next to each nucleotide sugar lists some of the compounds that contain the sugar. Iduronic acid, in the upper right corner of the diagram, is formed only after glucuronic acid is incorporated into a glycosaminoglycan (which is discussed in more detail in Chapter 49).

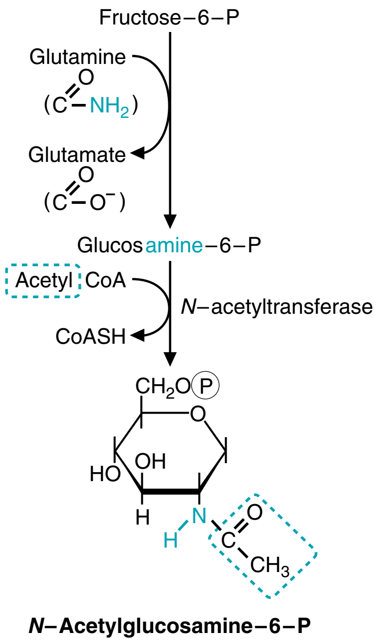


Fig. 30.9. The formation of *N*-acetylglucosamine 6-phosphate. The amino sugar is formed by a transfer of the amino group from the amide of glutamine to a carbon of the sugar. The amino group is acetylated by the transfer of an acetyl group from acetyl CoA.

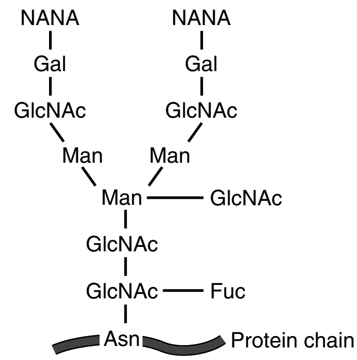


Fig. 30.10. An example of a branched glycoprotein. NANA = *N*-acetylneuraminic acid; Gal= galactose; GlcNAc = *N*-acetylglucosamine; Man = mannose; Fuc = fucose.

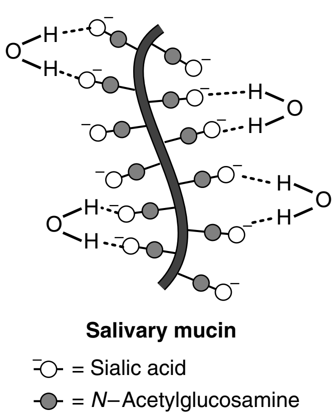


Fig. 30.11. Structure of salivary mucin. The sugars form hydrogen bonds with water. Sialic acid provides a negatively charged carboxylate group. The protein is extremely large, and the negatively charged sialic acids extend the carbohydrate chains so the molecules occupy a large space. All of the salivary glycoproteins are *O*-linked. NANA is a sialic acid.

N-Acetylmannosamine is the precursor of N-acetylneuraminic acid (NANA, a sialic acid) and GDP-mannose is the precursor of GDP-fucose (see Fig. 30.8). The negative charge on NANA is obtained by the addition of a 3-carbon carboxyl moiety from phosphoenolpyruvate.

II. GLYCOPROTEINS

A. Structure and Function

Glycoproteins contain short carbohydrate chains covalently linked to either serine/threonine or asparagine residues in the protein. These oligosaccharide chains are often branched, and they do not contain repeating disaccharides (Fig. 30.10). Most proteins in the blood are glycoproteins. They serve as hormones, antibodies, enzymes (including those of the blood clotting cascade), and as structural components of the extracellular matrix. Collagen contains galactosyl units and disaccharides composed of galactosyl-glucose attached to hydroxylysine residues (see Chapter 49). The secretions of mucus-producing cells, such as salivary mucin, are glycoproteins (Fig. 30.11).

Although most glycoproteins are secreted from cells, some are segregated in lysosomes, where they serve as the lysosomal enzymes that degrade various types of cellular and extracellular material. Other glycoproteins are produced like secretory proteins, but hydrophobic regions of the protein remain attached to the cell membrane, and the carbohydrate portion extends into the extracellular space (Fig. 30.12)(also see Chapter 15, section I). These glycoproteins serve as receptors for compounds such as hormones, as transport proteins, and as cell attachment and cell–cell recognition sites. Bacteria and viruses also bind to these sites.

B. Synthesis

The protein portion of glycoproteins is synthesized on the endoplasmic reticulum (ER). The carbohydrate chains are attached to the protein in the lumen of the ER and the Golgi complex. In some cases, the initial sugar is added to a serine or a threonine residue in the protein, and the carbohydrate chain is extended by the sequential addition of sugar residues to the nonreducing end. As seen previously in Table 2, UDP-sugars are the precursors for the addition of four of the seven sugars that are usually found in glycoproteins—glucose, galactose, *N*-acetylglucosamine, and *N*-acetylgalactosamine. GDP-sugars are the precursors for the addition of mannose and L-fucose, and CMP-NANA is the precursor for NANA.

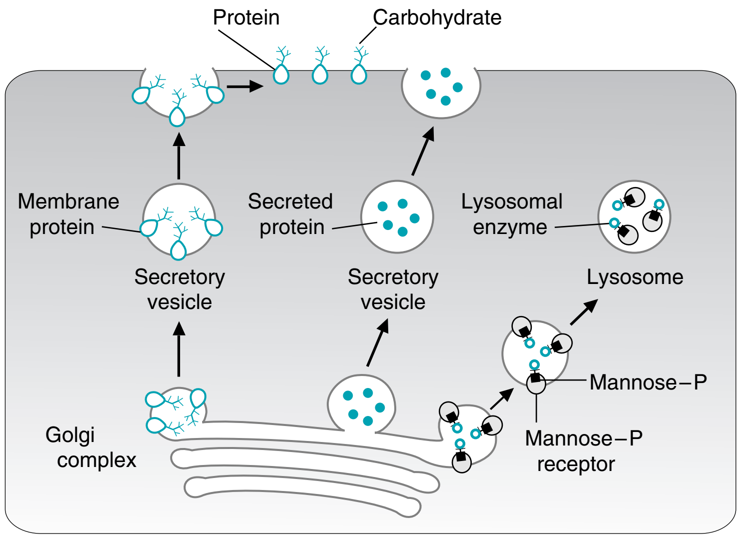


Fig. 30.12. Route from the Golgi complex to the final destination for lysosomal enzymes, cell membrane proteins, and secreted proteins, which include glycoproteins and proteoglycans.

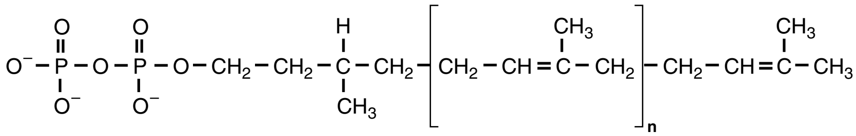


Fig. 30.13. Structure of dolichol phosphate. In humans, the isoprene unit (in brackets) is repeated approximately 17 times (n = ~17).

Dolichol phosphate (Fig. 30.13) (which is synthesized from isoprene units, as discussed in Chapter 34) is involved in transferring branched sugar chains to the amide nitrogen of asparagine residues. Sugars are removed and added as the glycoprotein moves from the ER through the Golgi complex (Fig. 30.14). As discussed in Chapter 10, the carbohydrate chain is used as a targeting marker for lysosomal enzymes.

I-cell (inclusion cell) disease is a rare condition in which lysosomal enzymes lack the mannosephosphate marker that targets them to lysosomes. Consequently, lysosomal enzymes are secreted from the cells. Because lysosomes lack their normal complement of enzymes, undegraded molecules accumulate within membranes inside these cells, forming inclusion bodies.

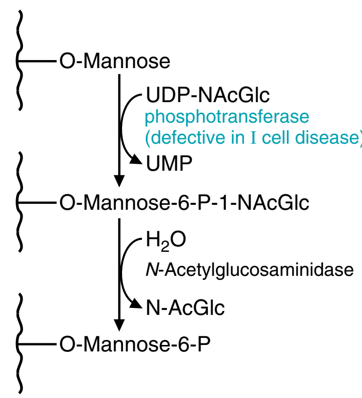
III. GLYCOLIPIDS

A. Function and Structure

Glycolipids are derivatives of the lipid sphingosine. These sphingolipids include the cerebrosides and the gangliosides (Fig. 30.15; see also Fig. 5.22). They contain ceramide, with carbohydrate moieties attached to its hydroxymethyl group.

Glycolipids are involved in intercellular communication. Oligosaccharides of identical composition are present in both the glycolipids and glycoproteins associated with the cell membrane, where they serve as cell recognition factors. For example, carbohydrate residues in these oligosaccharides are the antigens of the ABO blood group substances (Fig. 30.16).

The enzyme that is deficient in I-cell disease is a phosphotransferase located in the Golgi apparatus.



The phosphotransferase has the unique ability to recognize lysosomal proteins because of their three-dimensional structure, such that they can all be appropriately tagged for transport to the lysosomes.

By identifying the nature of antigenic determinants on the surface of the donor's red blood cells, **Erna Nemdy** is able to classify the donor's blood as belonging to certain specific blood groups. These antigenic determinants are located in the oligosaccharides of the glycoproteins and glycolipids of the cell membranes. The most important blood group in humans is the ABO group, which comprises two antigens, A and B. Individuals with the A antigen on their cells belong to blood group A. Those with B belong to group B, and those with both A and B belong to group AB. The absence of both the A and the B antigen results in blood type O (see Fig. 30.16).

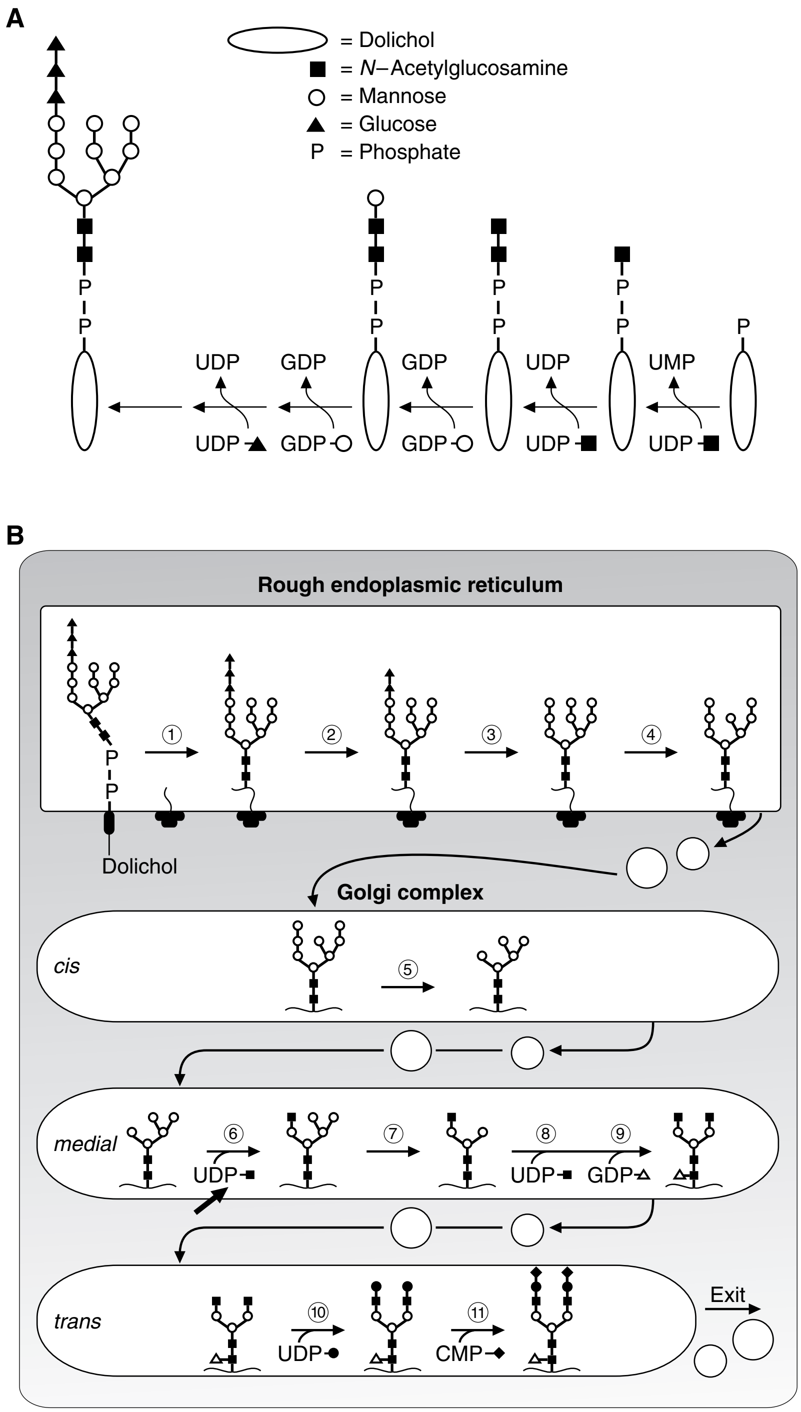


Fig. 30.14. Action of dolichol phosphate in transferring carbohydrate groups to proteins (A) and processing of these carbohydrate groups (B). Transfer of the branched oligosaccharide from dolichol phosphate to a protein in the lumen of the rough endoplasmic reticulum (RER) (step 1) and processing of the oligosaccharide (steps 2–11). Steps 1 through 4 occur in the RER. The glycoprotein is transferred in vesicles to the Golgi complex, where further modifications of the oligosaccharides occur (steps 5–11). B modified with permission from Kornfeld R, Kornfeld S. Annu Rev Biochem 1985;54:640. © 1985 by Annual Reviews, Inc. P = phosphate.

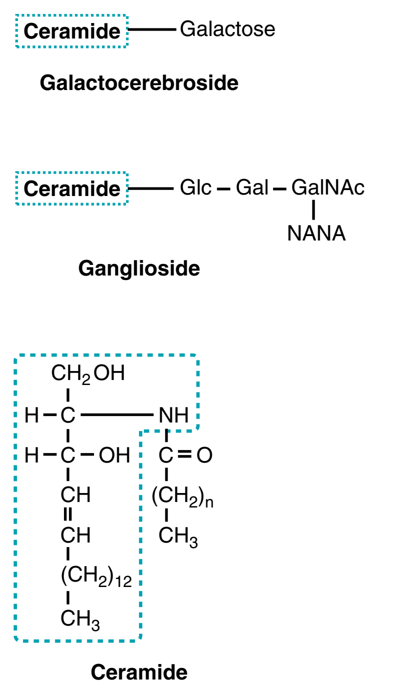


Fig. 30.15. Structures of cerebrosides and gangliosides. In these glycolipids, sugars are attached to ceramide (shown below the glycolipids). The boxed portion of ceramide is sphingosine, from which the name *sphingolipids* is derived.